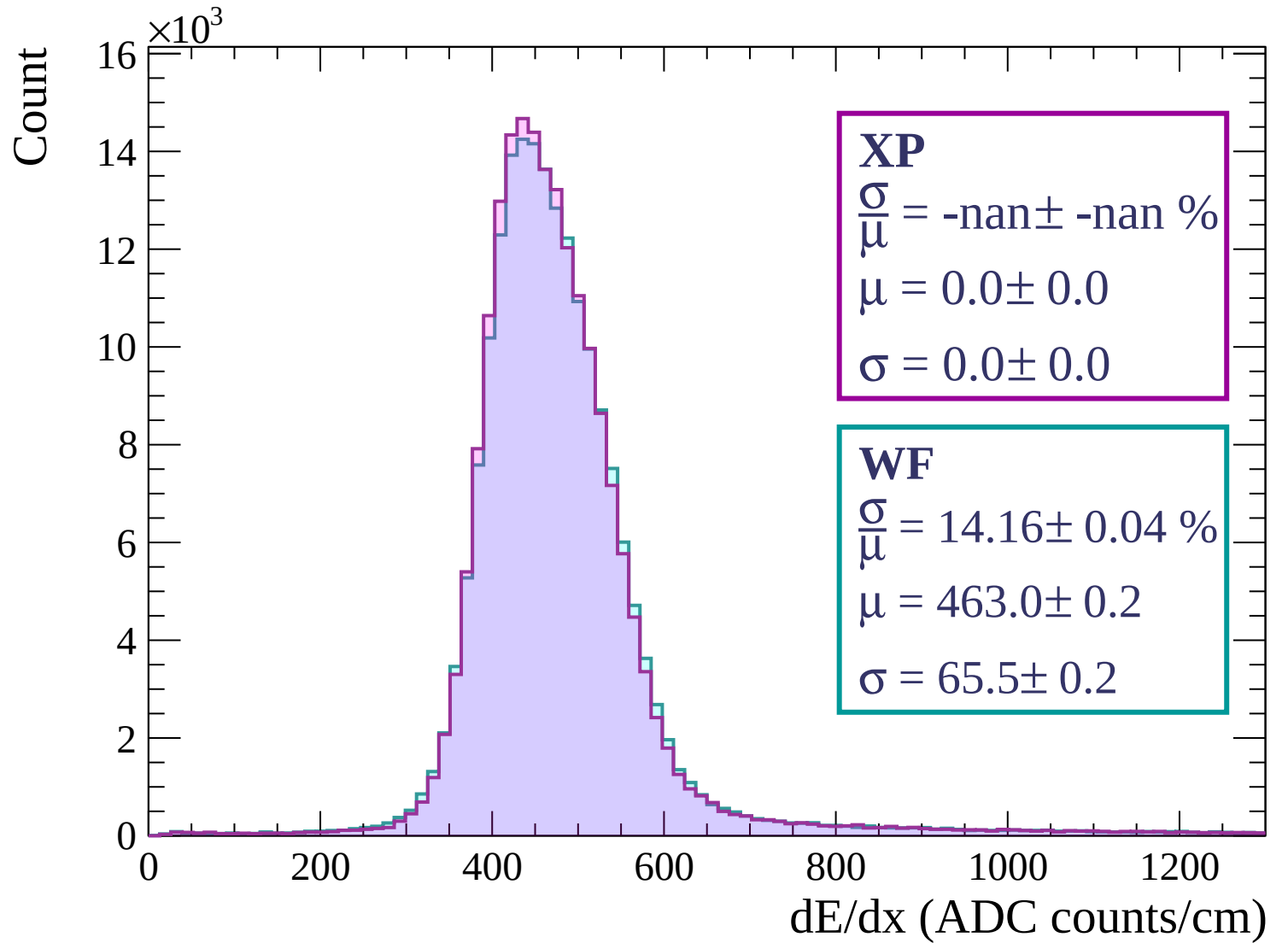
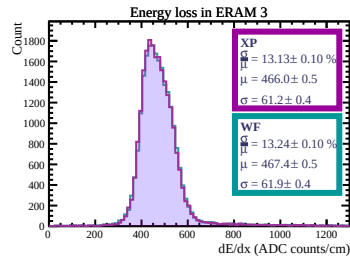
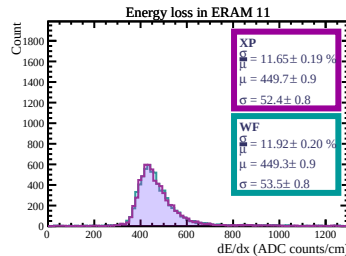
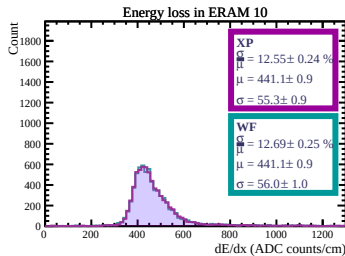
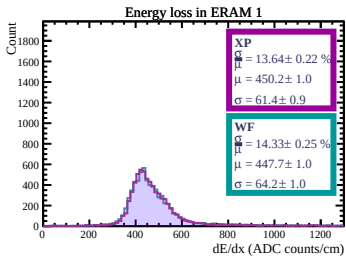
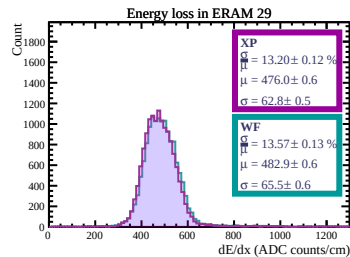
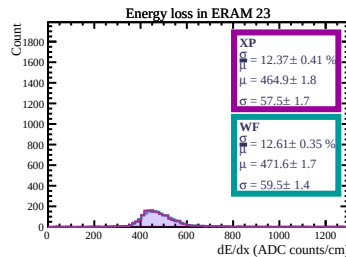
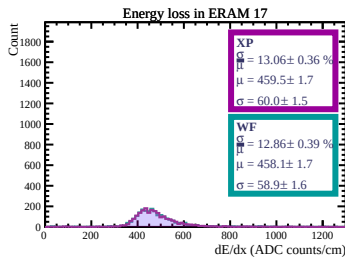
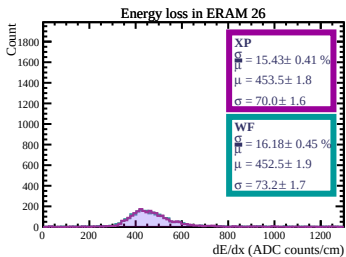
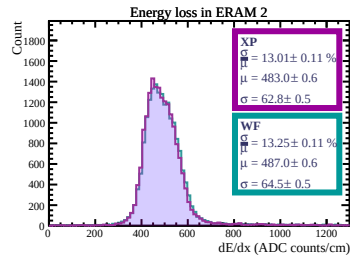
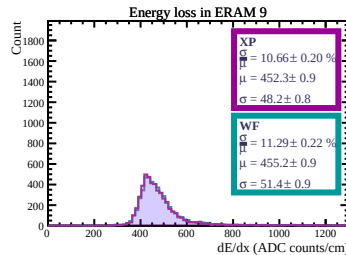
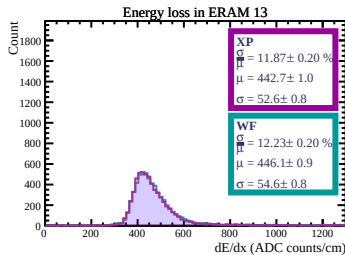
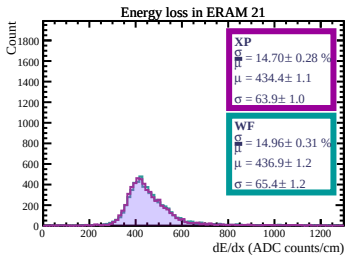
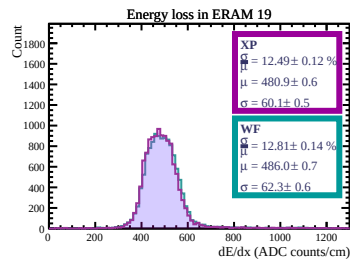
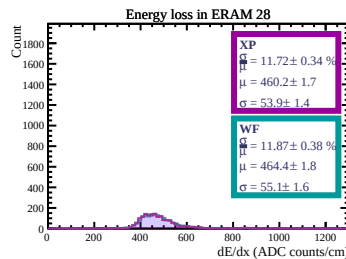
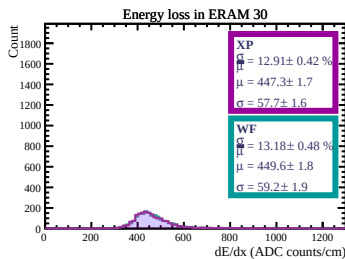
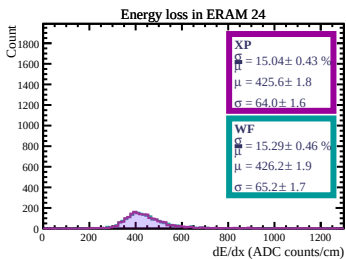
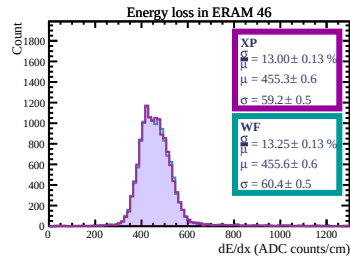
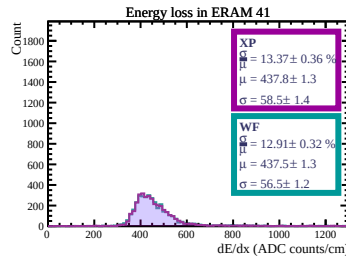
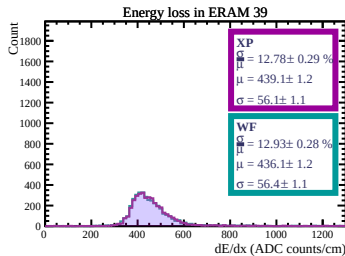
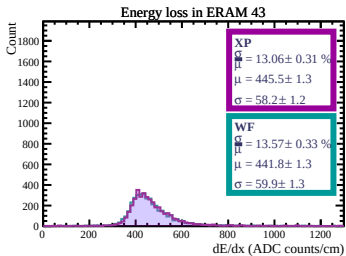
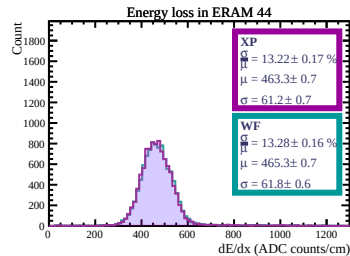
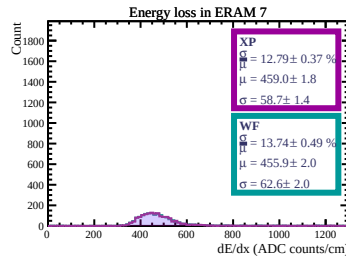
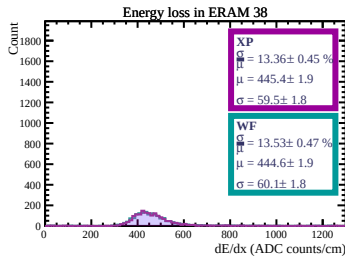
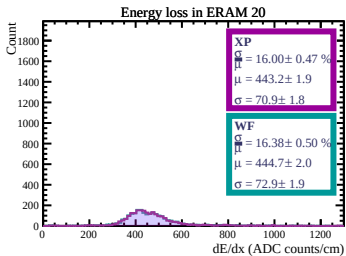
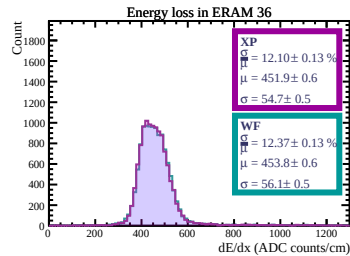
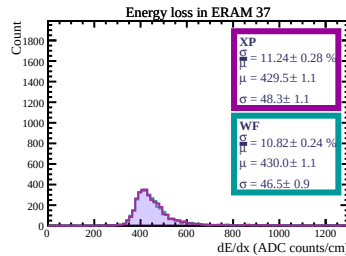
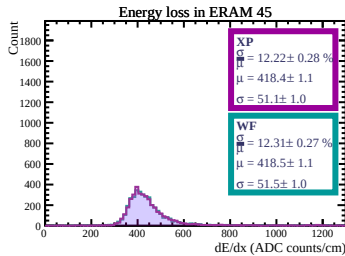
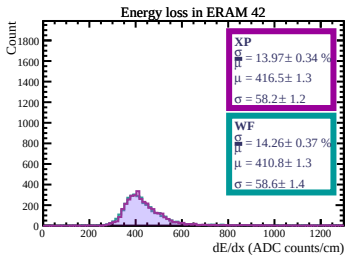
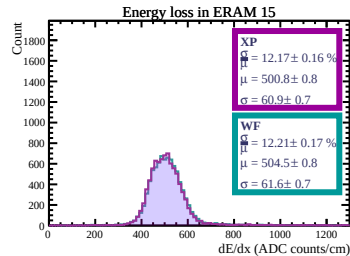
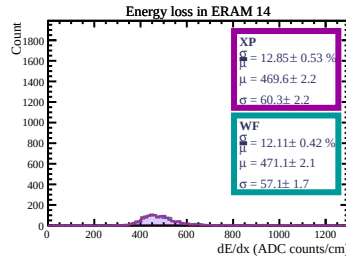
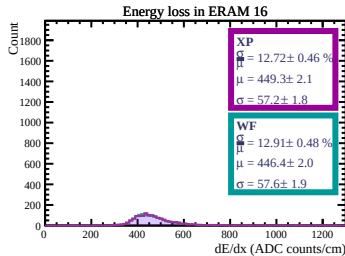
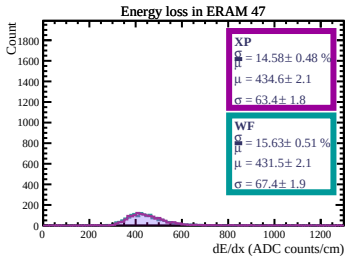
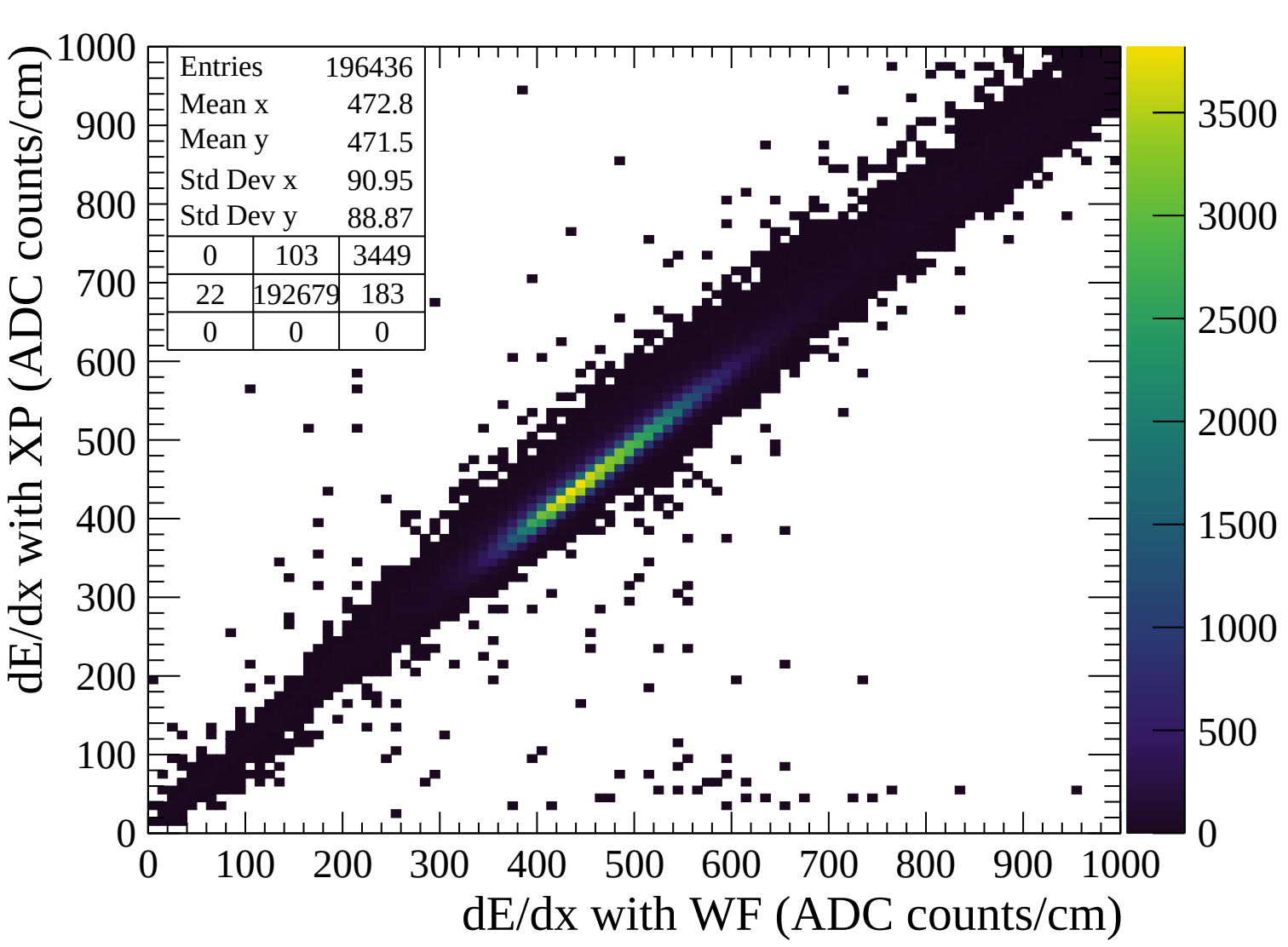
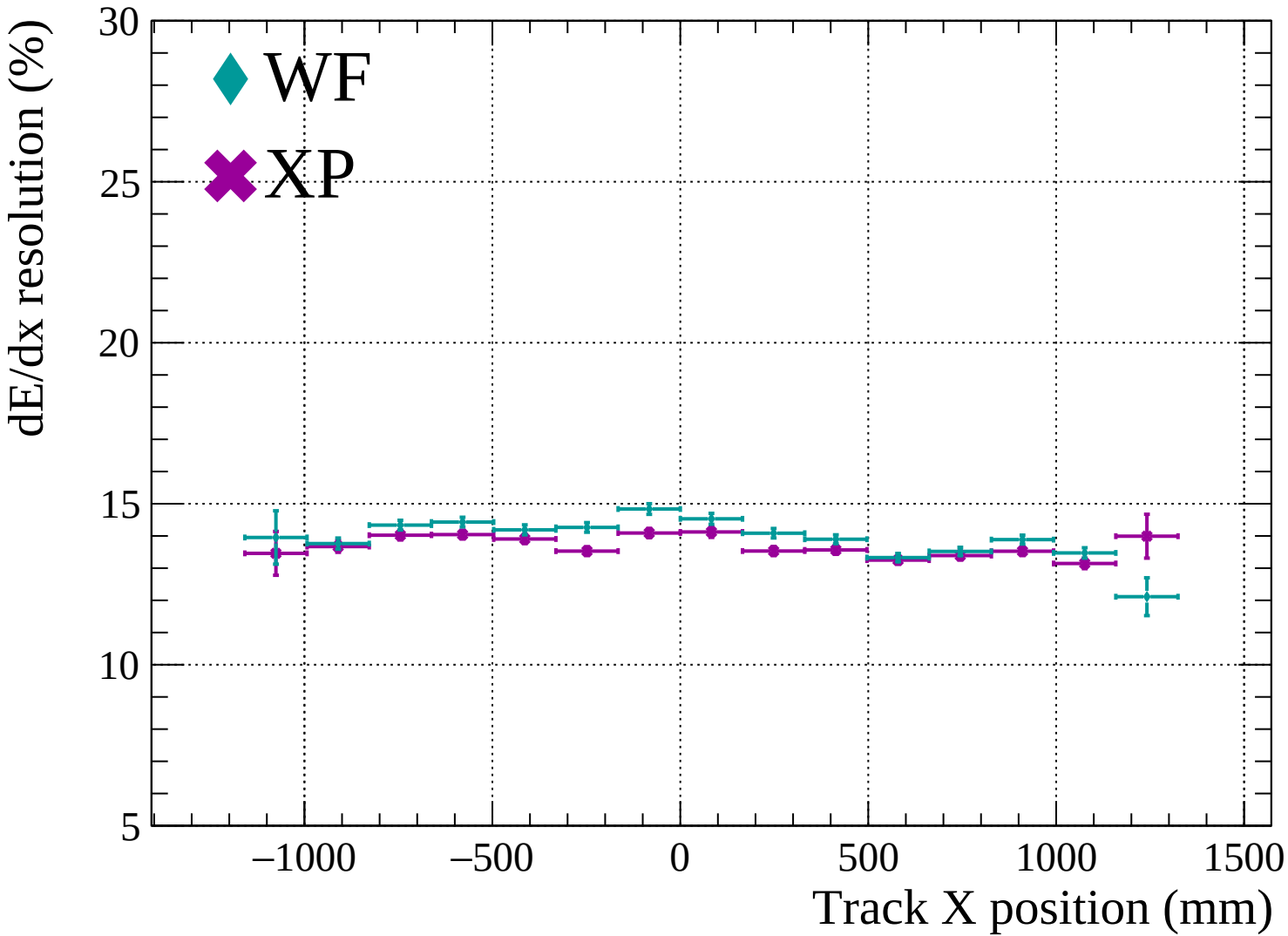


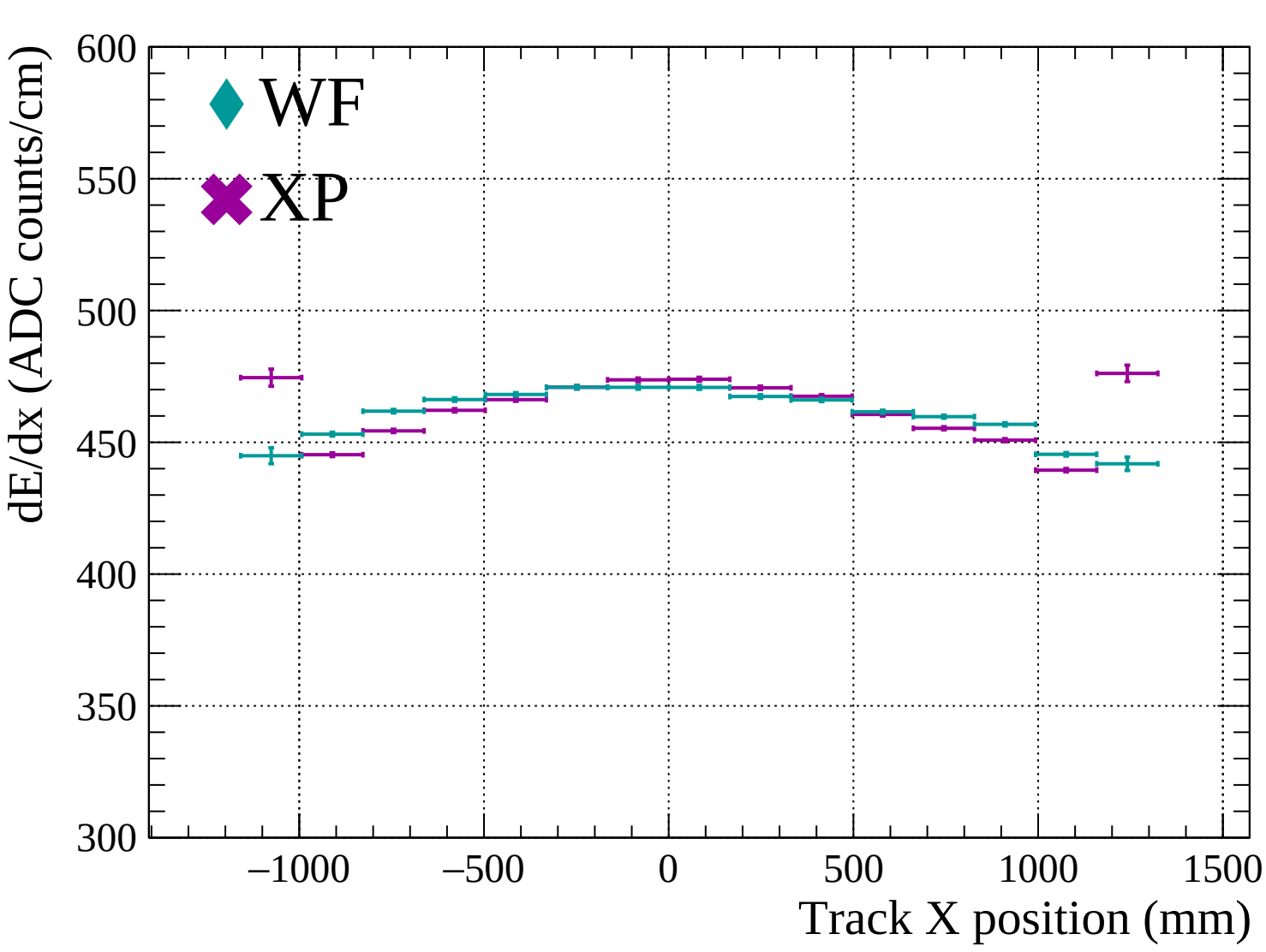


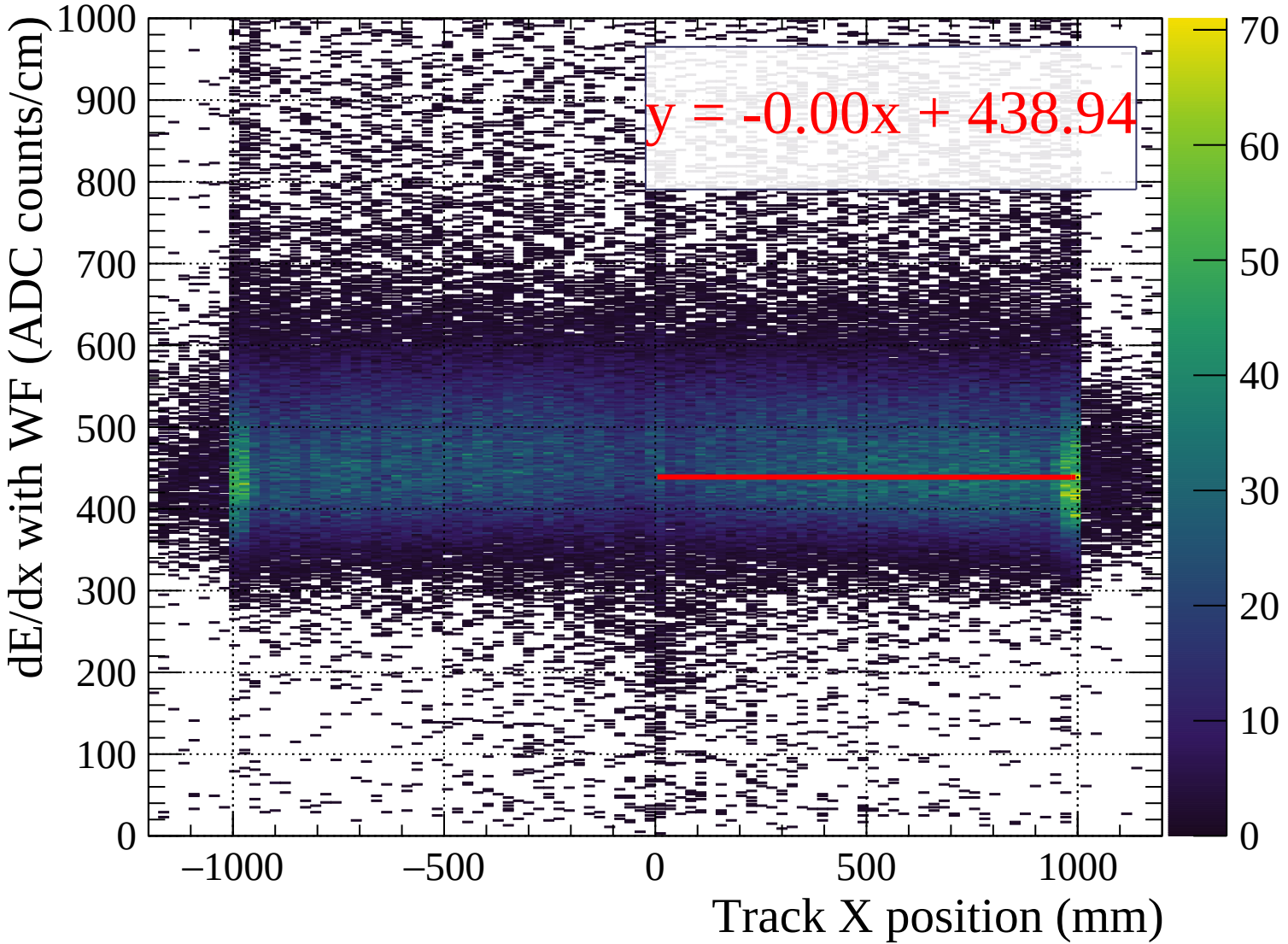


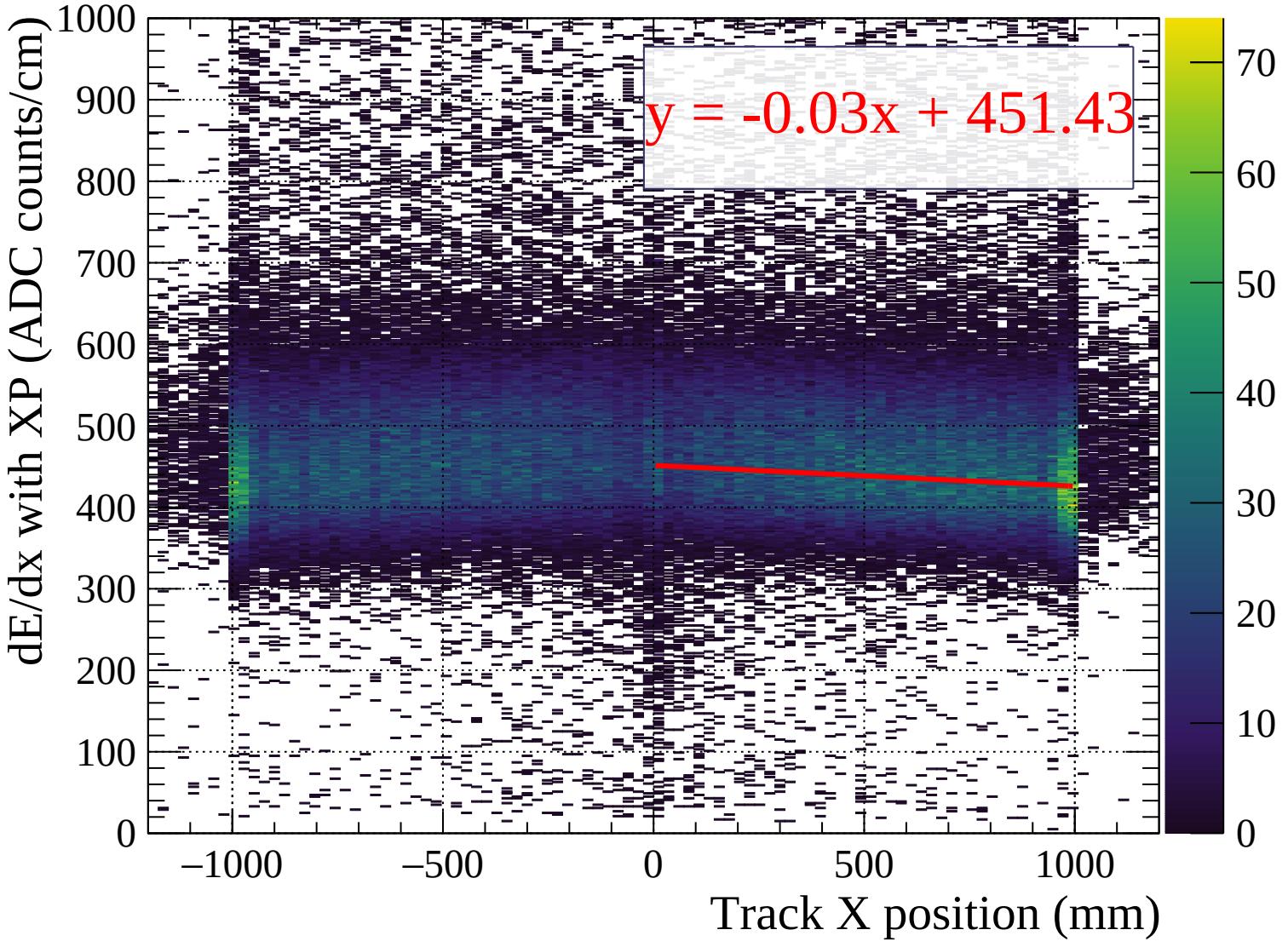


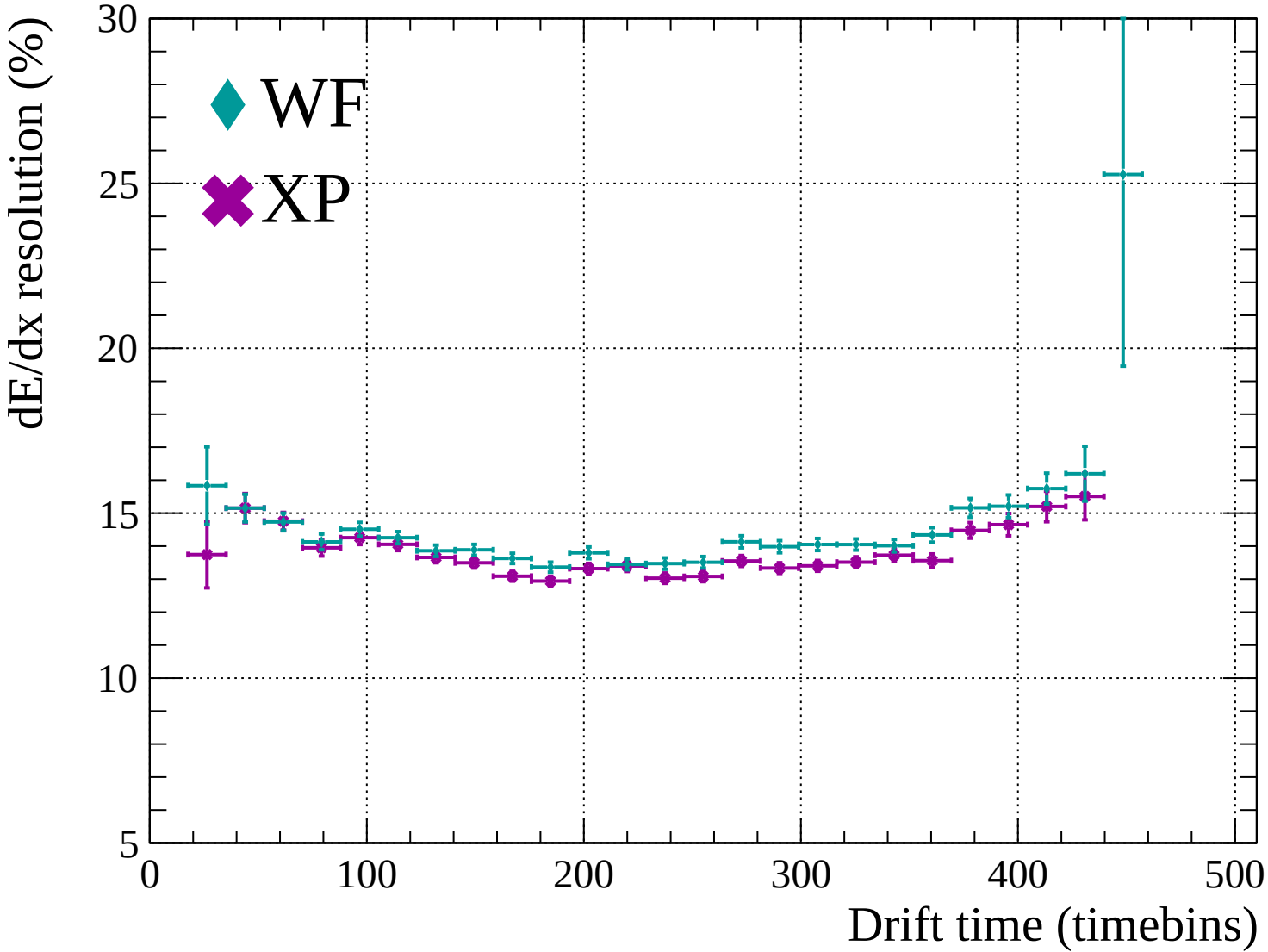


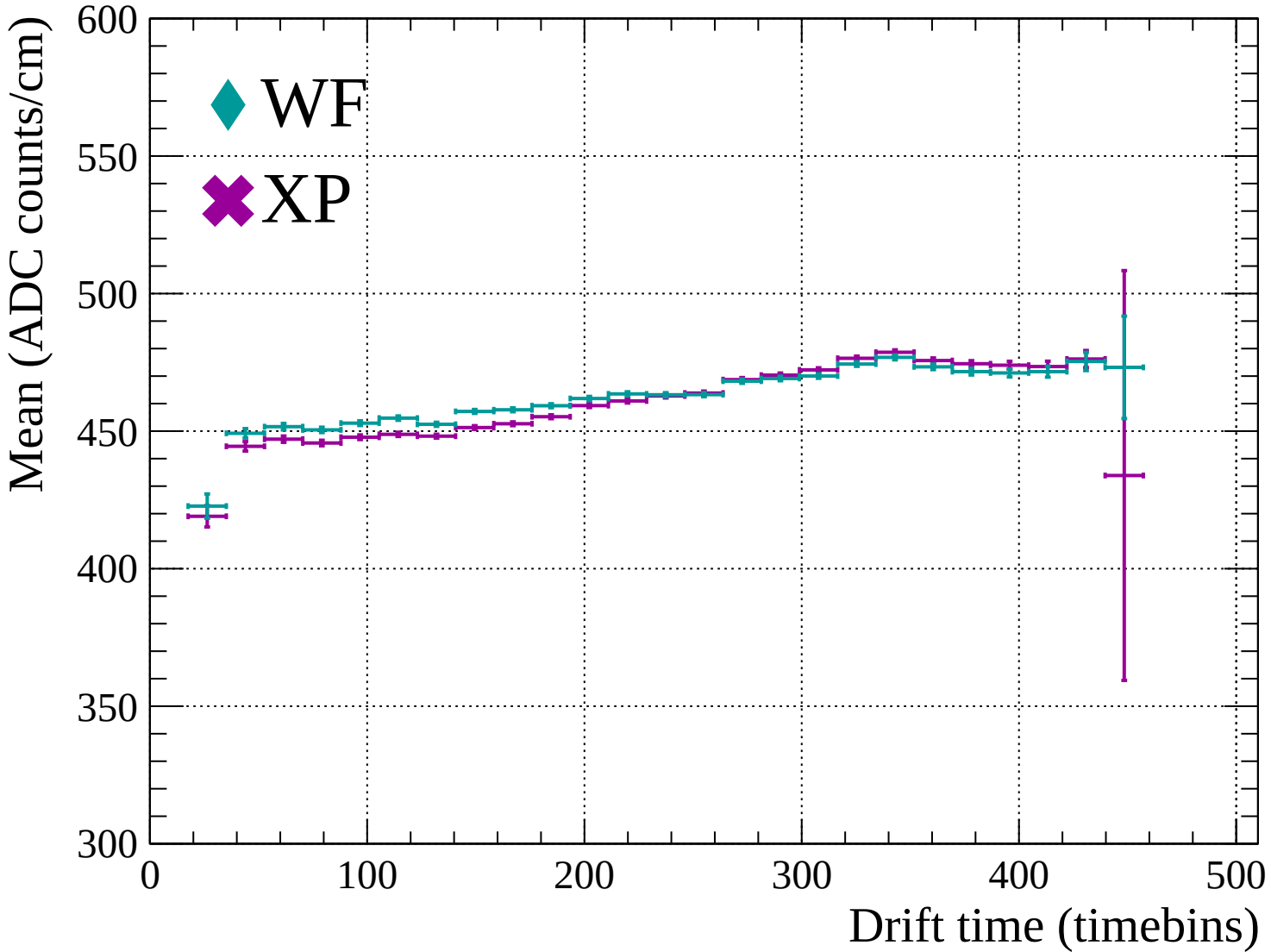


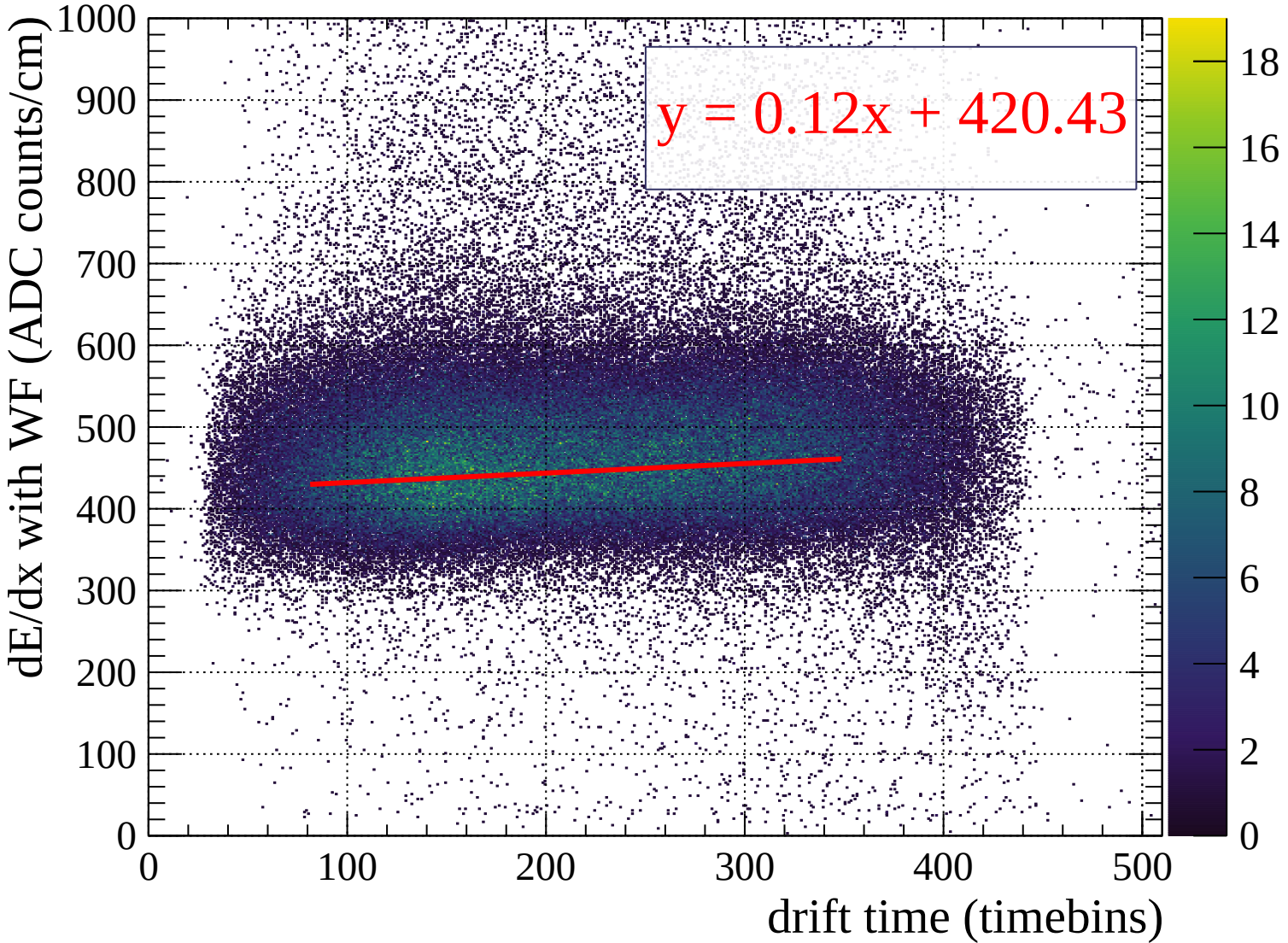


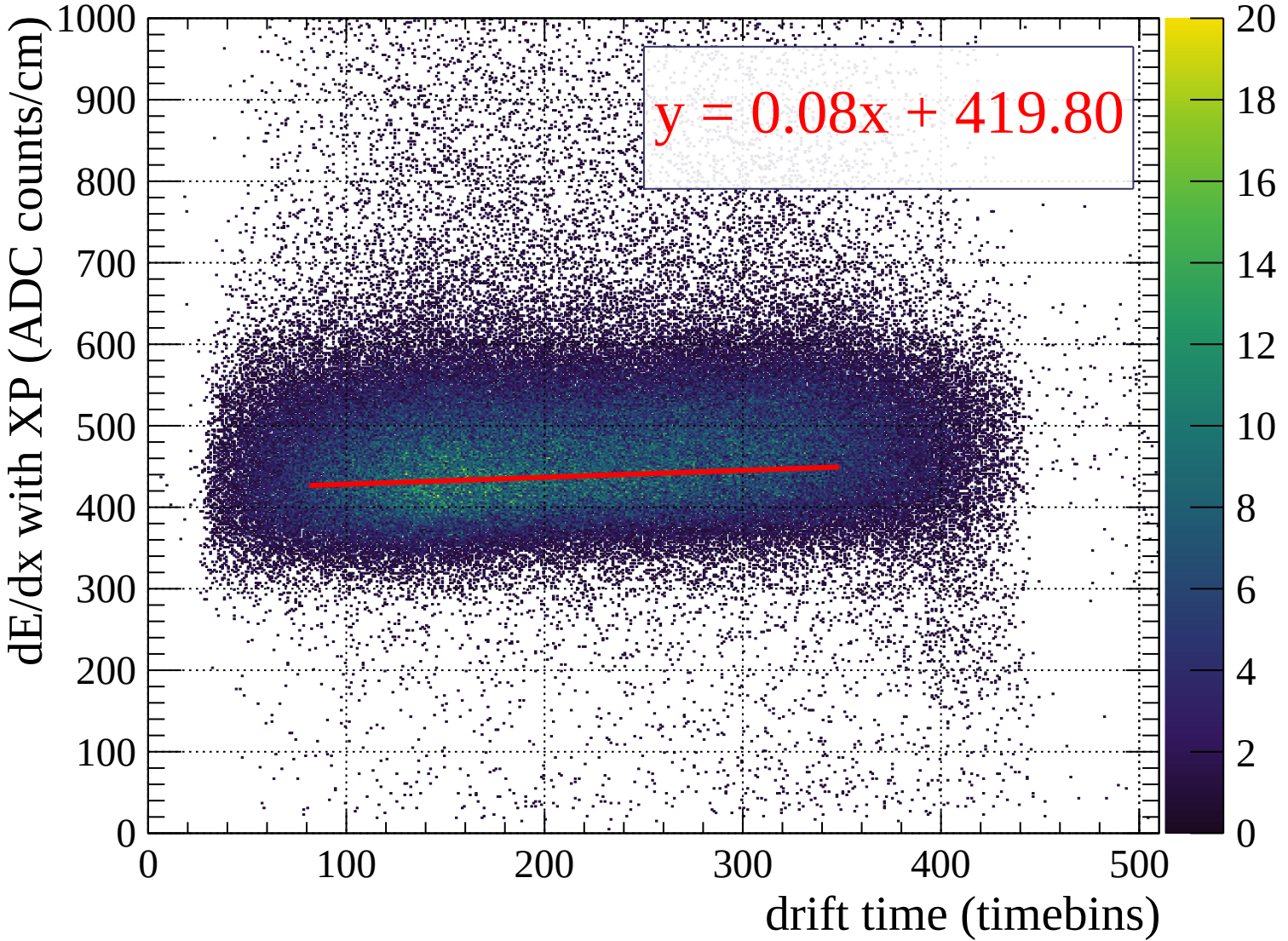


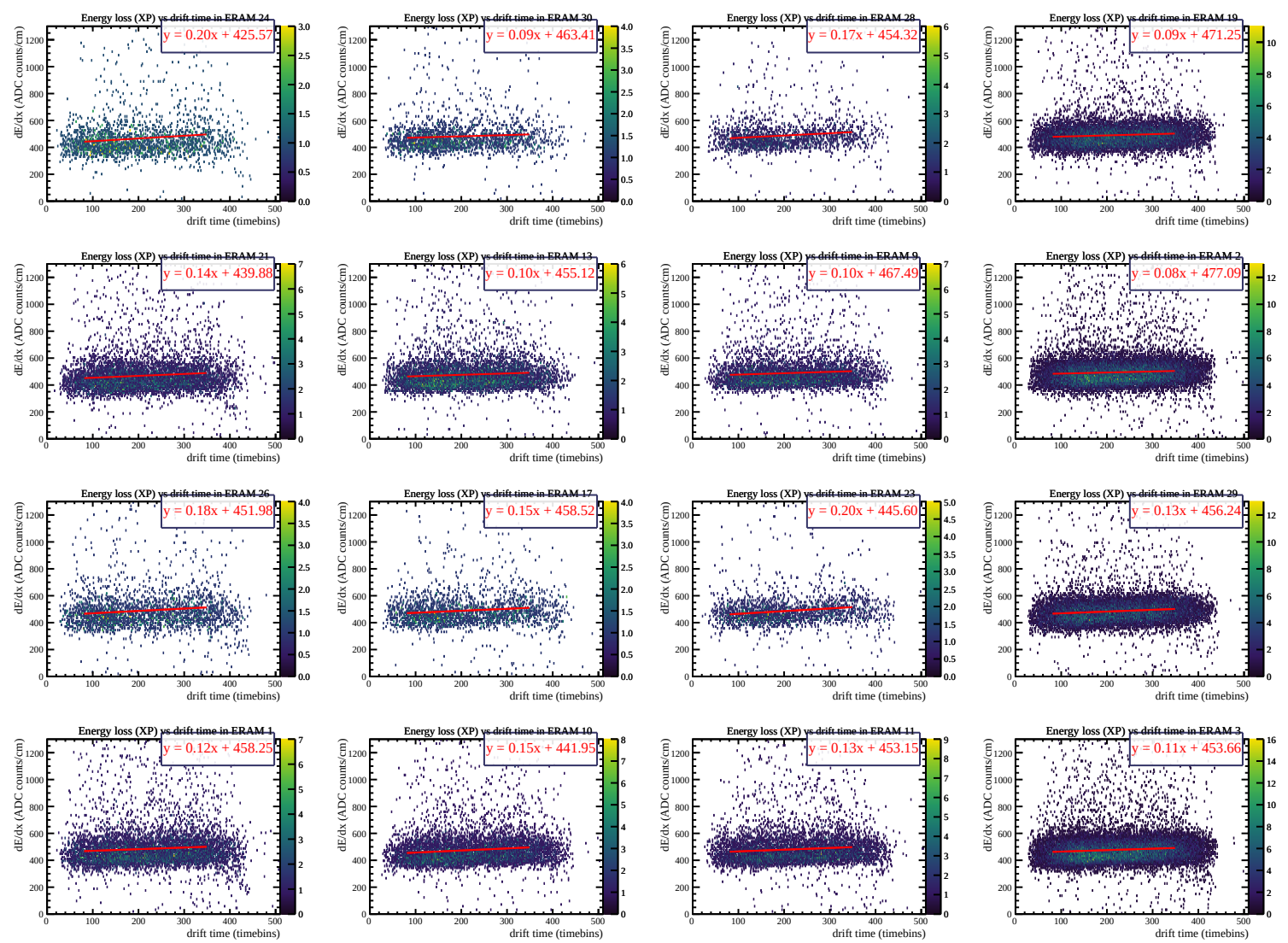


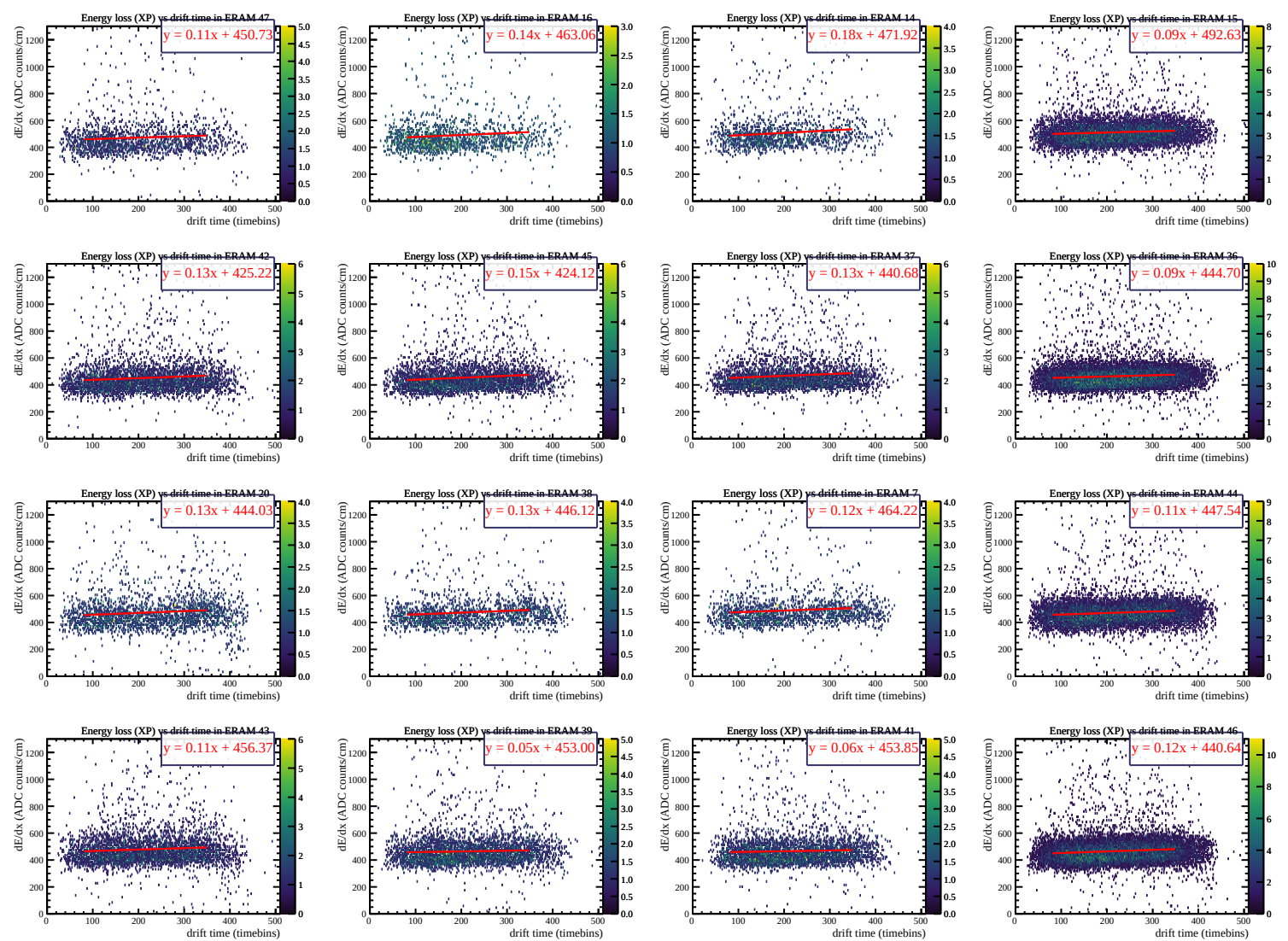


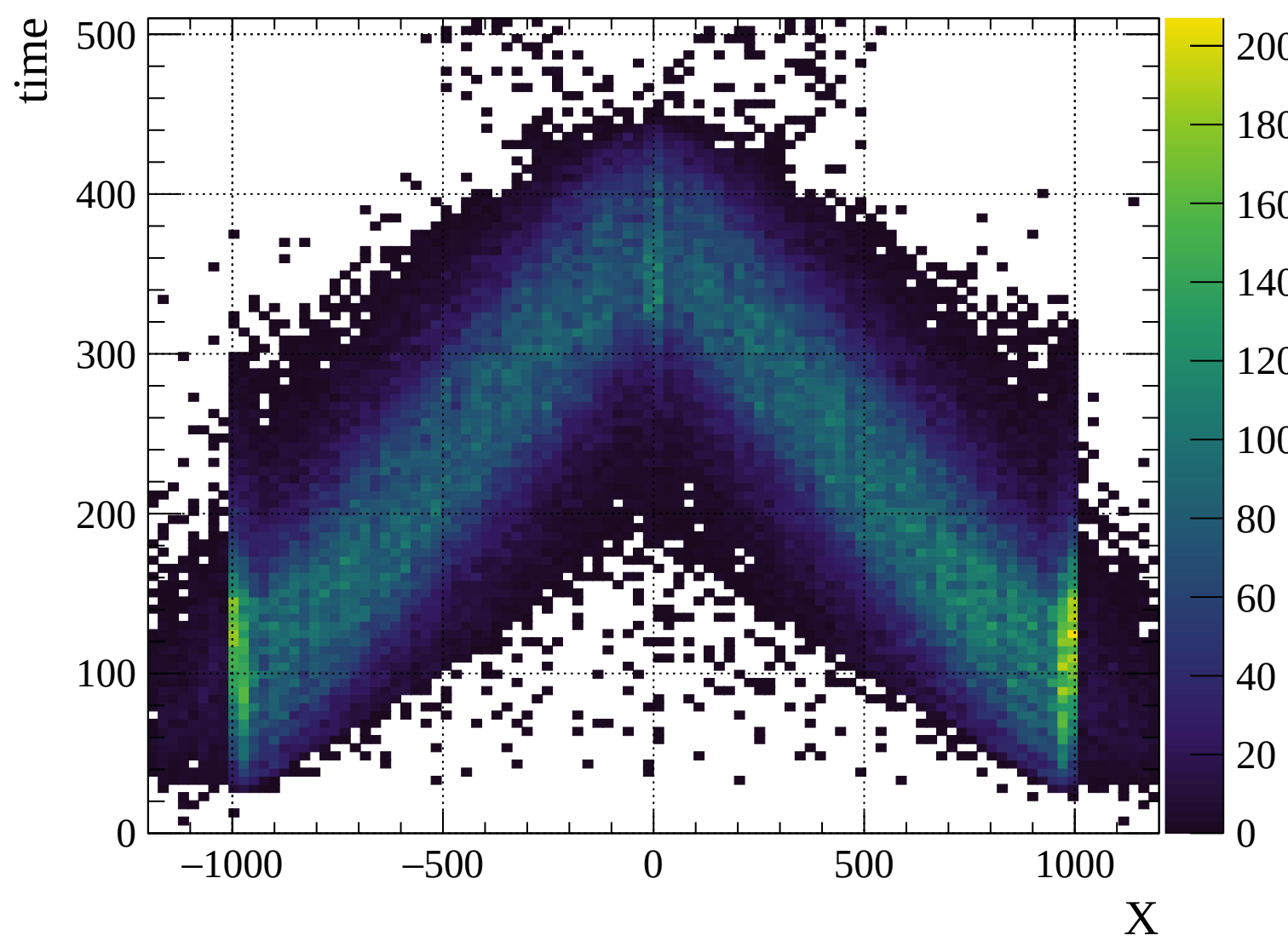


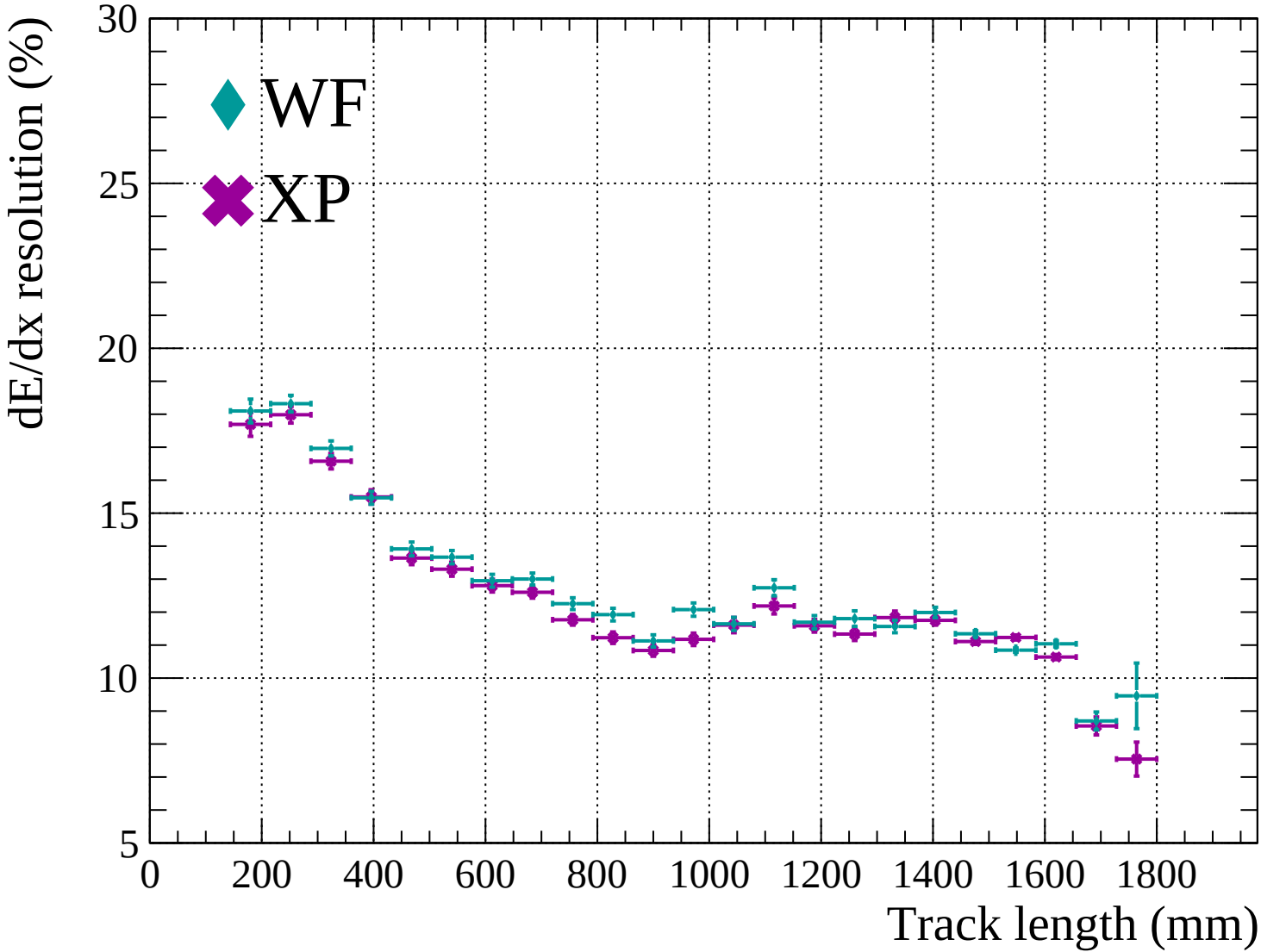


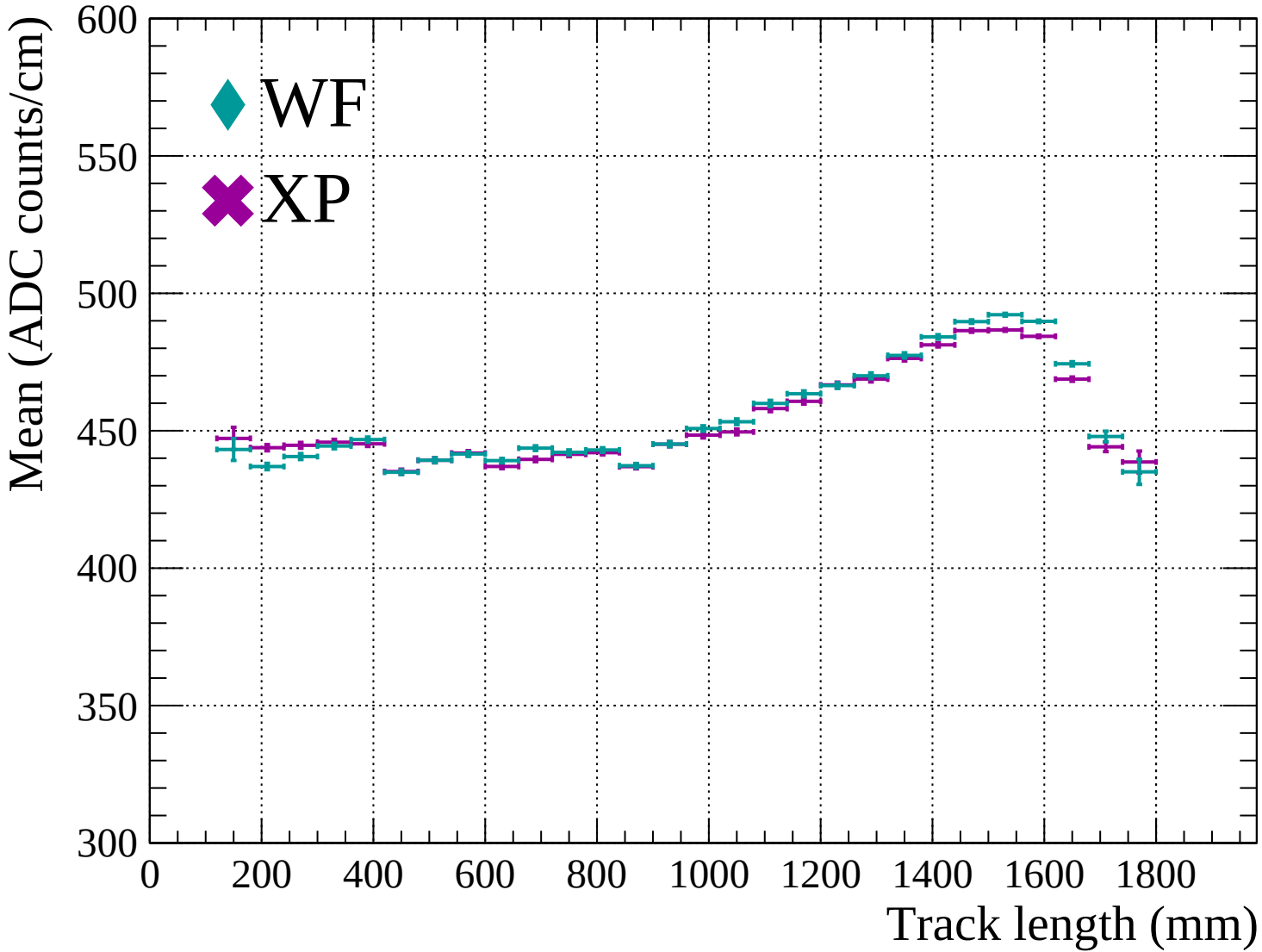


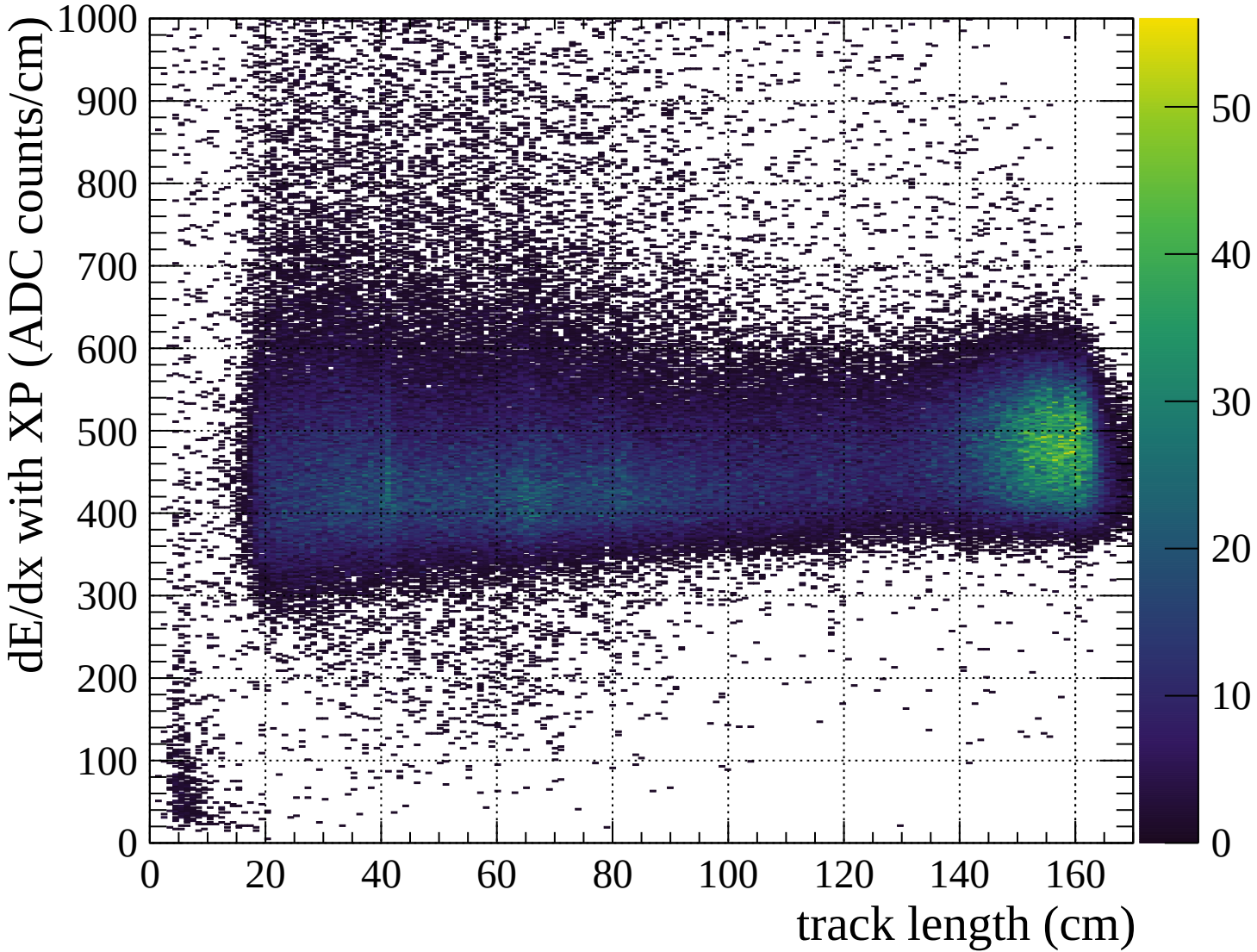


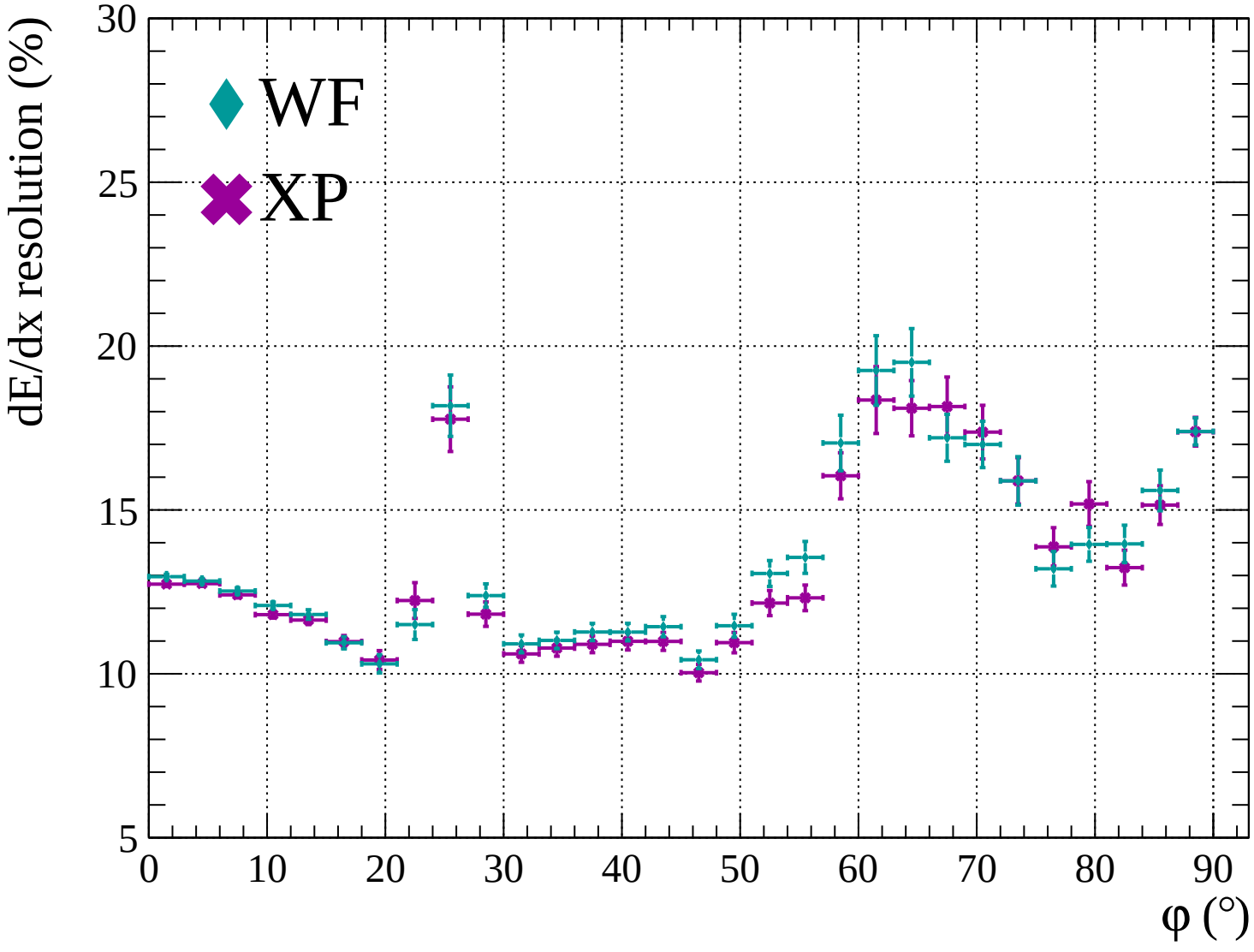


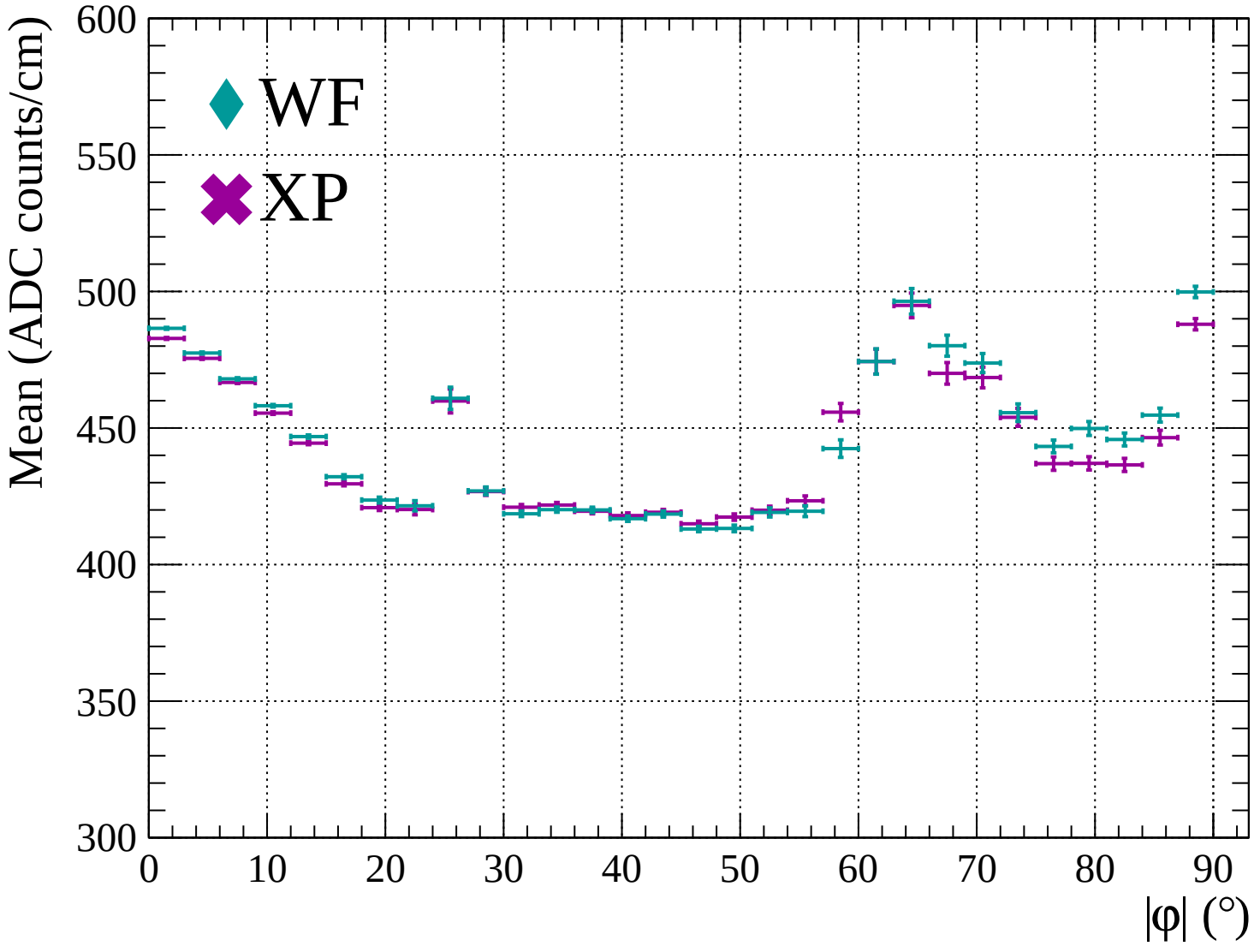


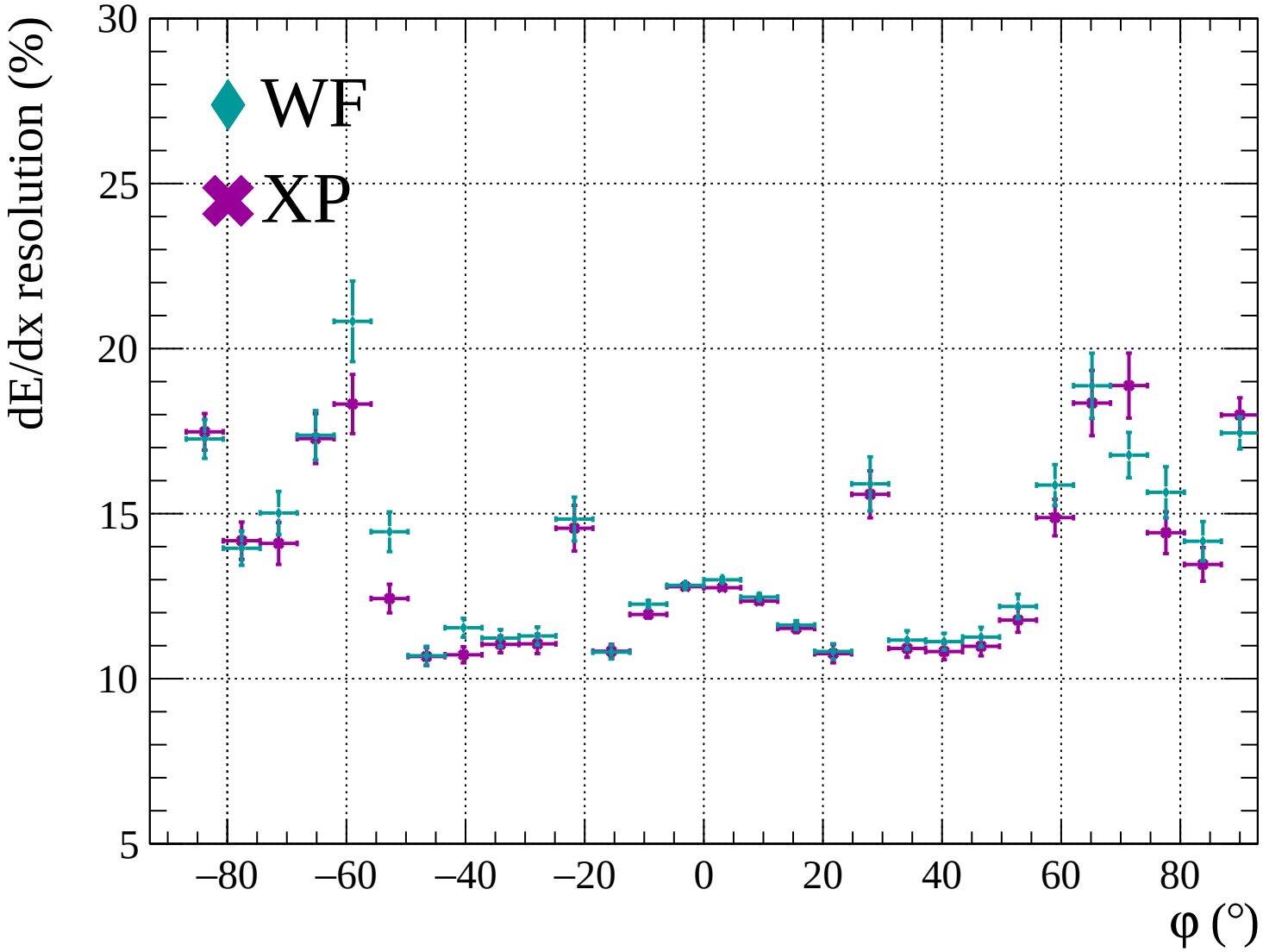


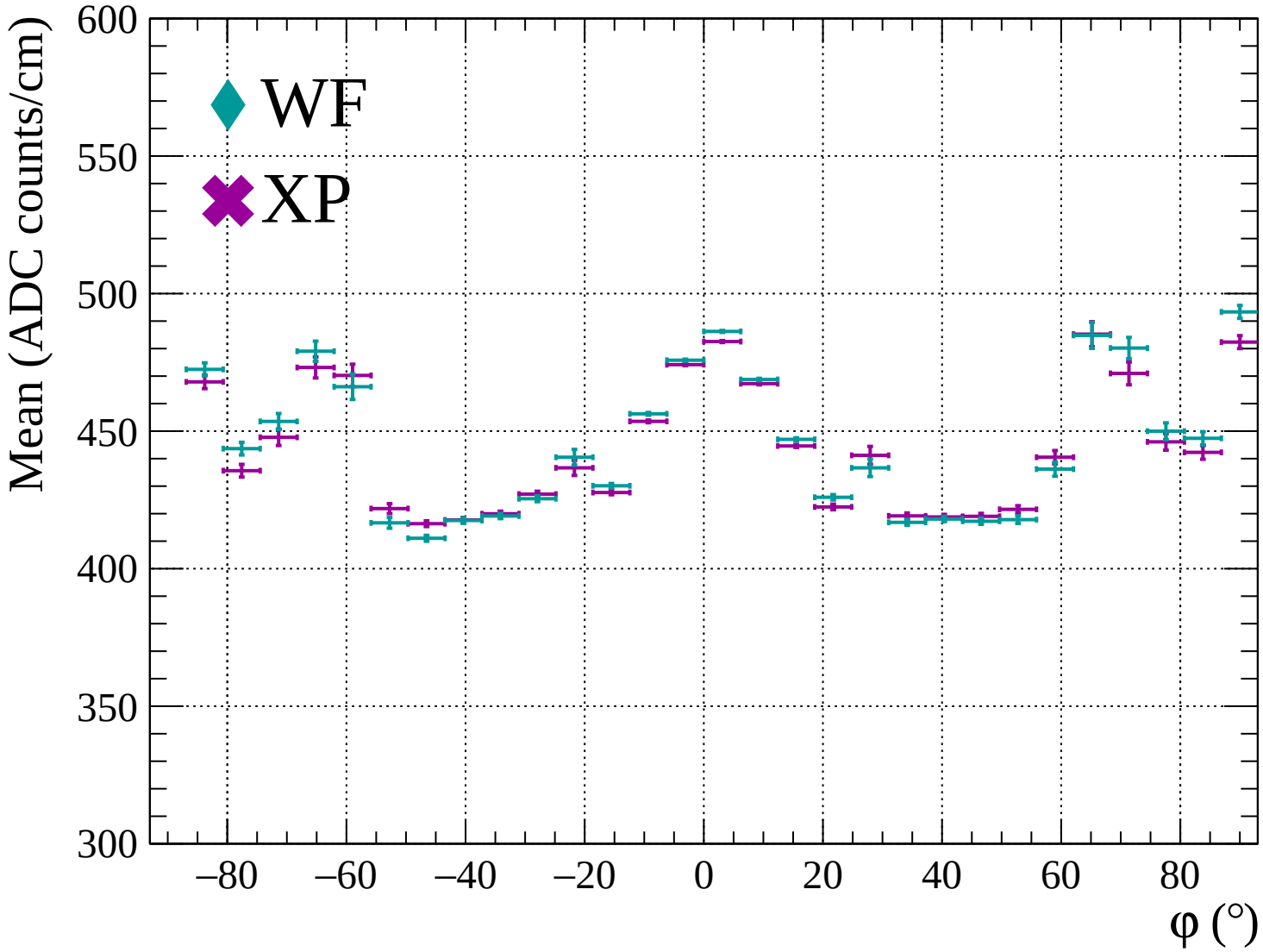


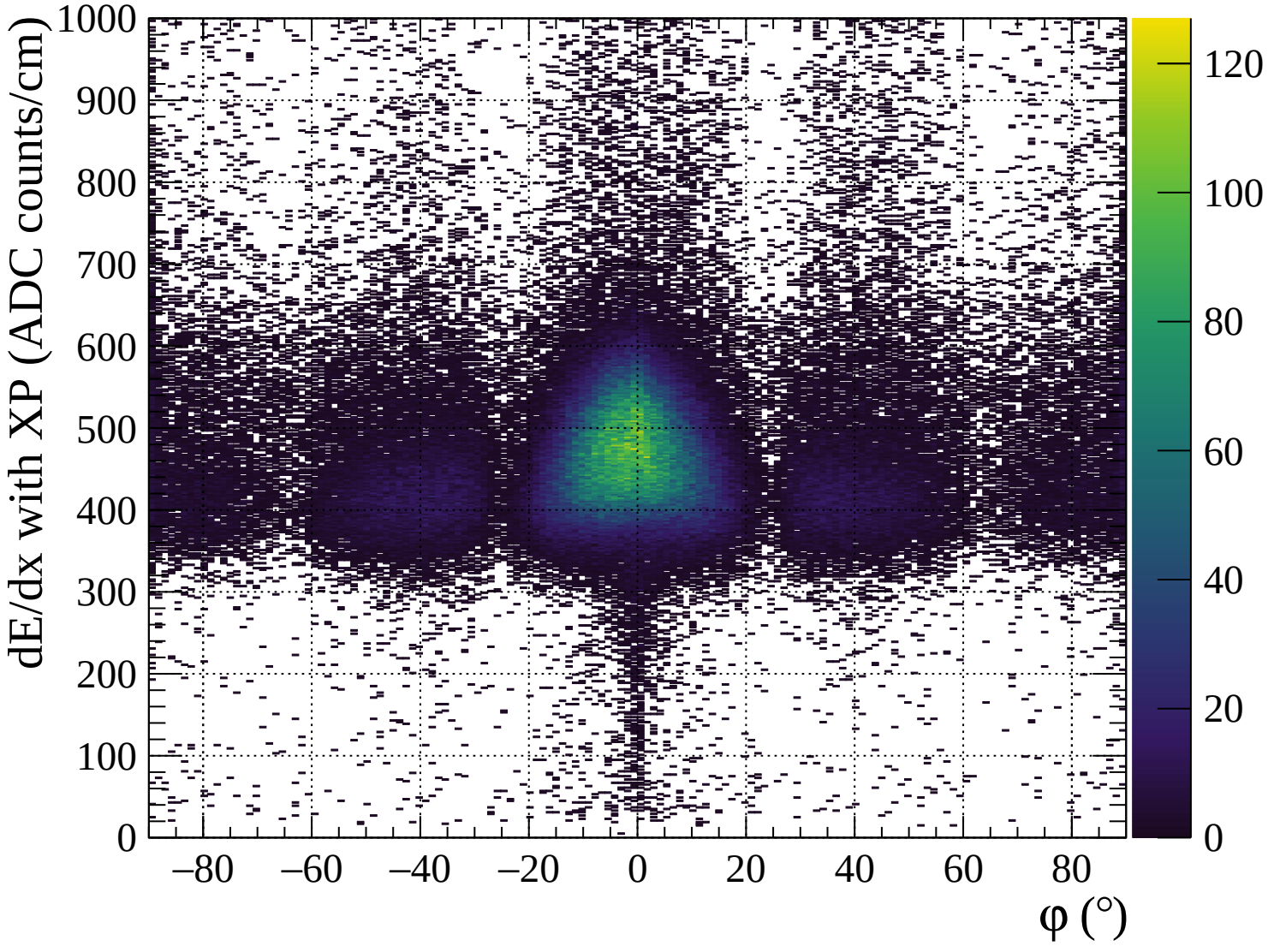


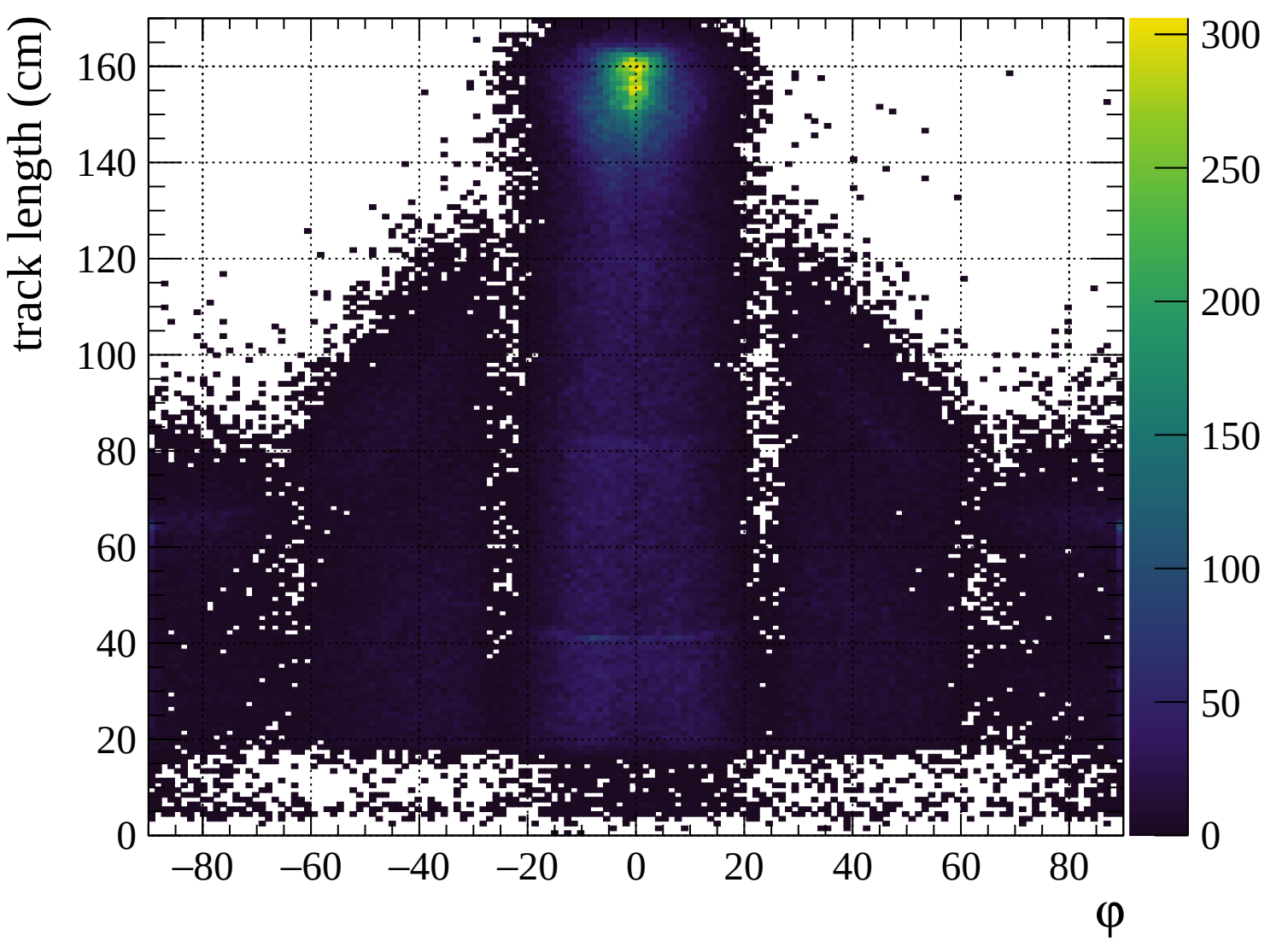


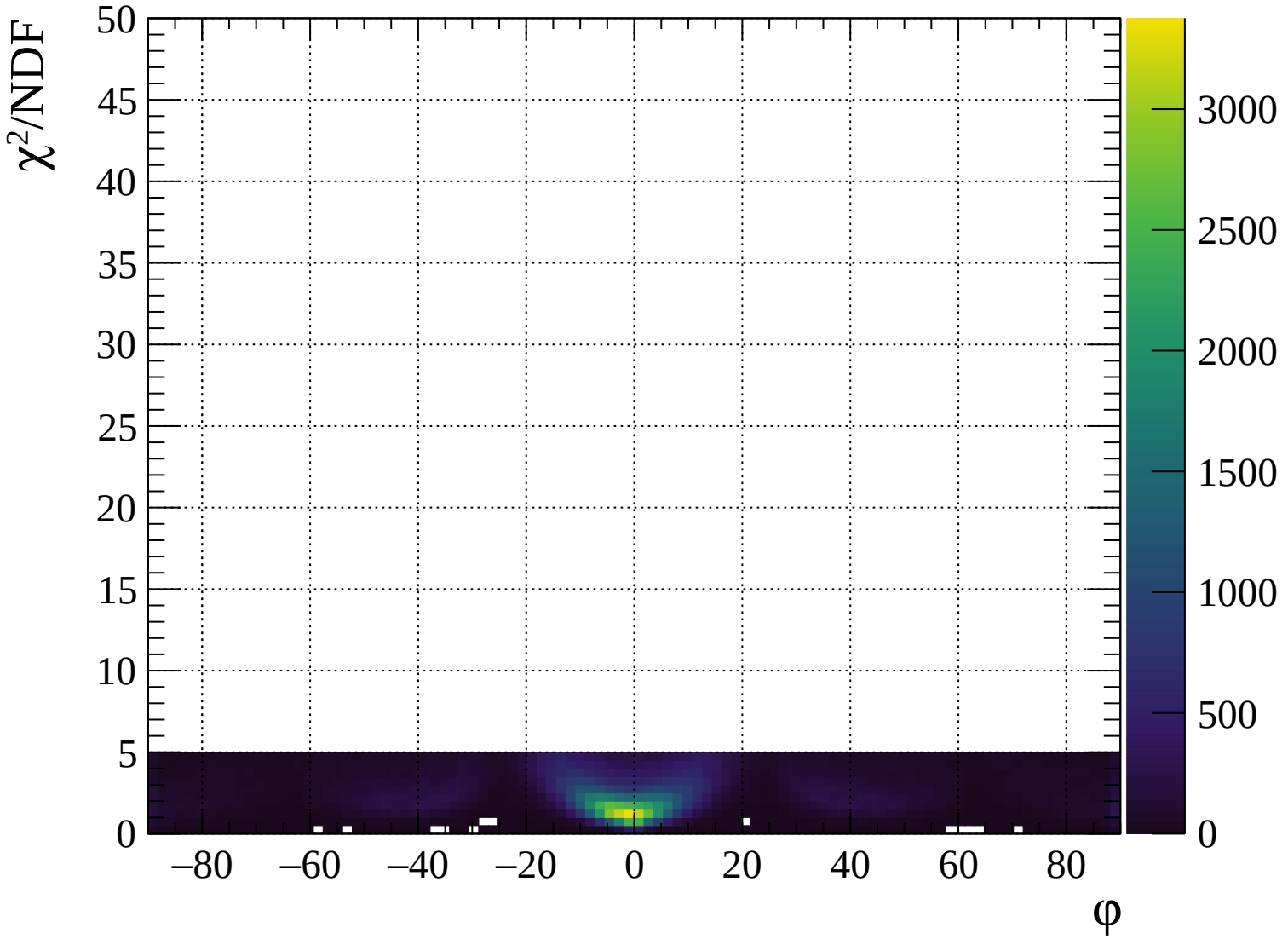


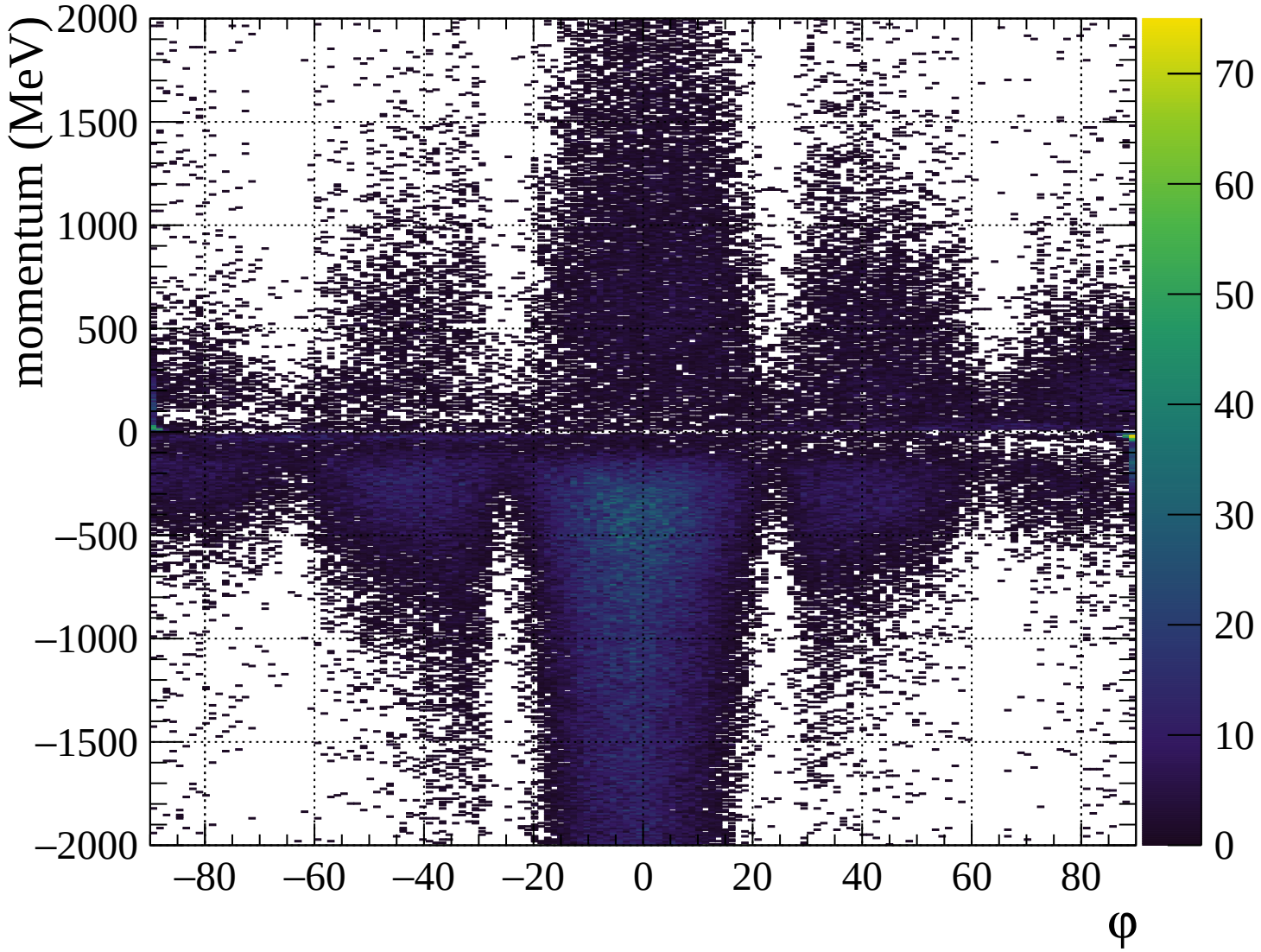


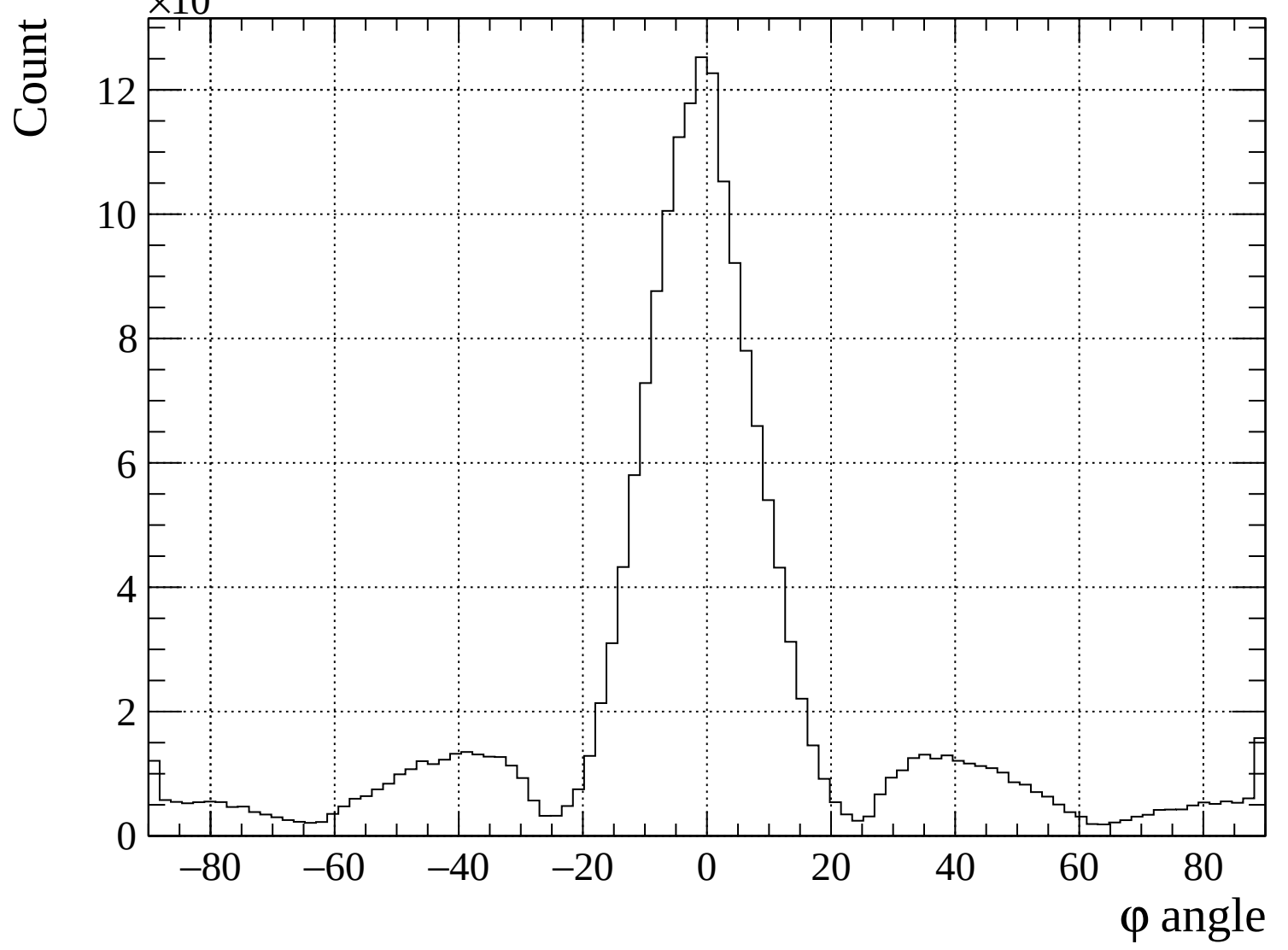


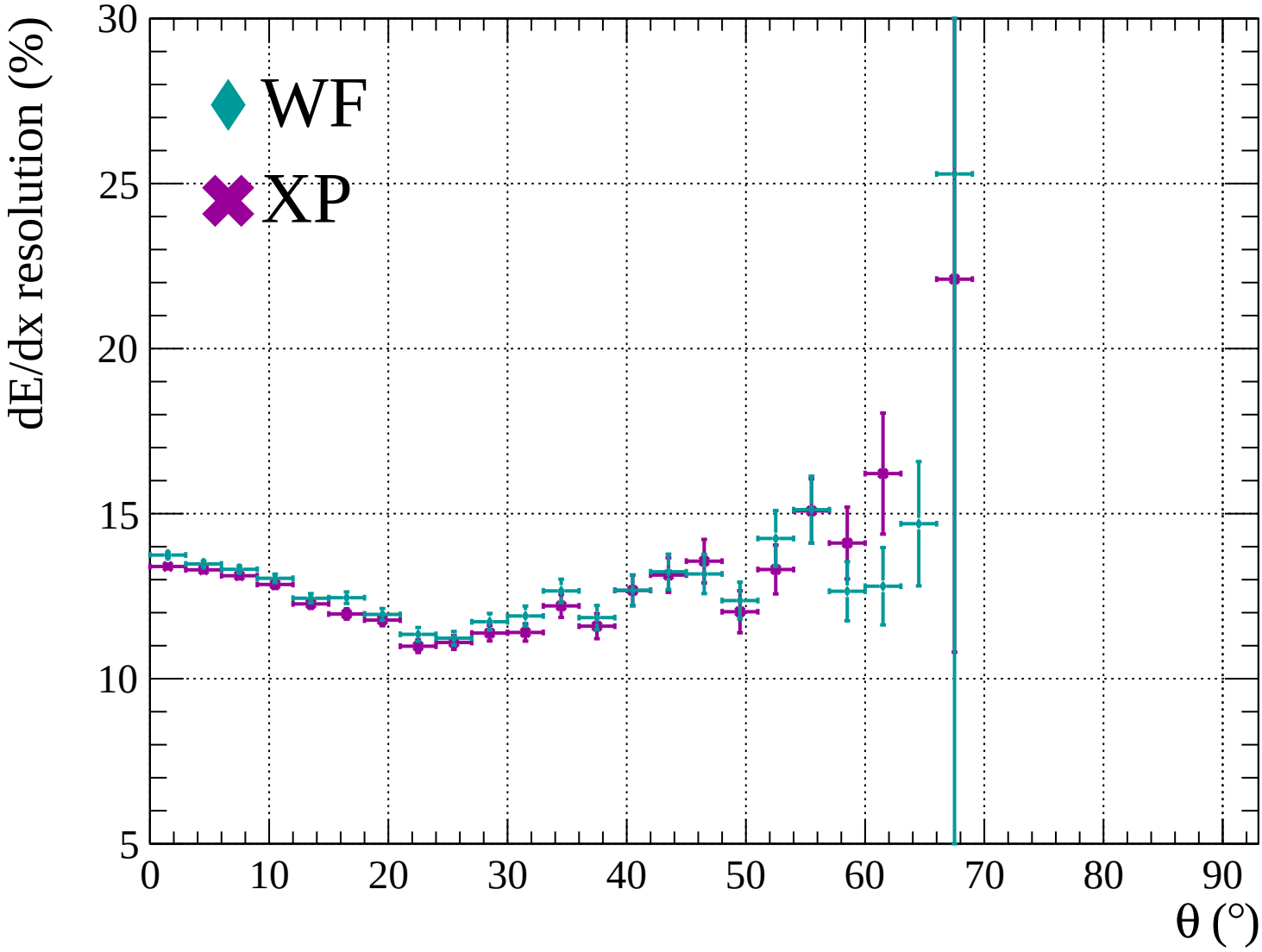


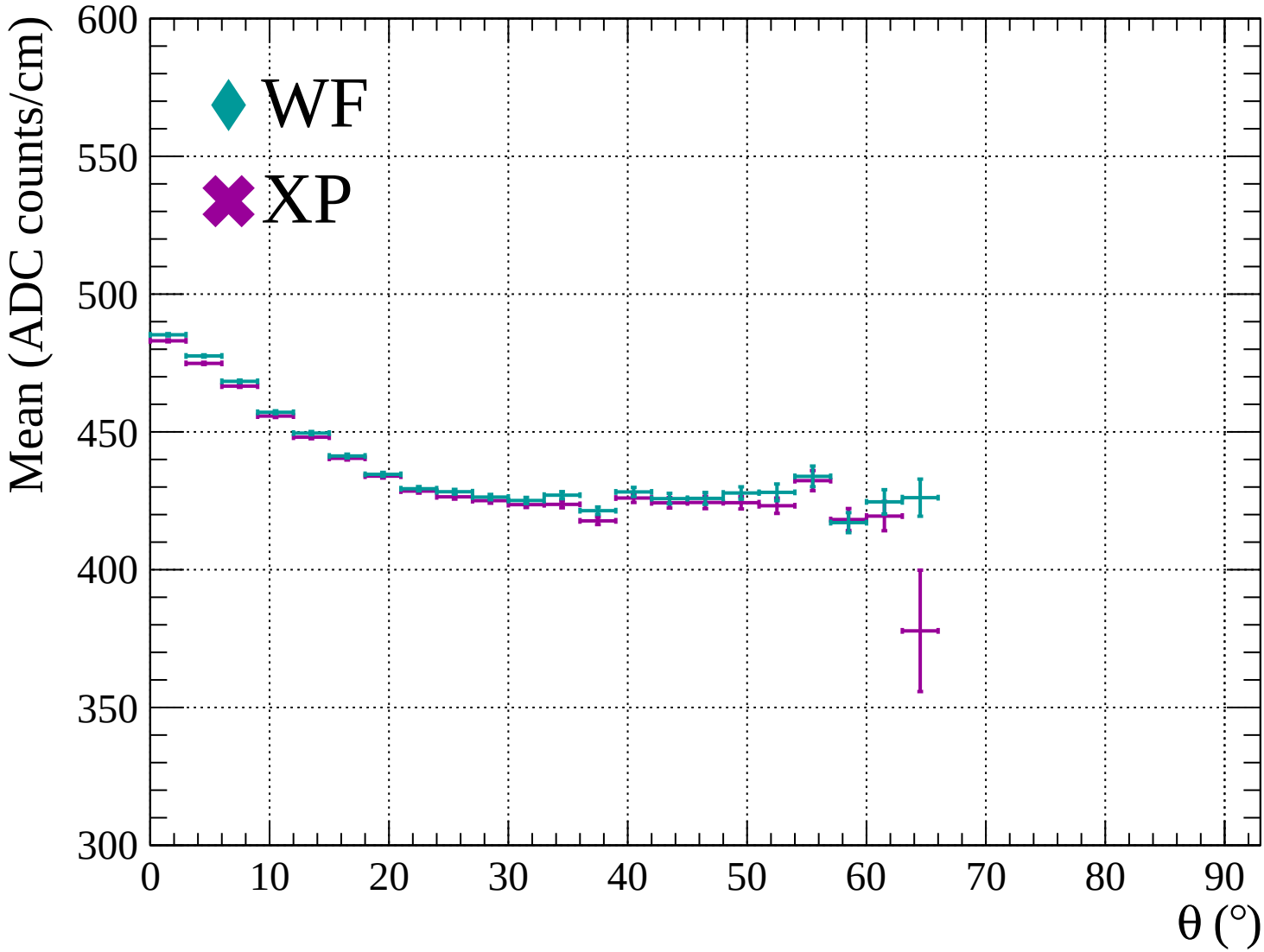


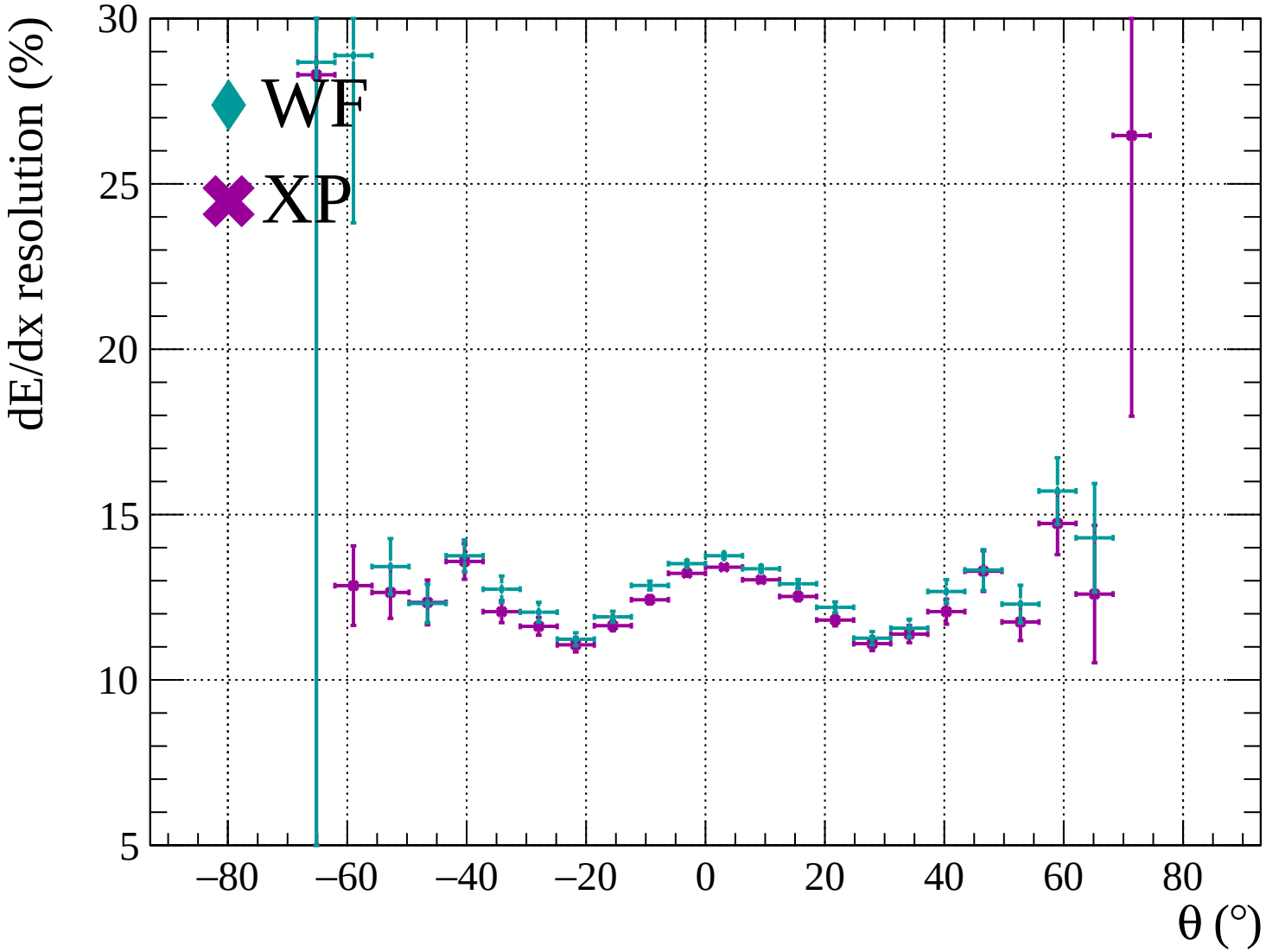


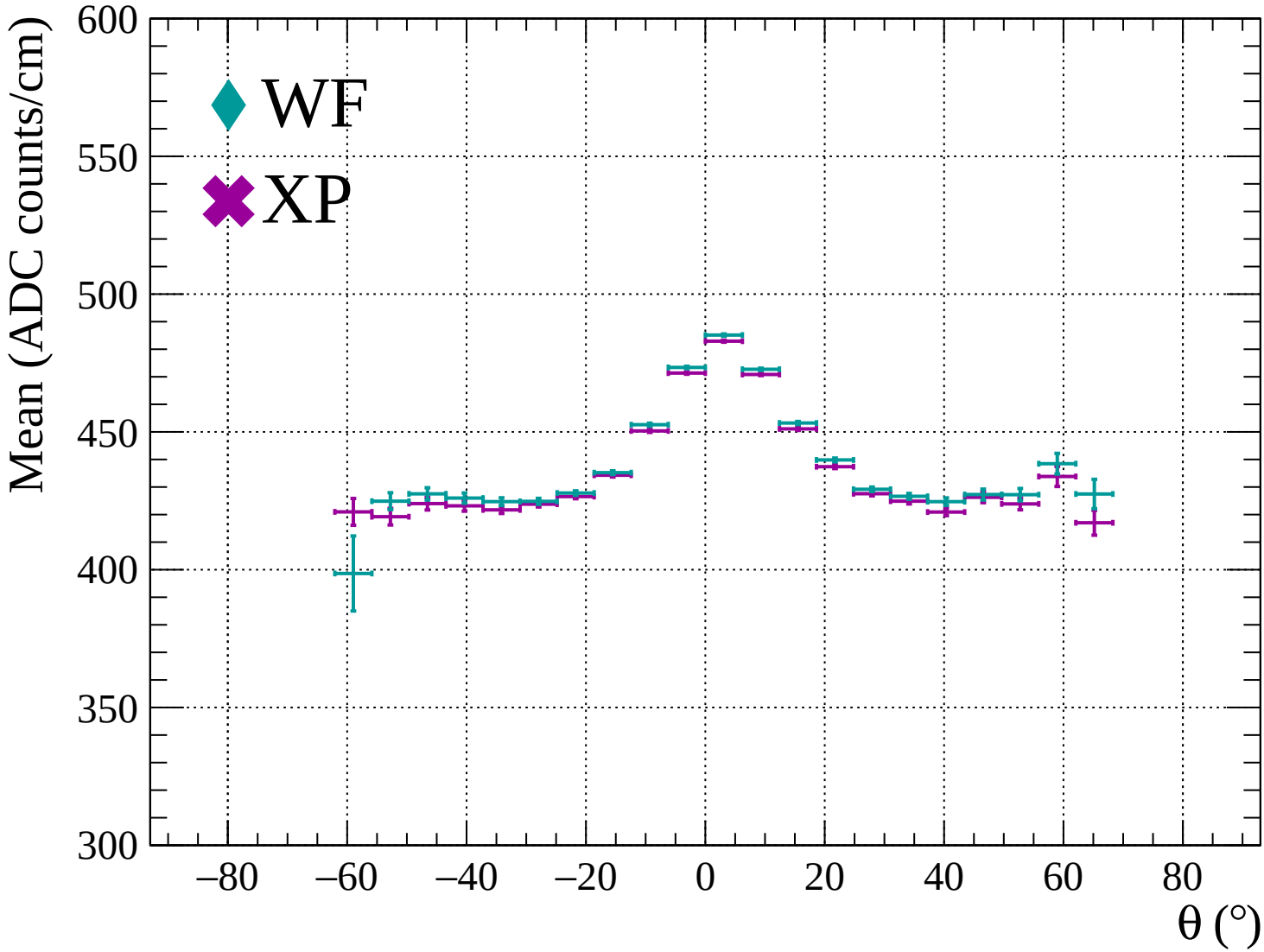


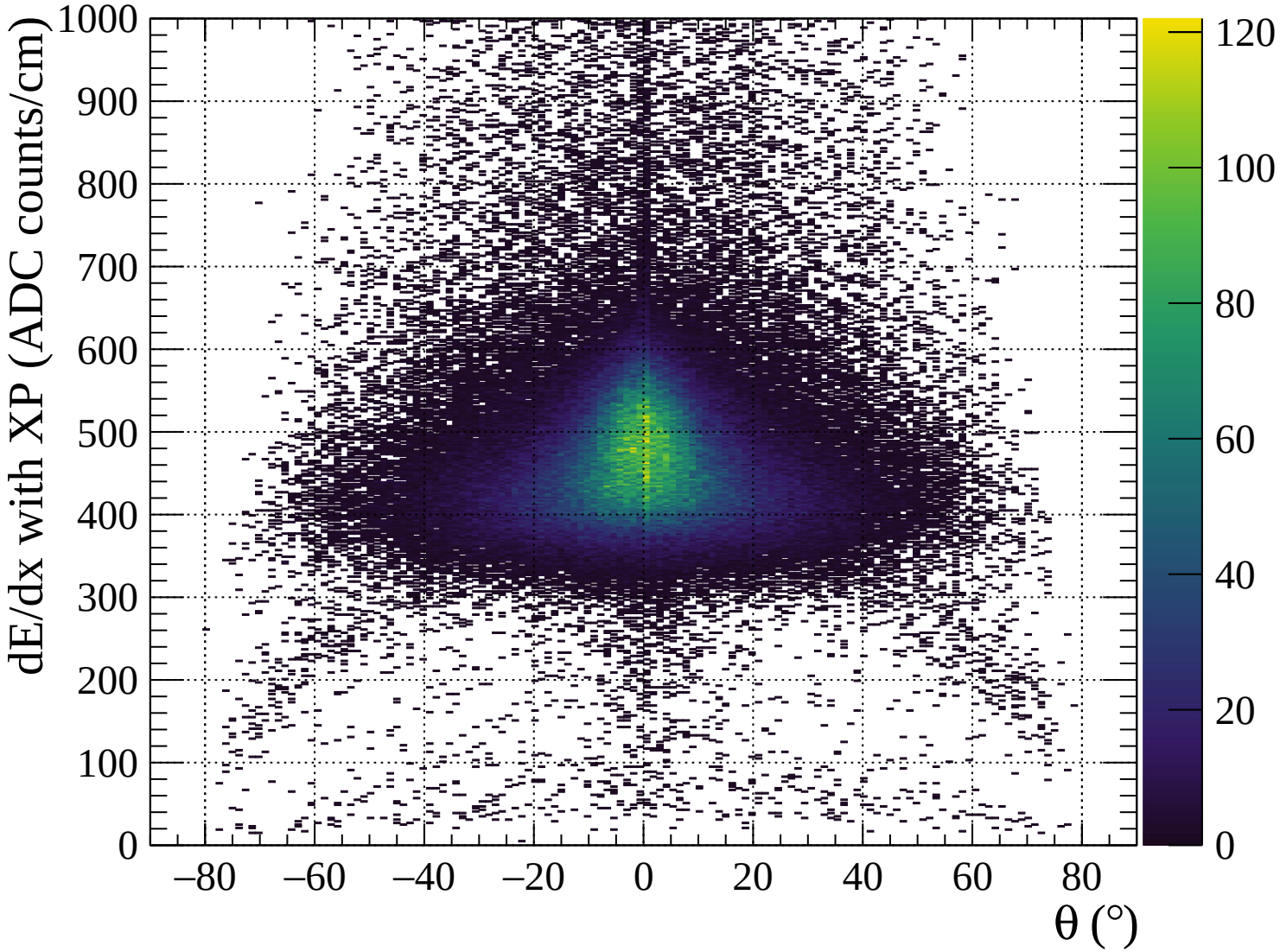


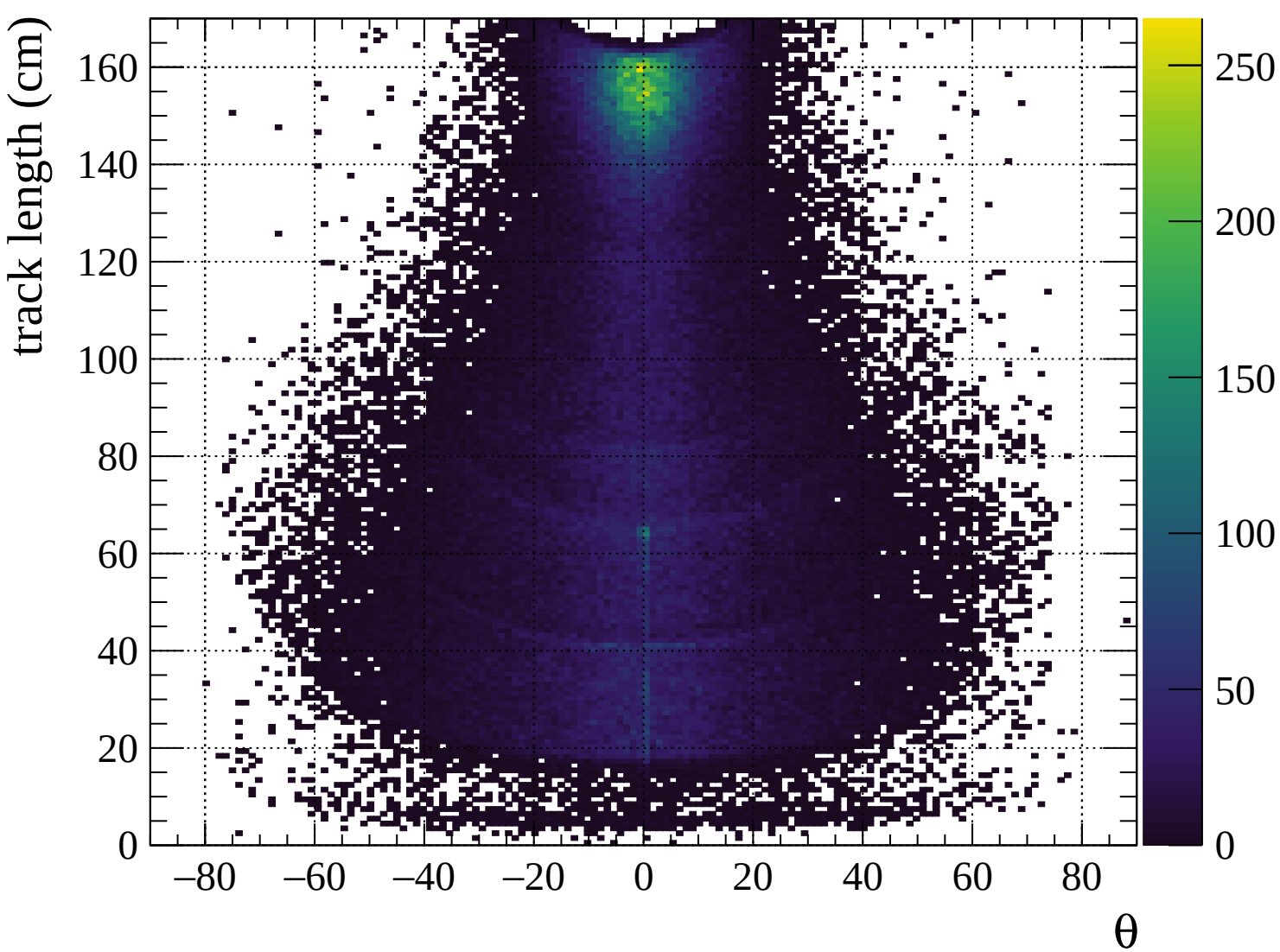


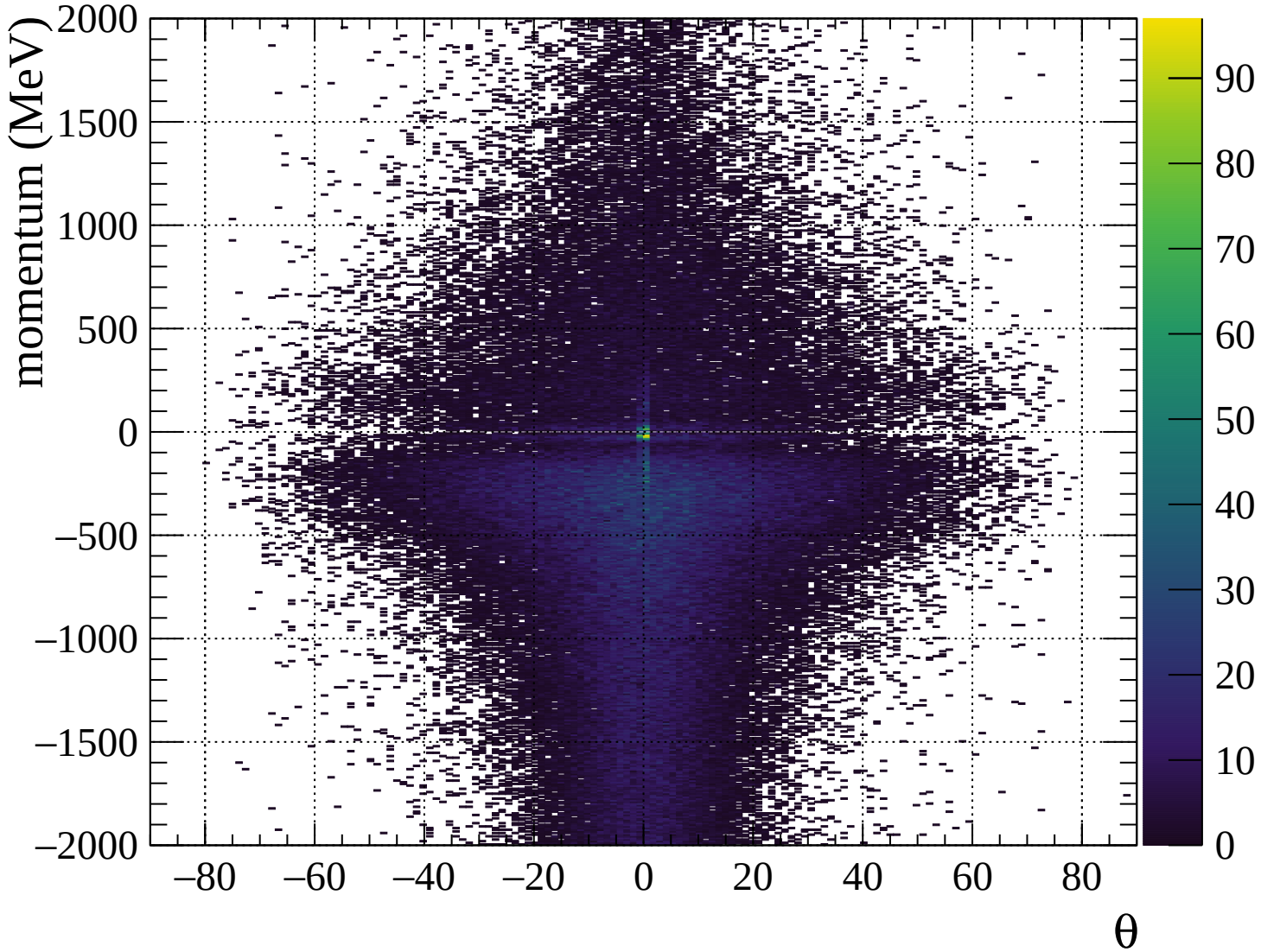


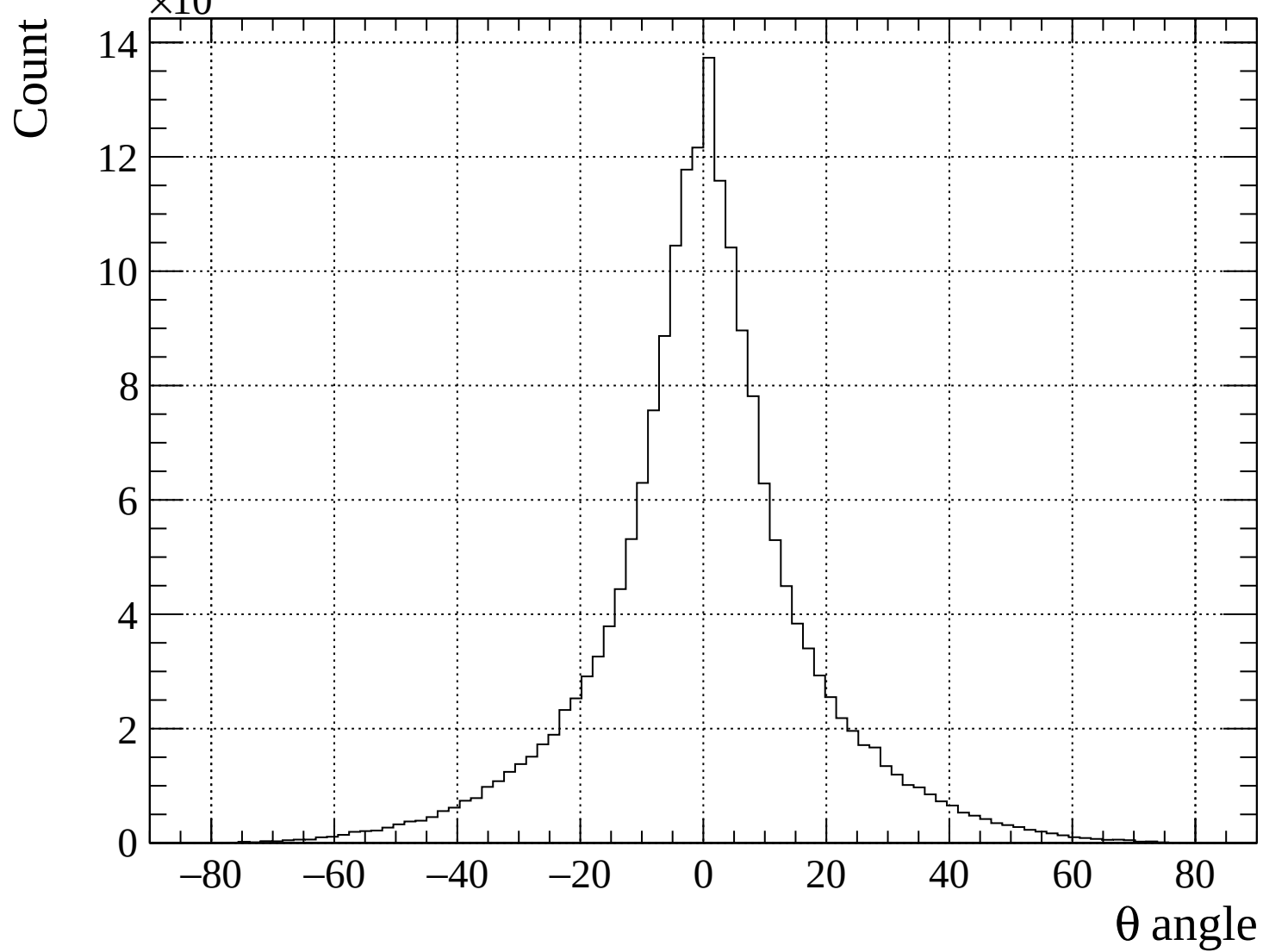


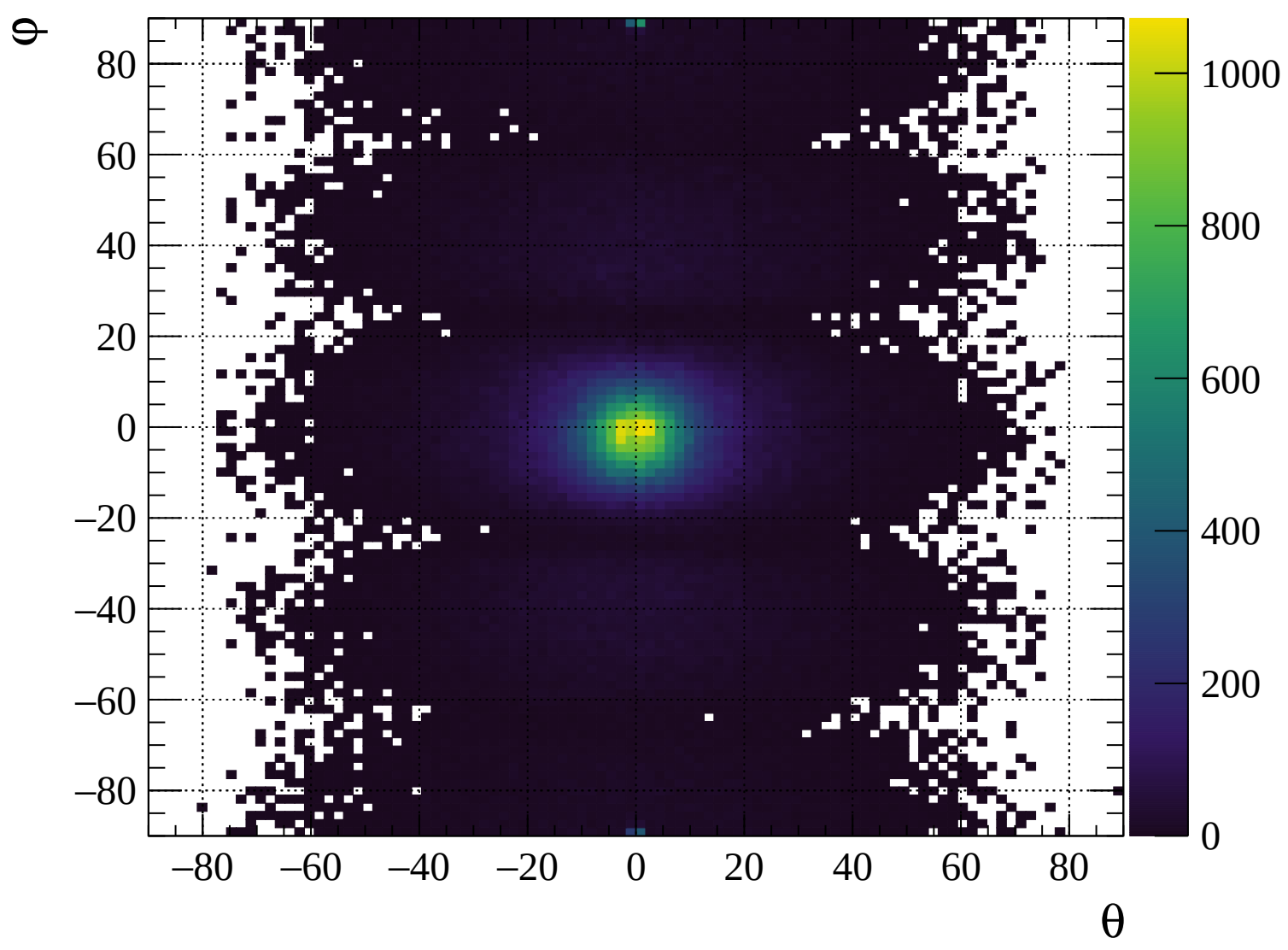


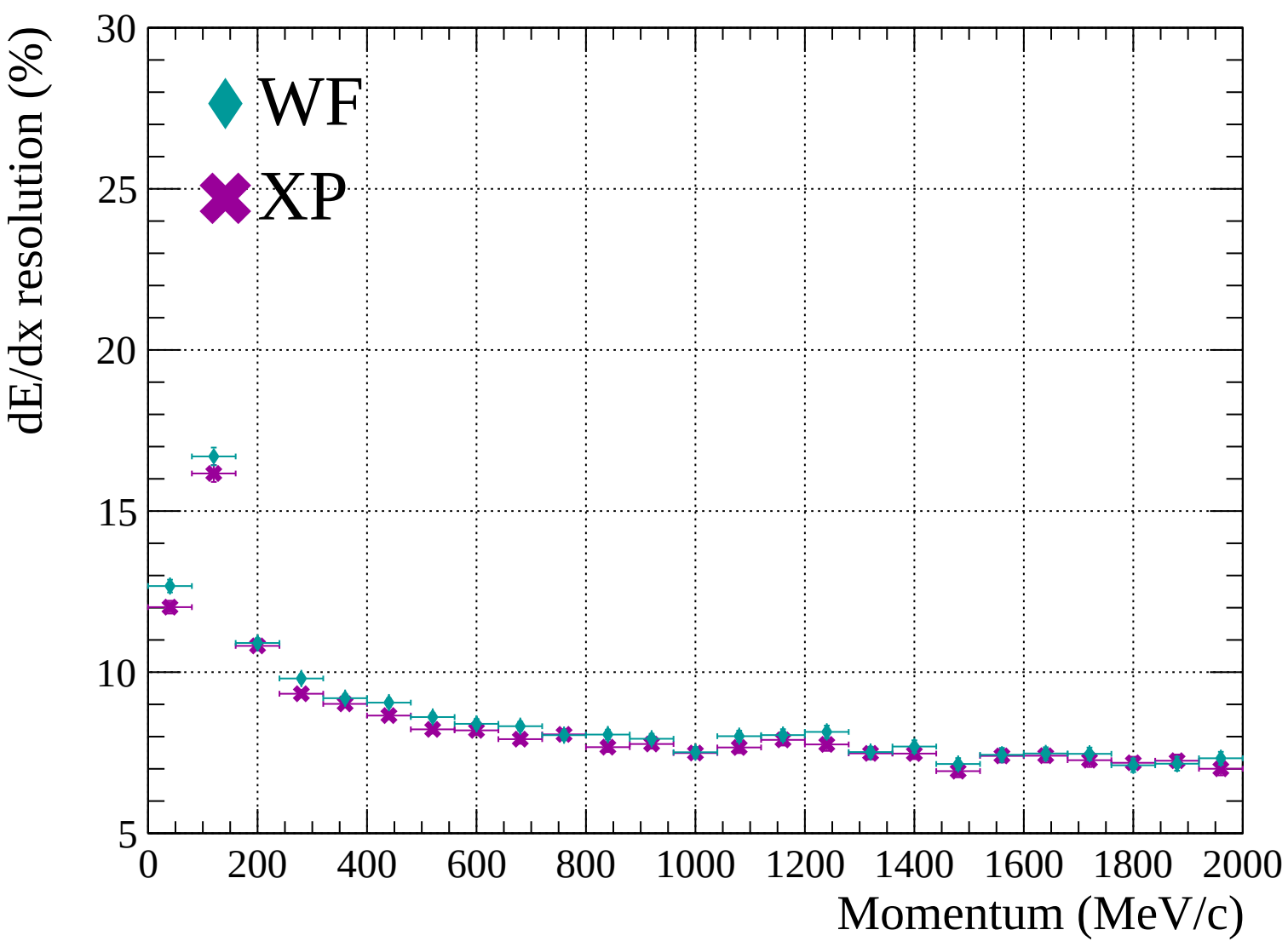


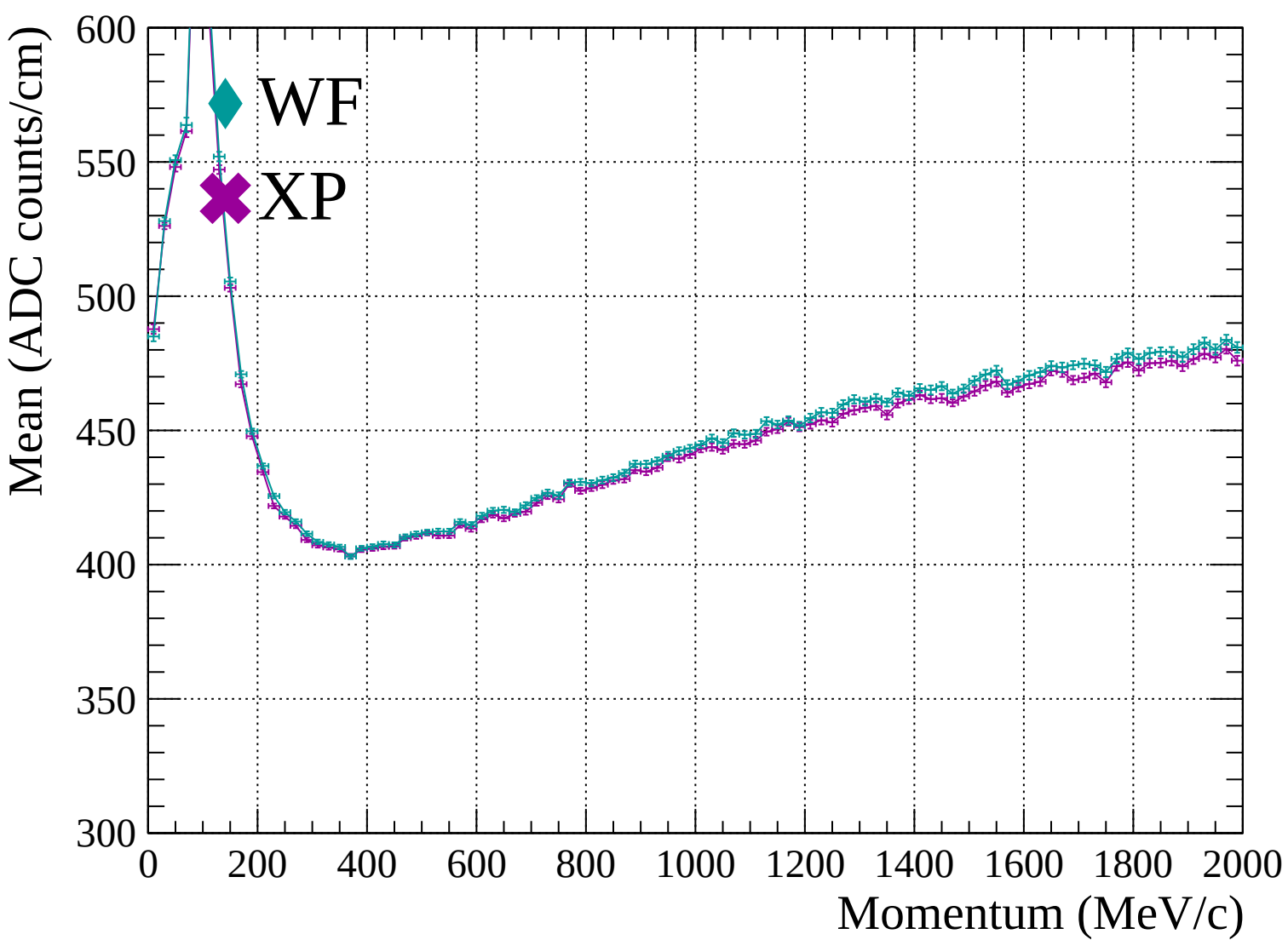


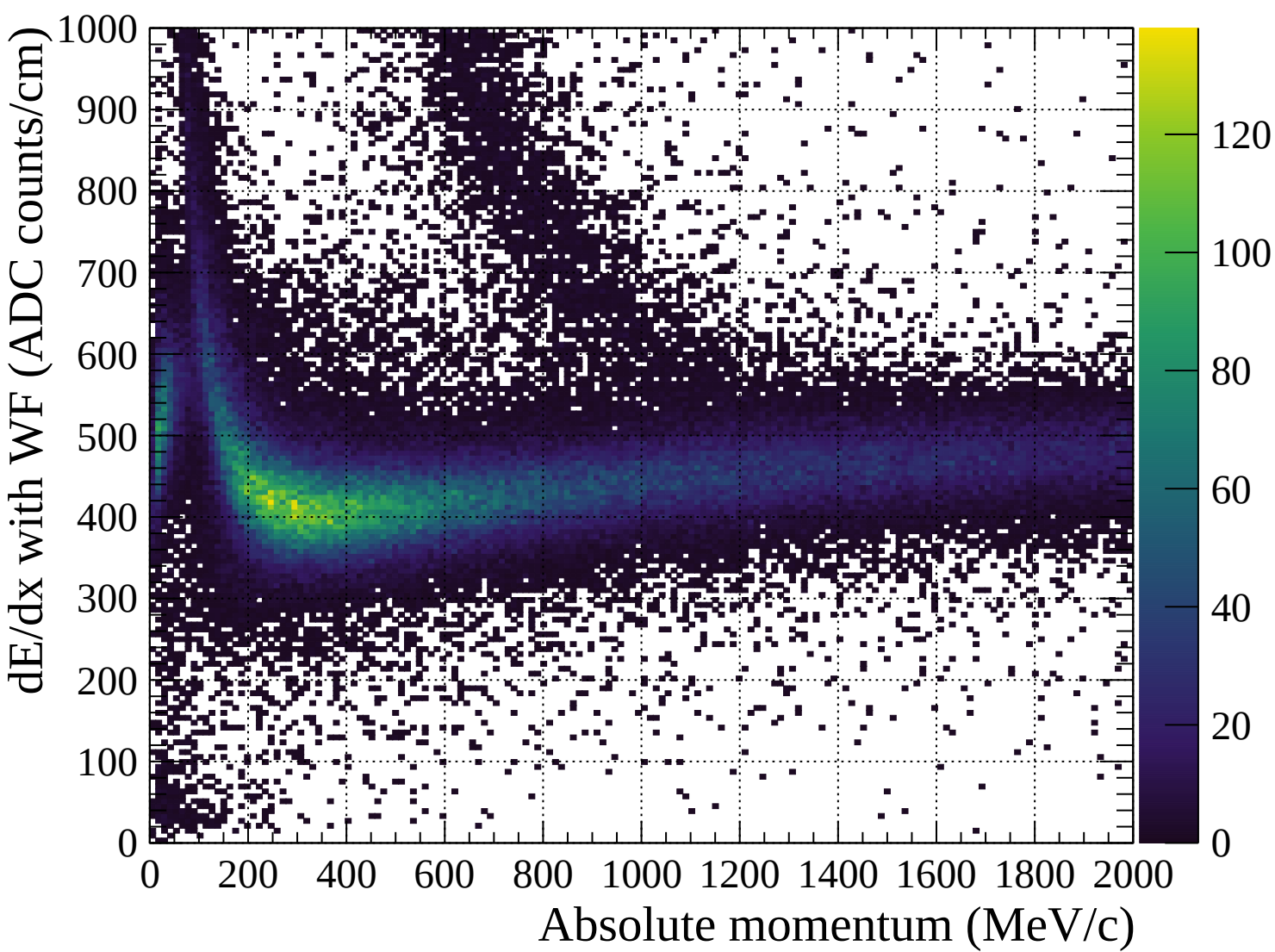


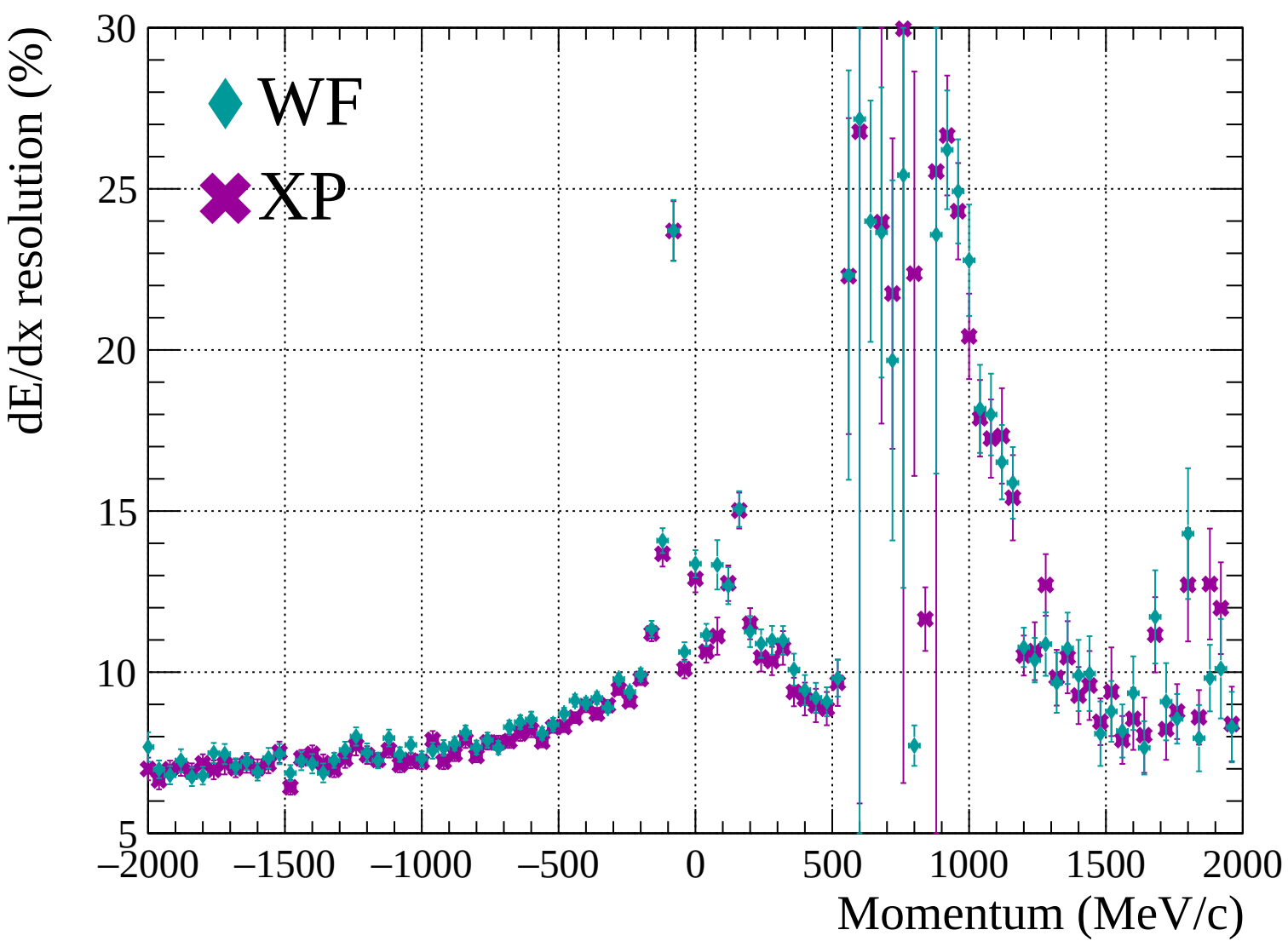


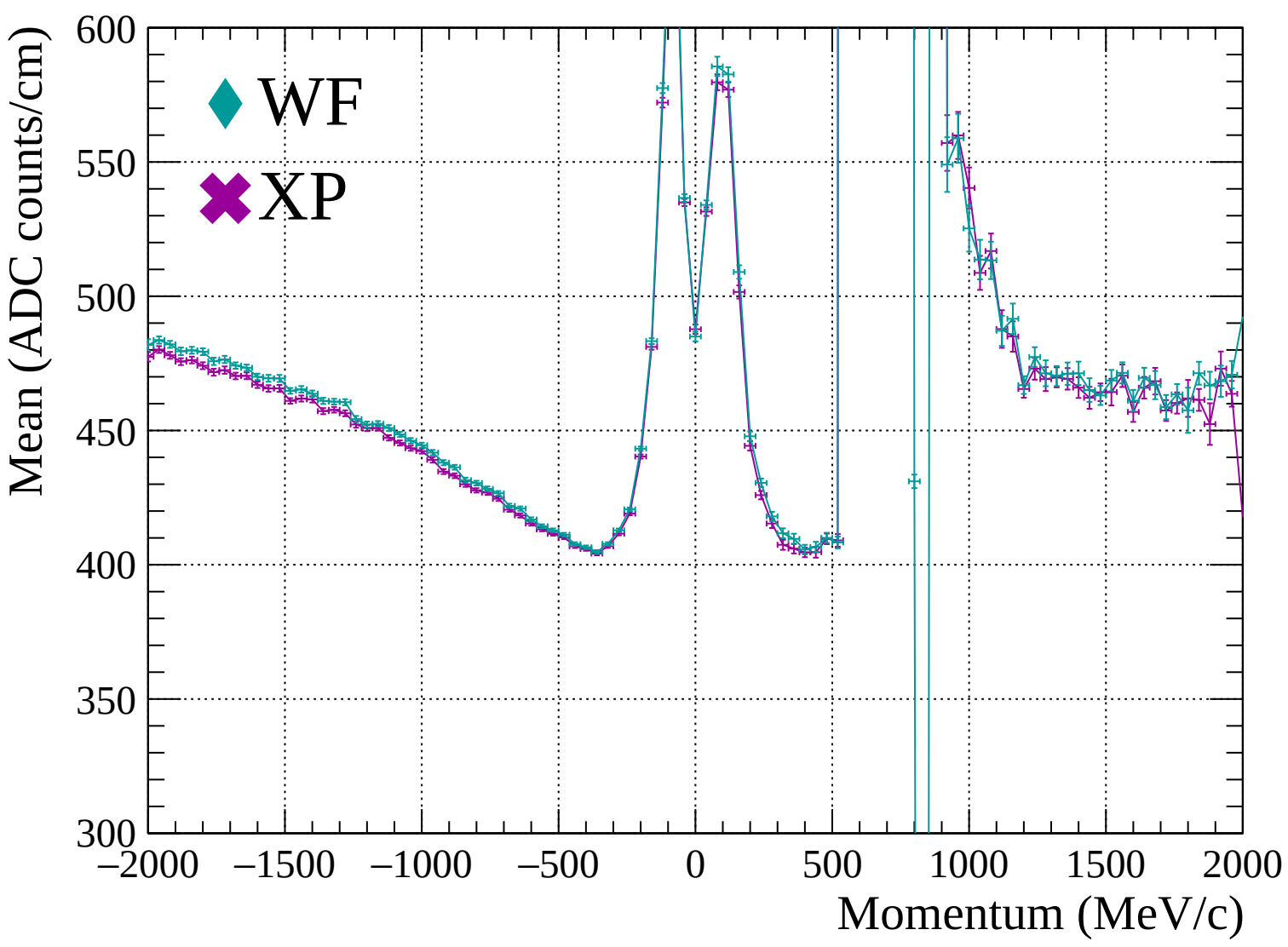


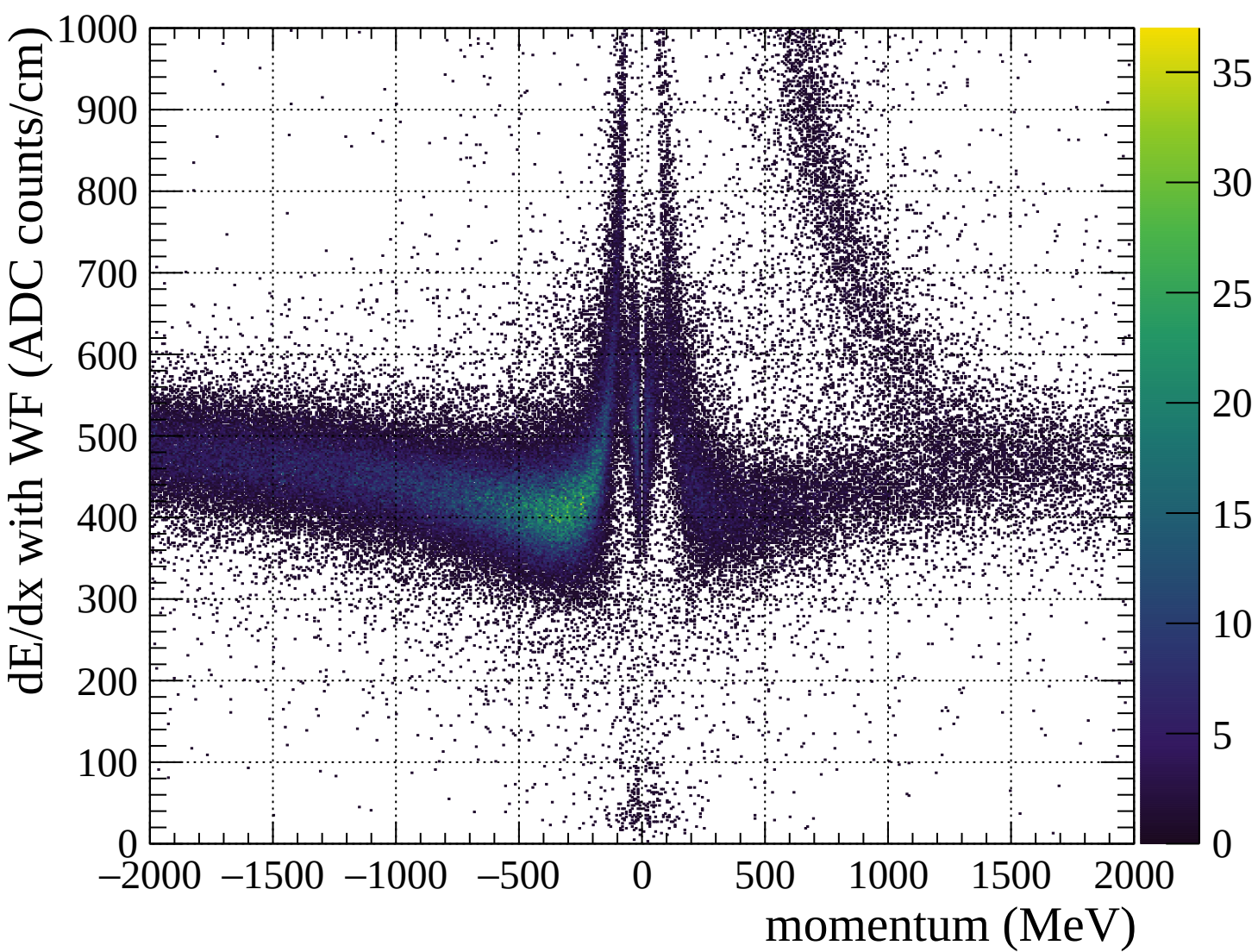


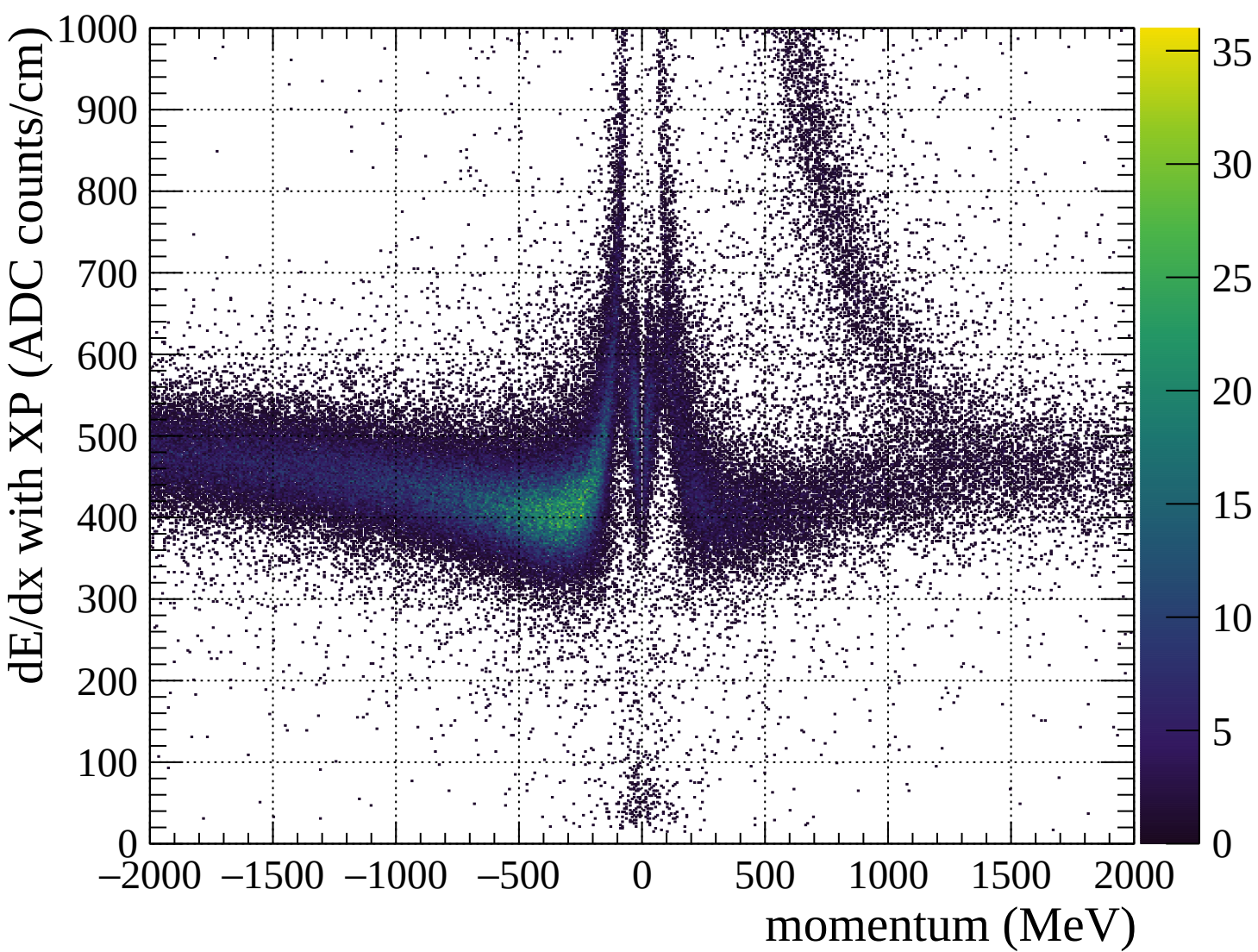


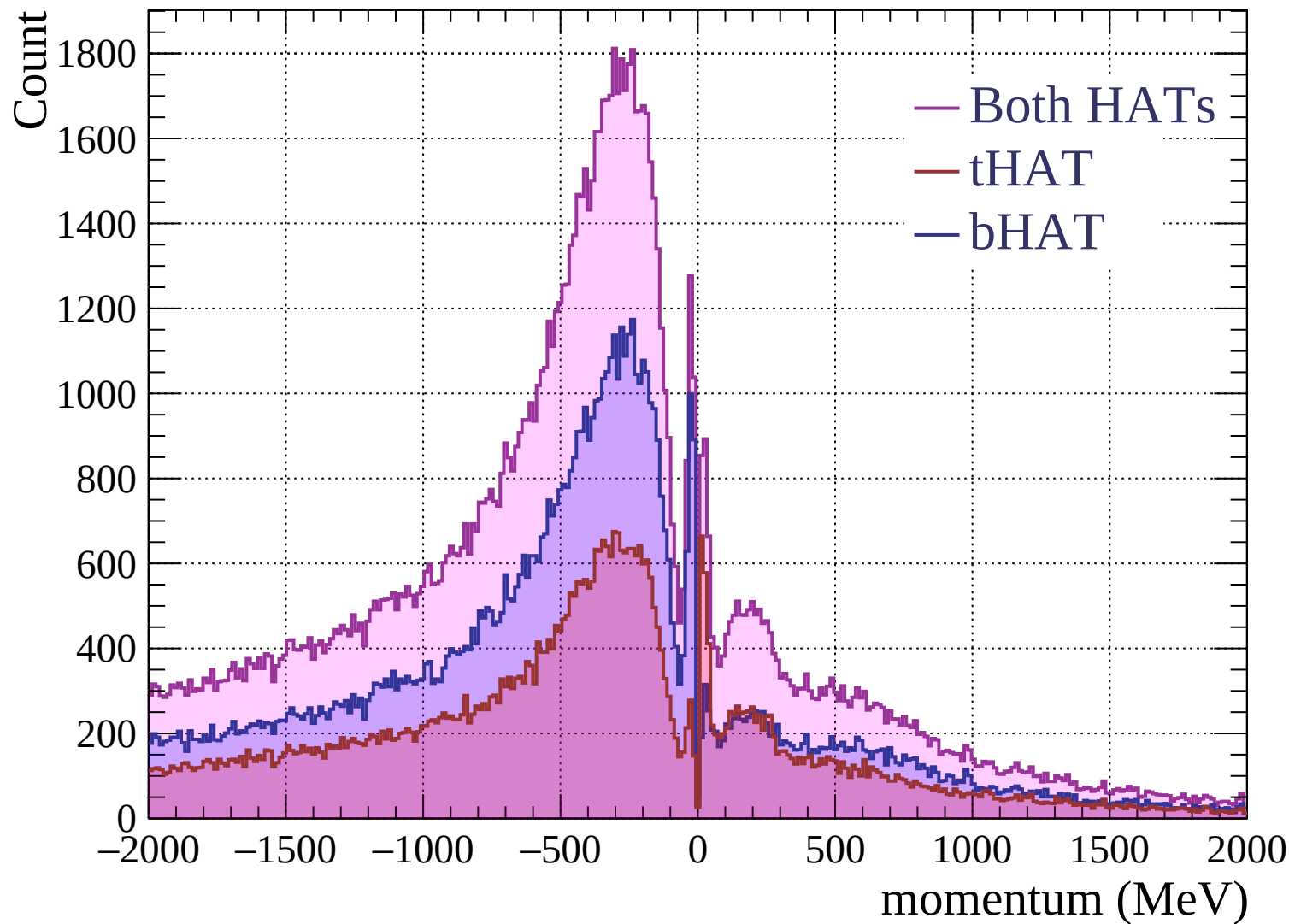


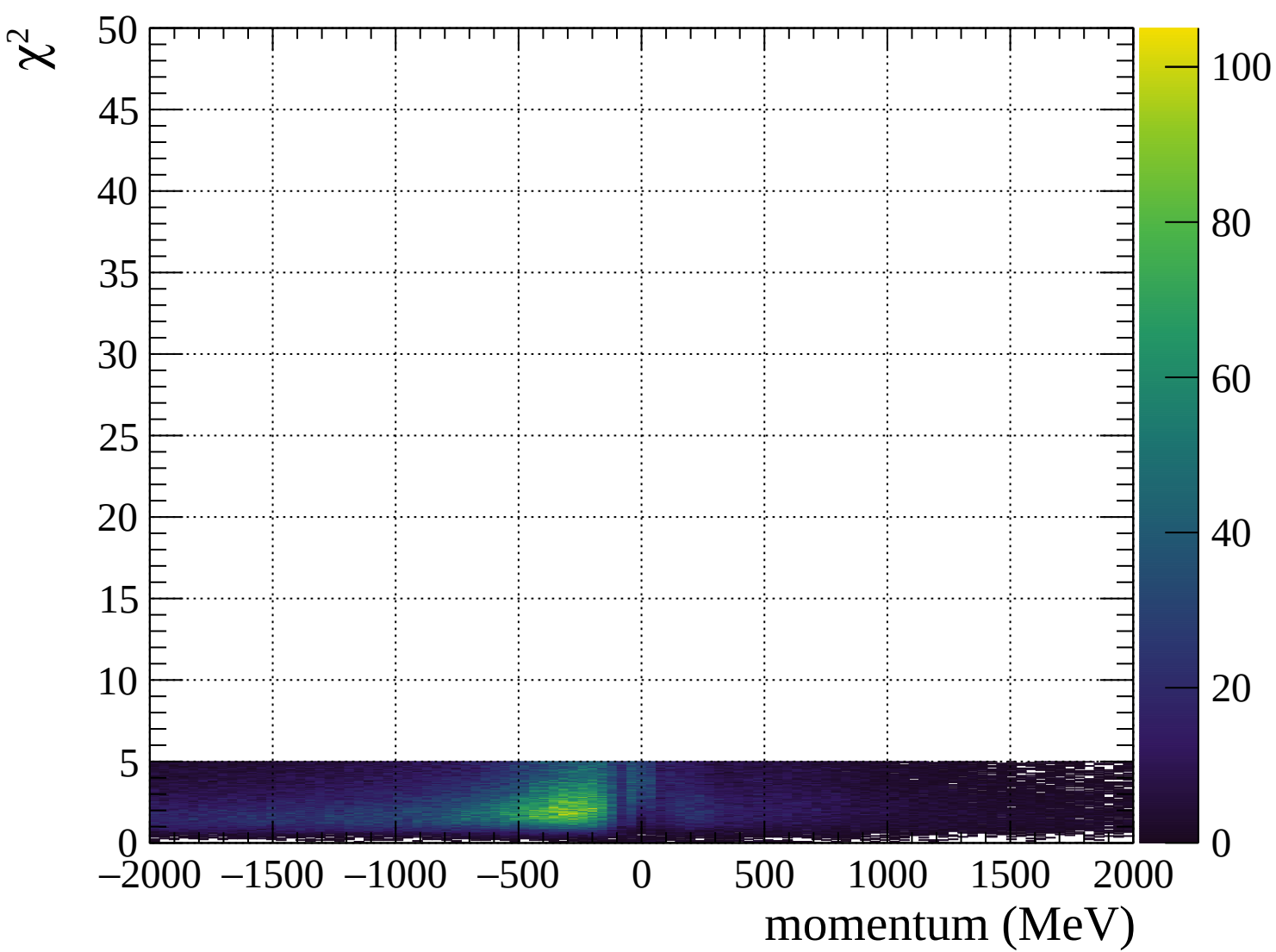




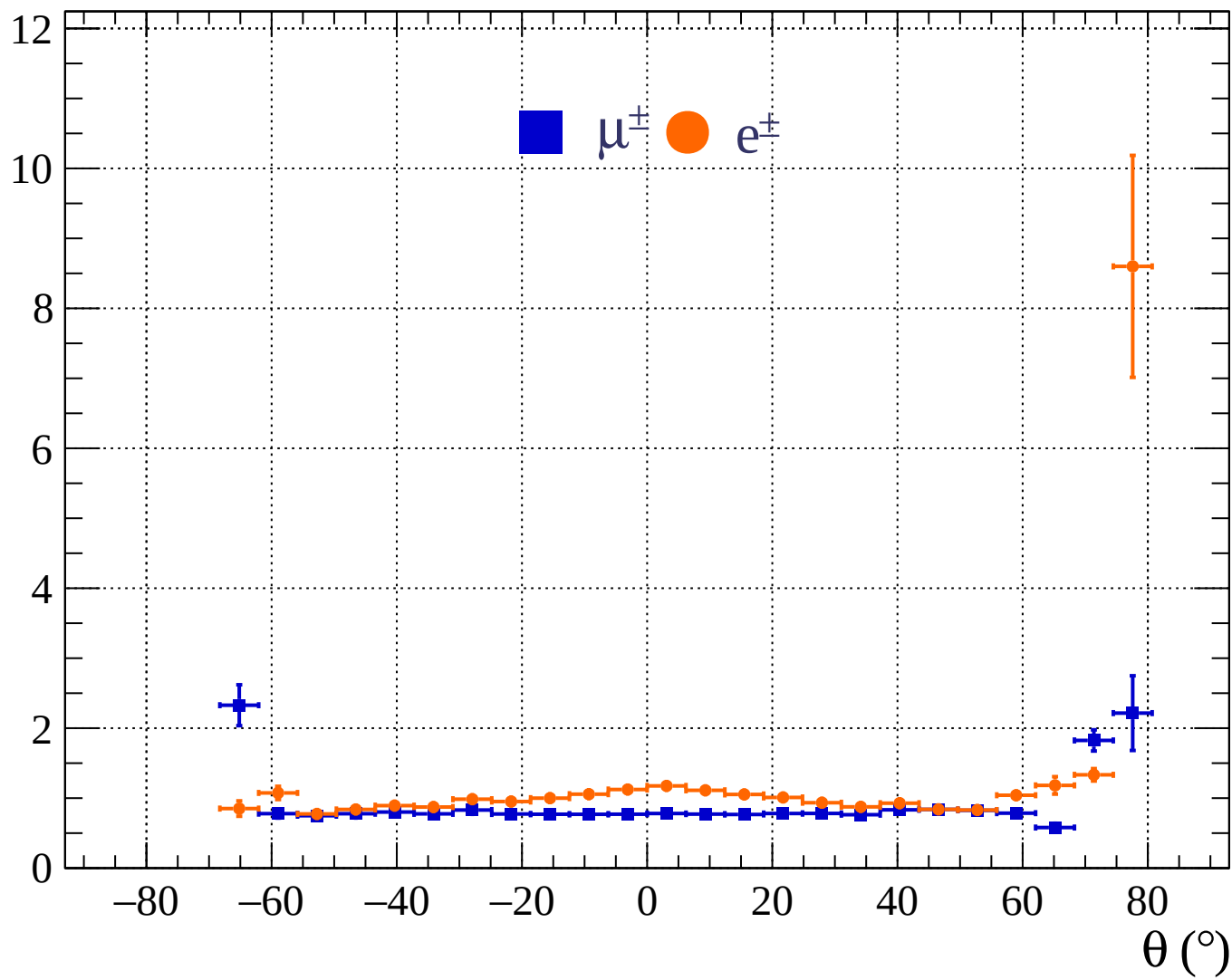




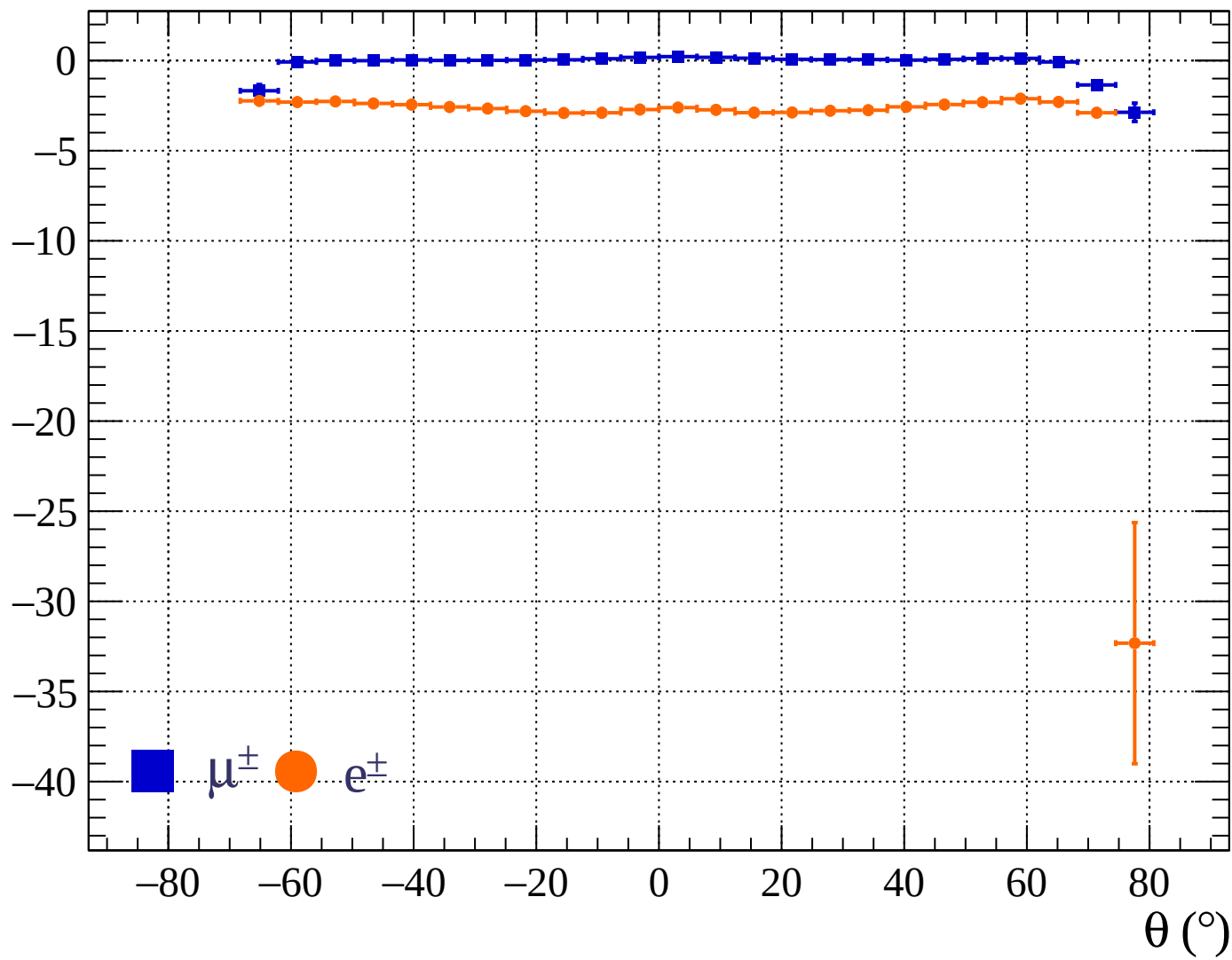




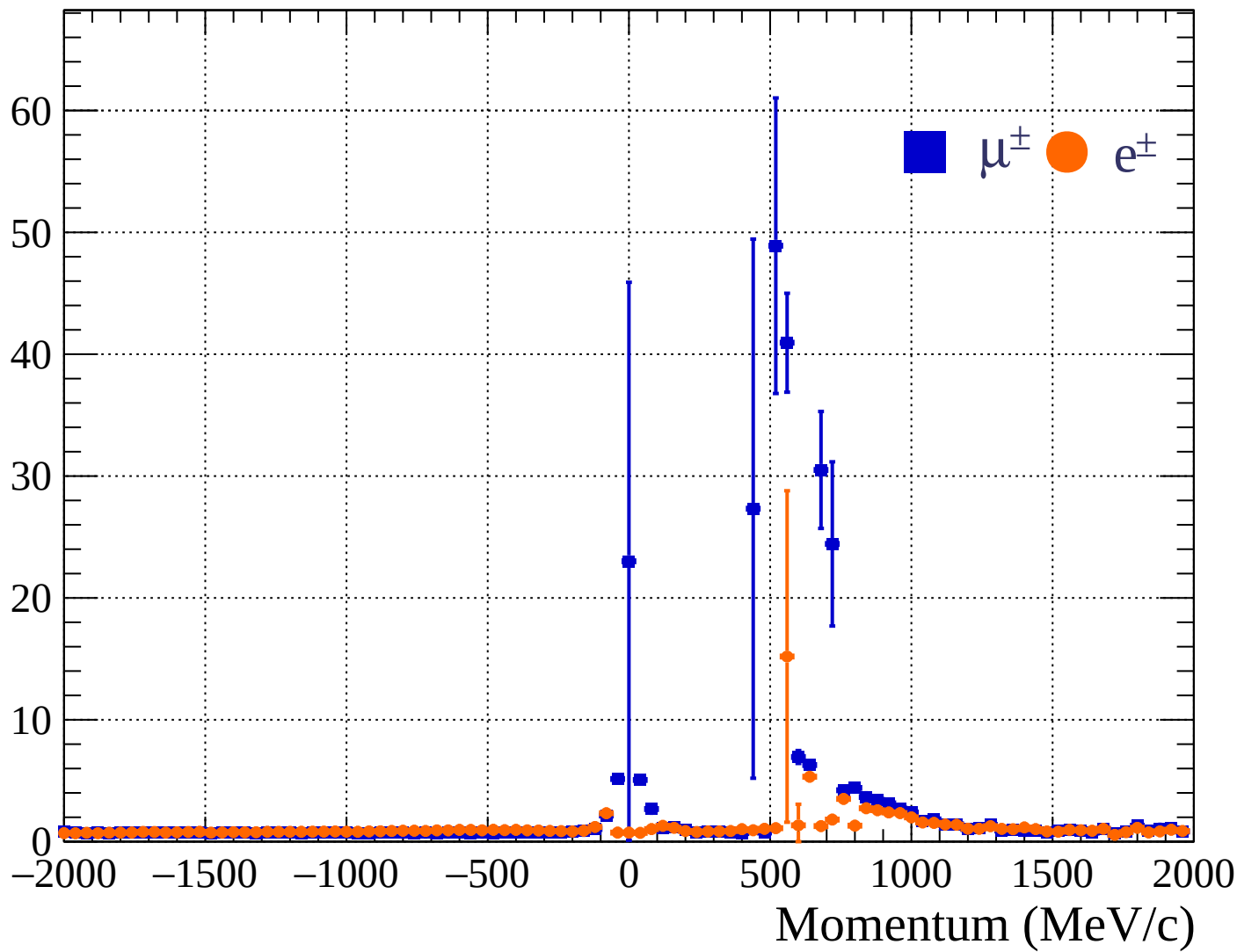
Pulls standard deviation

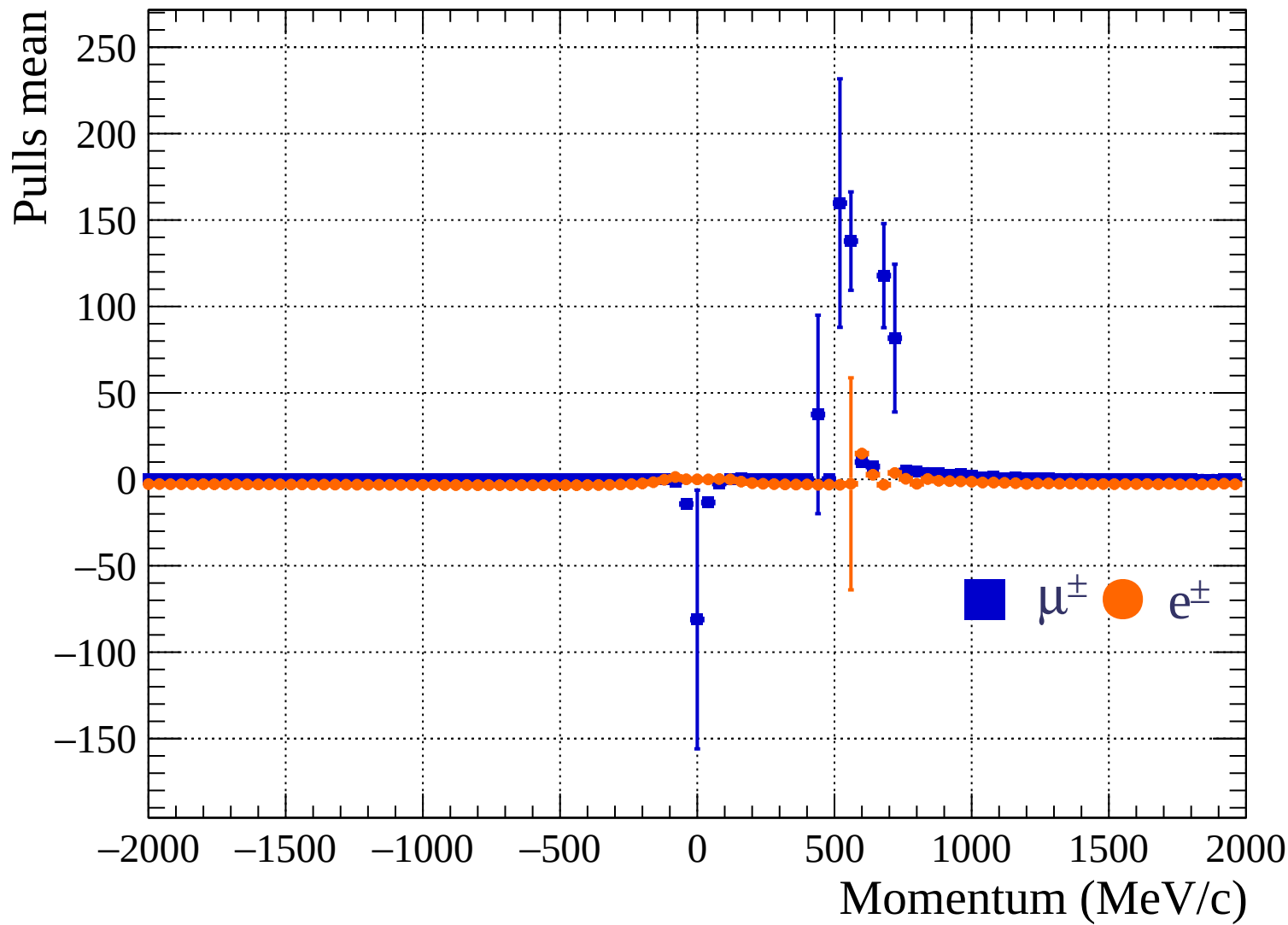


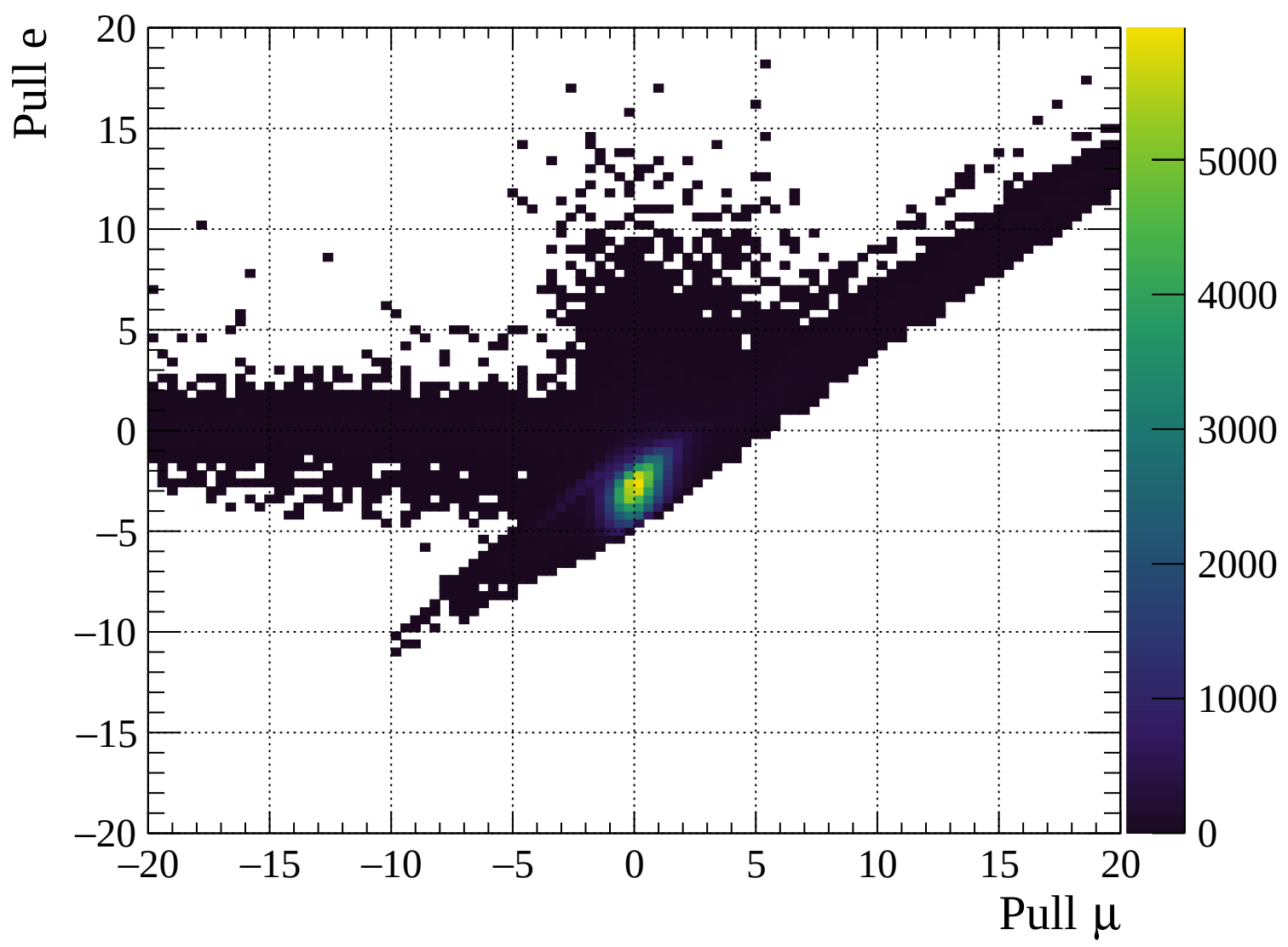
Pulls mean

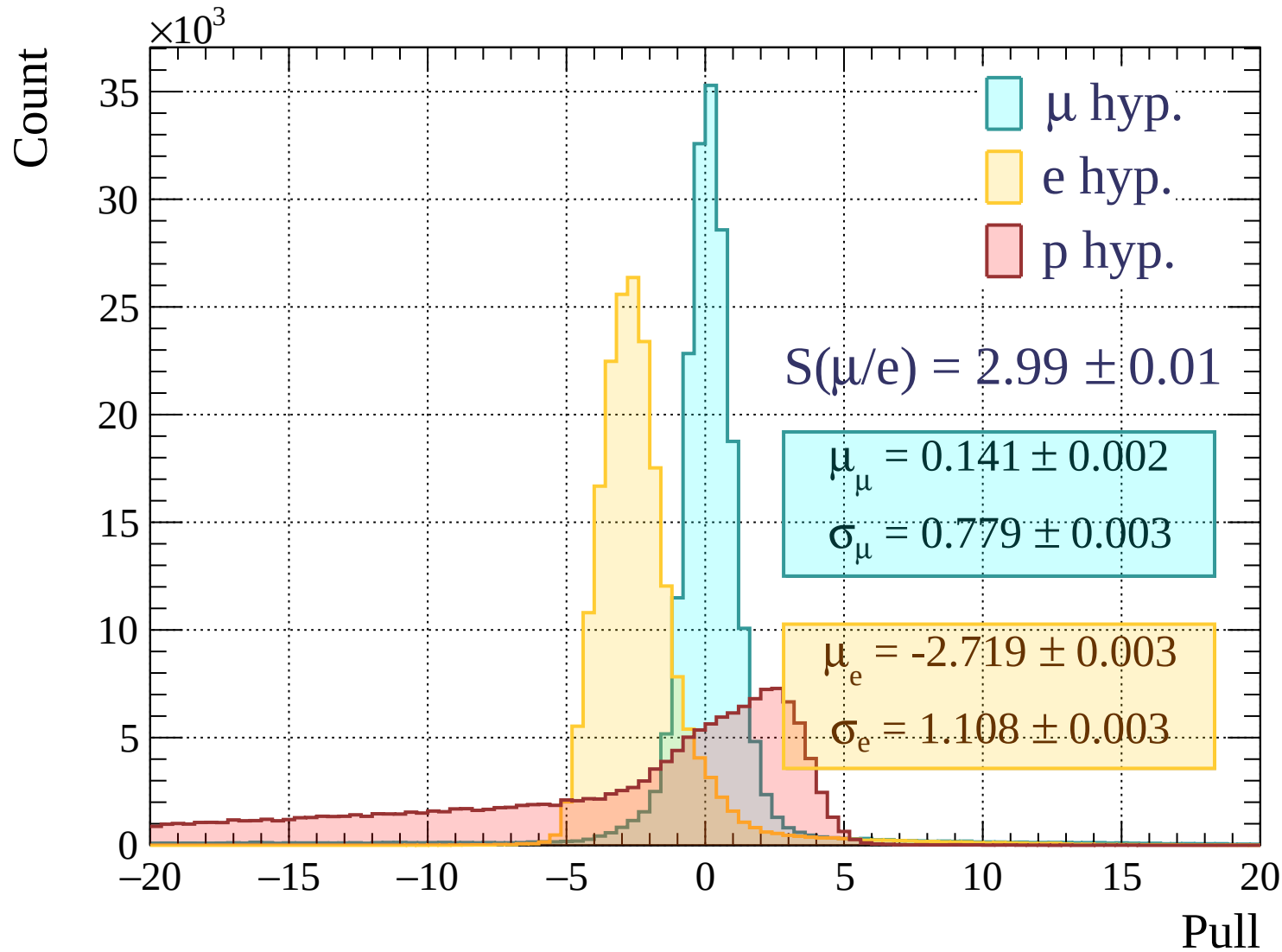


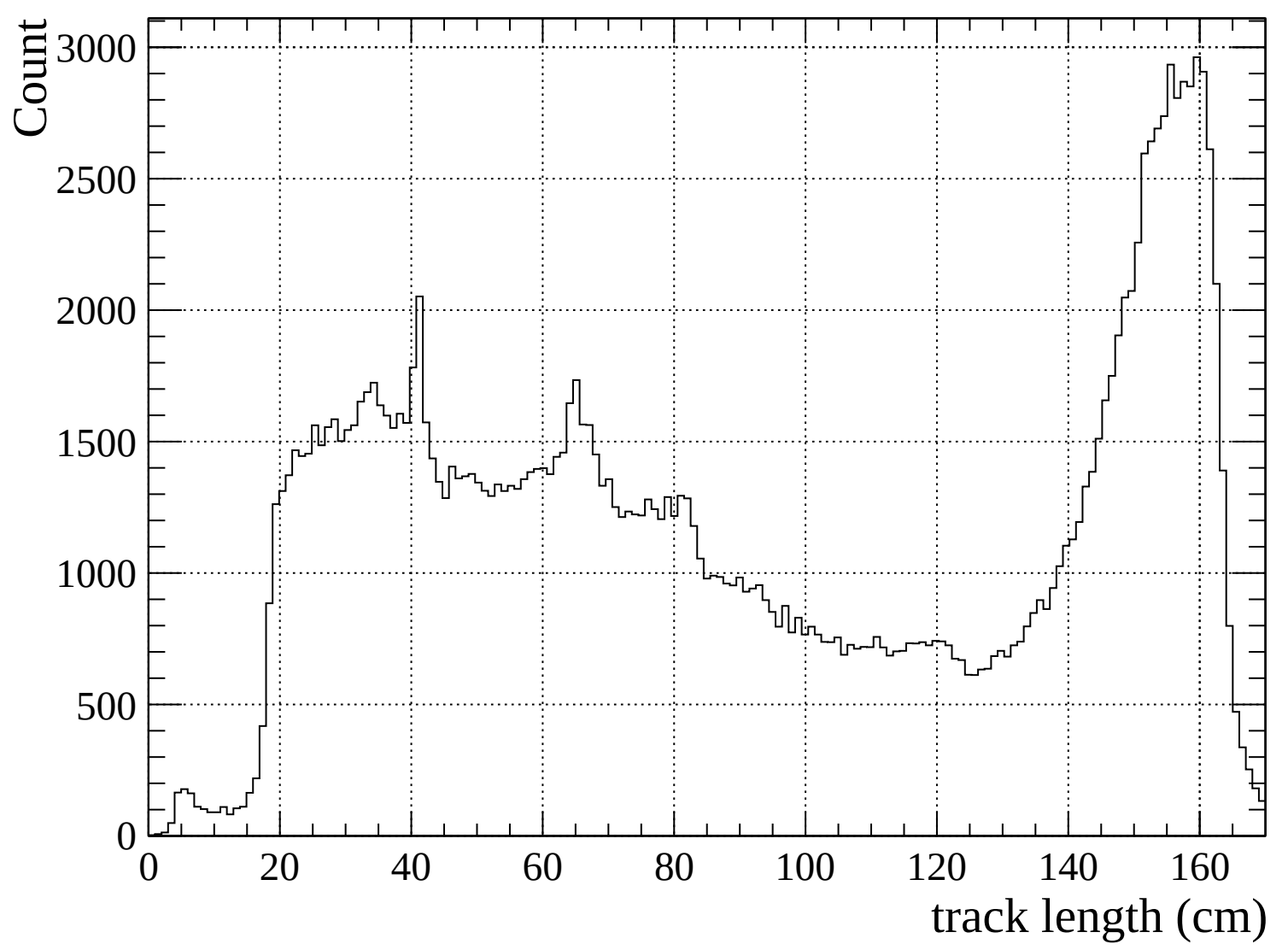
Pulls standard deviation

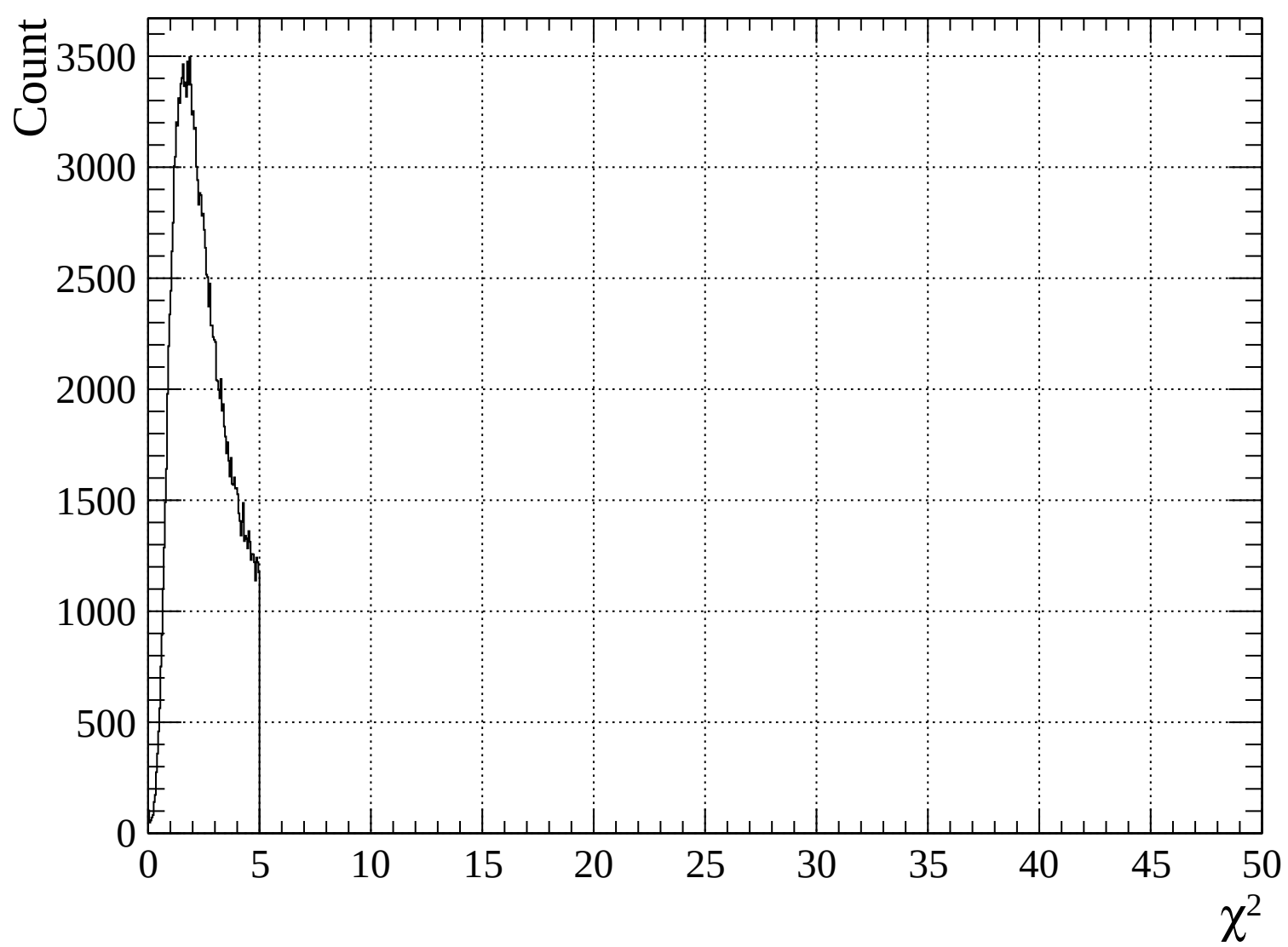


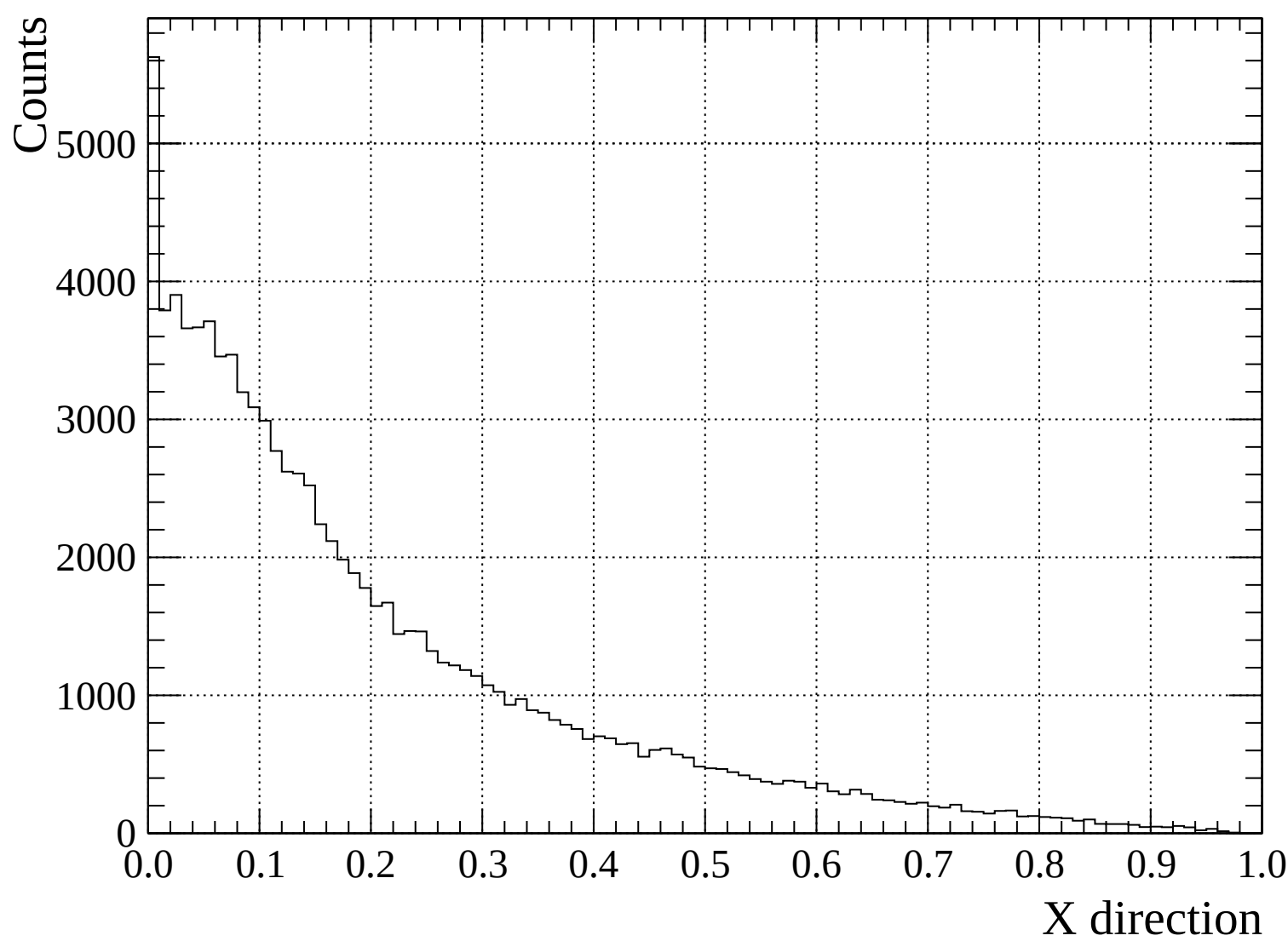


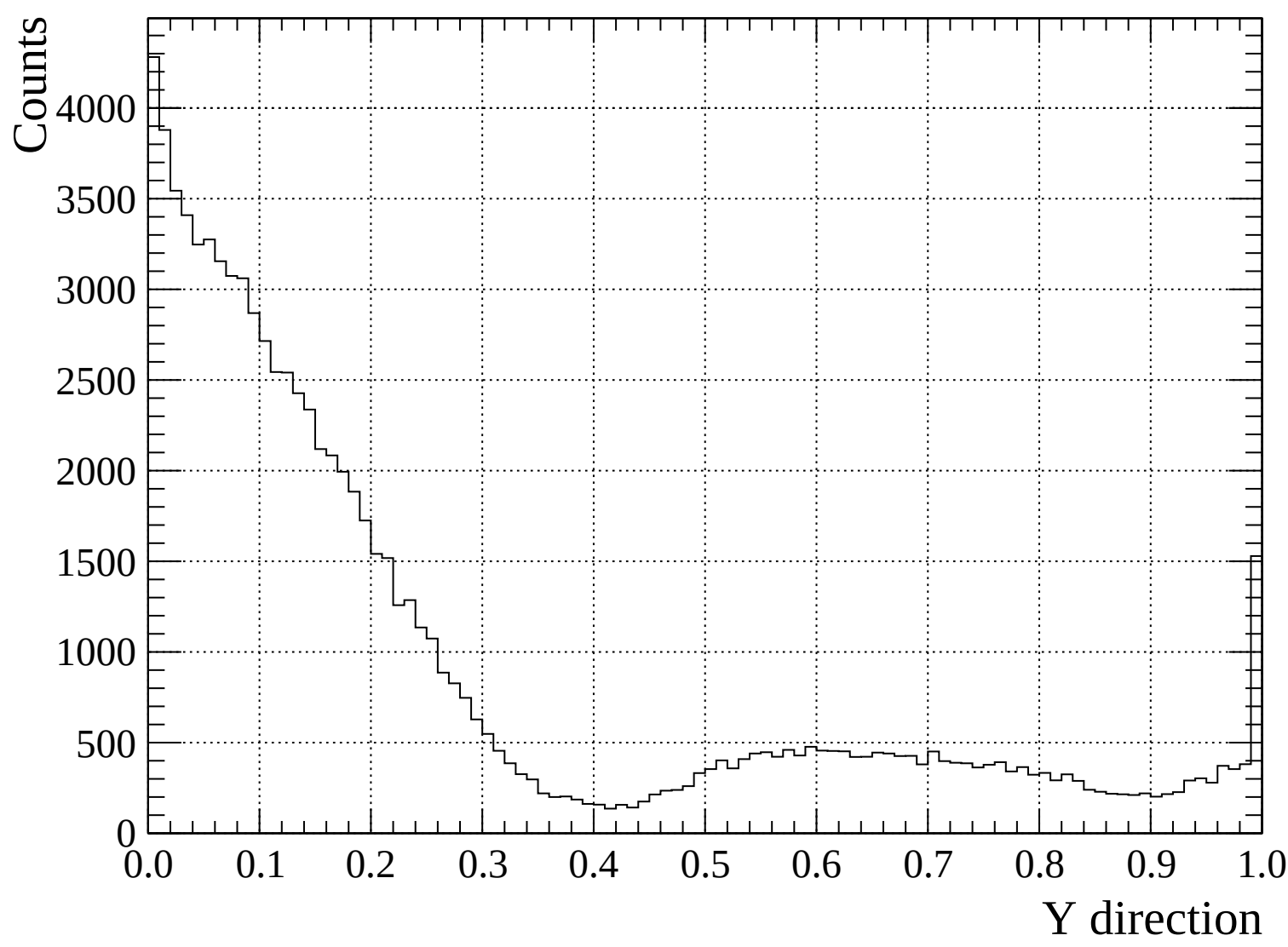


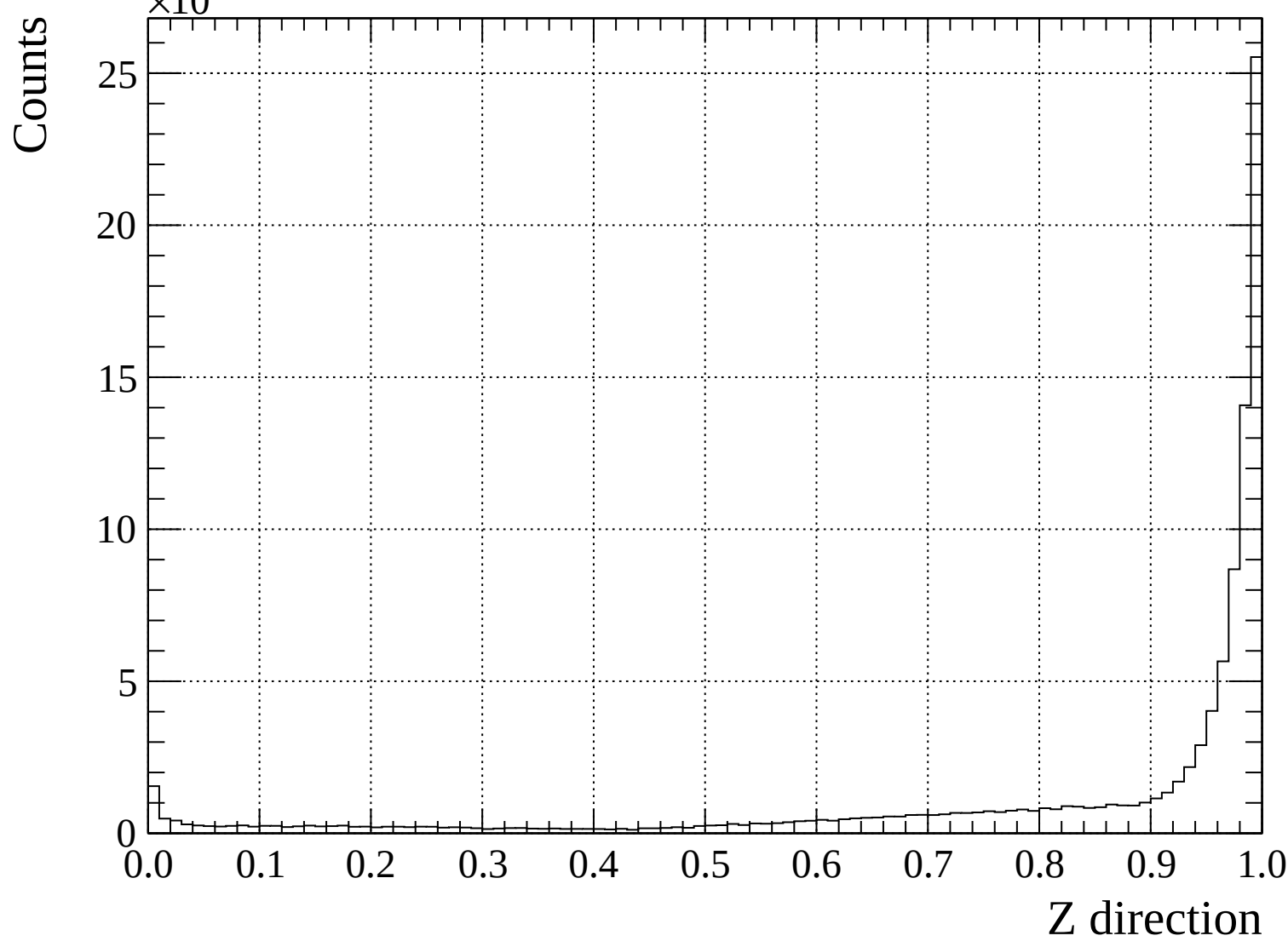


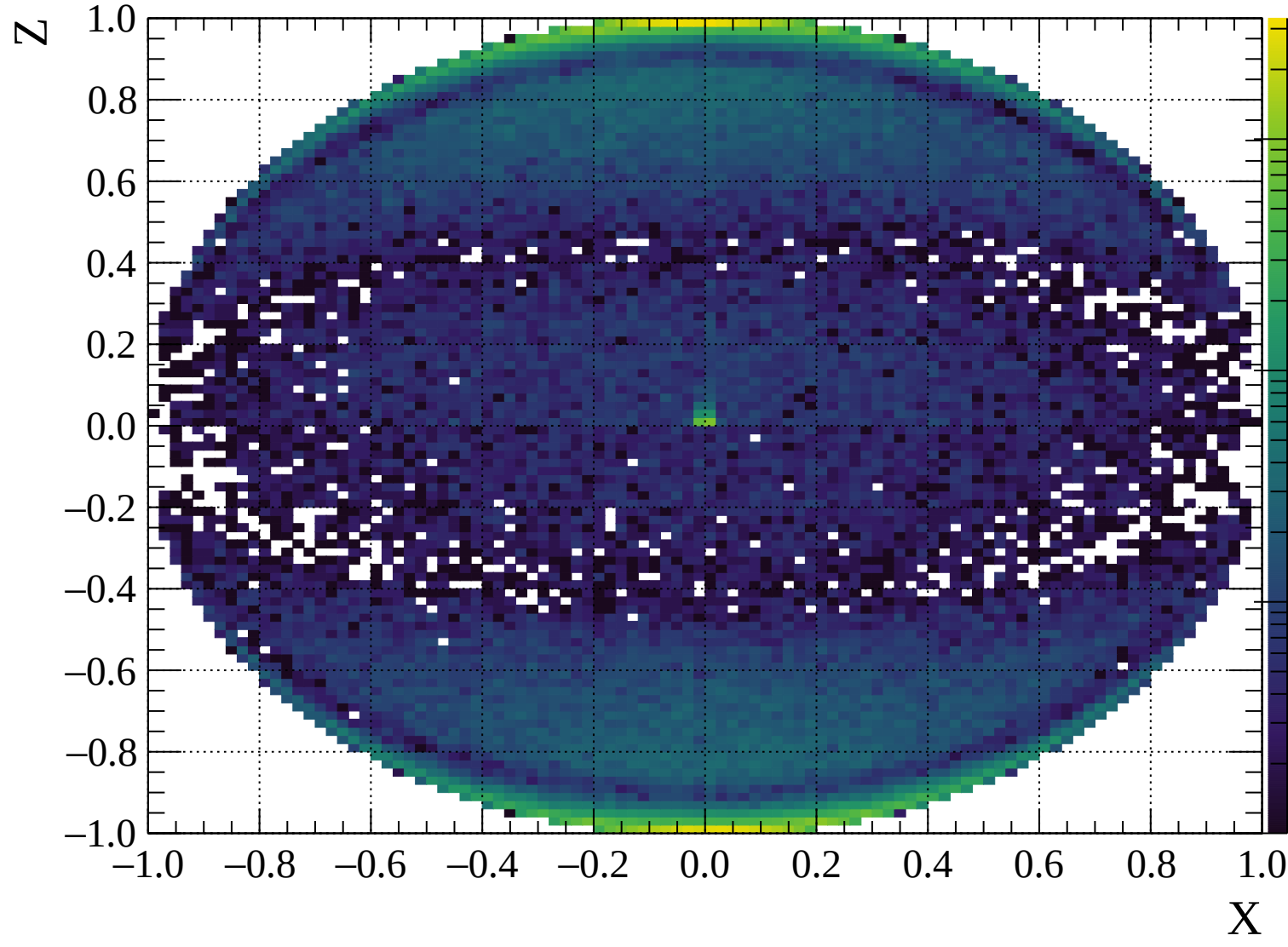


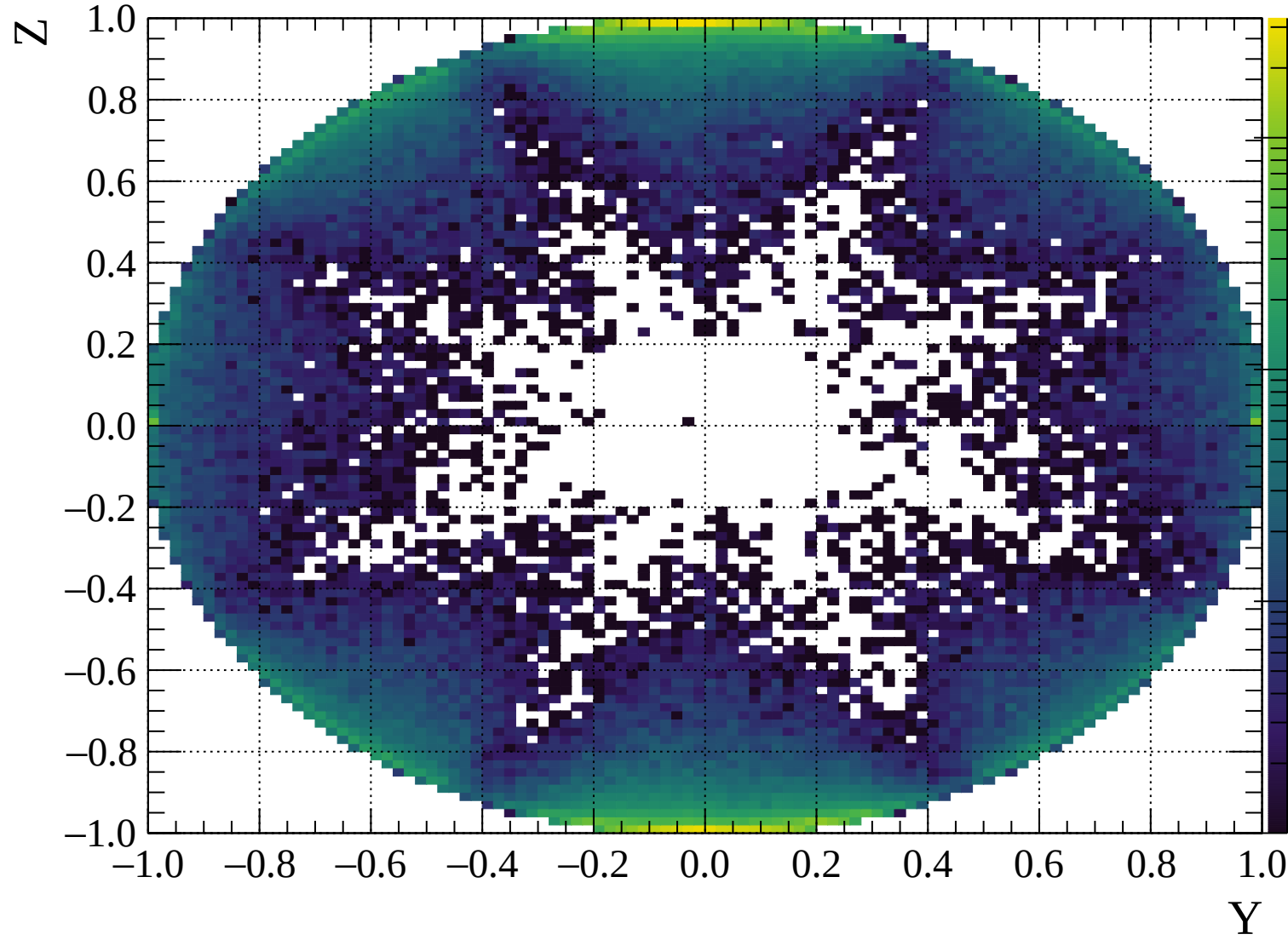


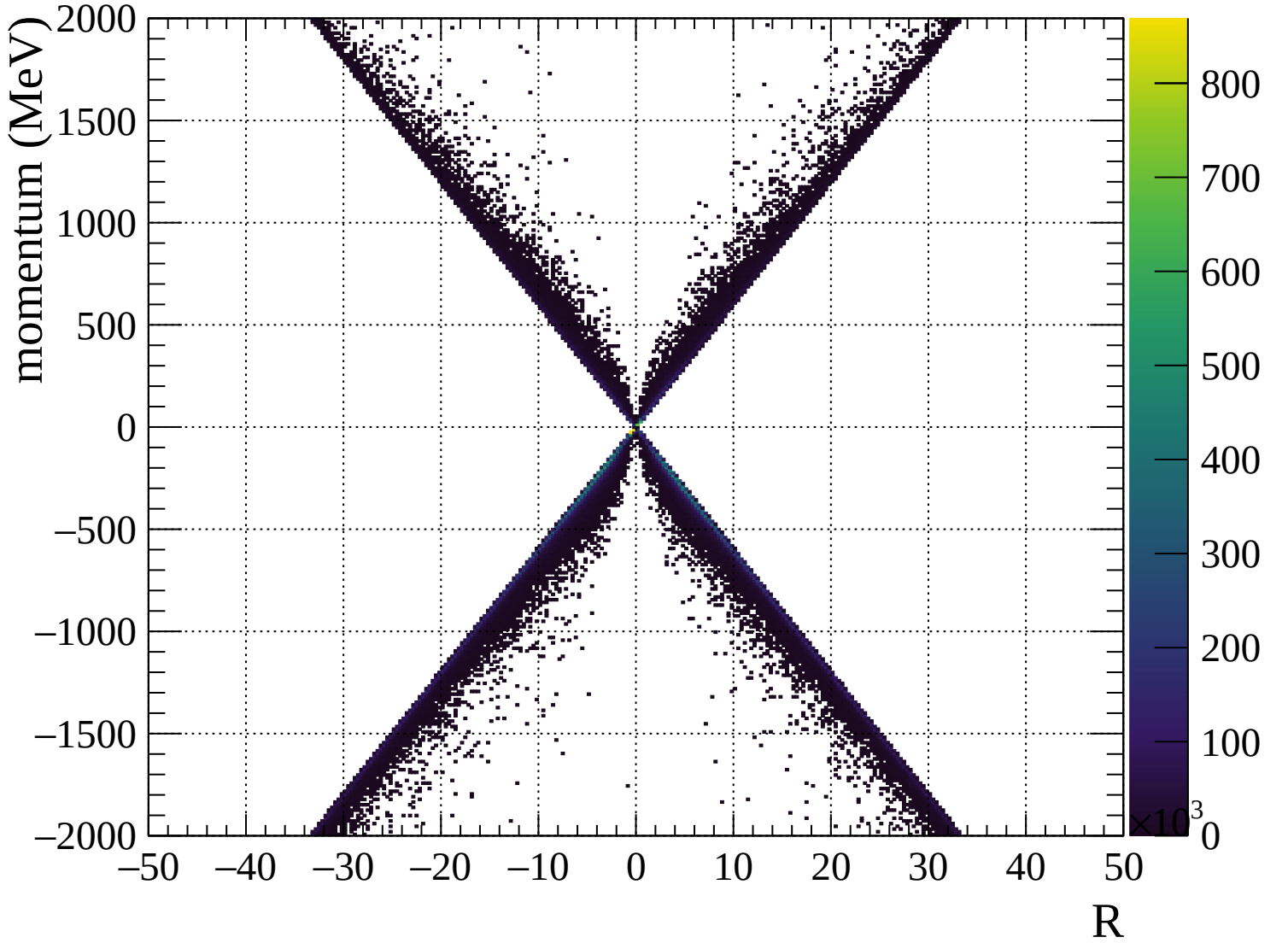


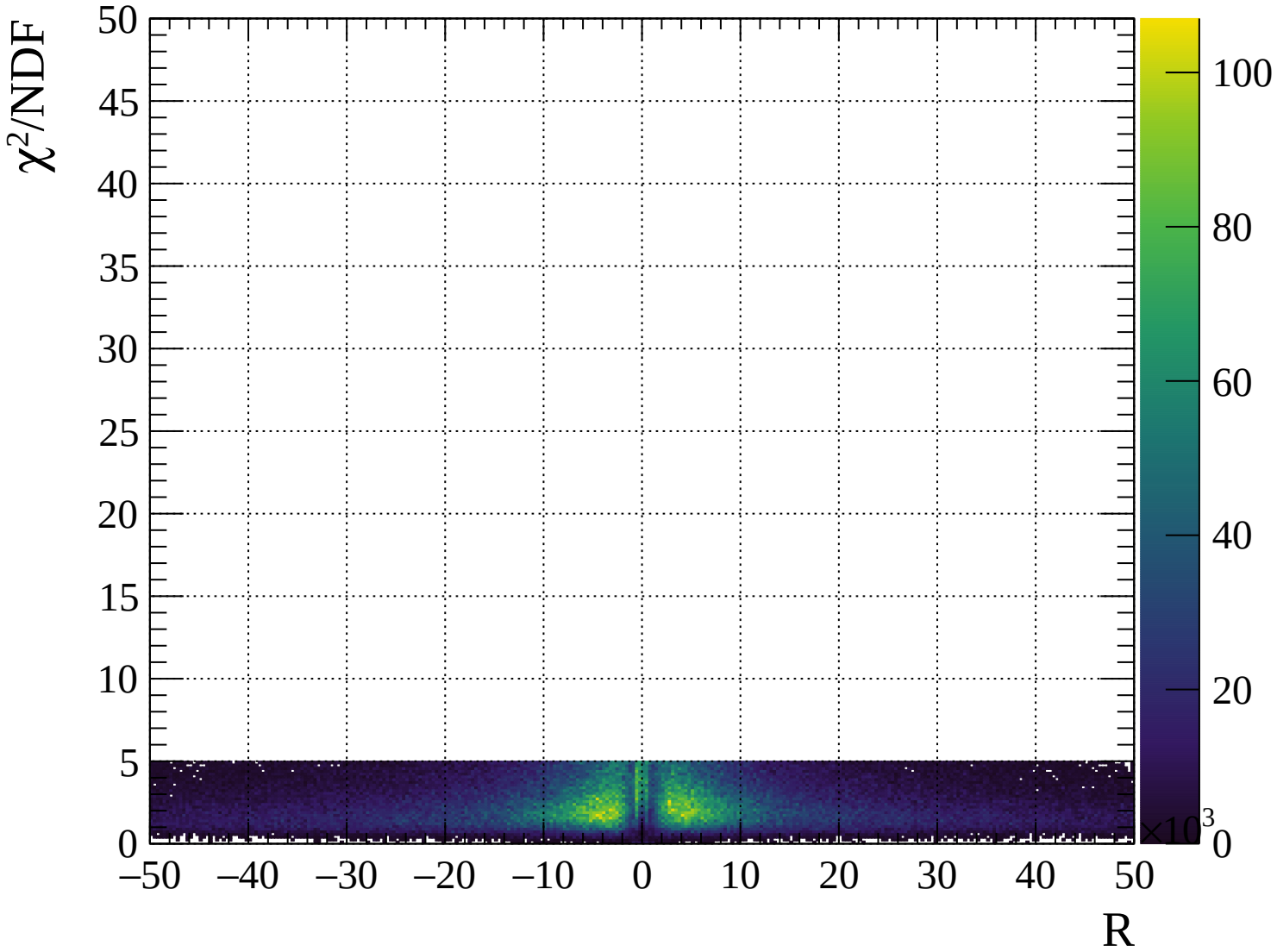




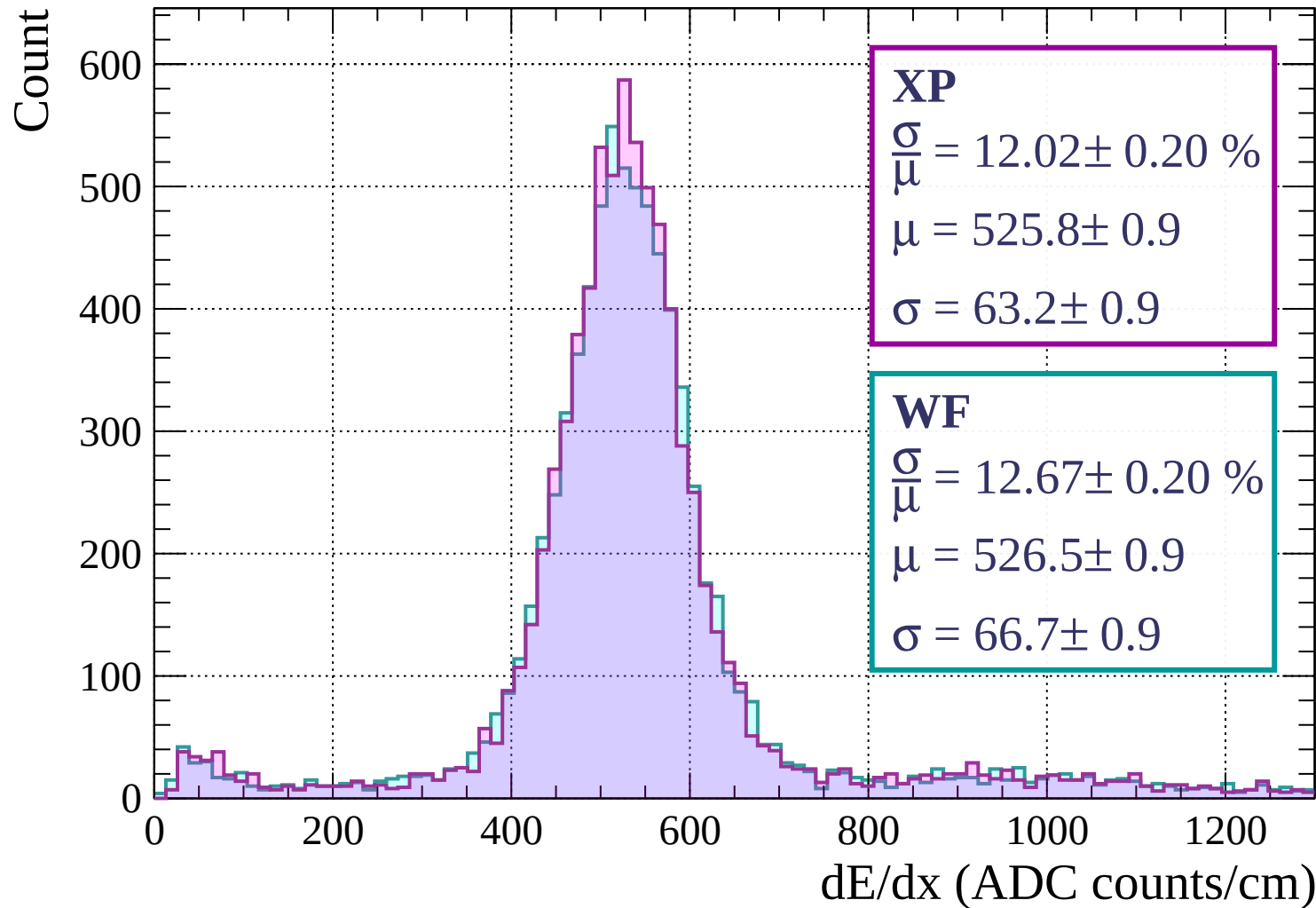






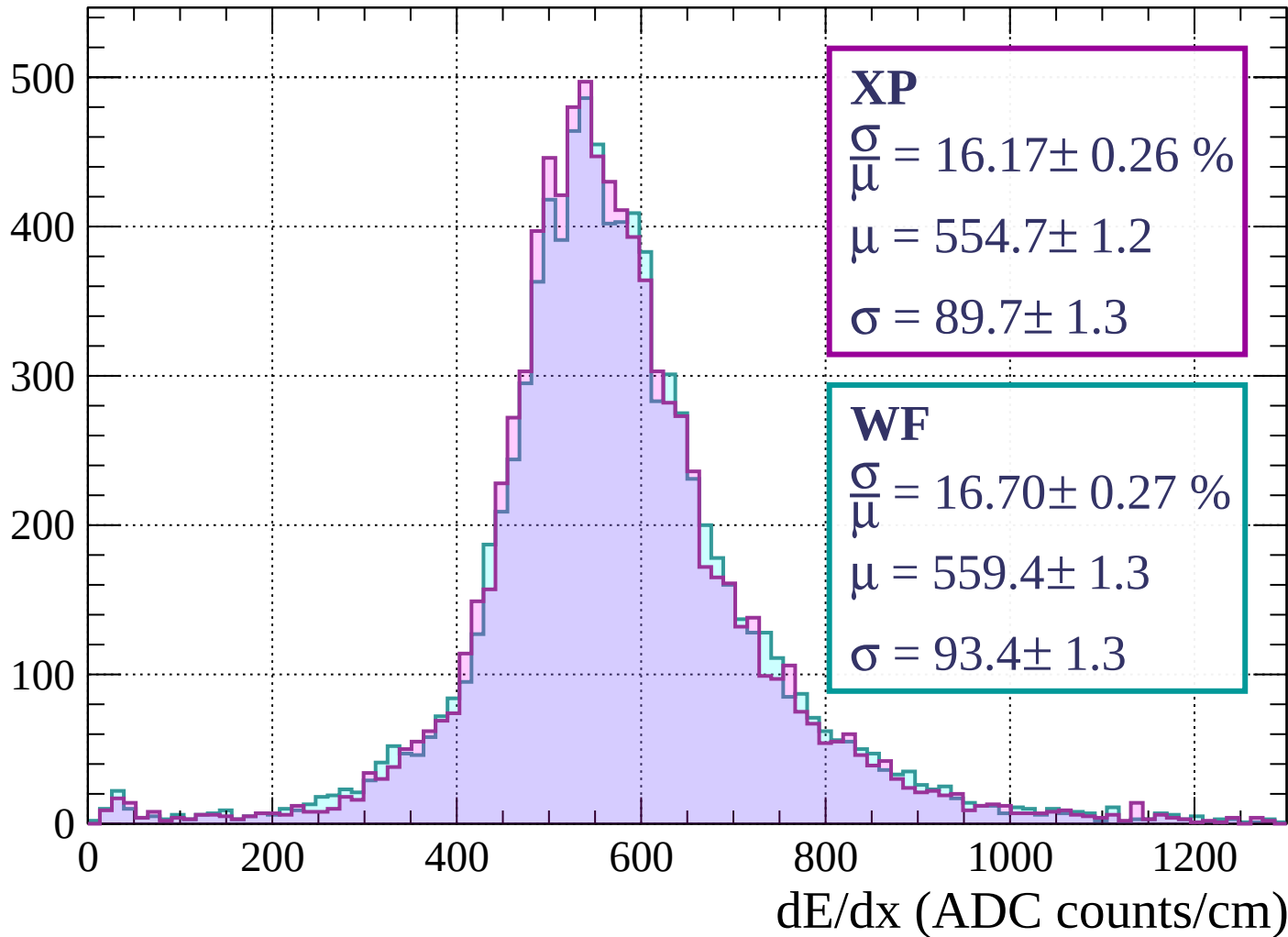


Energy loss | $0 < p < 80$

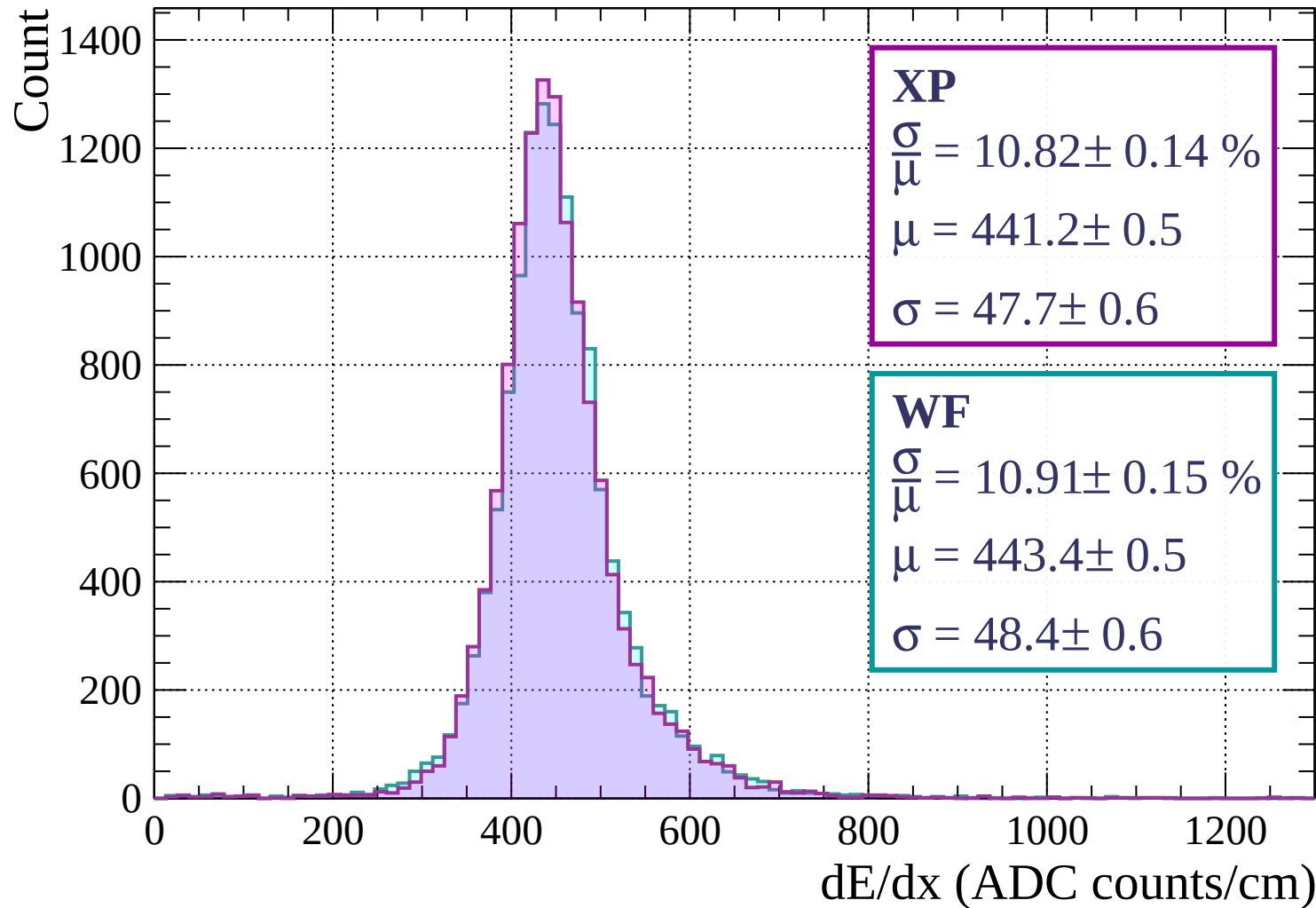


Energy loss | $80 < p < 160$

Count



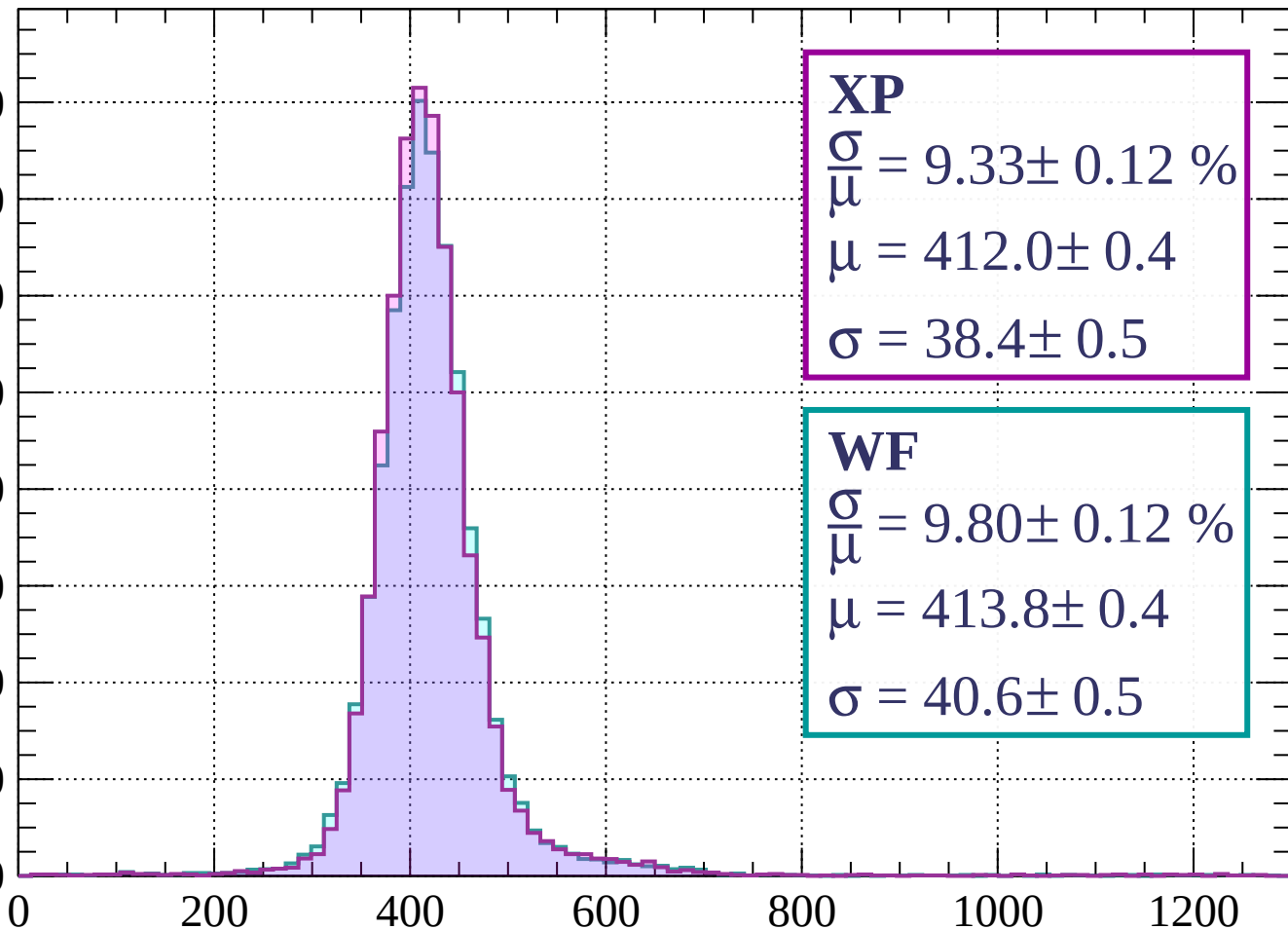
Energy loss | $160 < p < 240$



Energy loss | $240 < p < 320$

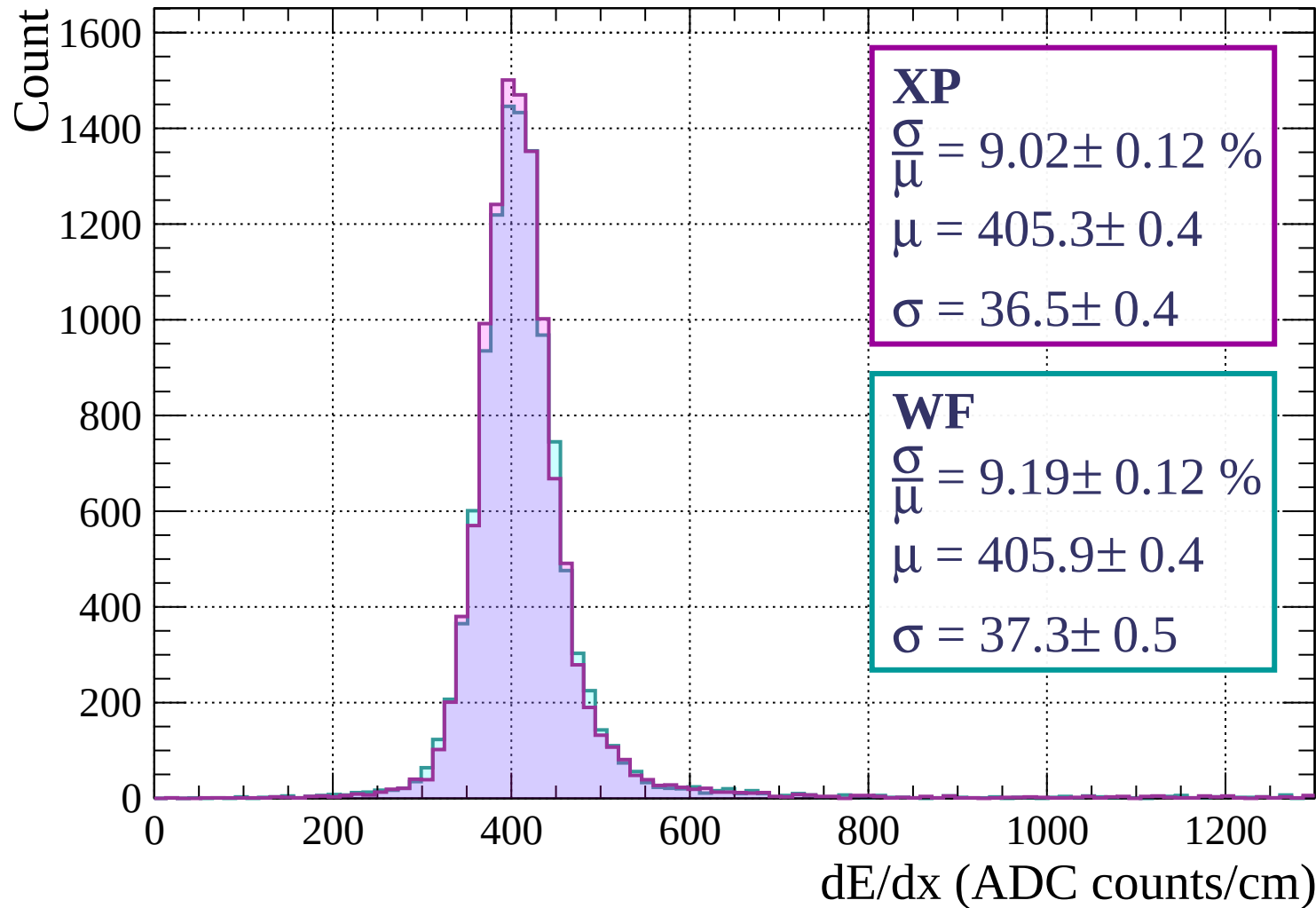
Count

1600
1400
1200
1000
800
600
400
200
0



dE/dx (ADC counts/cm)

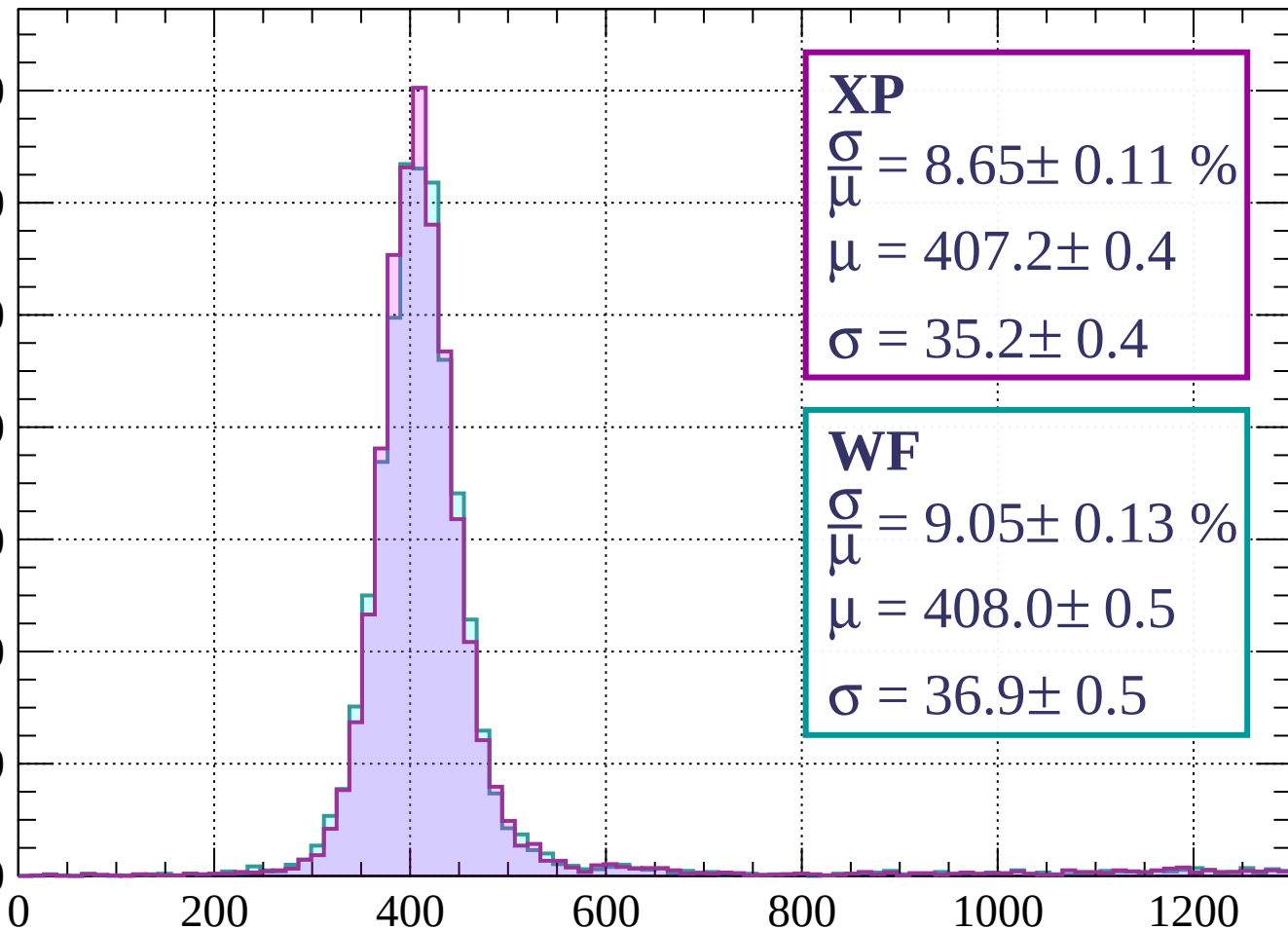
Energy loss | $320 < p < 400$



Energy loss | $400 < p < 480$

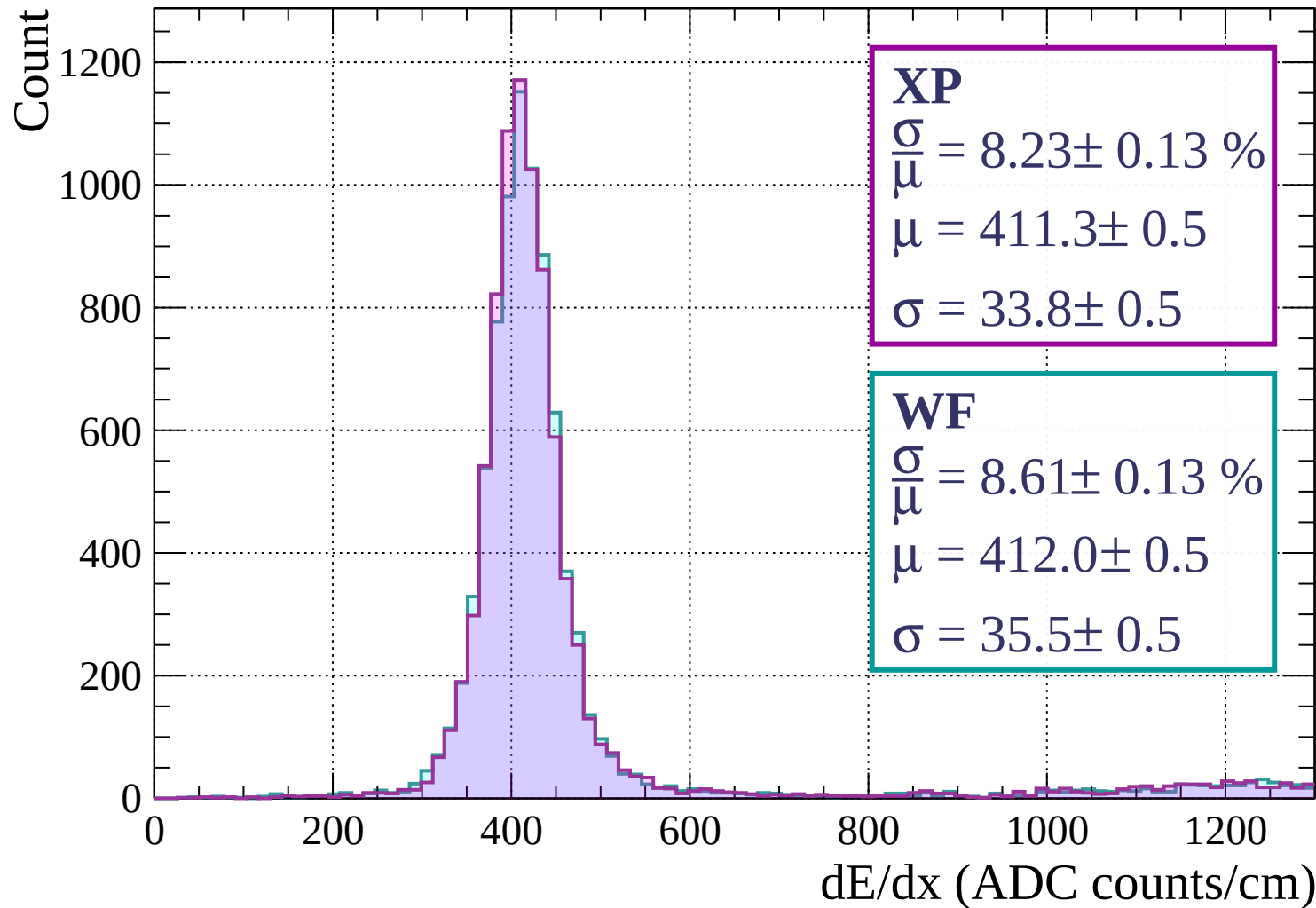
Count

1400
1200
1000
800
600
400
200
0



dE/dx (ADC counts/cm)

Energy loss | $480 < p < 560$



Energy loss | $560 < p < 640$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 8.19 \pm 0.14 \%$$

$$\mu = 415.9 \pm 0.5$$

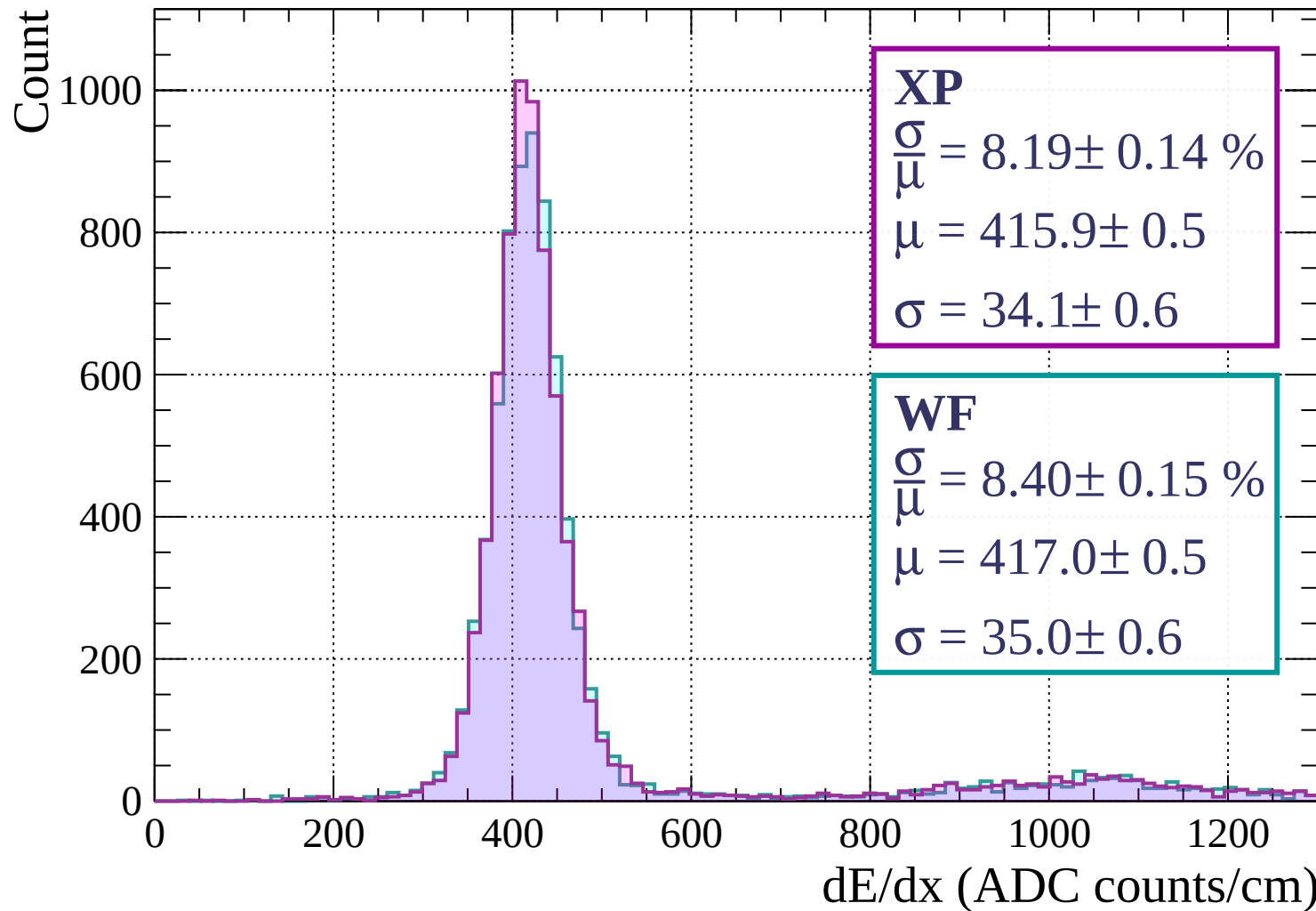
$$\sigma = 34.1 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 8.40 \pm 0.15 \%$$

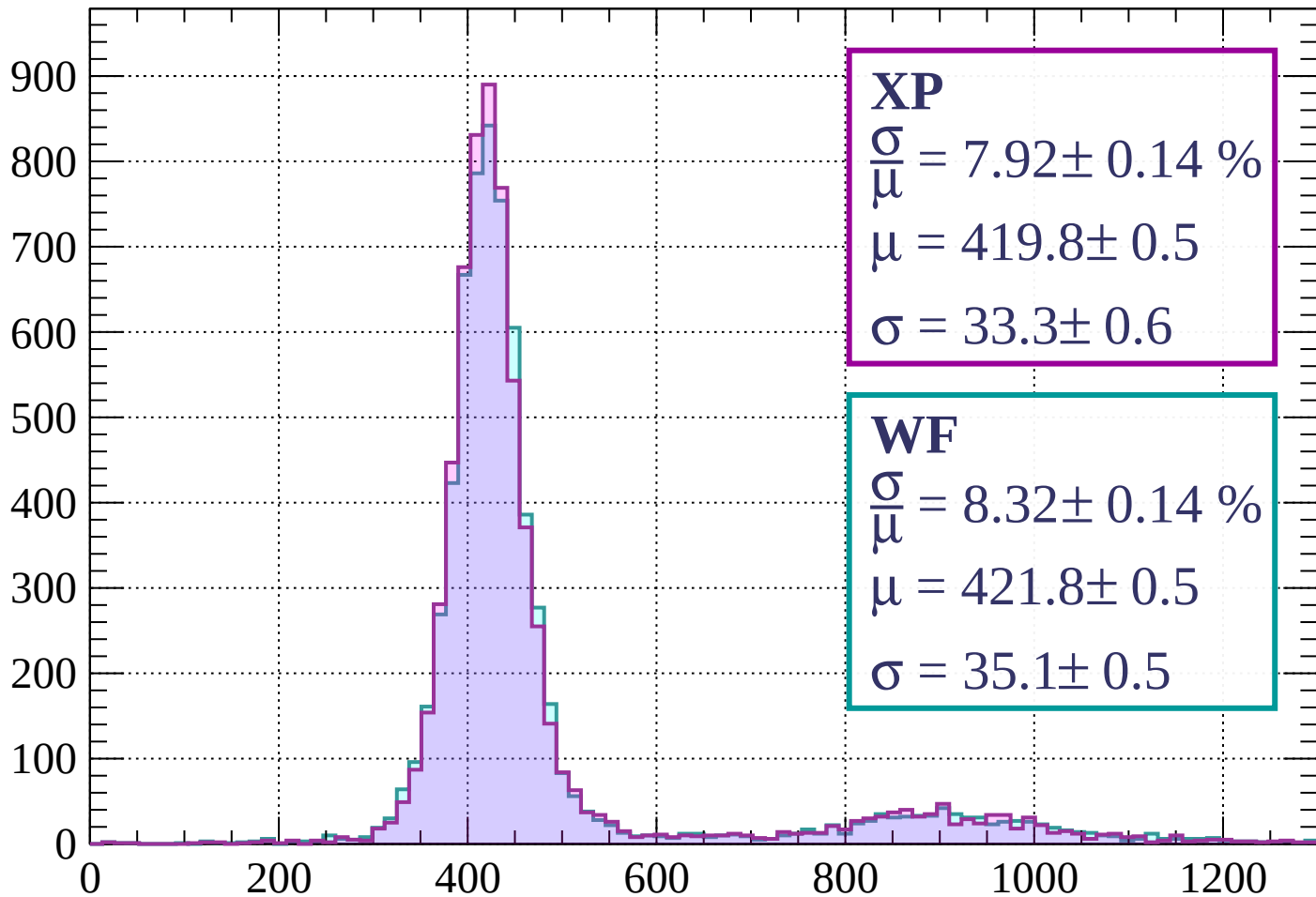
$$\mu = 417.0 \pm 0.5$$

$$\sigma = 35.0 \pm 0.6$$



Energy loss | $640 < p < 720$

Count



XP

$$\frac{\sigma}{\mu} = 7.92 \pm 0.14 \%$$

$$\mu = 419.8 \pm 0.5$$

$$\sigma = 33.3 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 8.32 \pm 0.14 \%$$

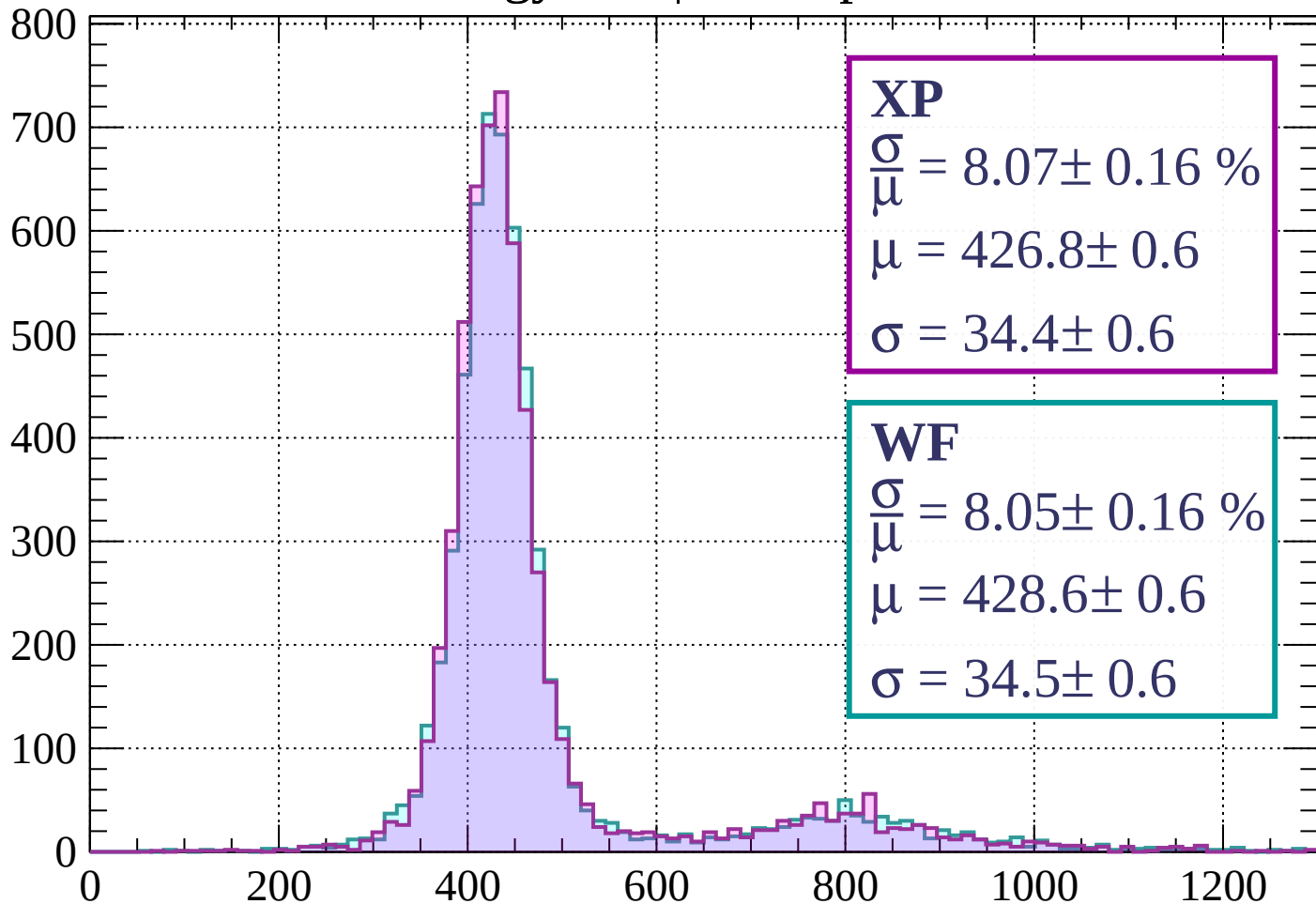
$$\mu = 421.8 \pm 0.5$$

$$\sigma = 35.1 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $720 < p < 800$

Count



XP

$$\frac{\sigma}{\mu} = 8.07 \pm 0.16 \%$$

$$\mu = 426.8 \pm 0.6$$

$$\sigma = 34.4 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 8.05 \pm 0.16 \%$$

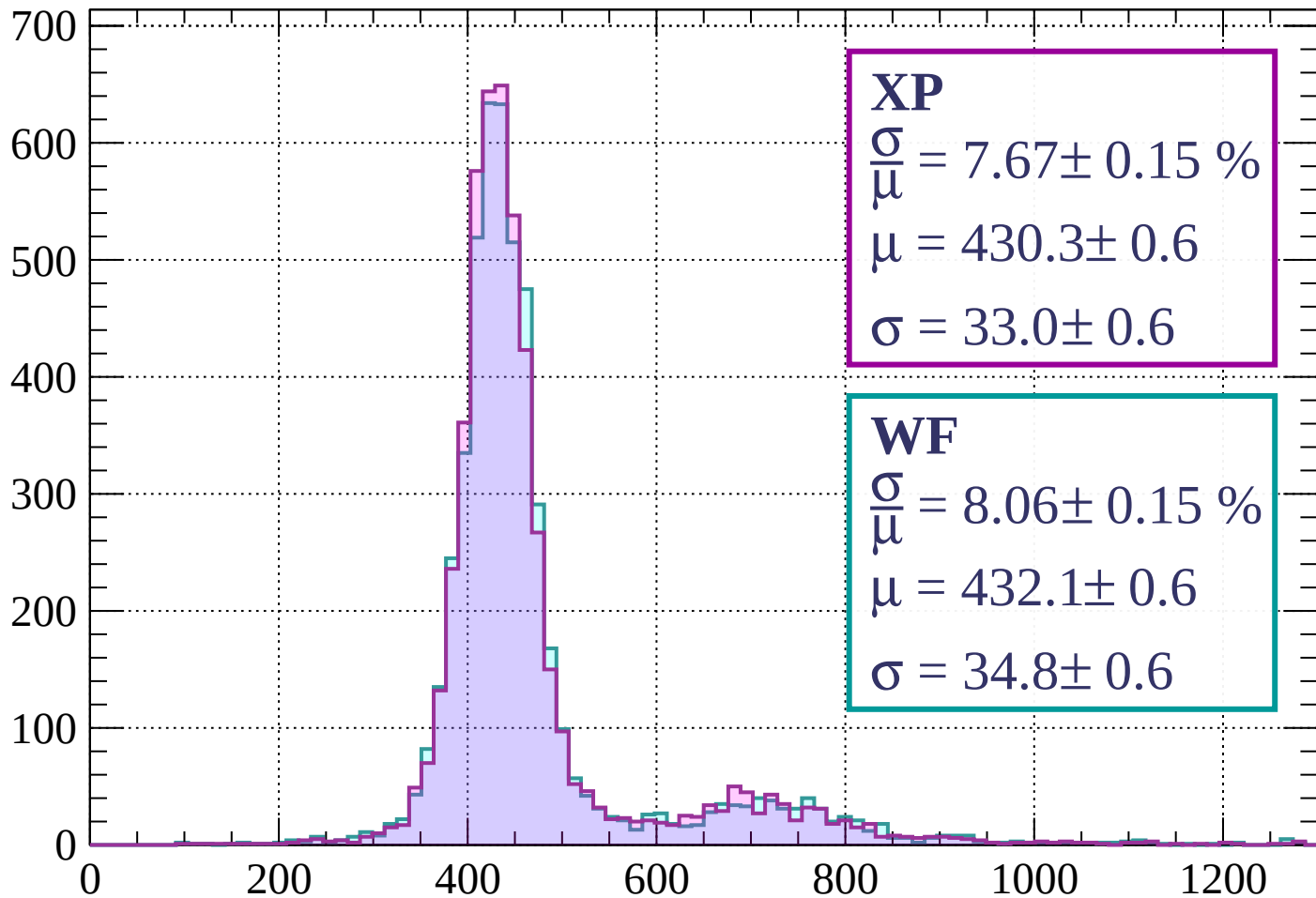
$$\mu = 428.6 \pm 0.6$$

$$\sigma = 34.5 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $800 < p < 880$

Count



XP

$$\frac{\sigma}{\mu} = 7.67 \pm 0.15 \%$$

$$\mu = 430.3 \pm 0.6$$

$$\sigma = 33.0 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 8.06 \pm 0.15 \%$$

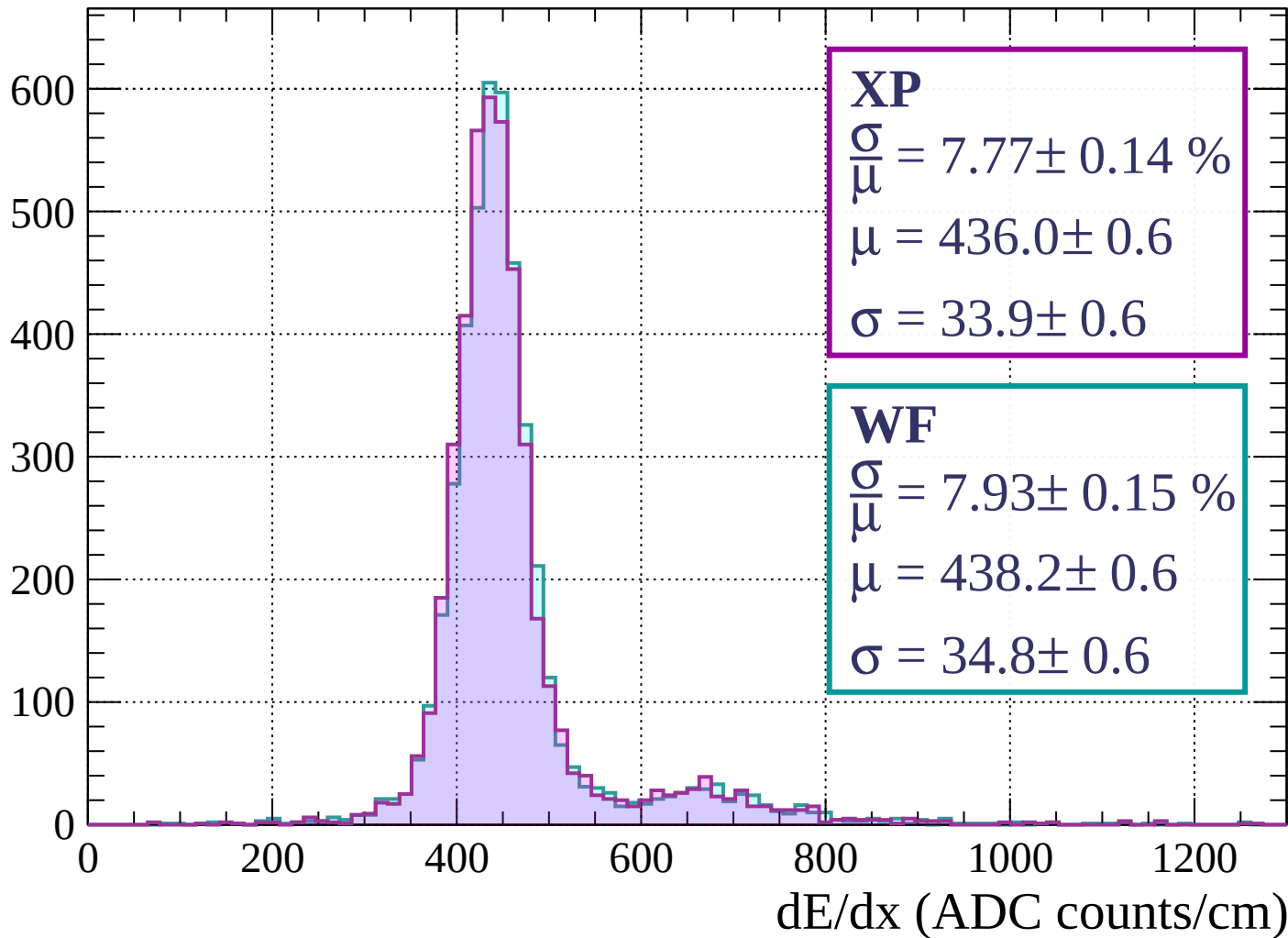
$$\mu = 432.1 \pm 0.6$$

$$\sigma = 34.8 \pm 0.6$$

dE/dx (ADC counts/cm)

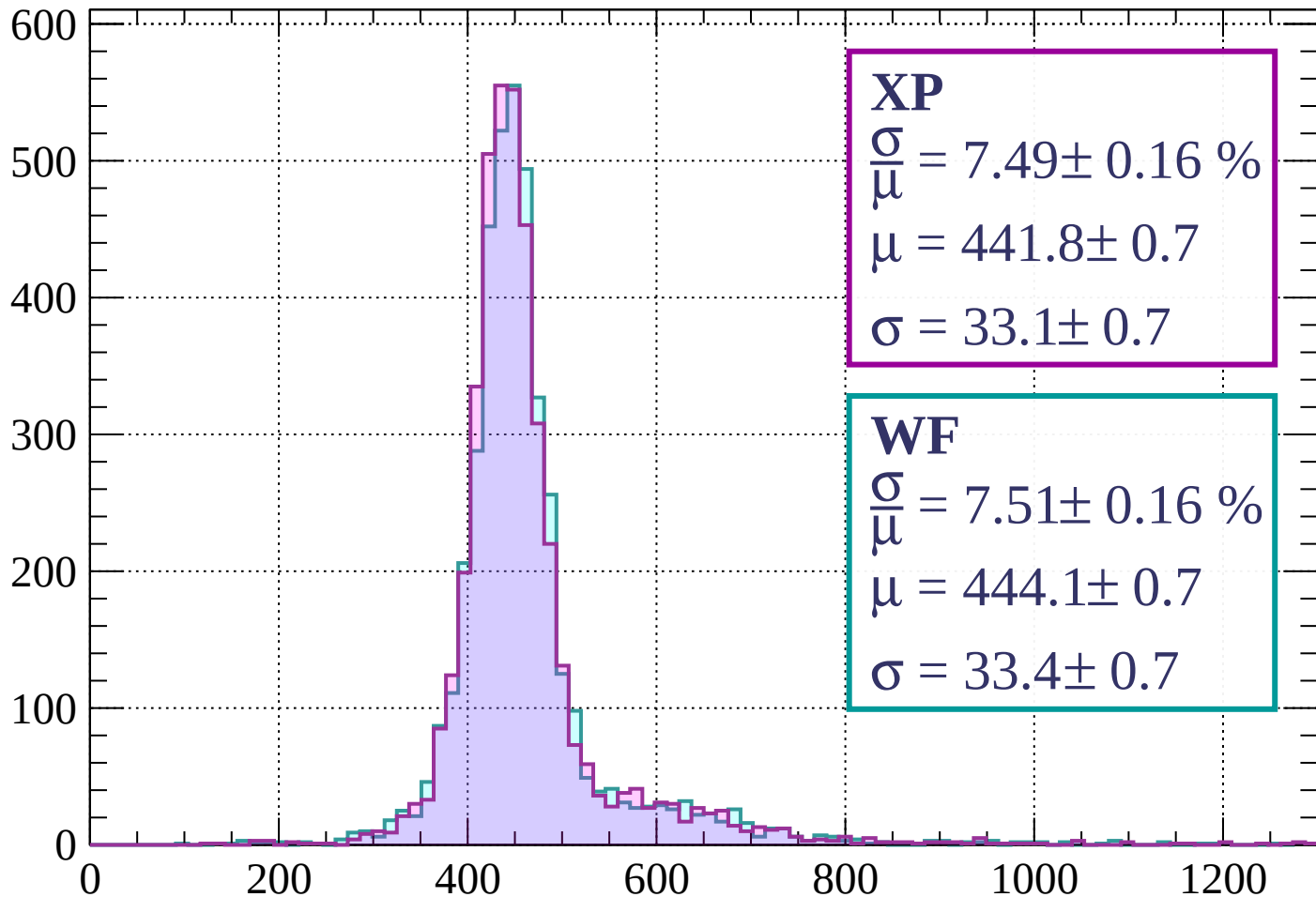
Energy loss | $880 < p < 960$

Count



Energy loss | $960 < p < 1040$

Count



XP

$$\frac{\sigma}{\mu} = 7.49 \pm 0.16 \%$$

$$\mu = 441.8 \pm 0.7$$

$$\sigma = 33.1 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 7.51 \pm 0.16 \%$$

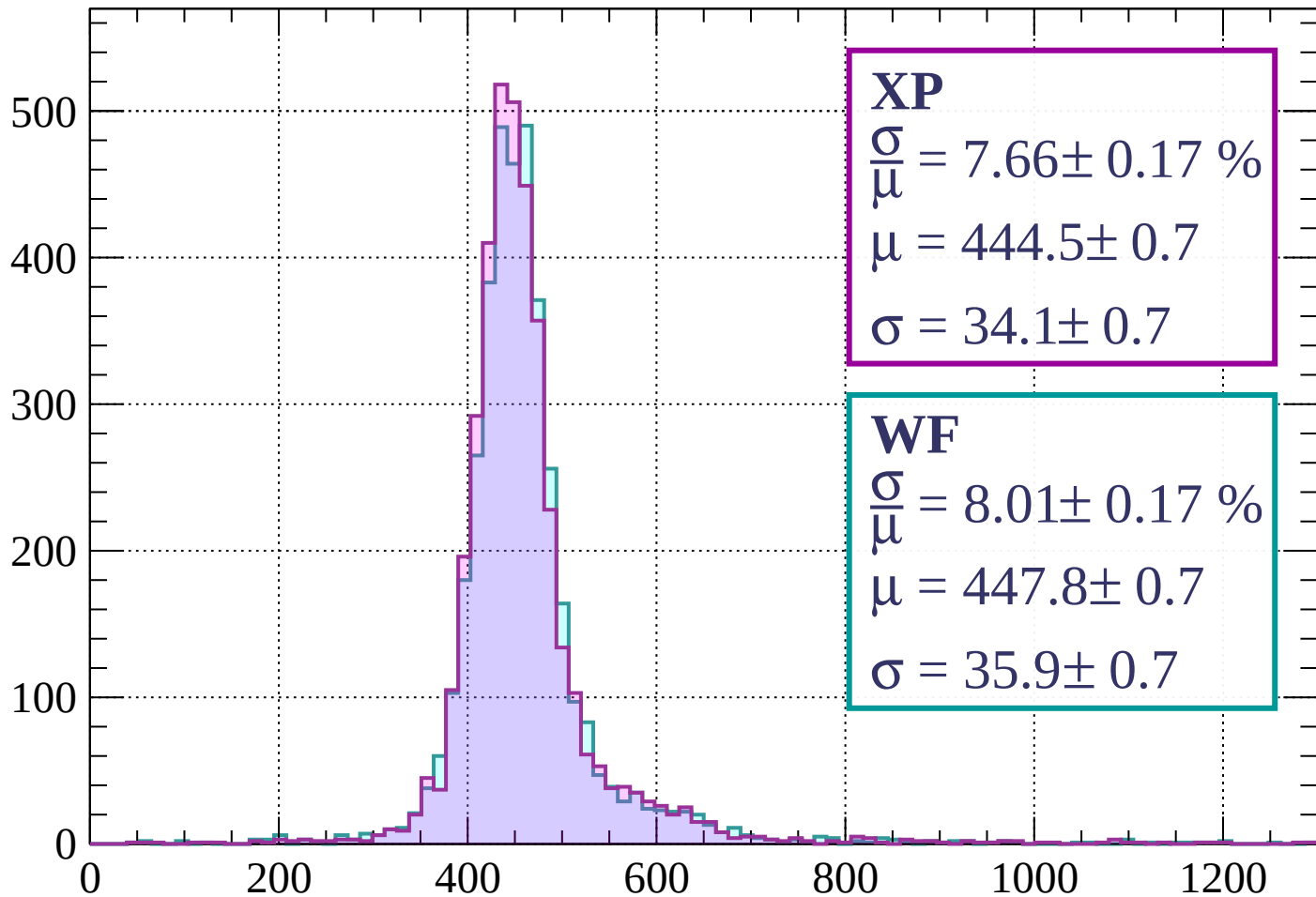
$$\mu = 444.1 \pm 0.7$$

$$\sigma = 33.4 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1040 < p < 1120$

Count



XP

$$\frac{\sigma}{\mu} = 7.66 \pm 0.17 \%$$

$$\mu = 444.5 \pm 0.7$$

$$\sigma = 34.1 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.01 \pm 0.17 \%$$

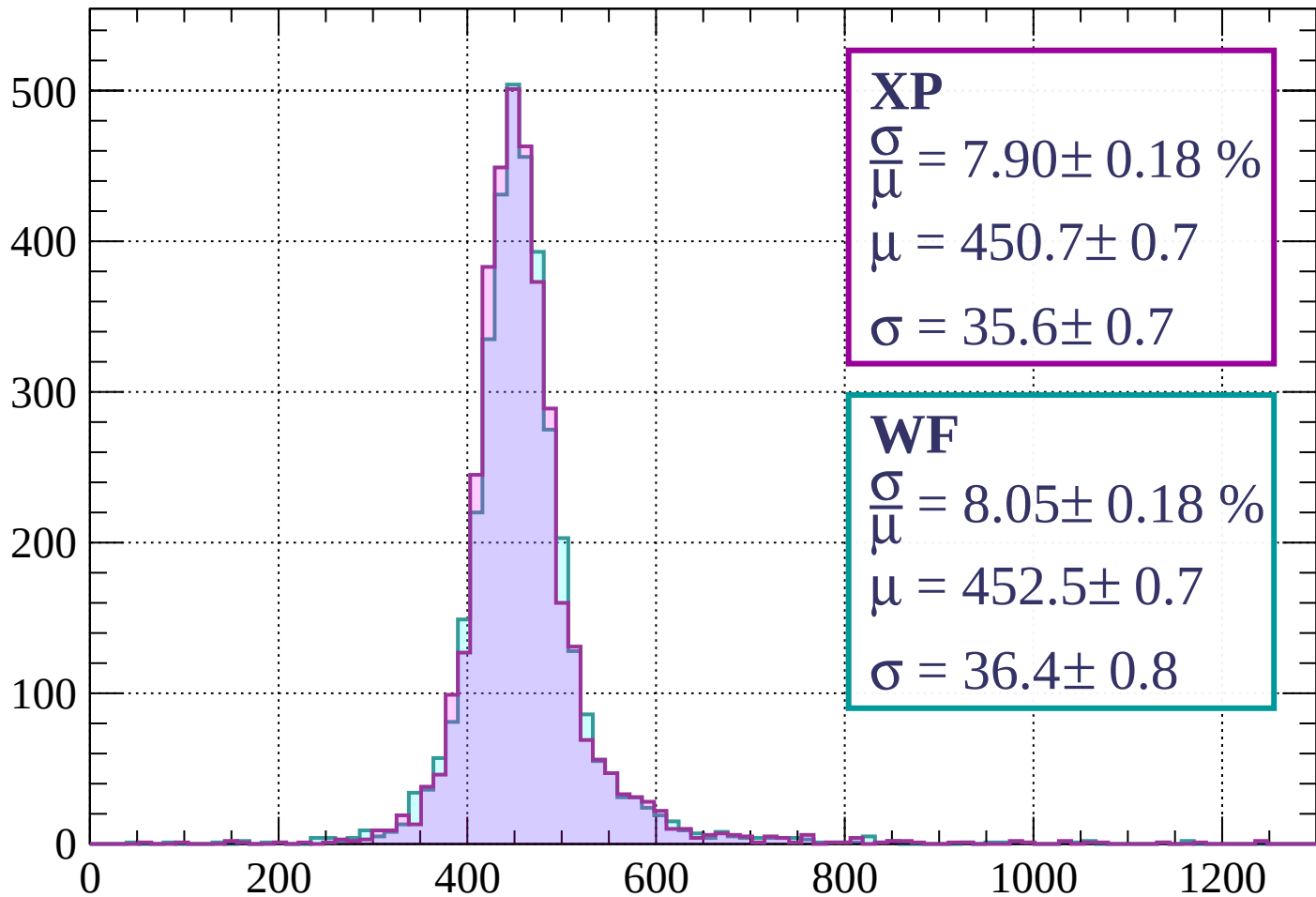
$$\mu = 447.8 \pm 0.7$$

$$\sigma = 35.9 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1120 < p < 1200$

Count



XP

$$\frac{\sigma}{\mu} = 7.90 \pm 0.18 \%$$

$$\mu = 450.7 \pm 0.7$$

$$\sigma = 35.6 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.05 \pm 0.18 \%$$

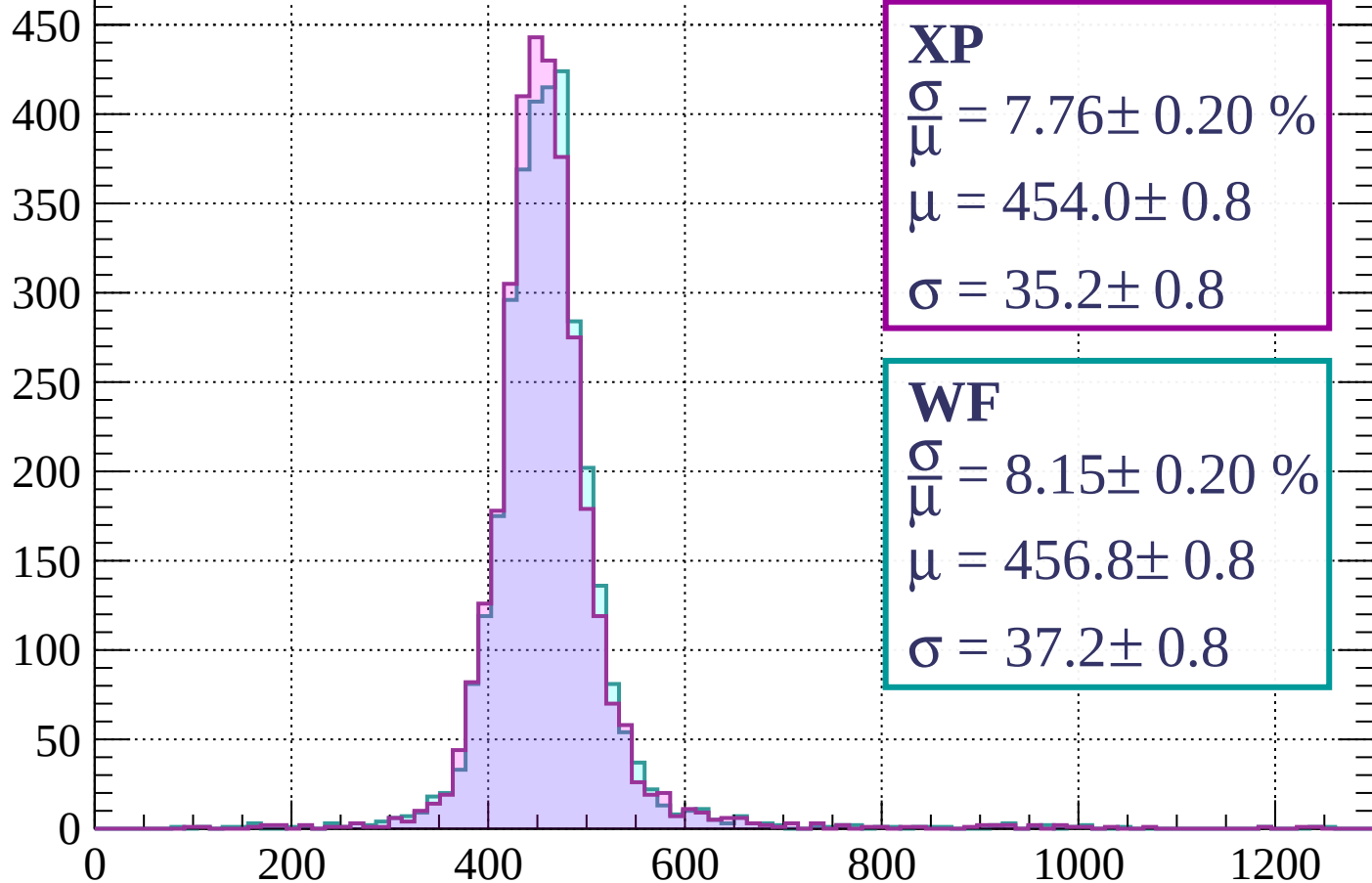
$$\mu = 452.5 \pm 0.7$$

$$\sigma = 36.4 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1200 < p < 1280$

Count



XP

$$\frac{\sigma}{\mu} = 7.76 \pm 0.20 \%$$

$$\mu = 454.0 \pm 0.8$$

$$\sigma = 35.2 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 8.15 \pm 0.20 \%$$

$$\mu = 456.8 \pm 0.8$$

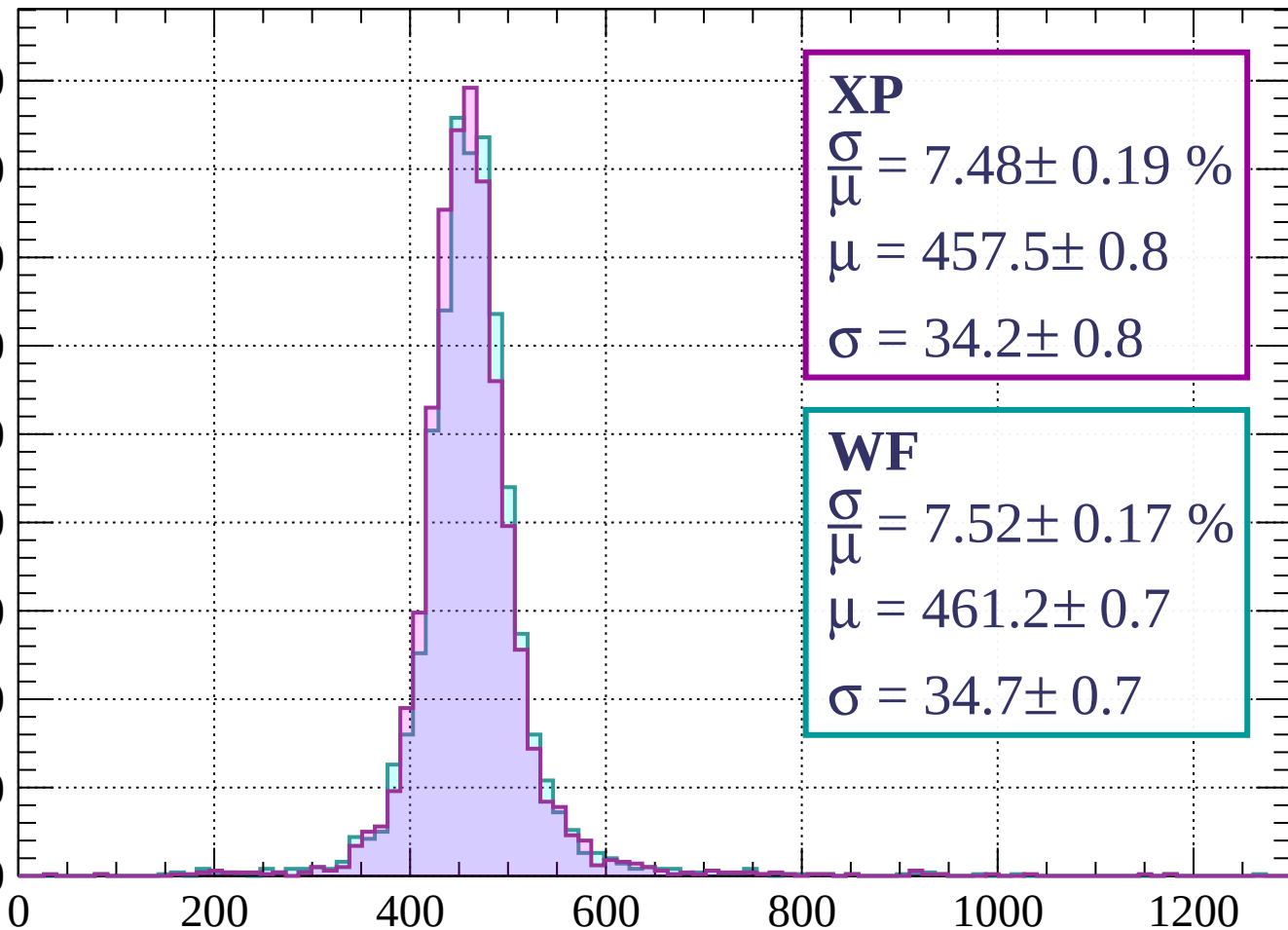
$$\sigma = 37.2 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1280 < p < 1360$

Count

450
400
350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 7.48 \pm 0.19 \%$$

$$\mu = 457.5 \pm 0.8$$

$$\sigma = 34.2 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 7.52 \pm 0.17 \%$$

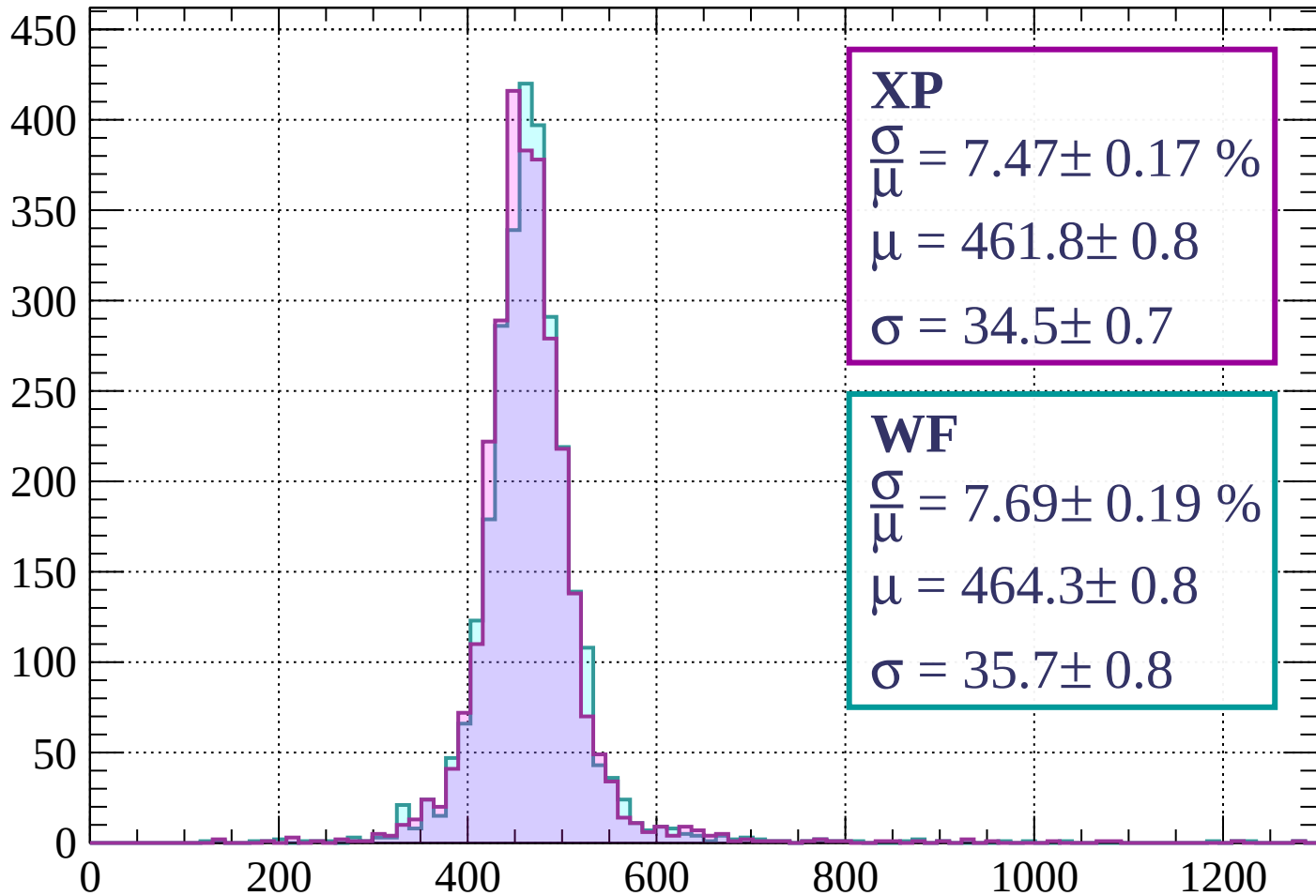
$$\mu = 461.2 \pm 0.7$$

$$\sigma = 34.7 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1360 < p < 1440$

Count



XP

$$\frac{\sigma}{\mu} = 7.47 \pm 0.17 \%$$

$$\mu = 461.8 \pm 0.8$$

$$\sigma = 34.5 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 7.69 \pm 0.19 \%$$

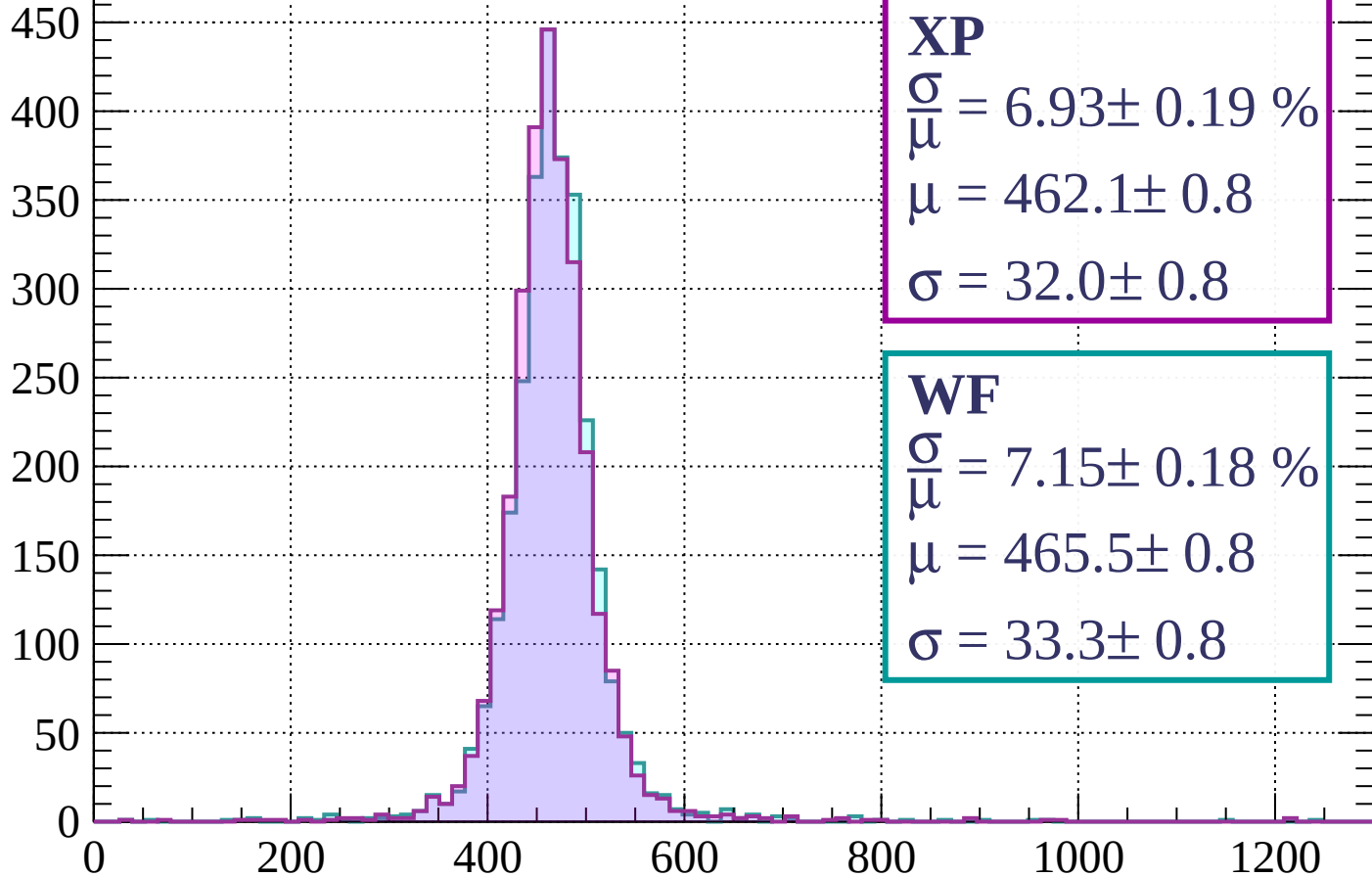
$$\mu = 464.3 \pm 0.8$$

$$\sigma = 35.7 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1440 < p < 1520$

Count



XP

$$\frac{\sigma}{\mu} = 6.93 \pm 0.19 \%$$

$$\mu = 462.1 \pm 0.8$$

$$\sigma = 32.0 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 7.15 \pm 0.18 \%$$

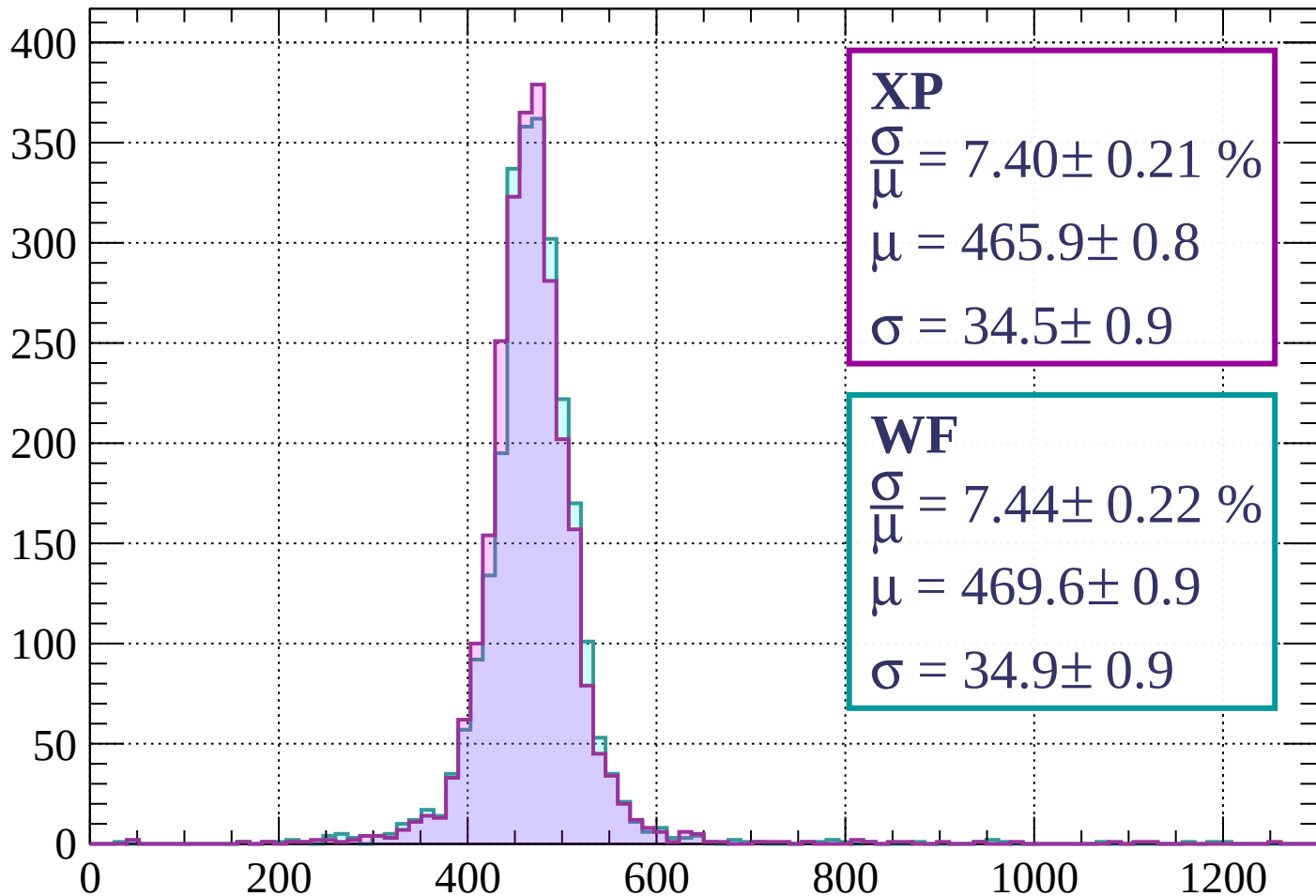
$$\mu = 465.5 \pm 0.8$$

$$\sigma = 33.3 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1520 < p < 1600$

Count



XP

$$\frac{\sigma}{\mu} = 7.40 \pm 0.21 \%$$

$$\mu = 465.9 \pm 0.8$$

$$\sigma = 34.5 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 7.44 \pm 0.22 \%$$

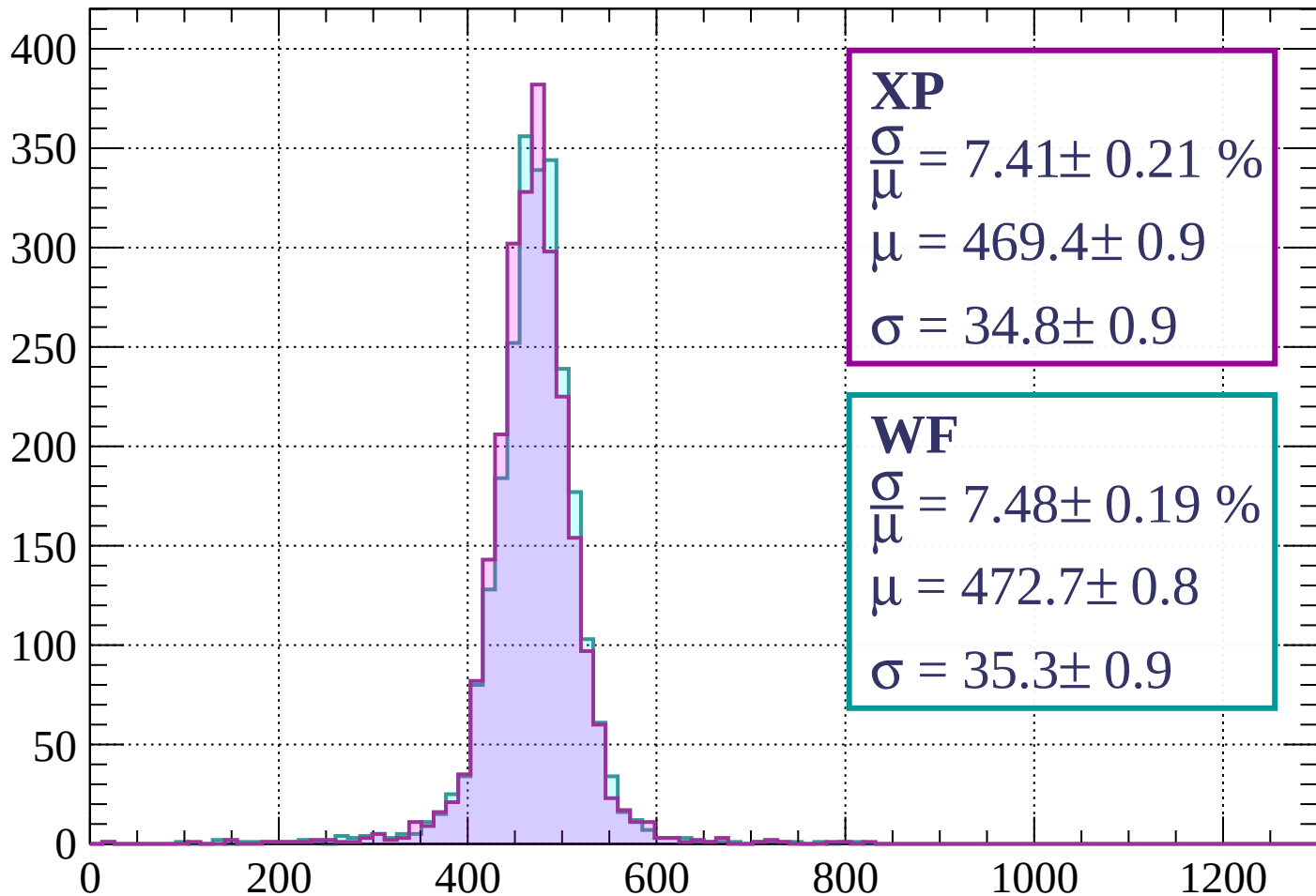
$$\mu = 469.6 \pm 0.9$$

$$\sigma = 34.9 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1600 < p < 1680$

Count



XP

$$\frac{\sigma}{\mu} = 7.41 \pm 0.21 \%$$

$$\mu = 469.4 \pm 0.9$$

$$\sigma = 34.8 \pm 0.9$$

WF

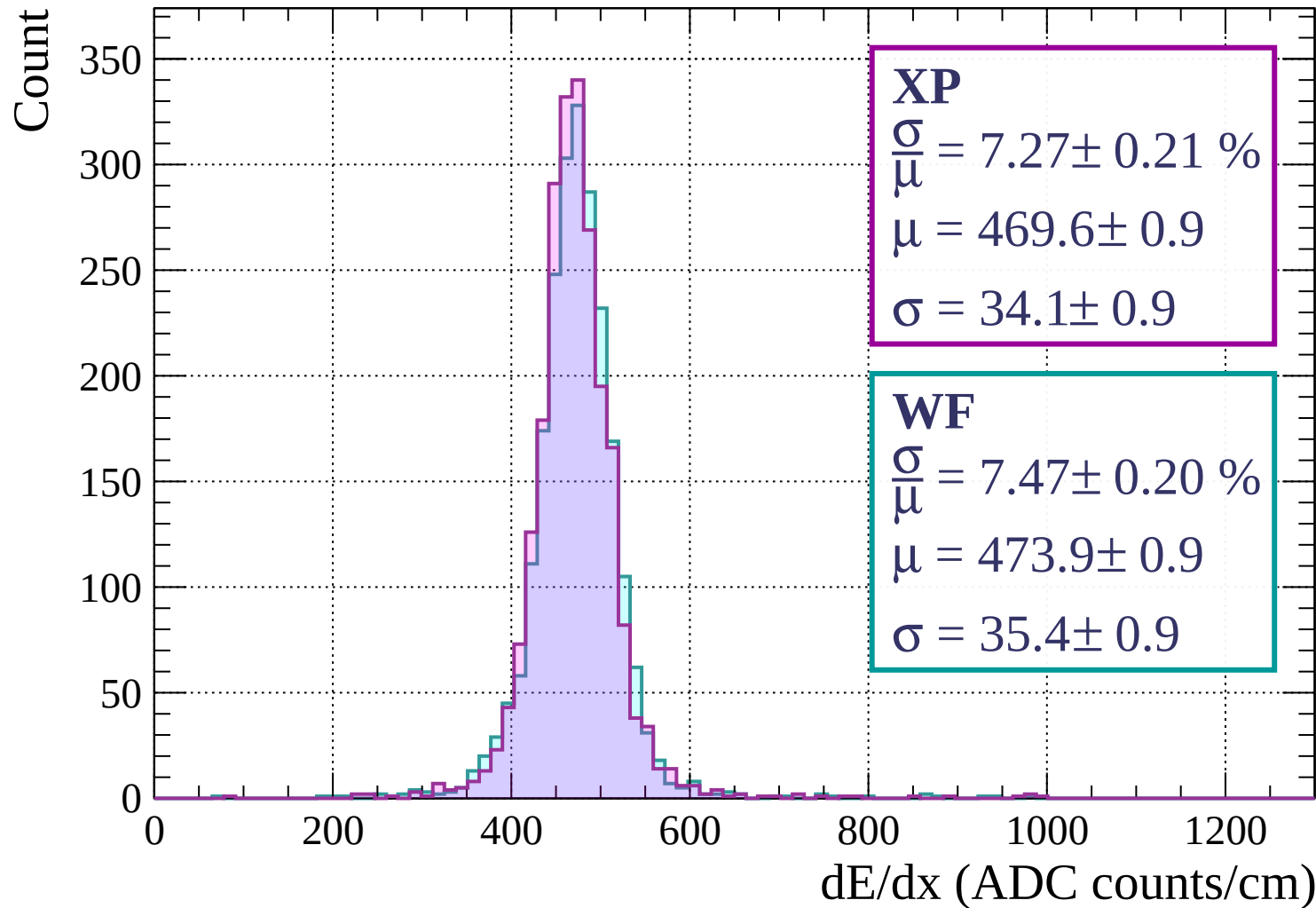
$$\frac{\sigma}{\mu} = 7.48 \pm 0.19 \%$$

$$\mu = 472.7 \pm 0.8$$

$$\sigma = 35.3 \pm 0.9$$

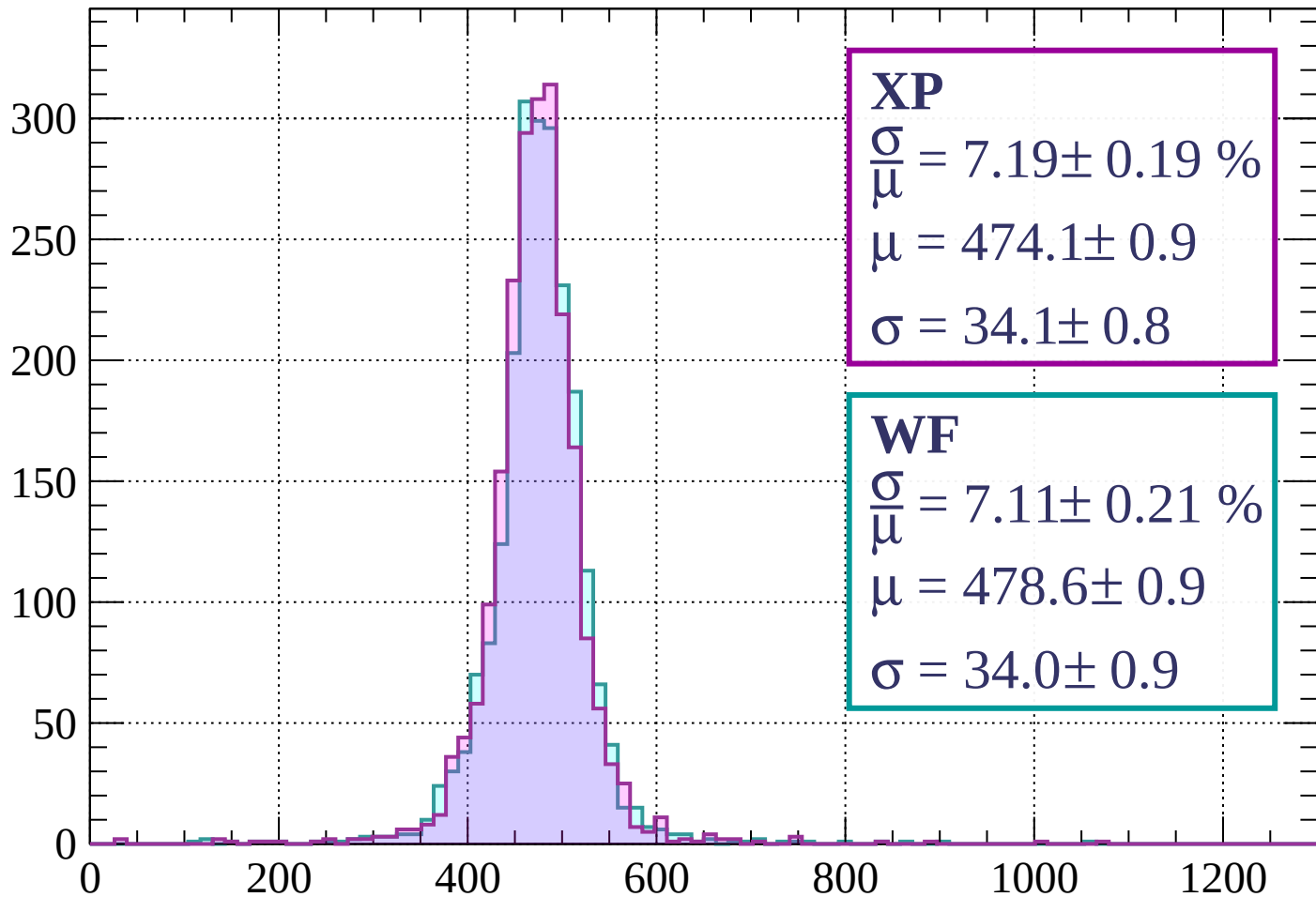
dE/dx (ADC counts/cm)

Energy loss | $1680 < p < 1760$



Energy loss | $1760 < p < 1840$

Count



XP

$$\frac{\sigma}{\mu} = 7.19 \pm 0.19 \%$$

$$\mu = 474.1 \pm 0.9$$

$$\sigma = 34.1 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 7.11 \pm 0.21 \%$$

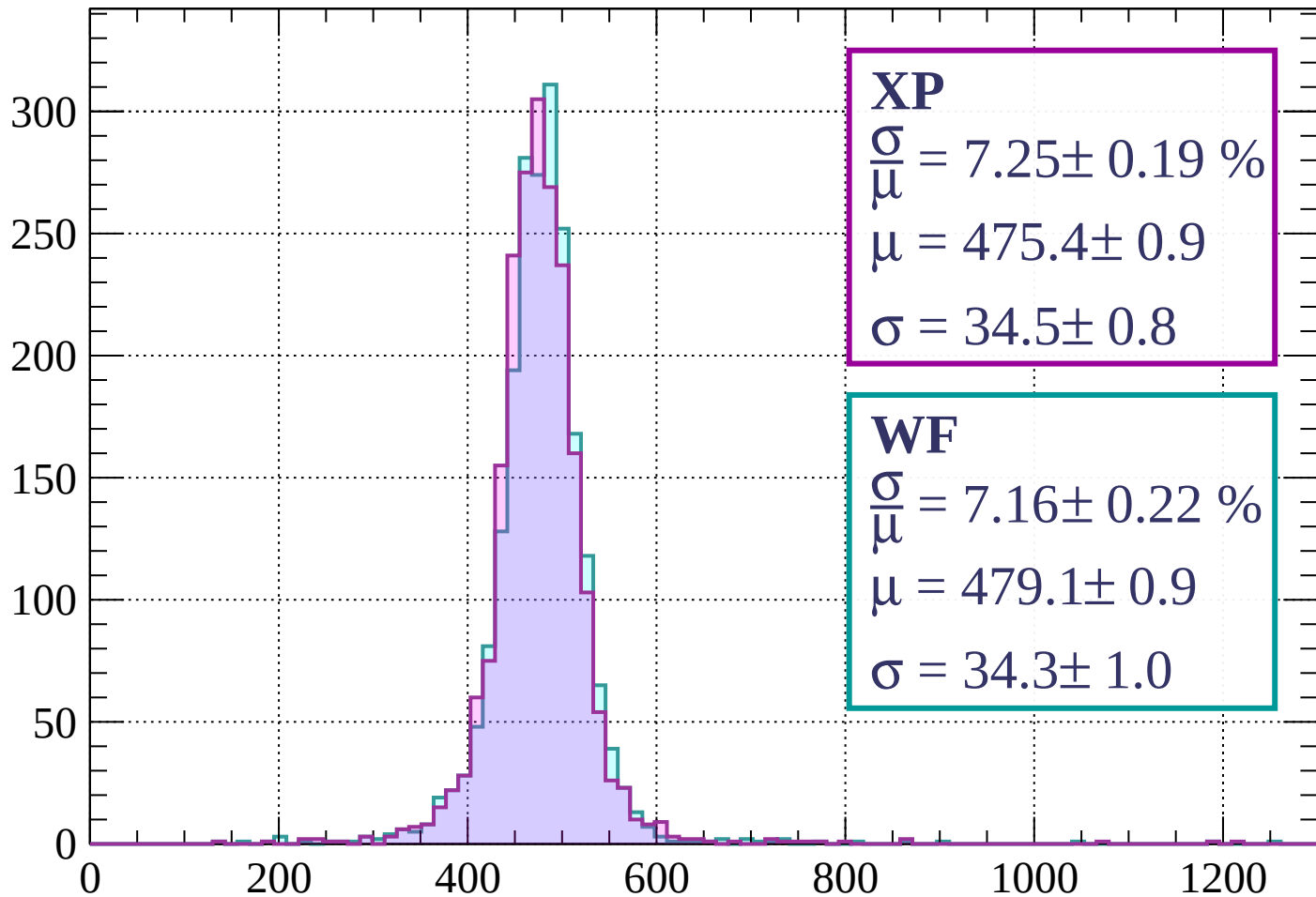
$$\mu = 478.6 \pm 0.9$$

$$\sigma = 34.0 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1840 < p < 1920$

Count



XP

$$\frac{\sigma}{\mu} = 7.25 \pm 0.19 \%$$

$$\mu = 475.4 \pm 0.9$$

$$\sigma = 34.5 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 7.16 \pm 0.22 \%$$

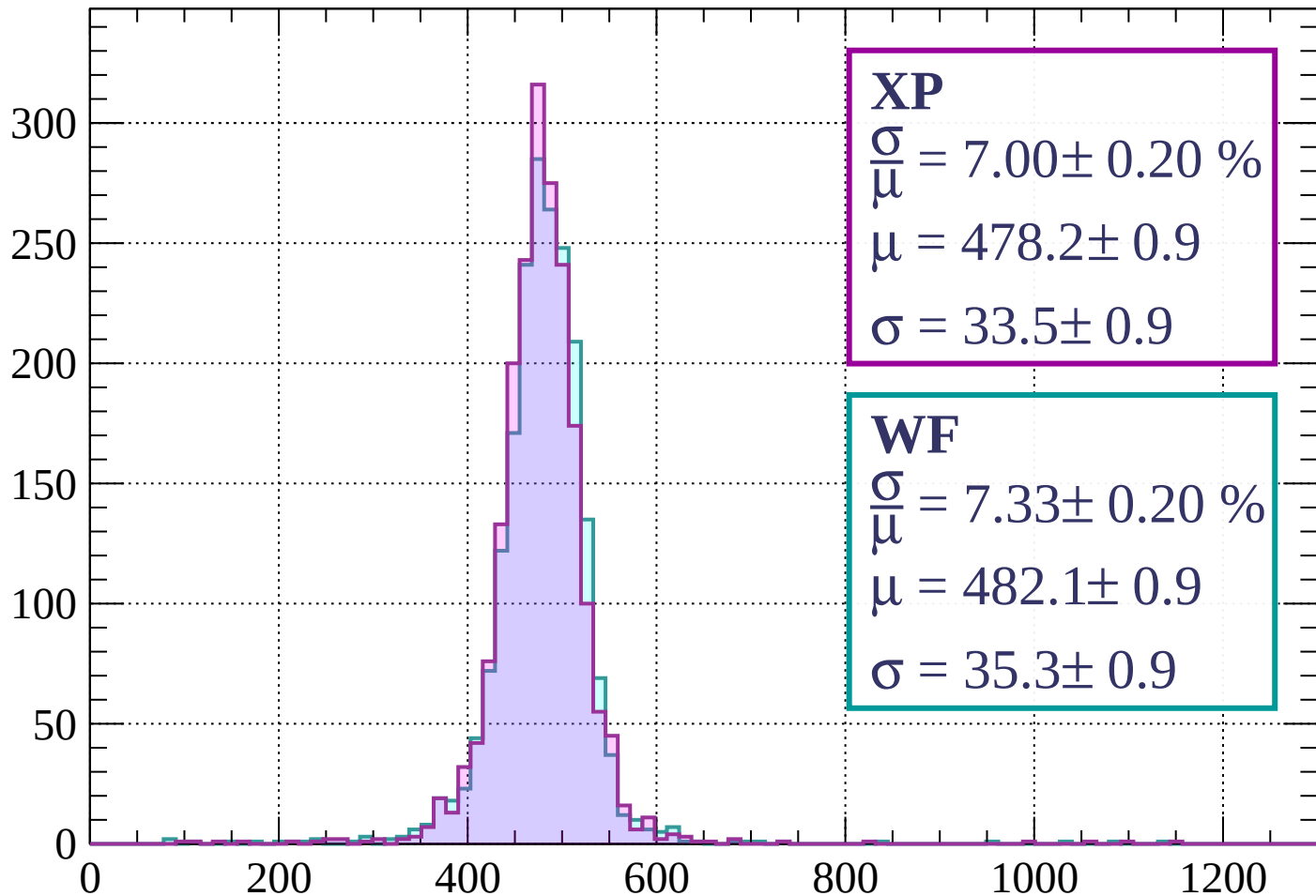
$$\mu = 479.1 \pm 0.9$$

$$\sigma = 34.3 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $1920 < p < 2000$

Count



XP

$$\frac{\sigma}{\mu} = 7.00 \pm 0.20 \%$$

$$\mu = 478.2 \pm 0.9$$

$$\sigma = 33.5 \pm 0.9$$

WF

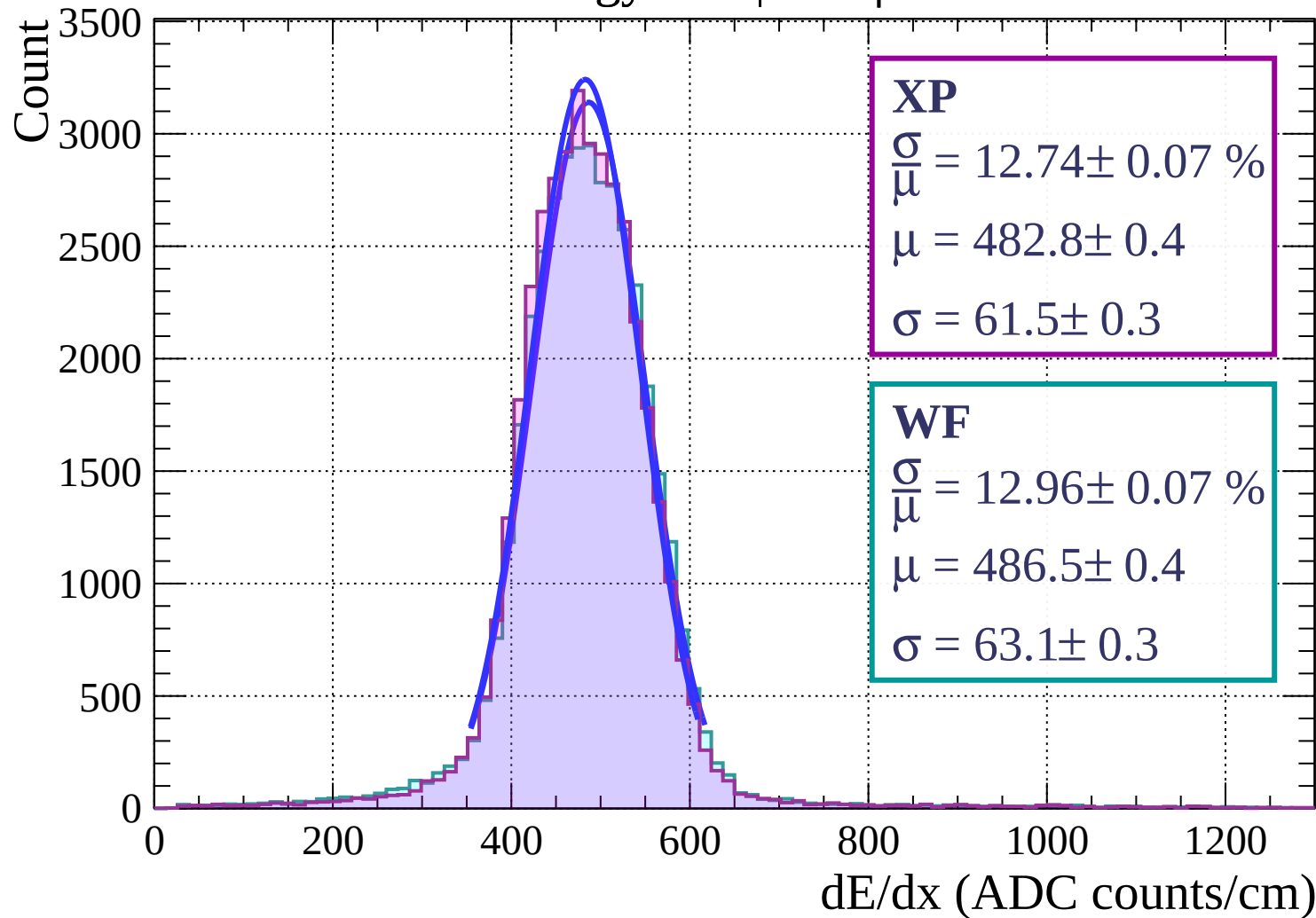
$$\frac{\sigma}{\mu} = 7.33 \pm 0.20 \%$$

$$\mu = 482.1 \pm 0.9$$

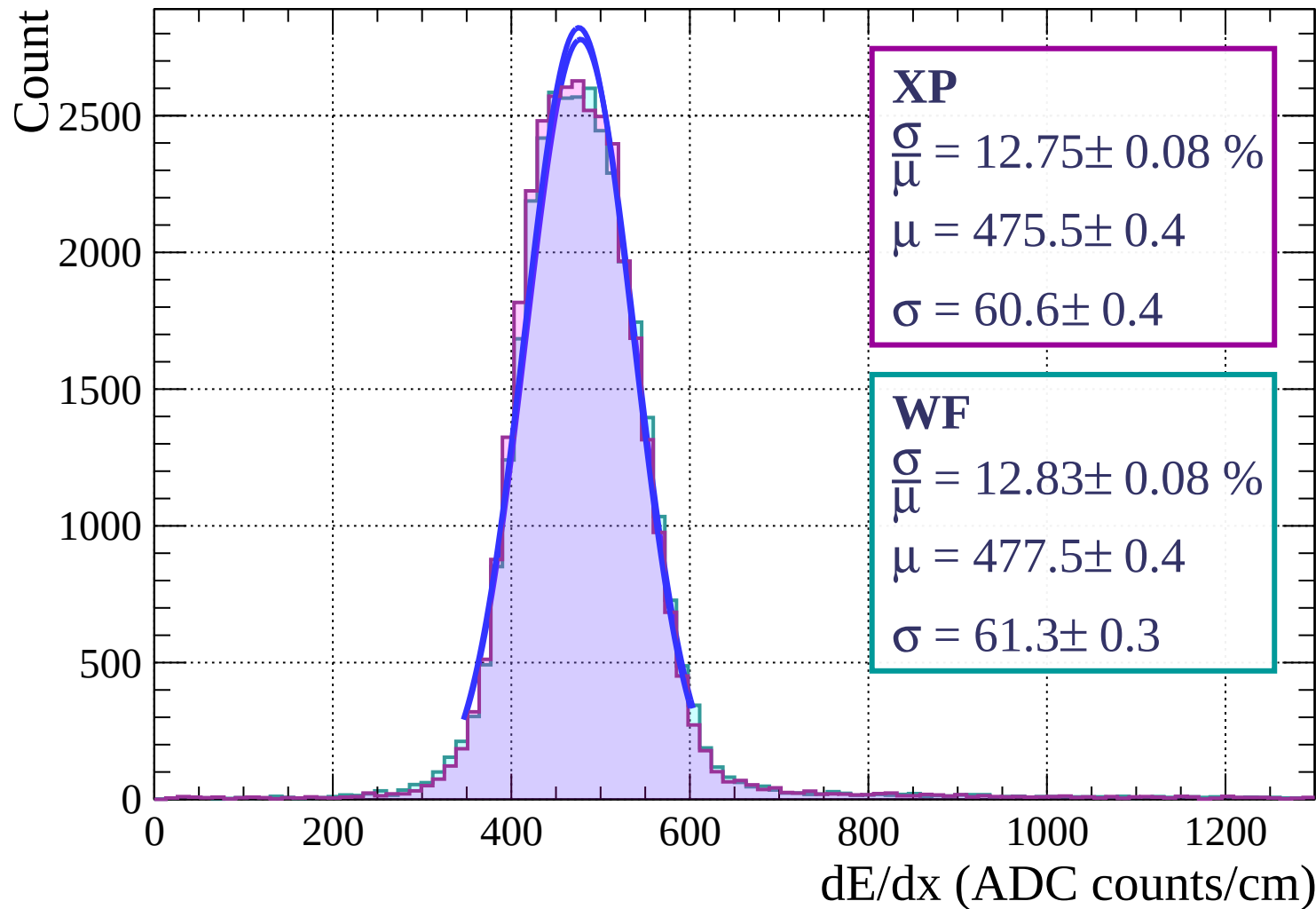
$$\sigma = 35.3 \pm 0.9$$

dE/dx (ADC counts/cm)

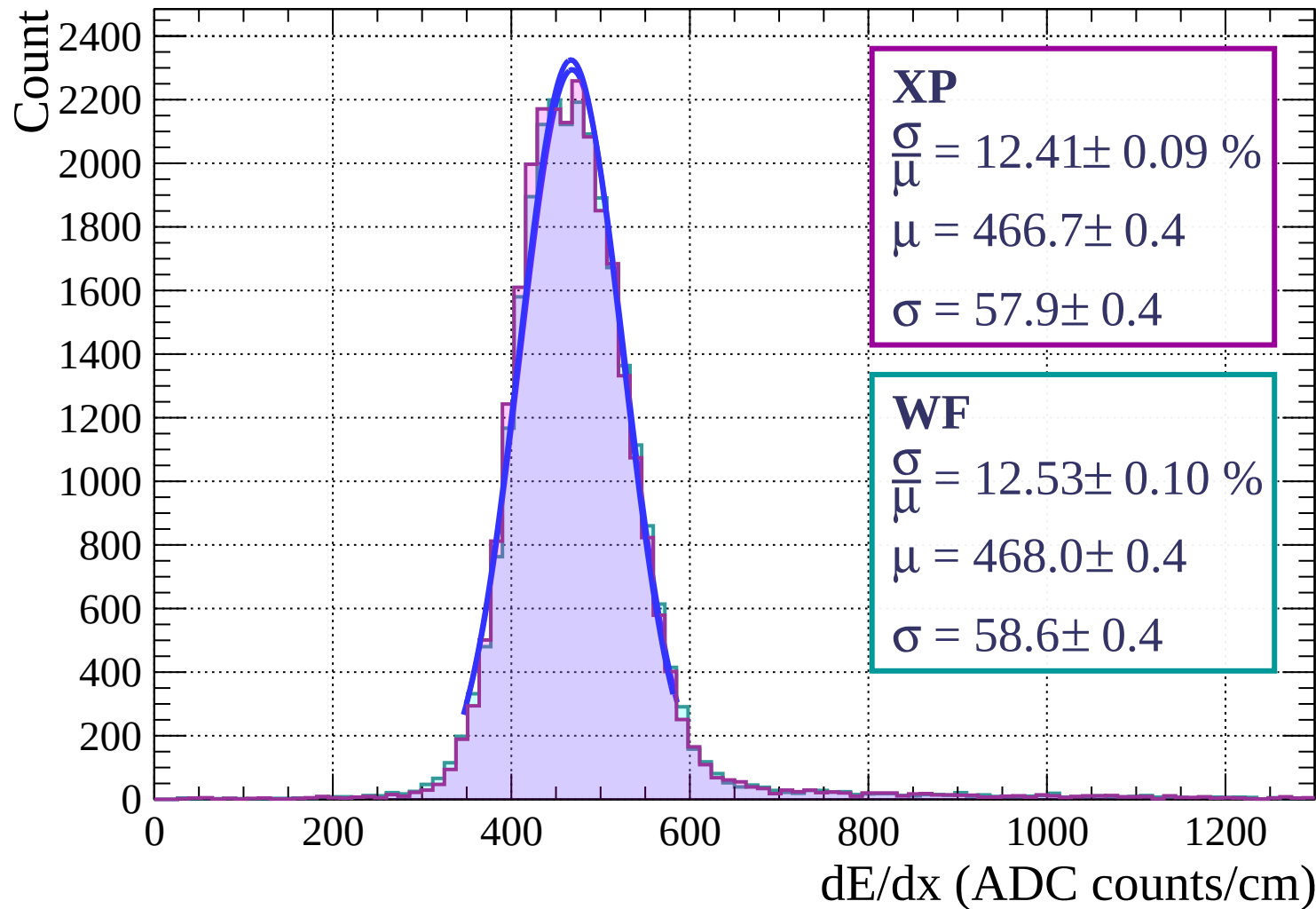
Energy loss | $0 < \phi < 3$



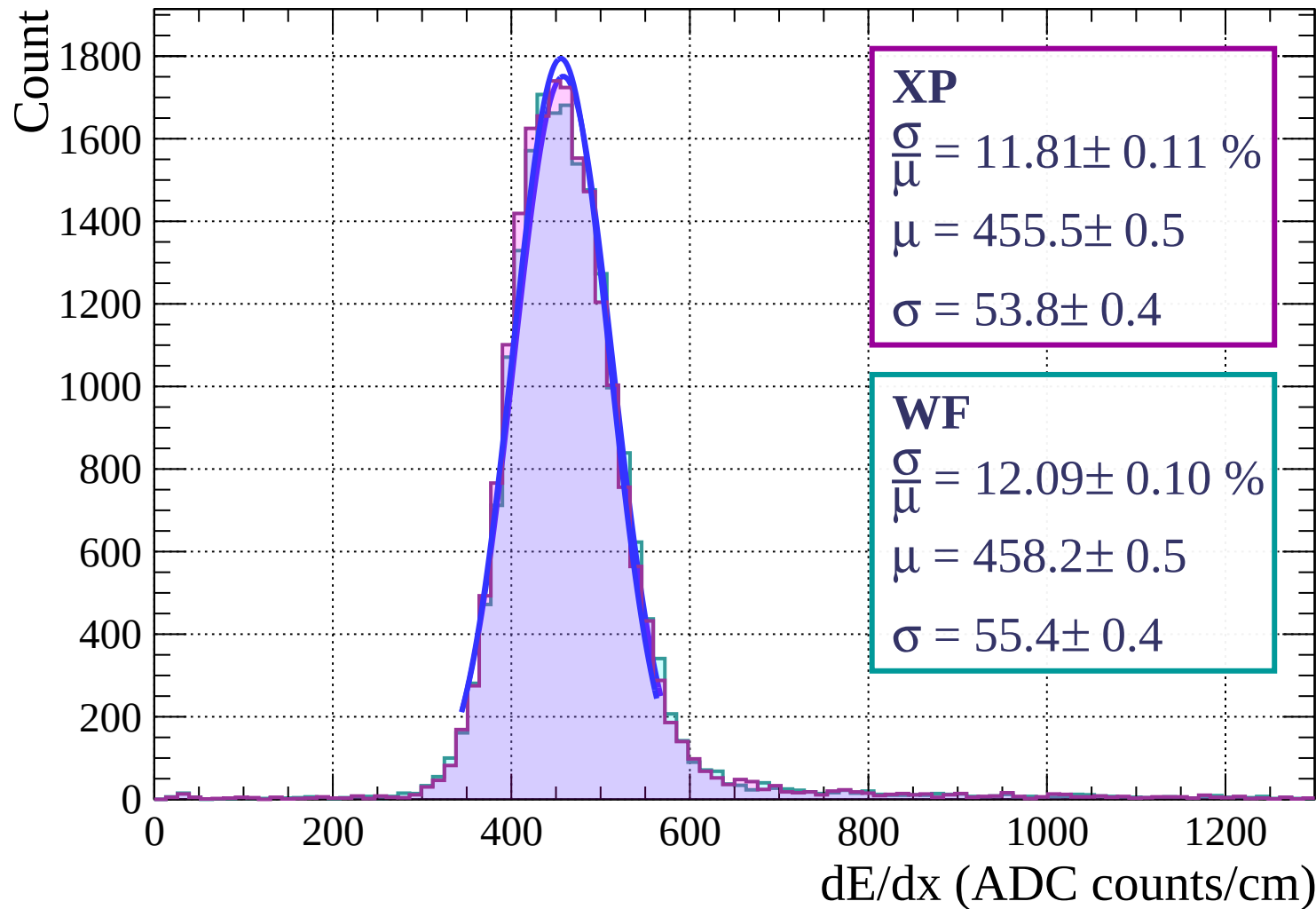
Energy loss | $3 < \phi < 6$



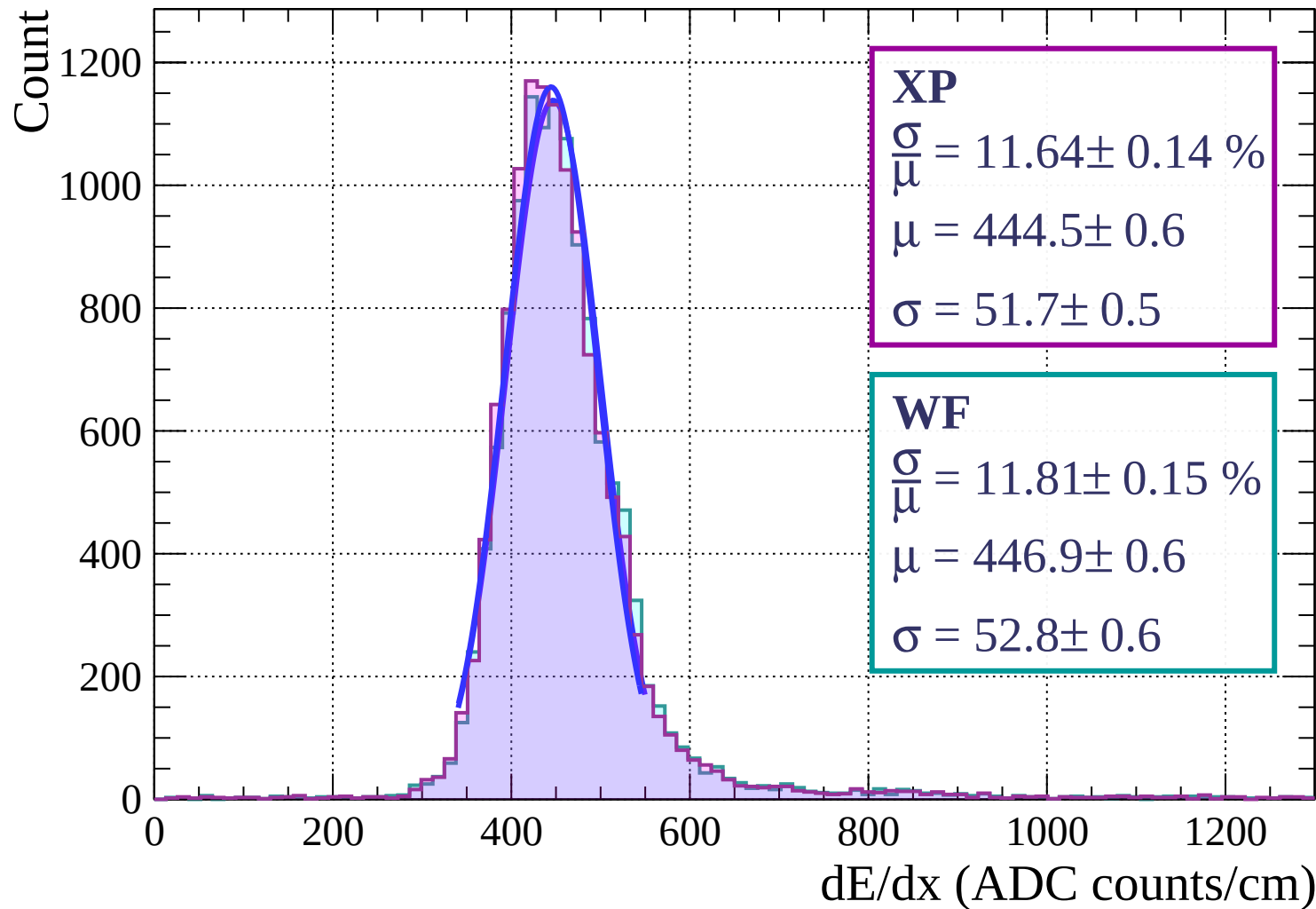
Energy loss | $6 < \phi < 9$



Energy loss | $9 < \phi < 12$

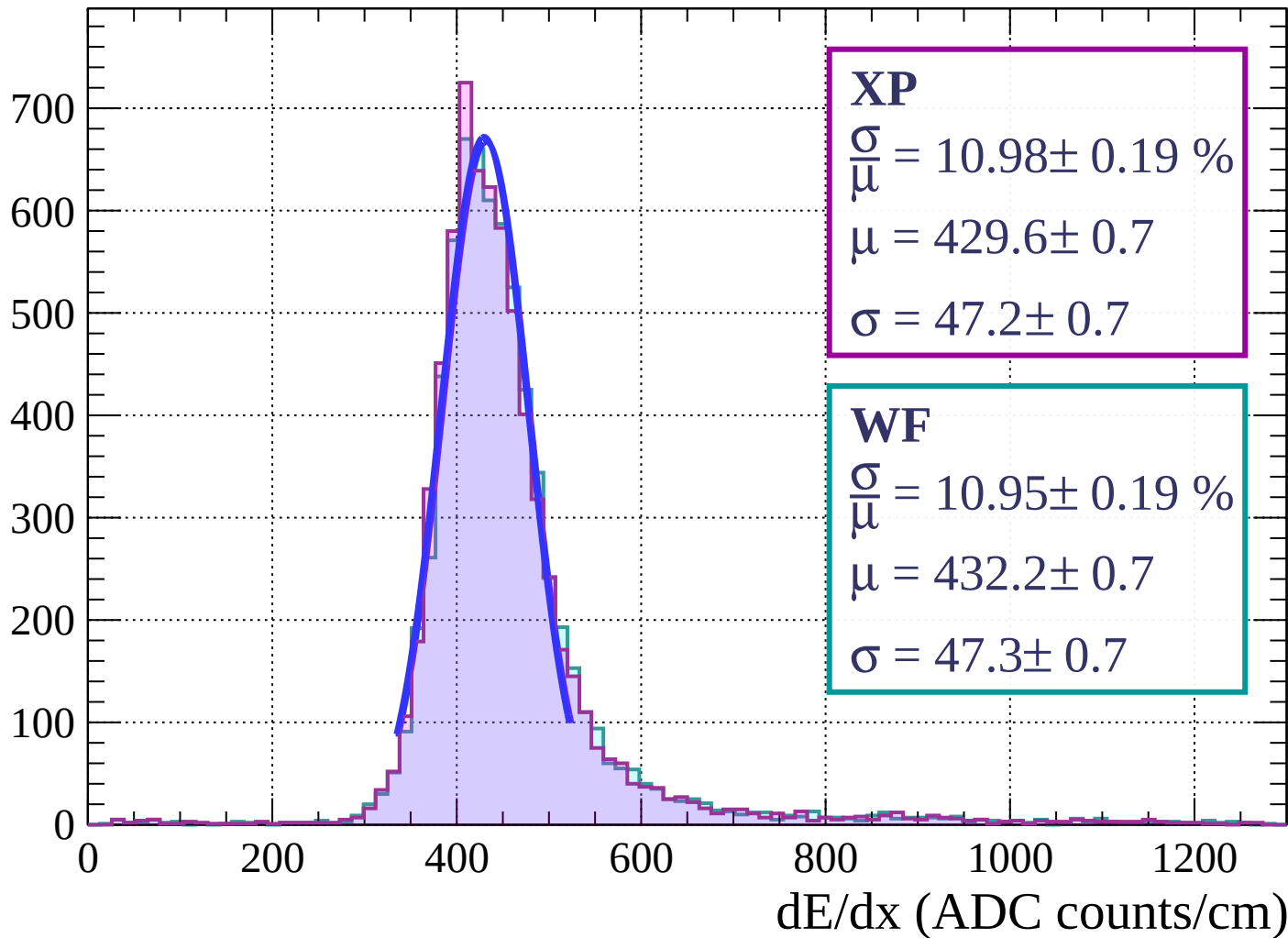


Energy loss | $12 < \phi < 15$



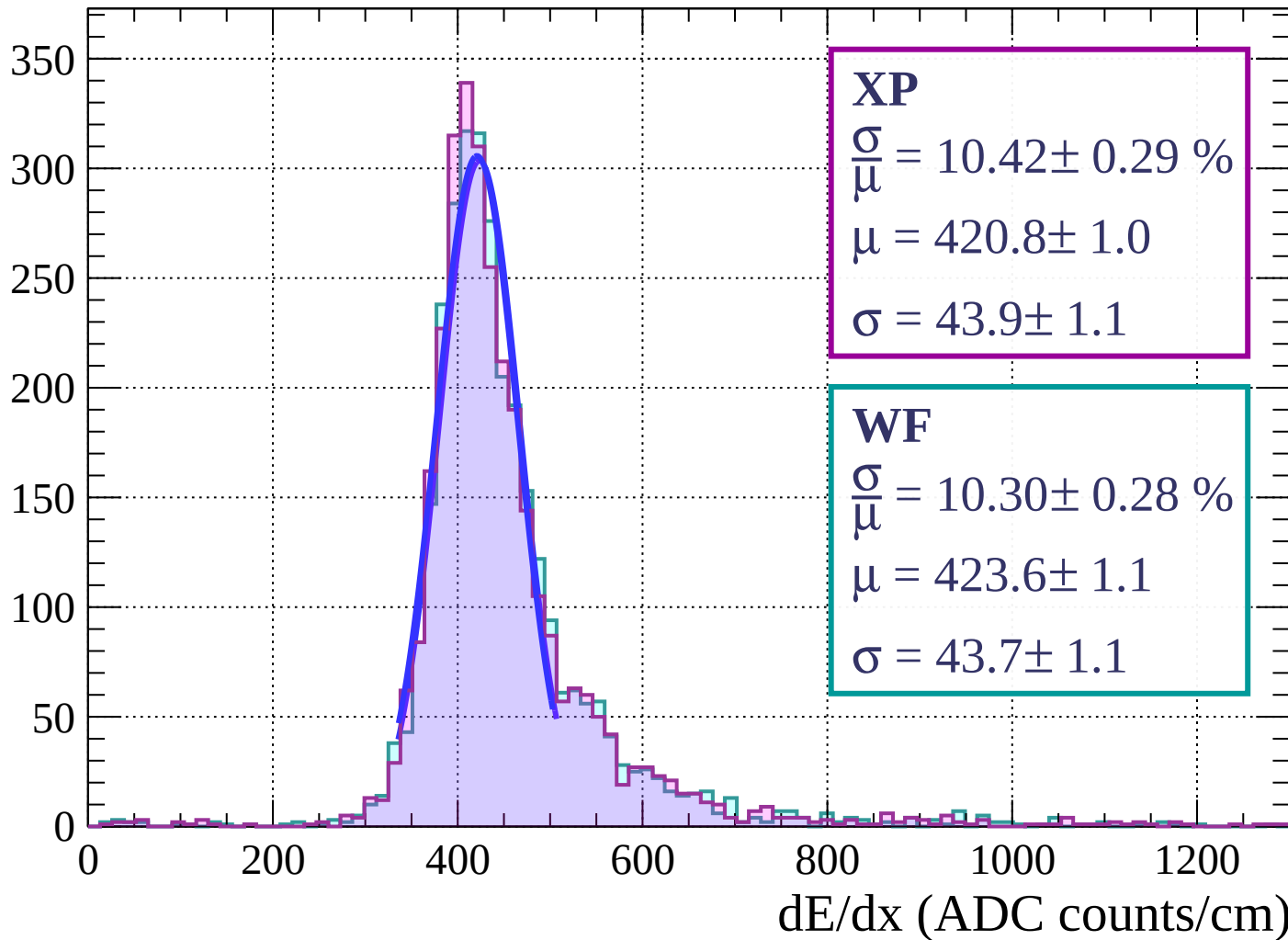
Energy loss | $15 < \phi < 18$

Count

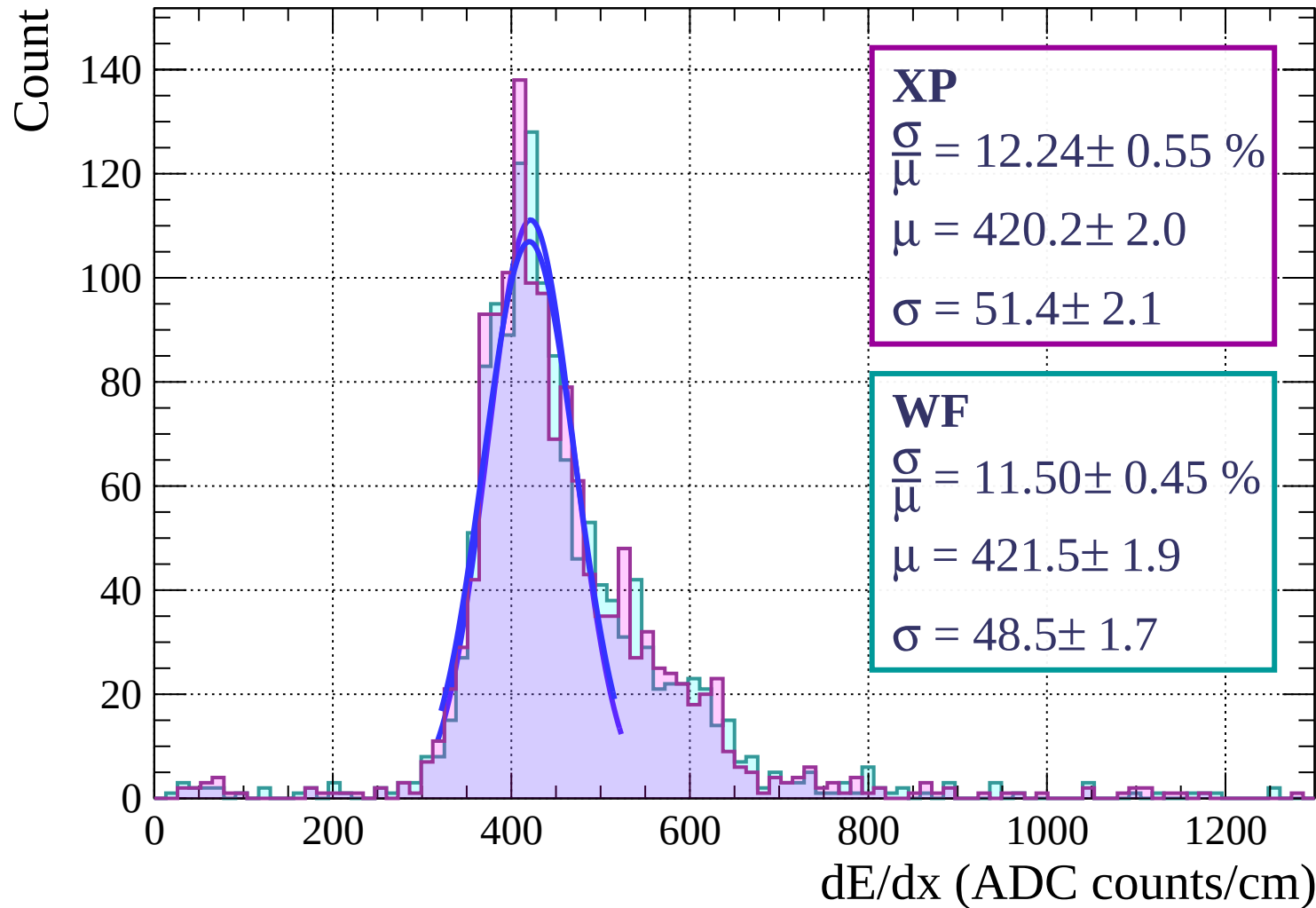


Energy loss | $18 < \phi < 21$

Count

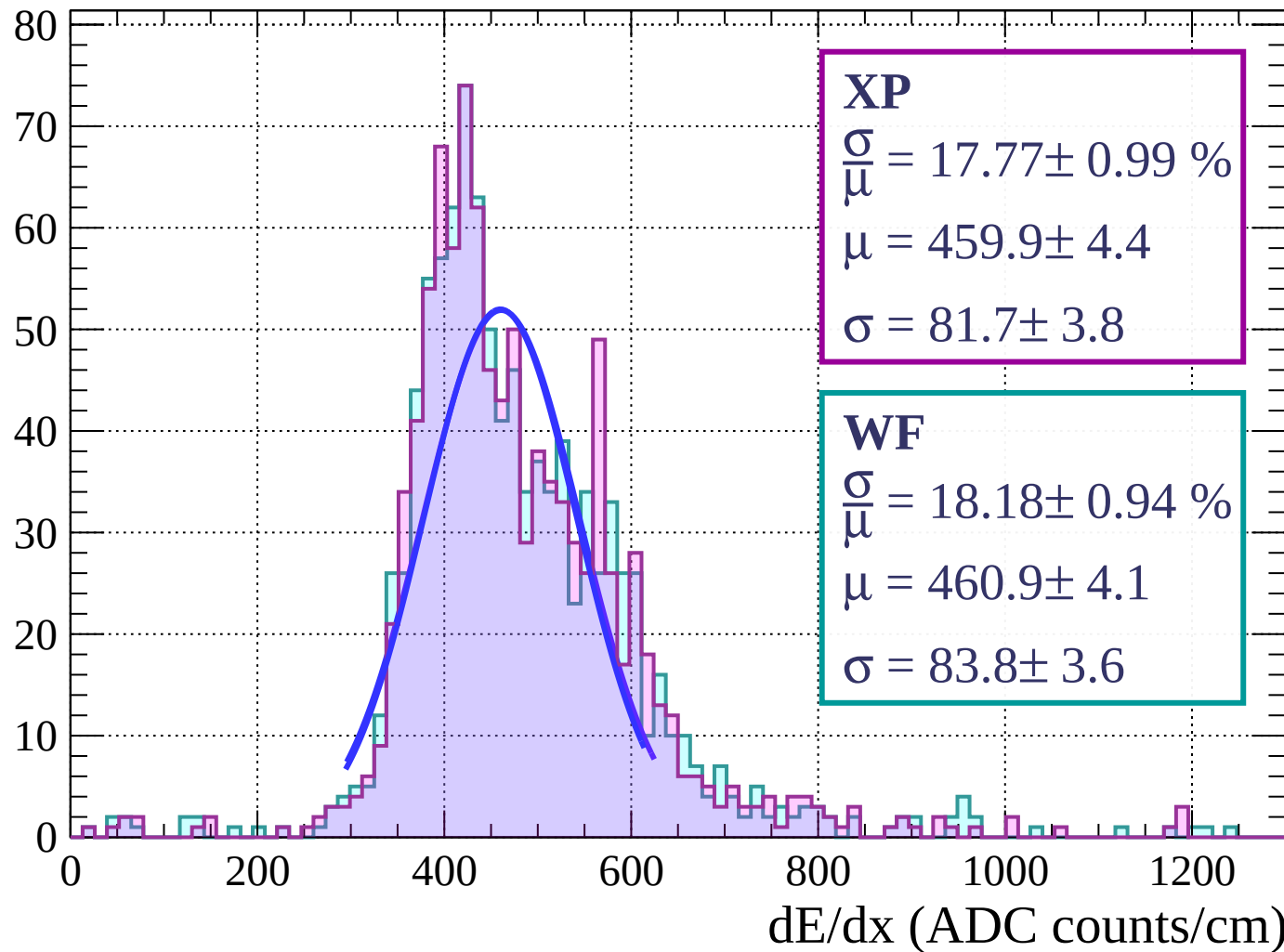


Energy loss | $21 < \phi < 24$

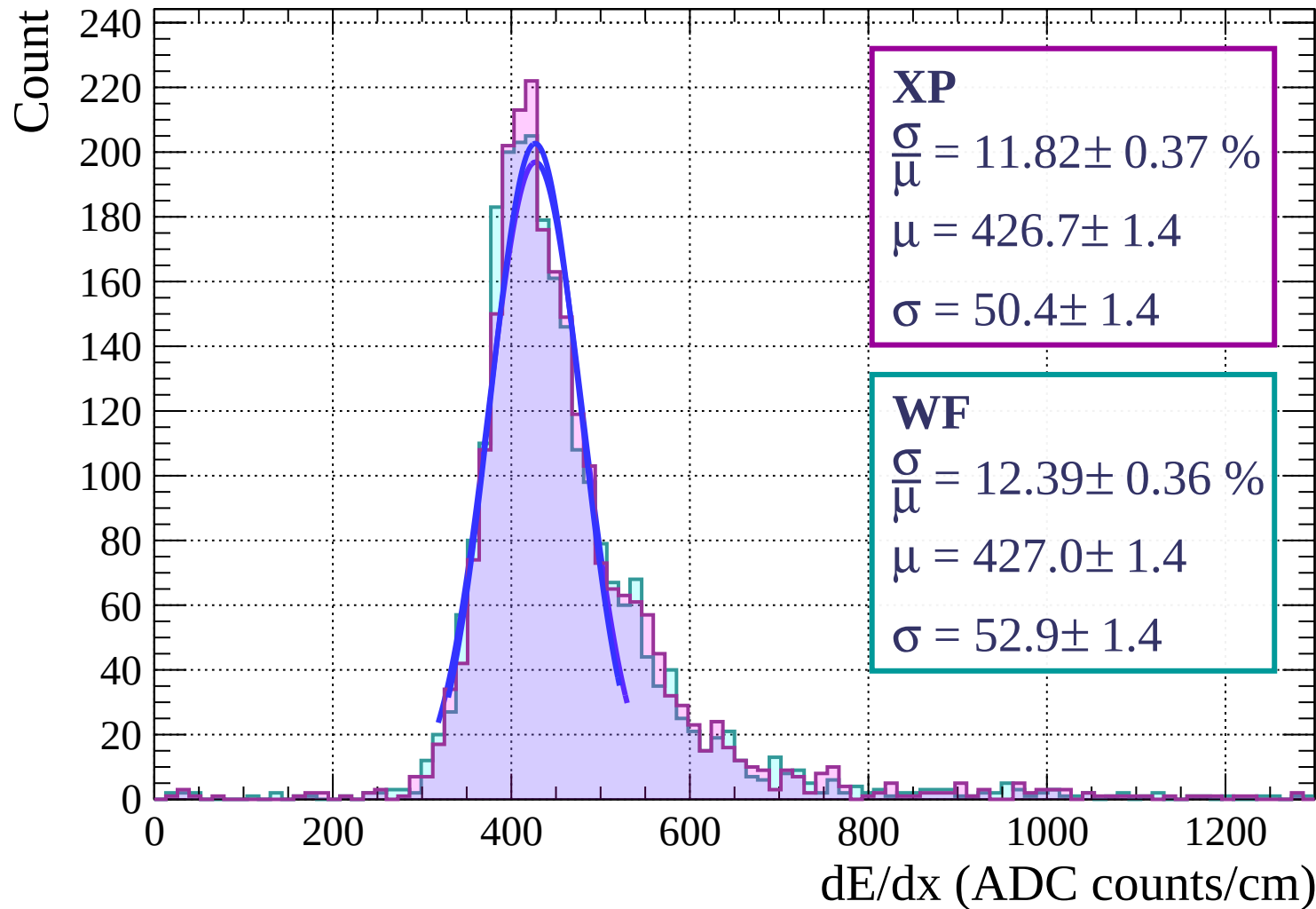


Energy loss | $24 < \phi < 27$

Count

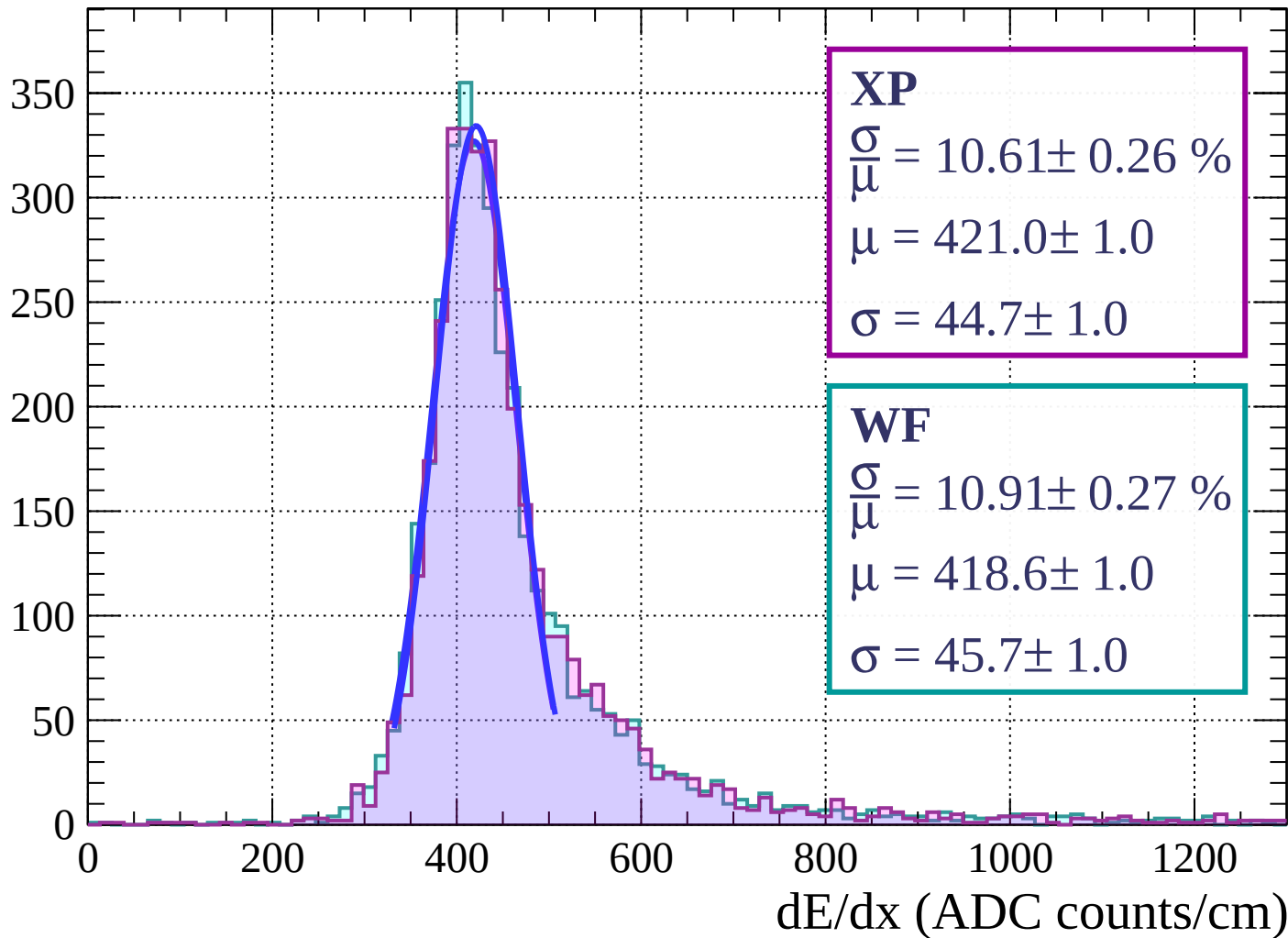


Energy loss | $27 < \phi < 30$

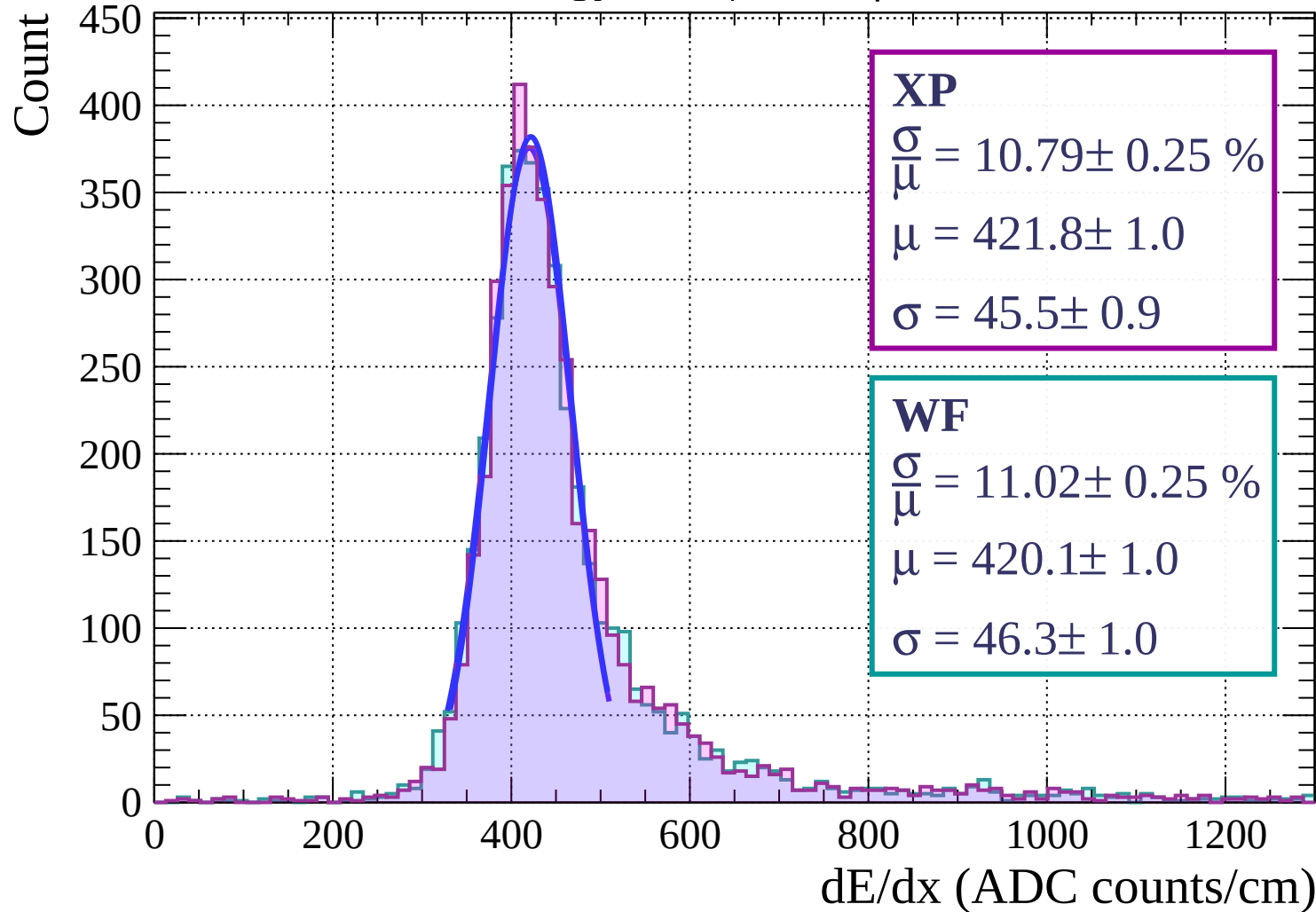


Energy loss | $30 < \phi < 33$

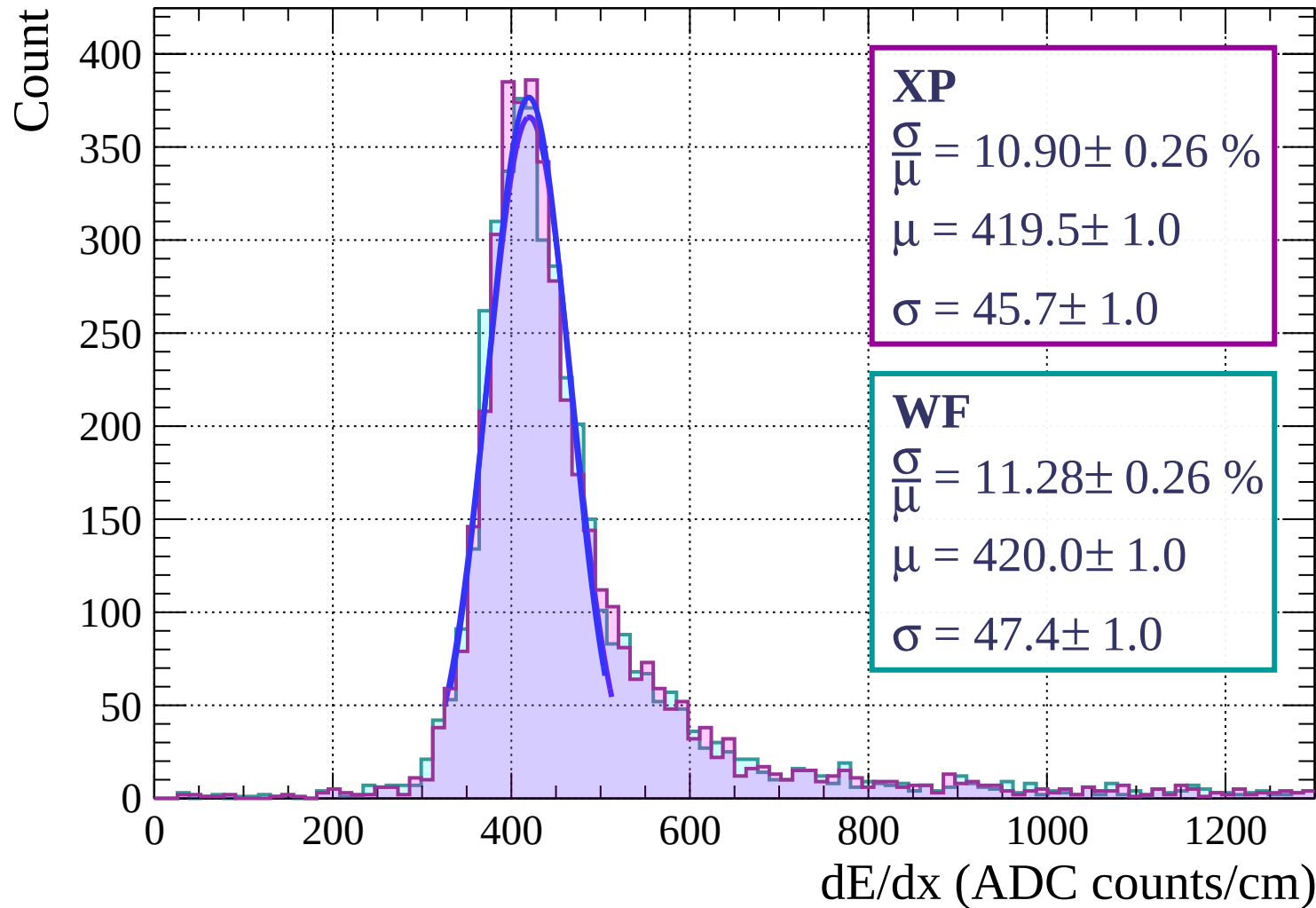
Count



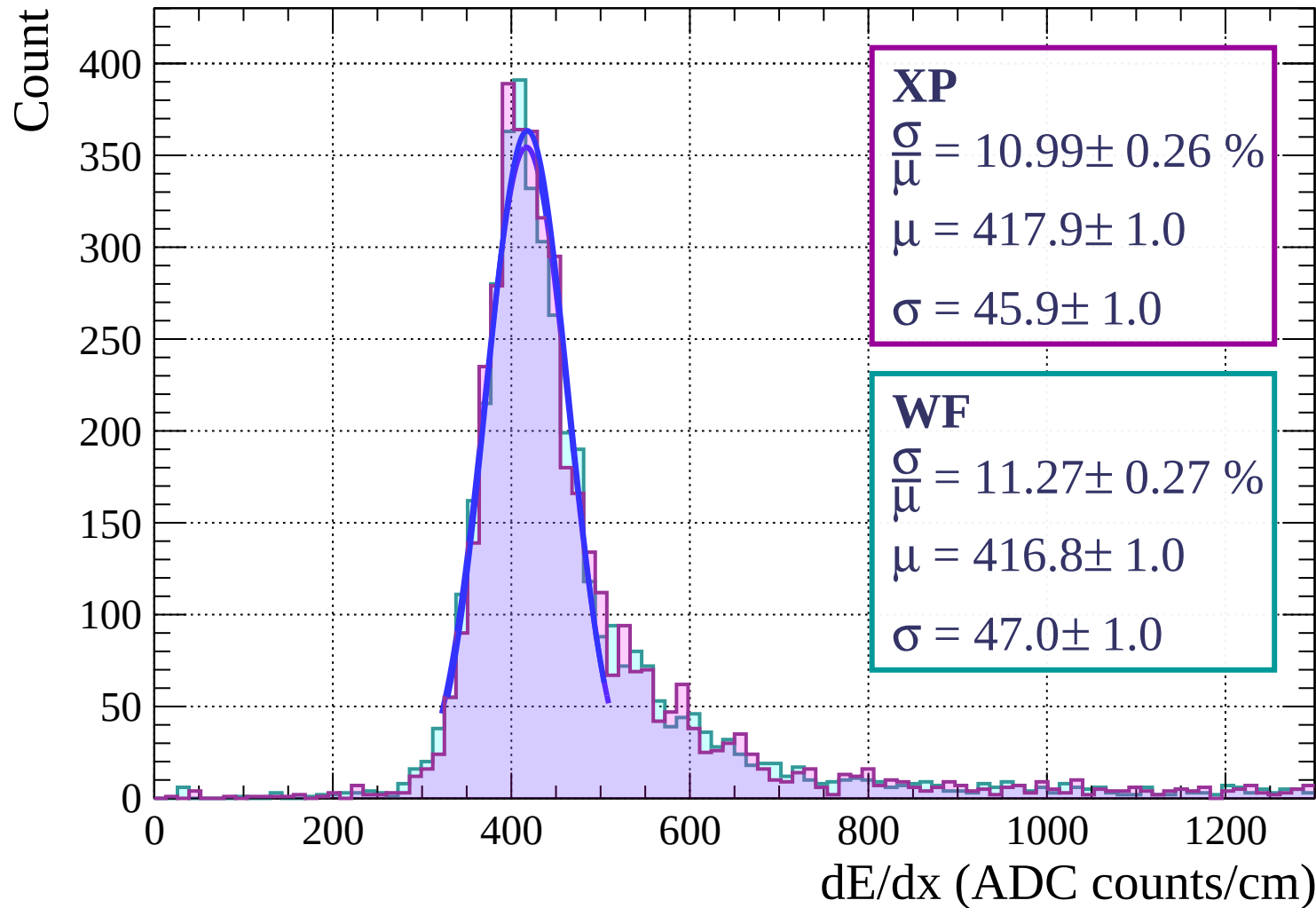
Energy loss | $33 < \phi < 36$



Energy loss | $36 < \phi < 39$

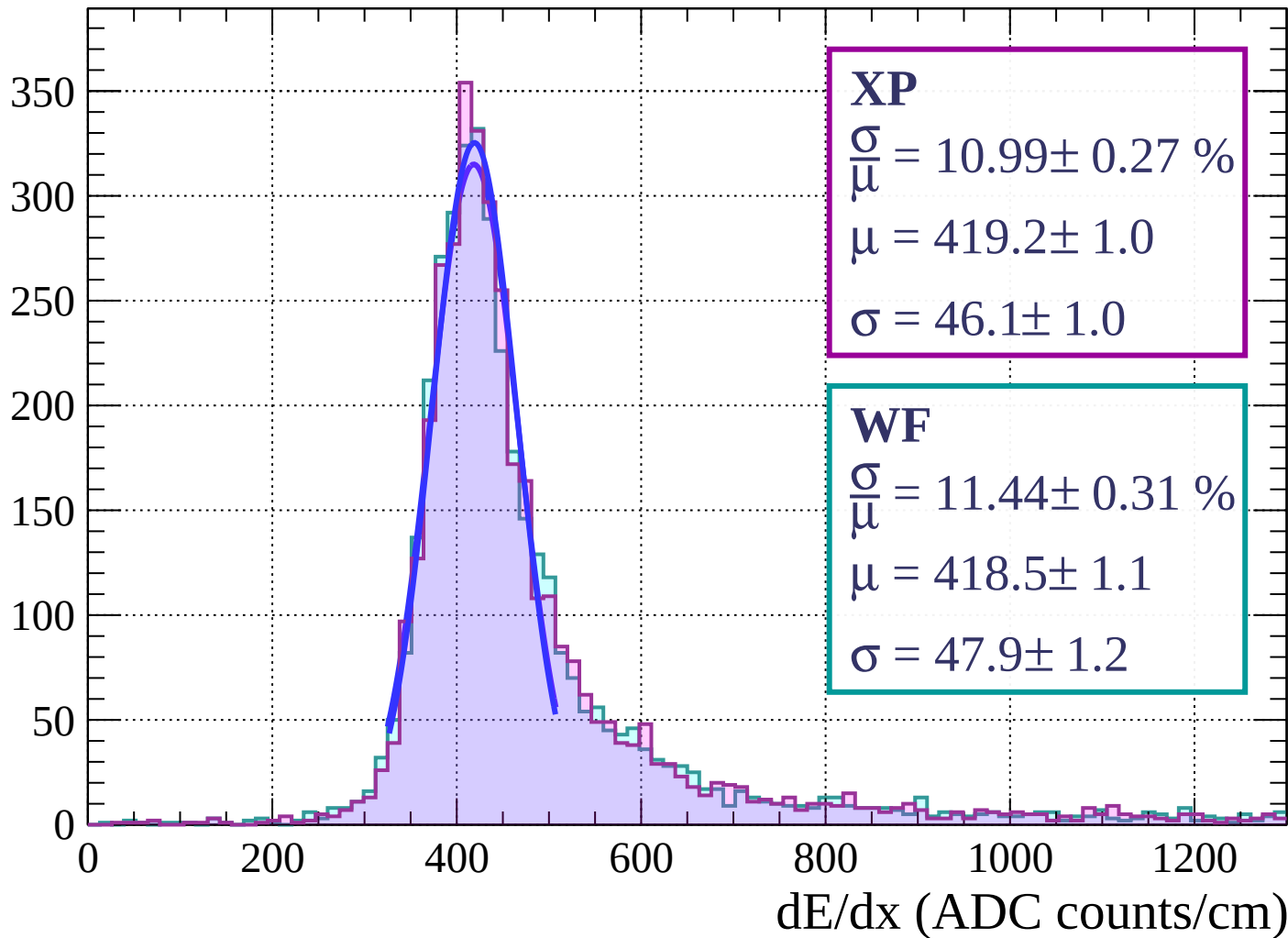


Energy loss | $39 < \phi < 42$



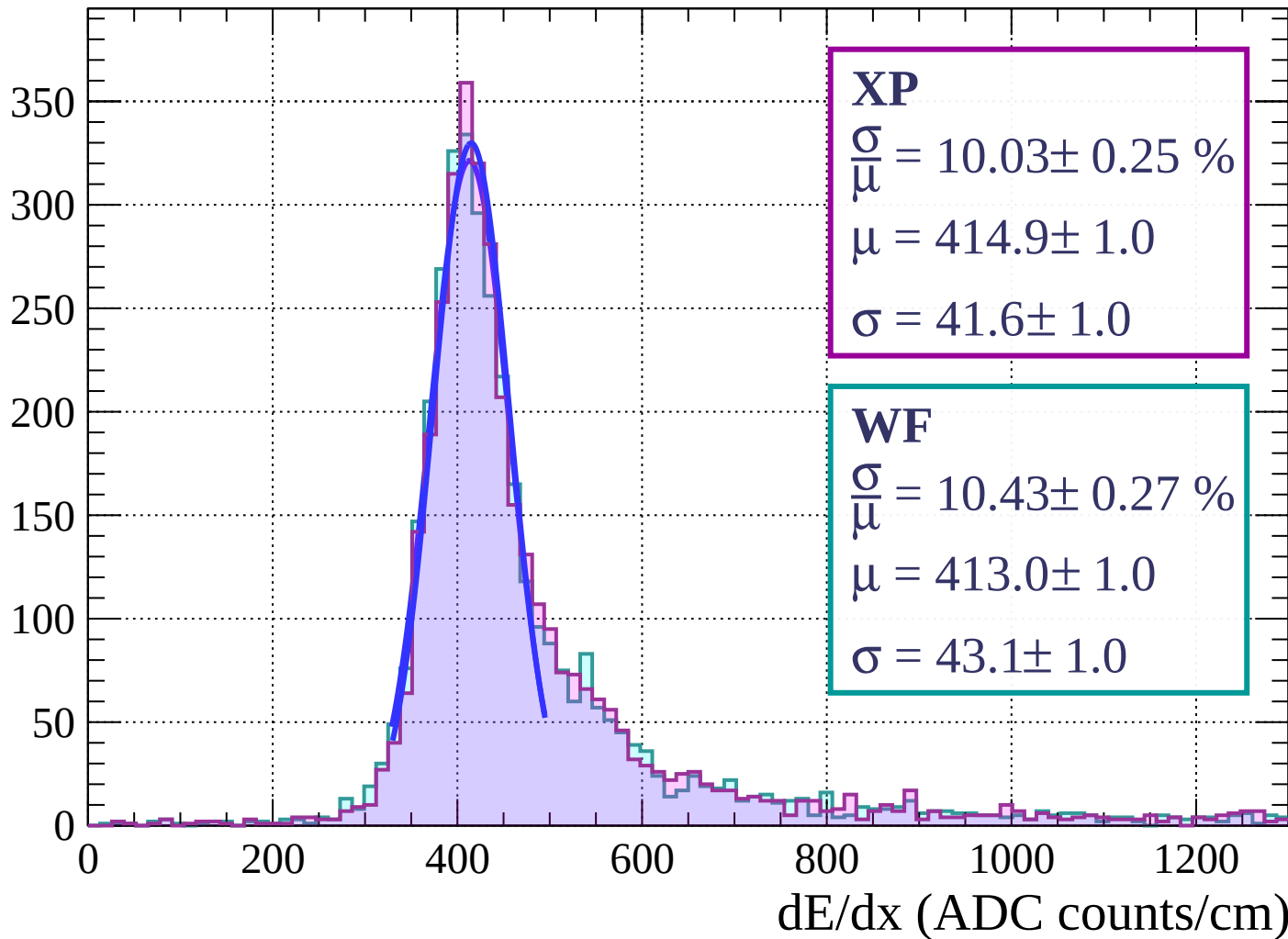
Energy loss | $42 < \phi < 45$

Count

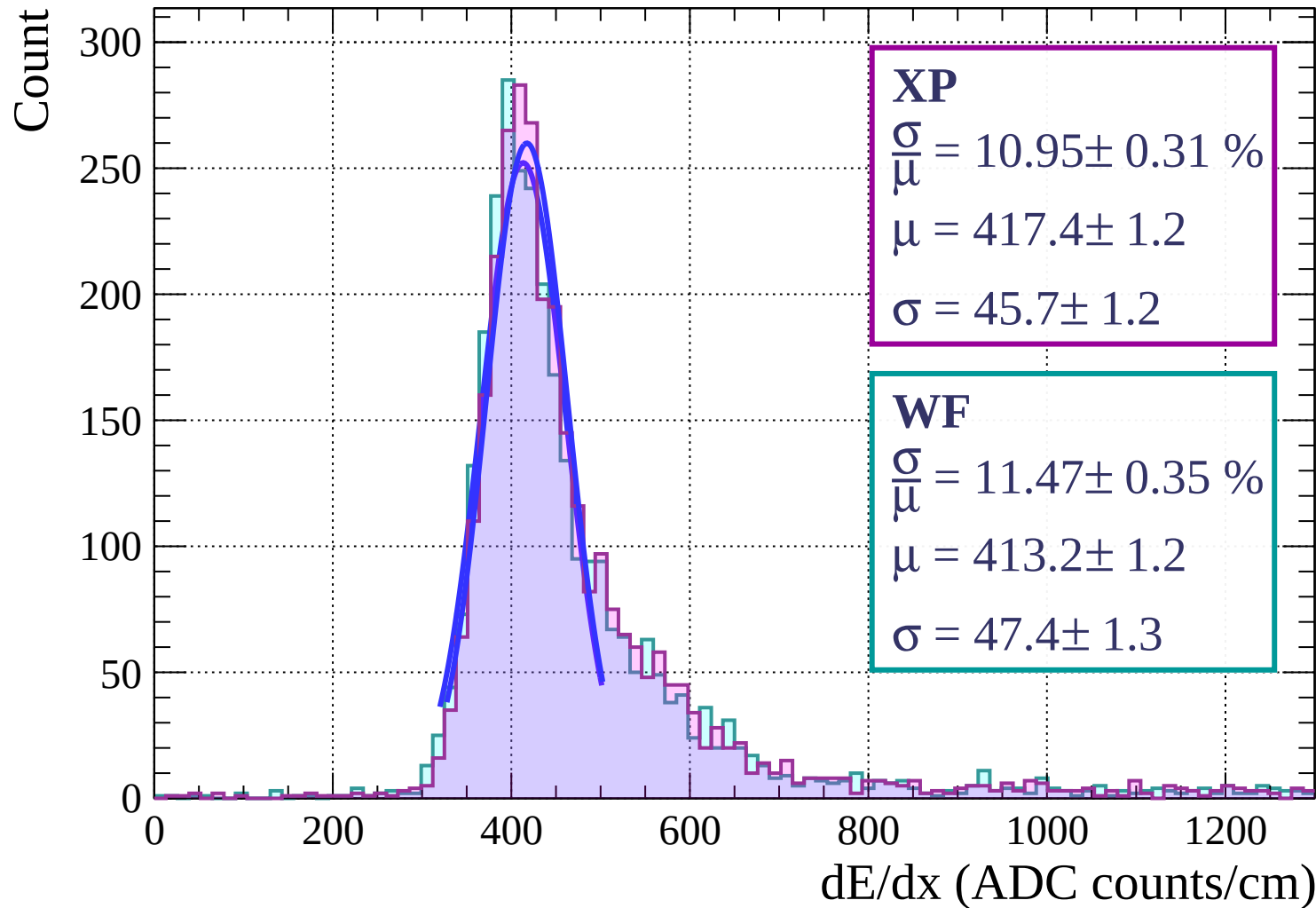


Energy loss | $45 < \phi < 48$

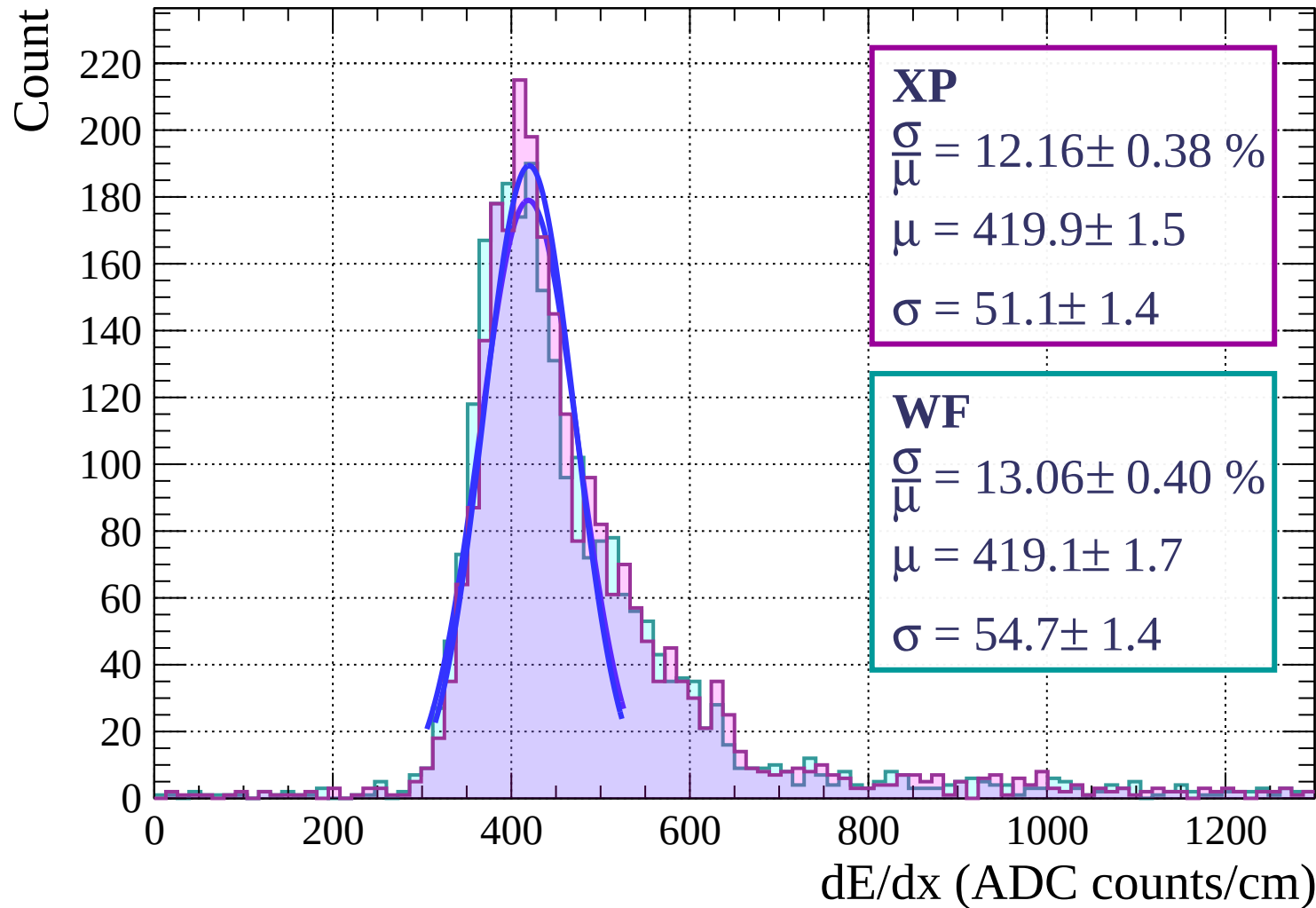
Count



Energy loss | $48 < \phi < 51$

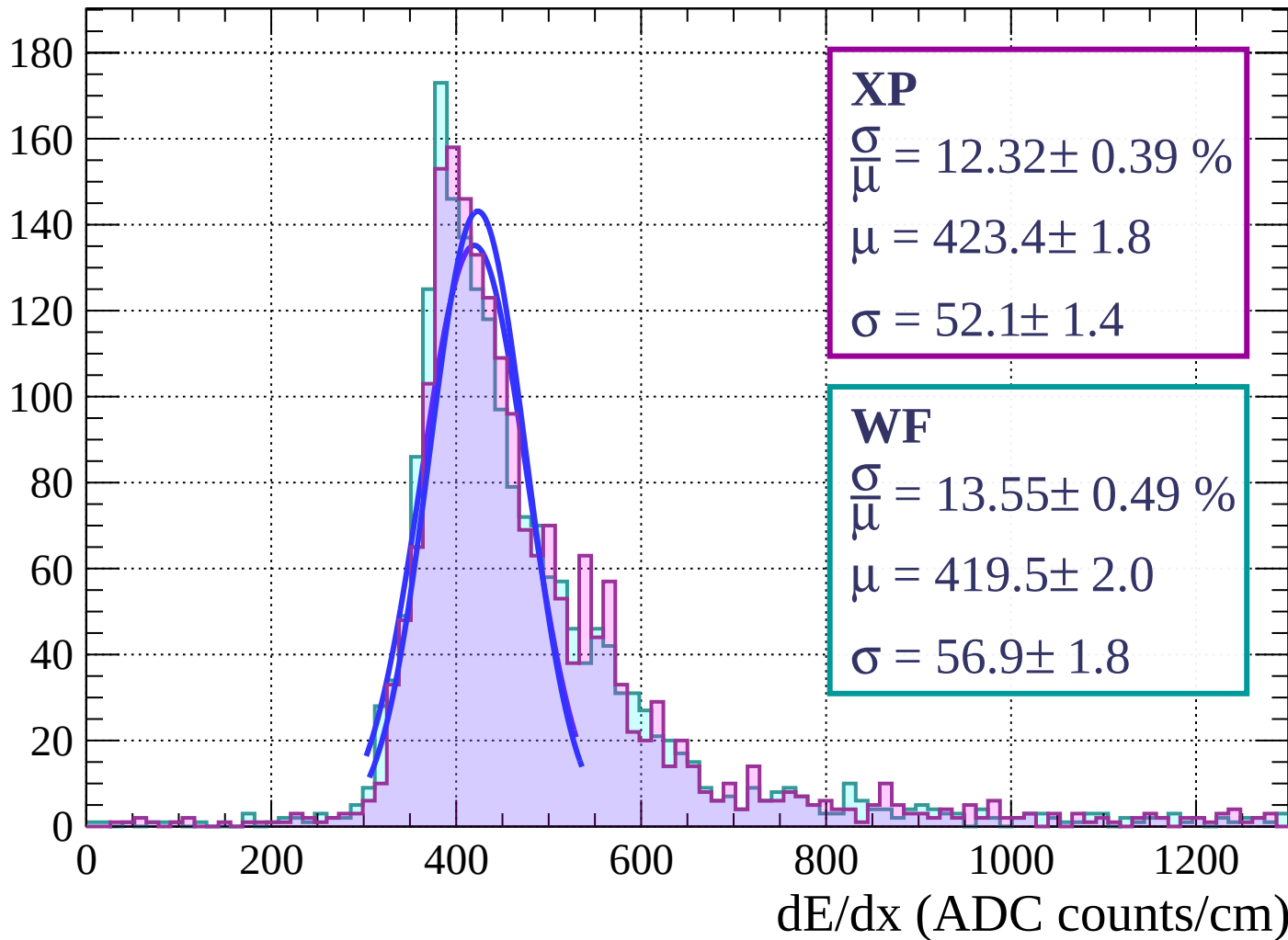


Energy loss | $51 < \phi < 54$



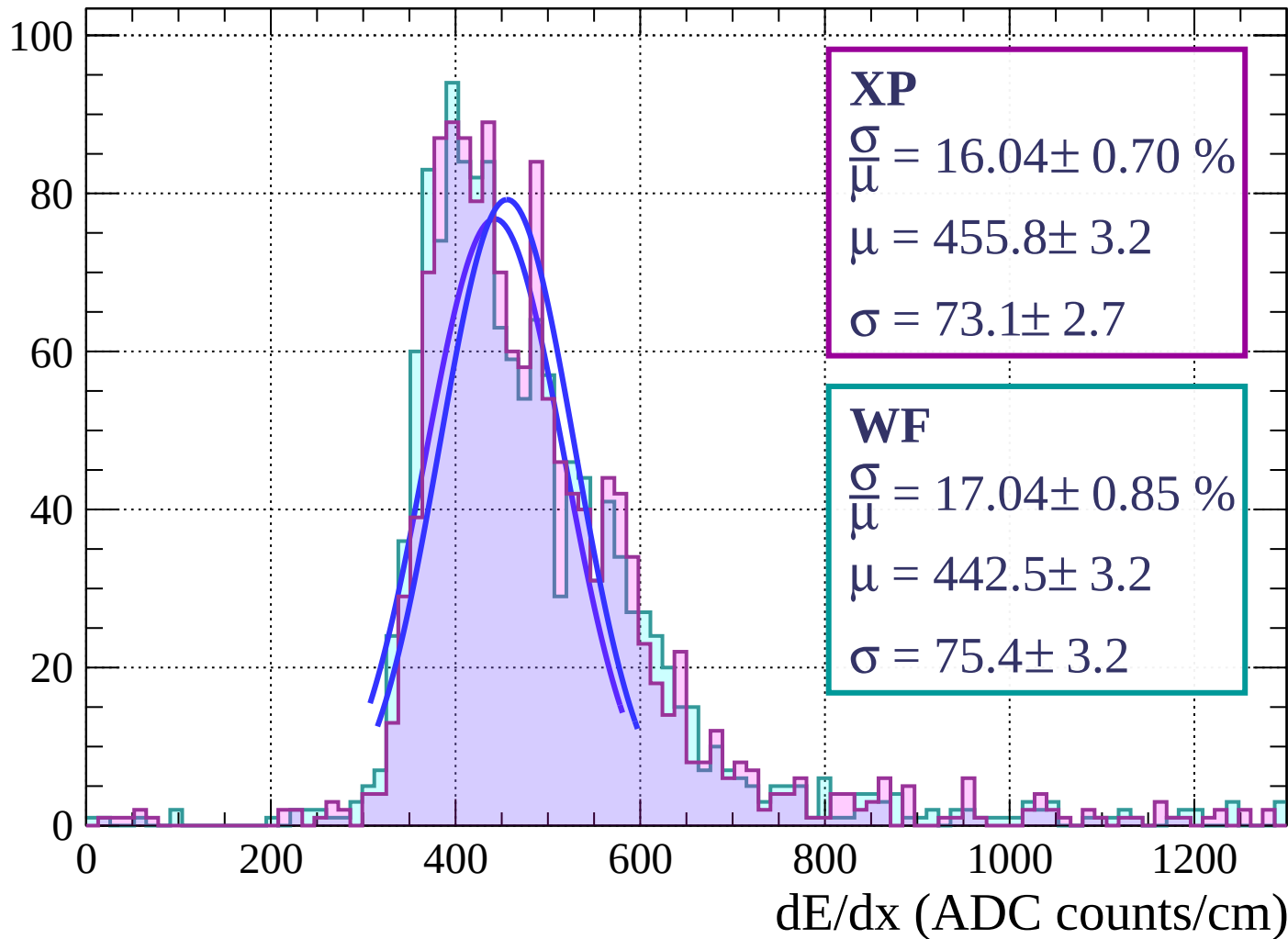
Energy loss | $54 < \phi < 57$

Count



Energy loss | $57 < \phi < 60$

Count



Energy loss | $60 < \phi < 63$

Count

XP

$$\frac{\sigma}{\mu} = 18.35 \pm 1.02 \%$$

$$\mu = 474.3 \pm 4.5$$

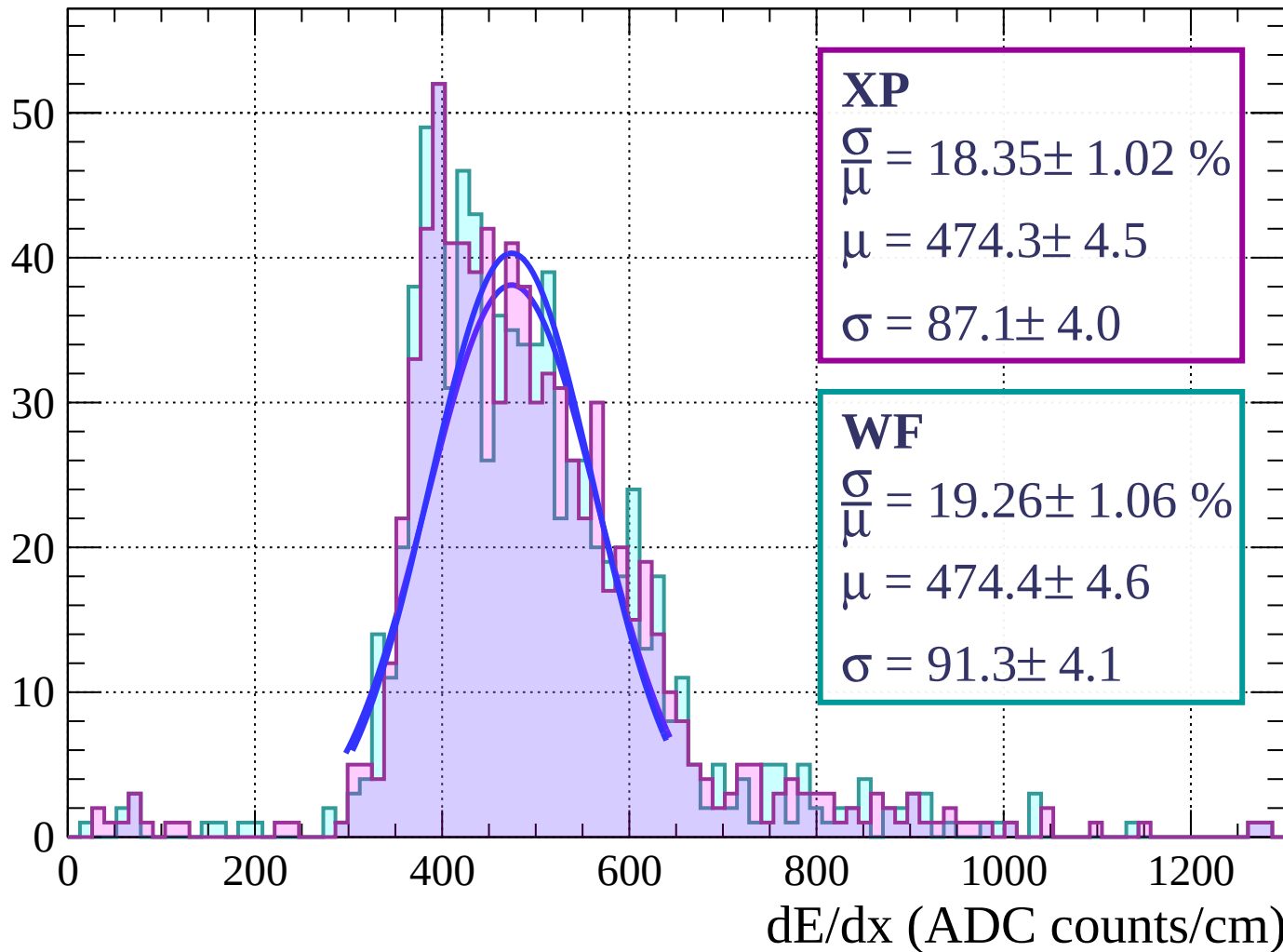
$$\sigma = 87.1 \pm 4.0$$

WF

$$\frac{\sigma}{\mu} = 19.26 \pm 1.06 \%$$

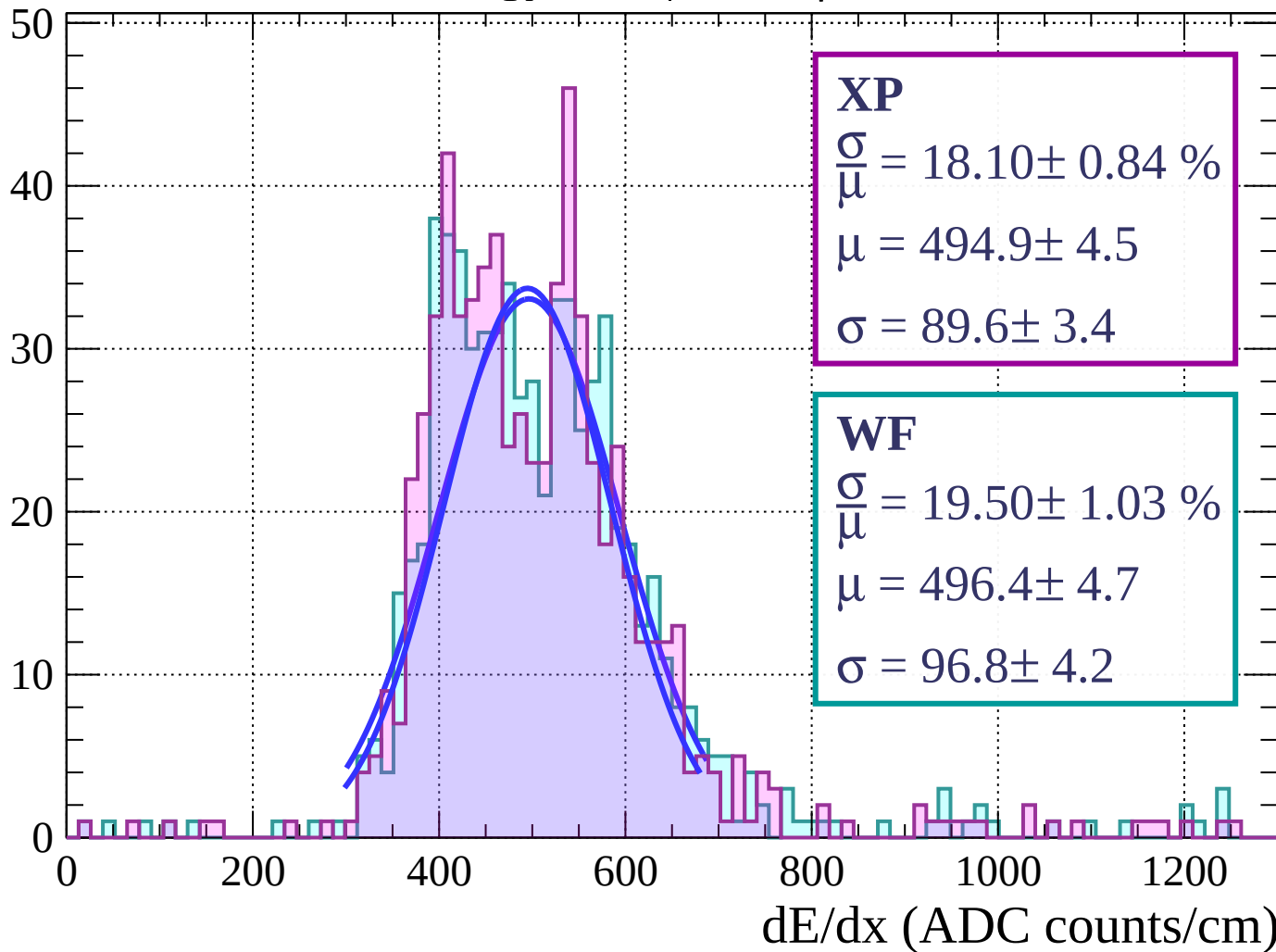
$$\mu = 474.4 \pm 4.6$$

$$\sigma = 91.3 \pm 4.1$$



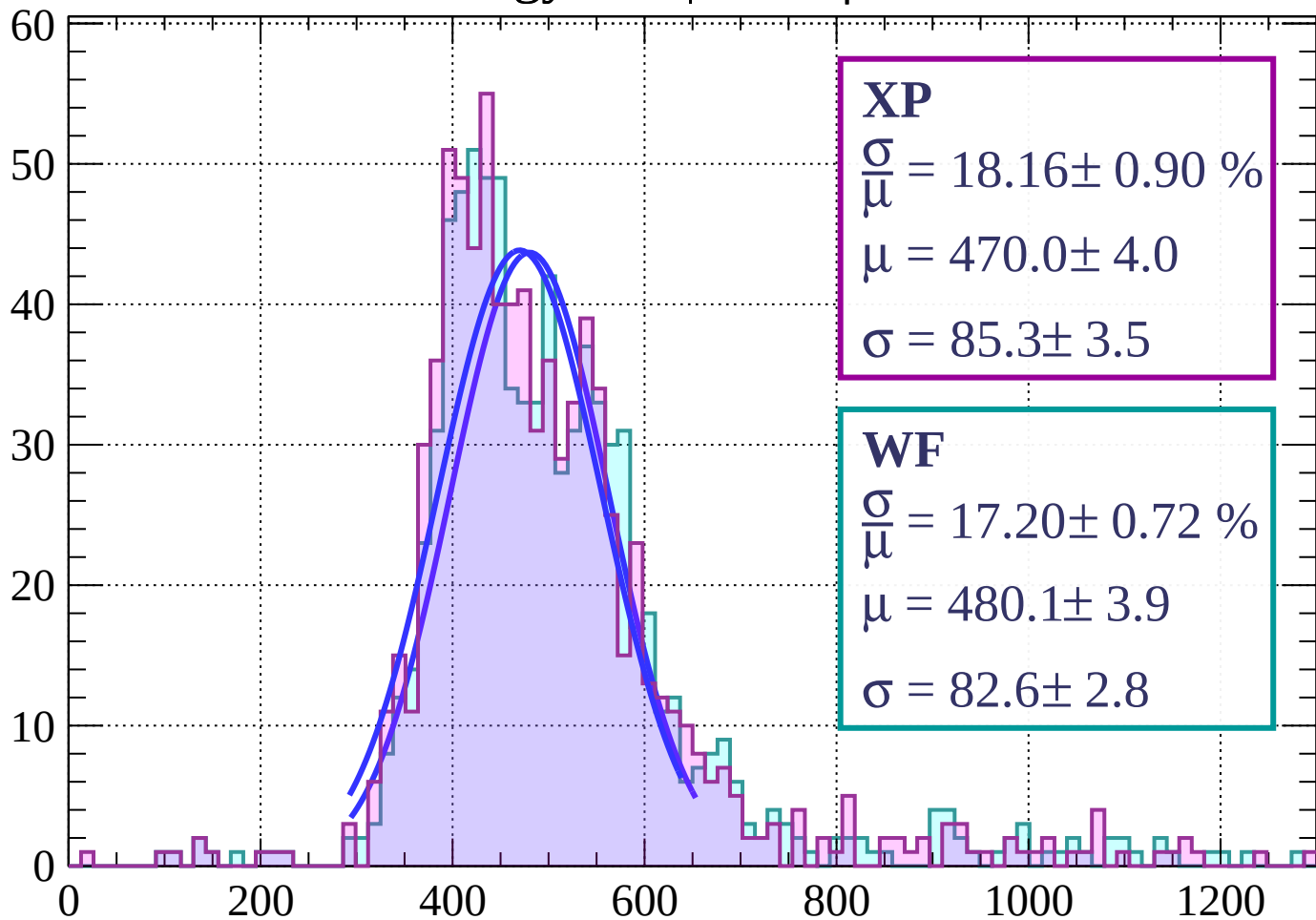
Energy loss | $63 < \phi < 66$

Count



Energy loss | $66 < \phi < 69$

Count



XP

$$\frac{\sigma}{\mu} = 18.16 \pm 0.90 \%$$

$$\mu = 470.0 \pm 4.0$$

$$\sigma = 85.3 \pm 3.5$$

WF

$$\frac{\sigma}{\mu} = 17.20 \pm 0.72 \%$$

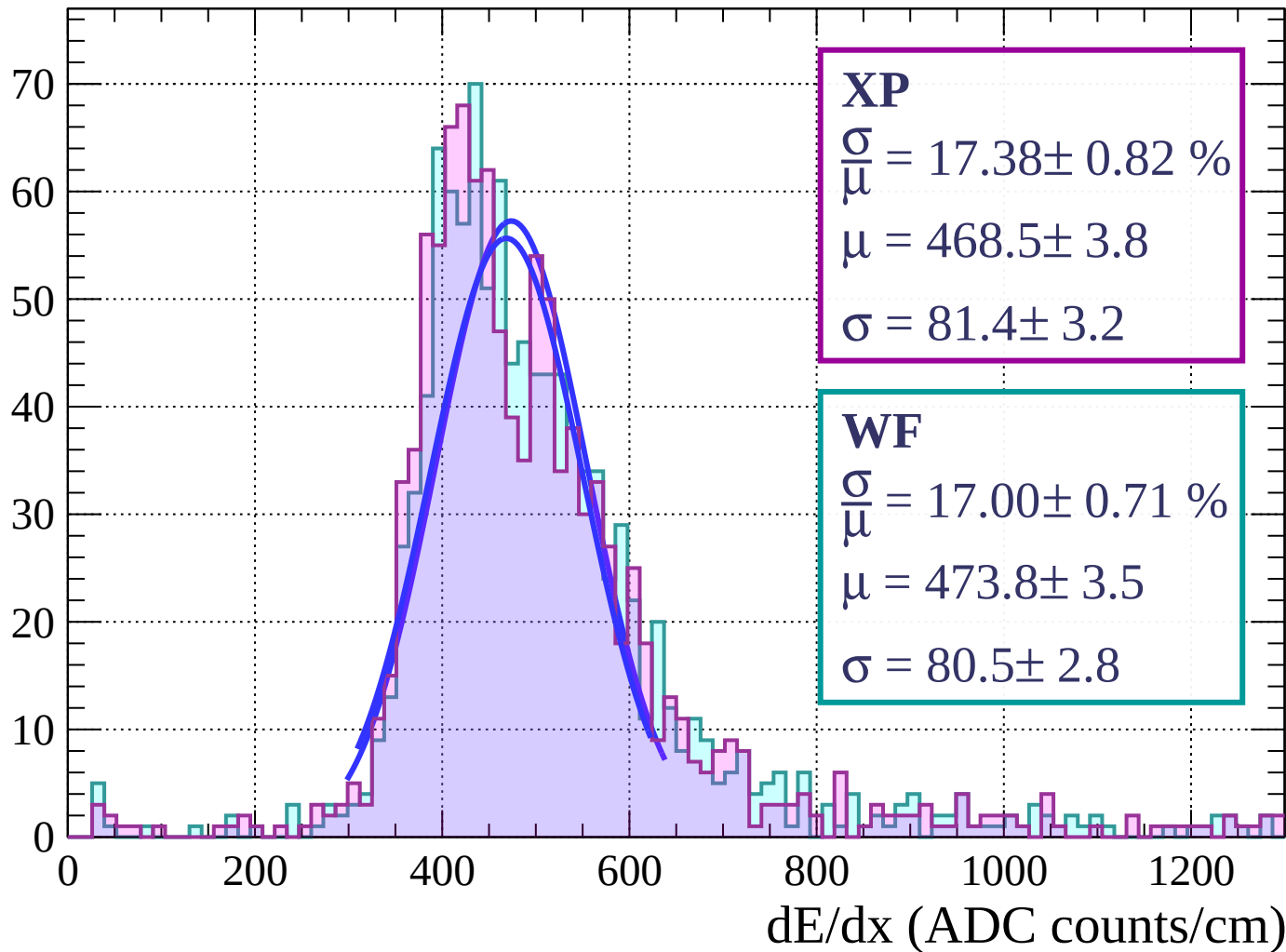
$$\mu = 480.1 \pm 3.9$$

$$\sigma = 82.6 \pm 2.8$$

dE/dx (ADC counts/cm)

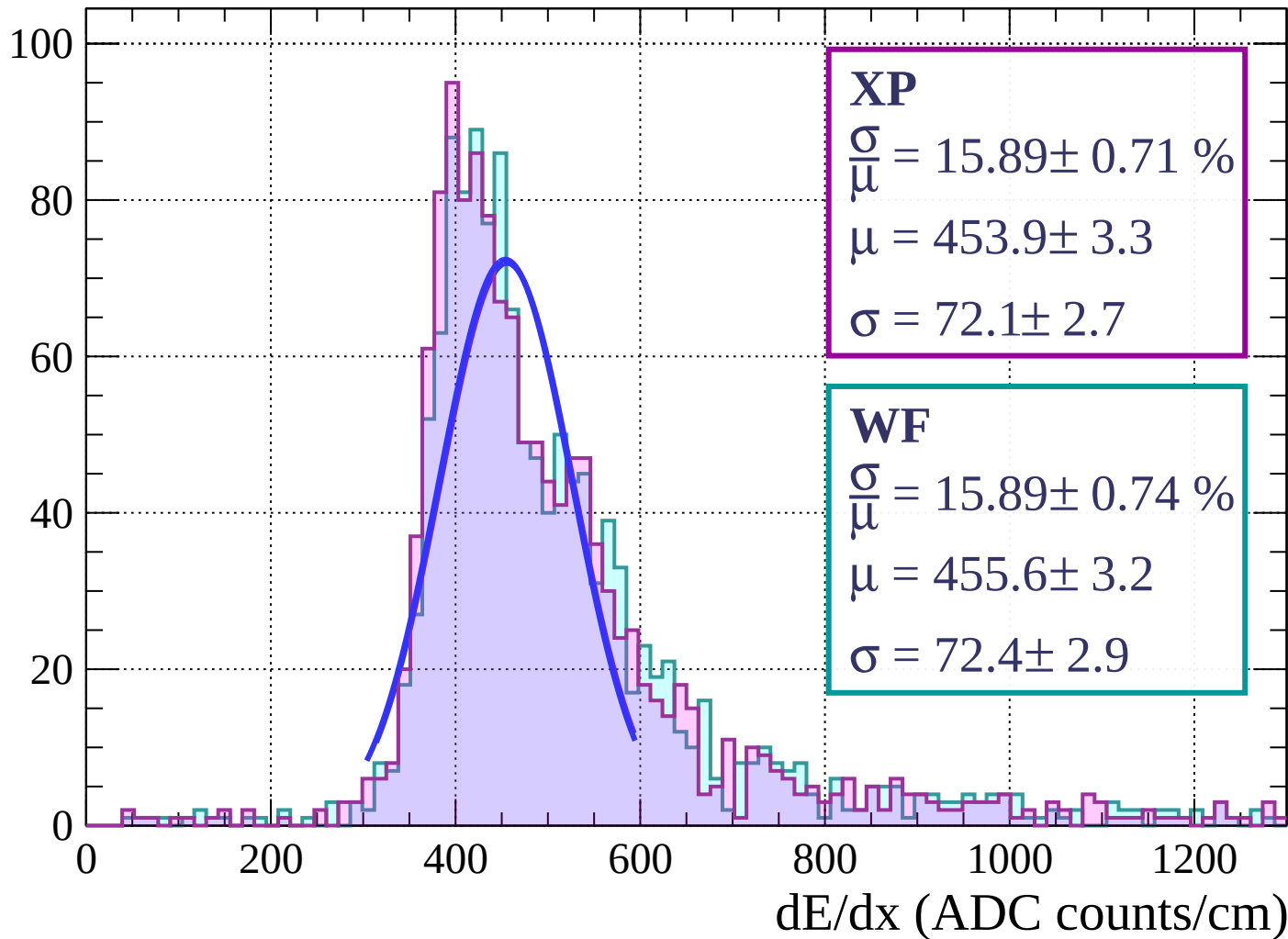
Energy loss | $69 < \phi < 72$

Count



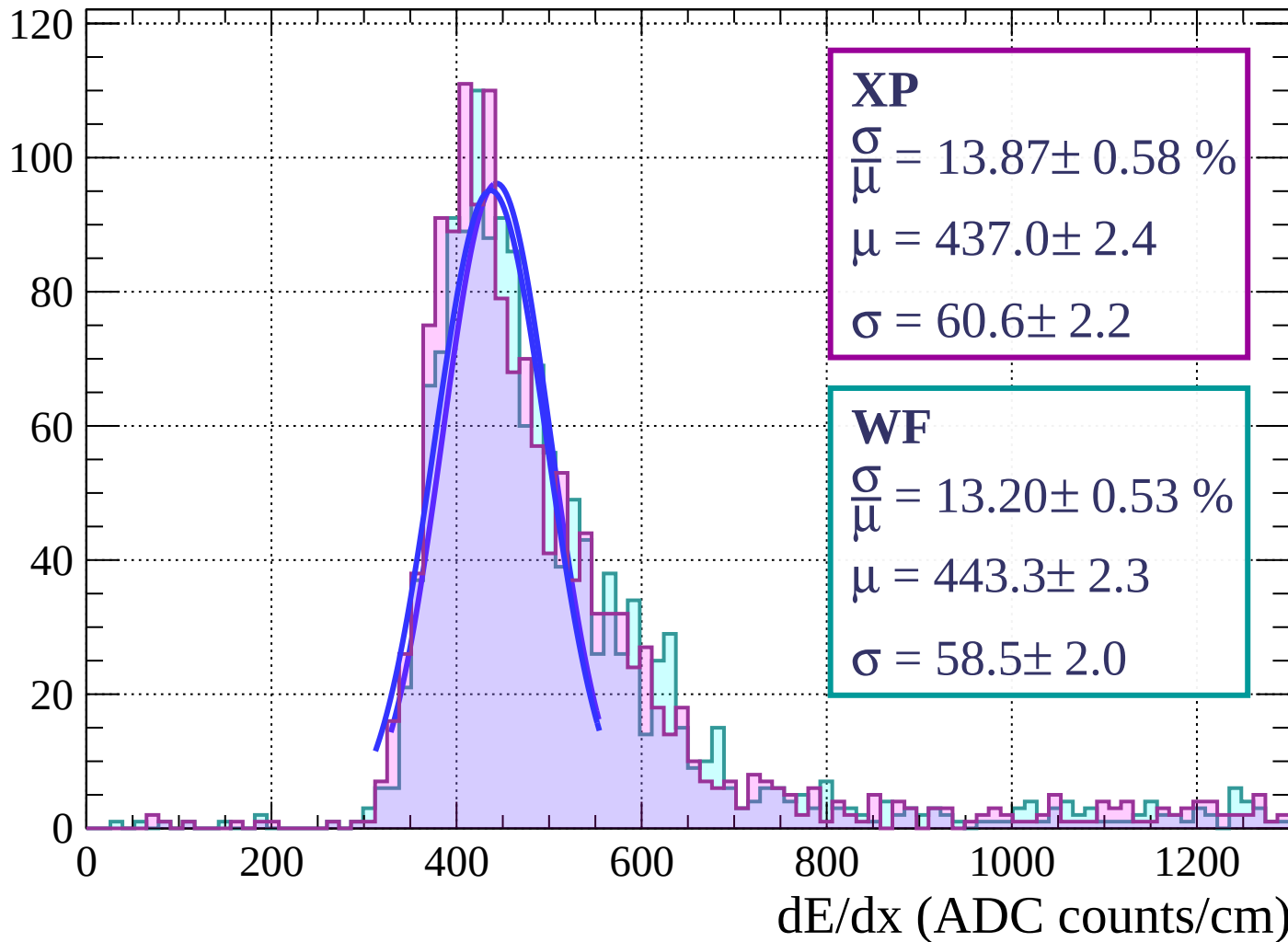
Energy loss | $72 < \phi < 75$

Count



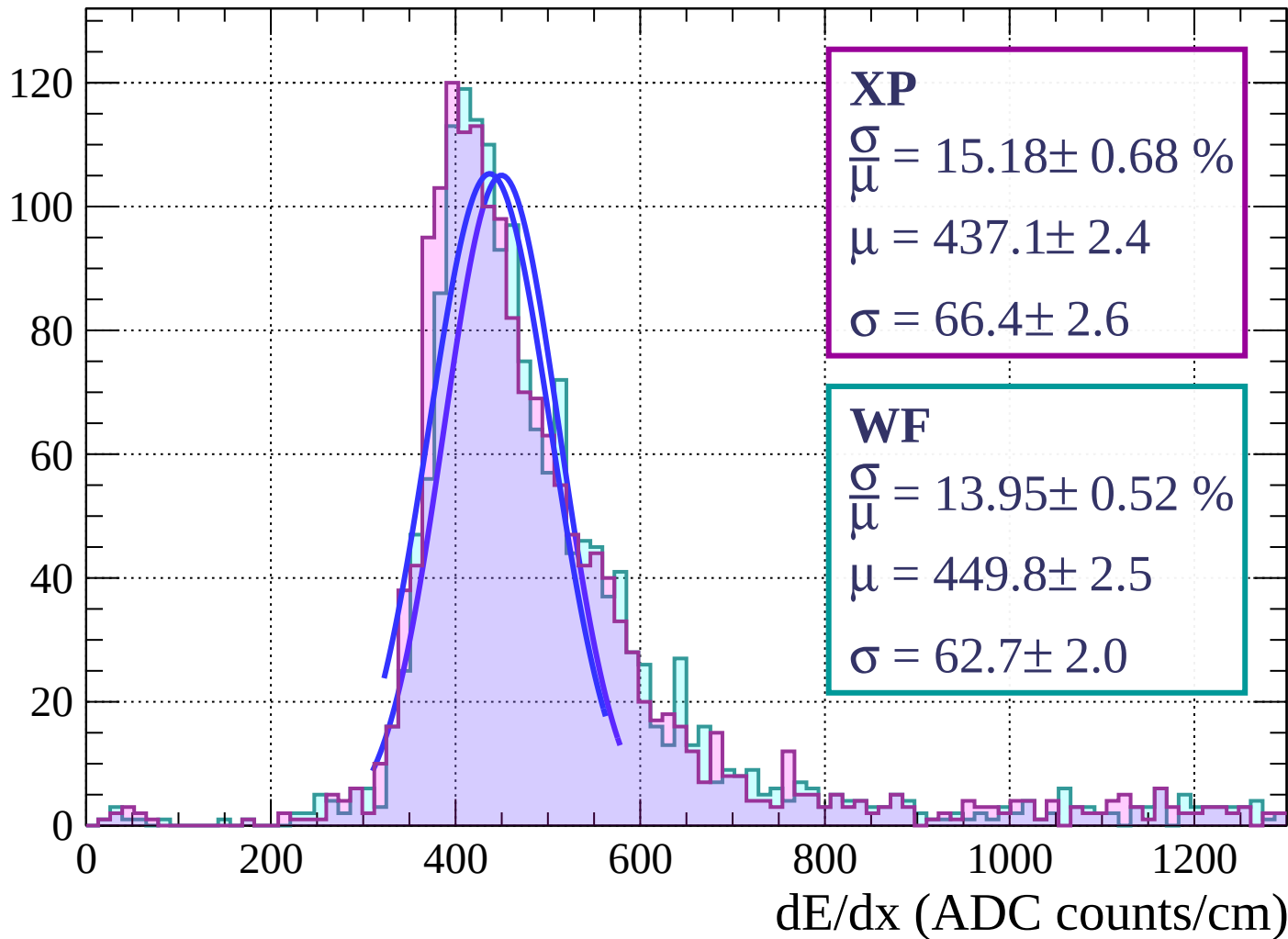
Energy loss | $75 < \phi < 78$

Count



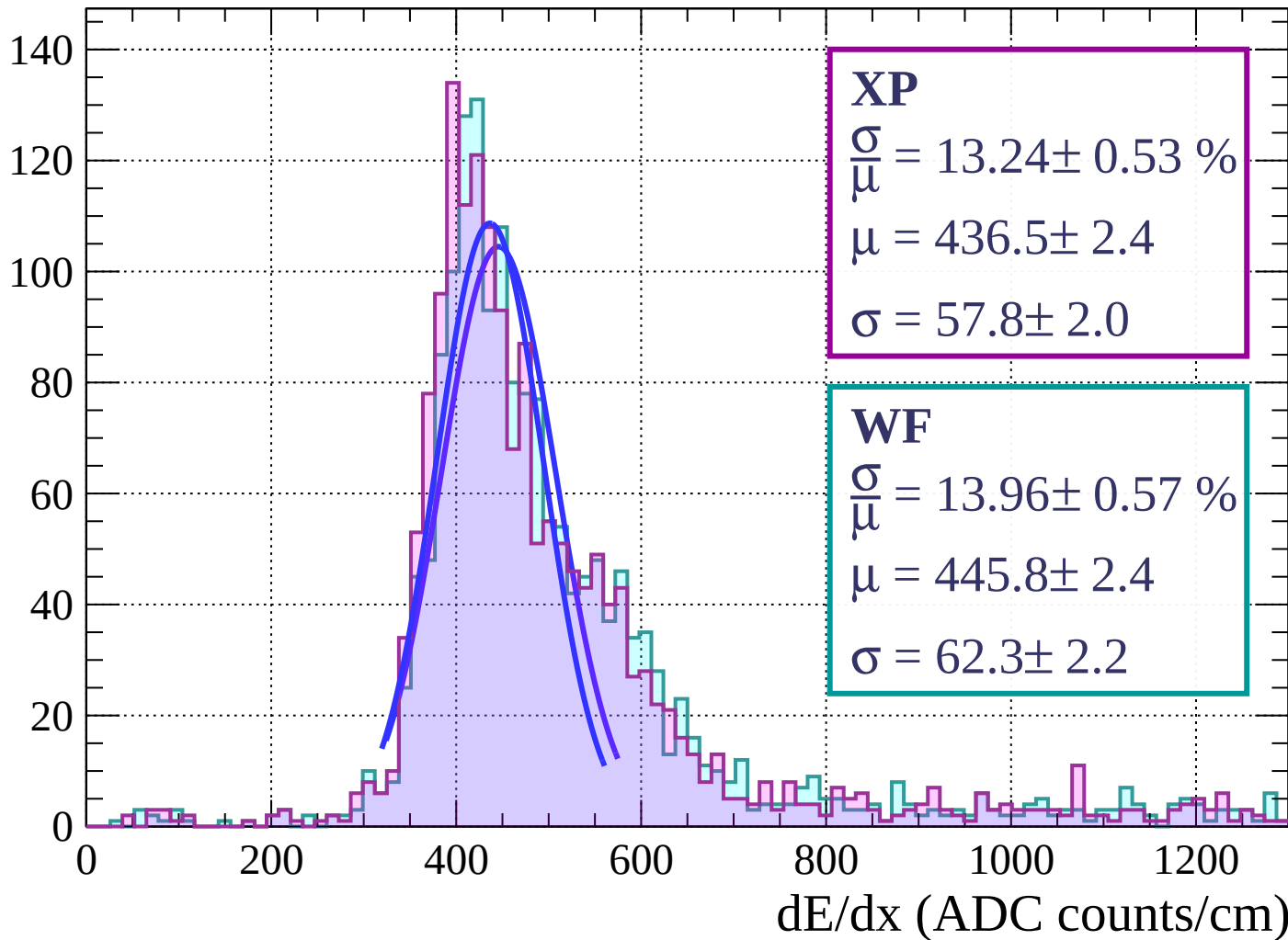
Energy loss | $78 < \phi < 81$

Count



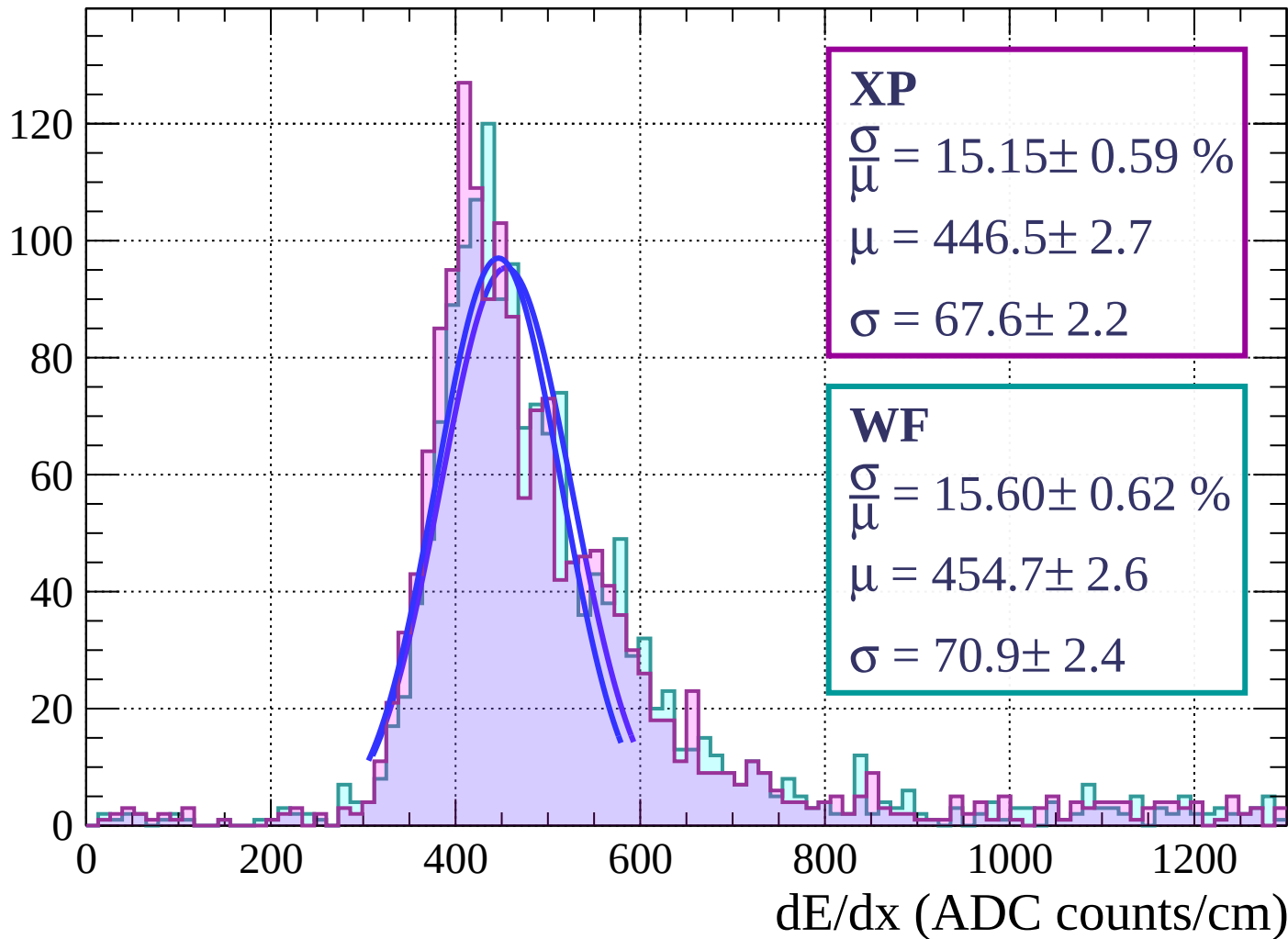
Energy loss | $81 < \phi < 84$

Count



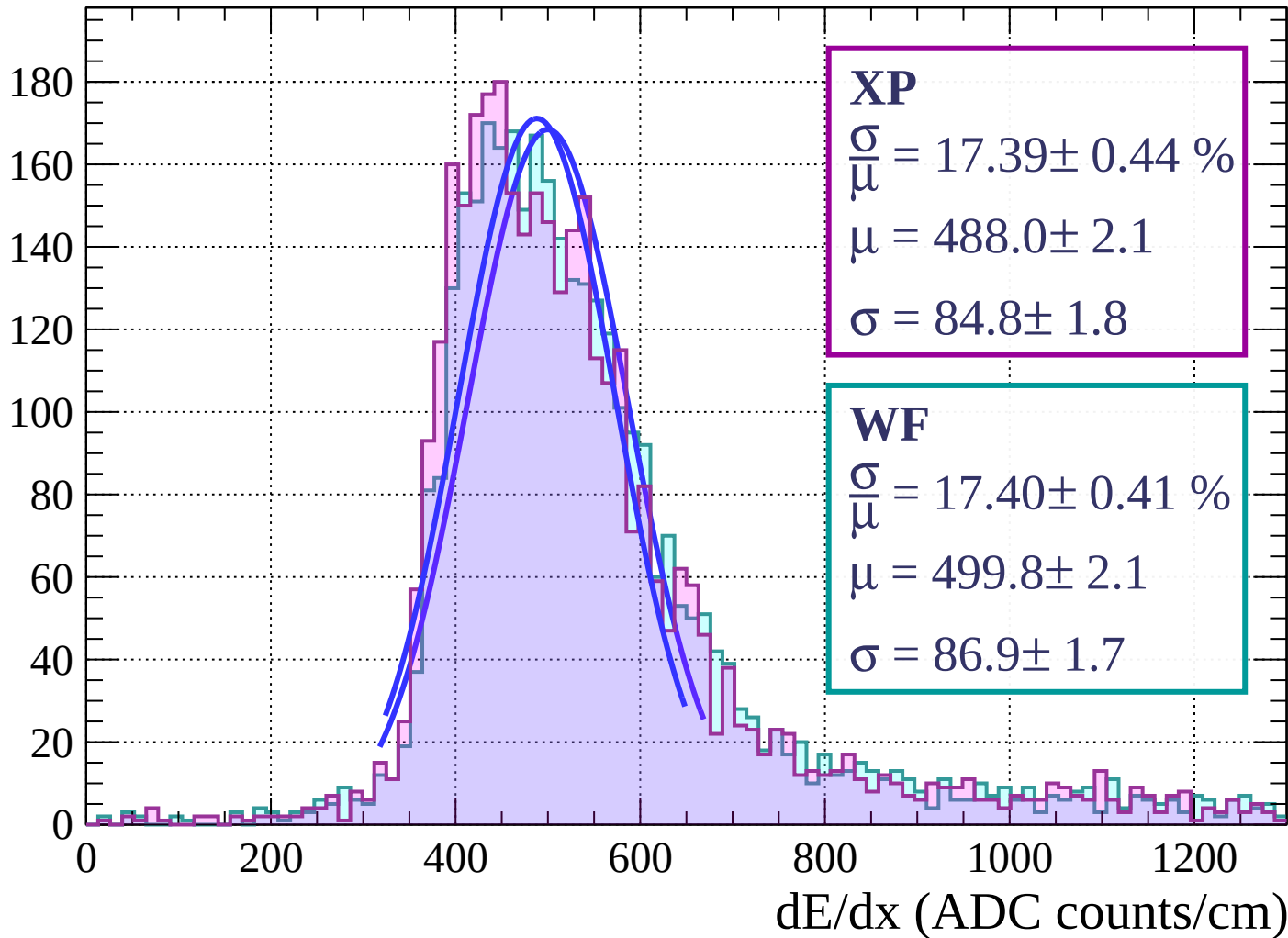
Energy loss | $84 < \phi < 87$

Count

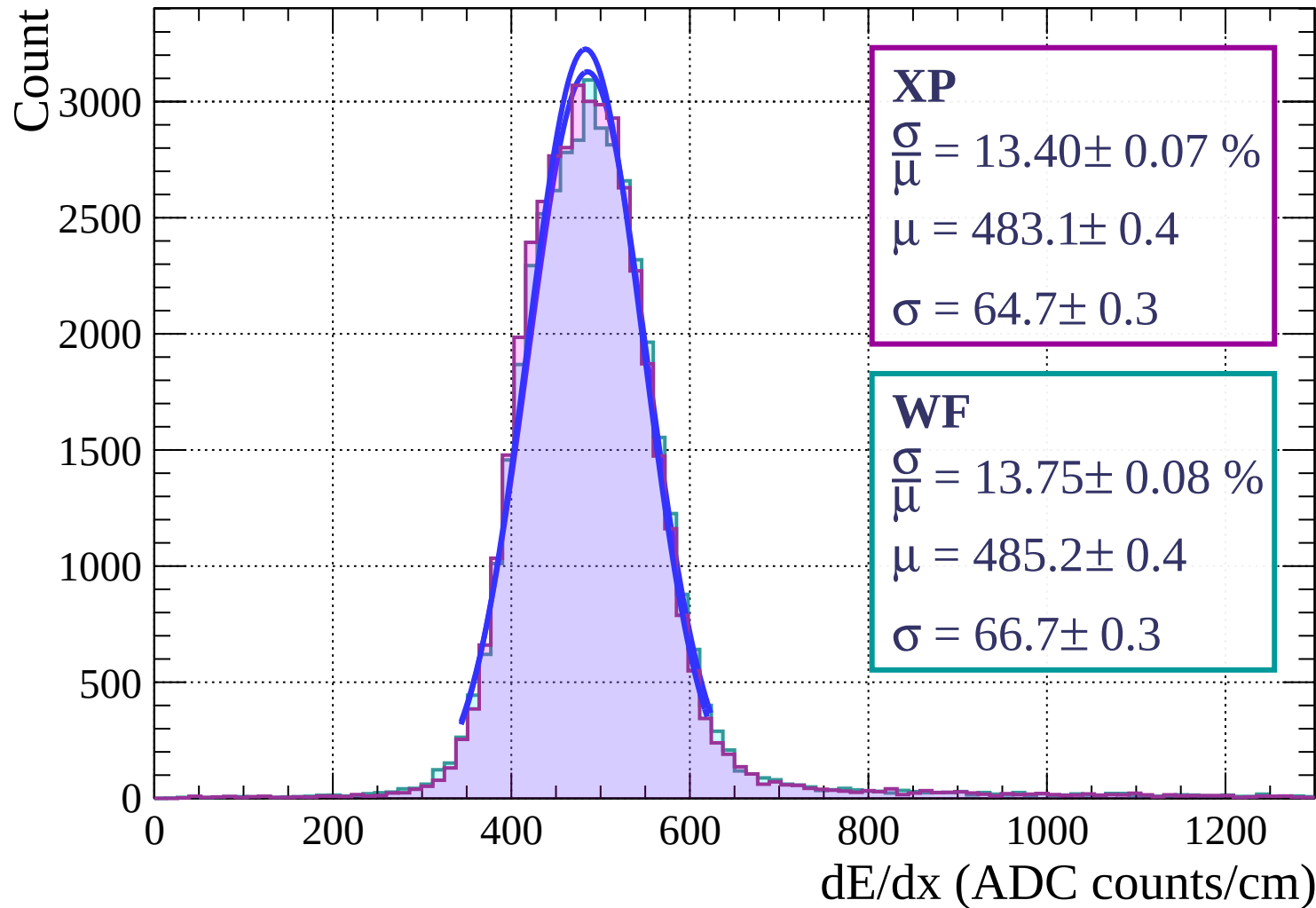


Energy loss | $87 < \phi < 90$

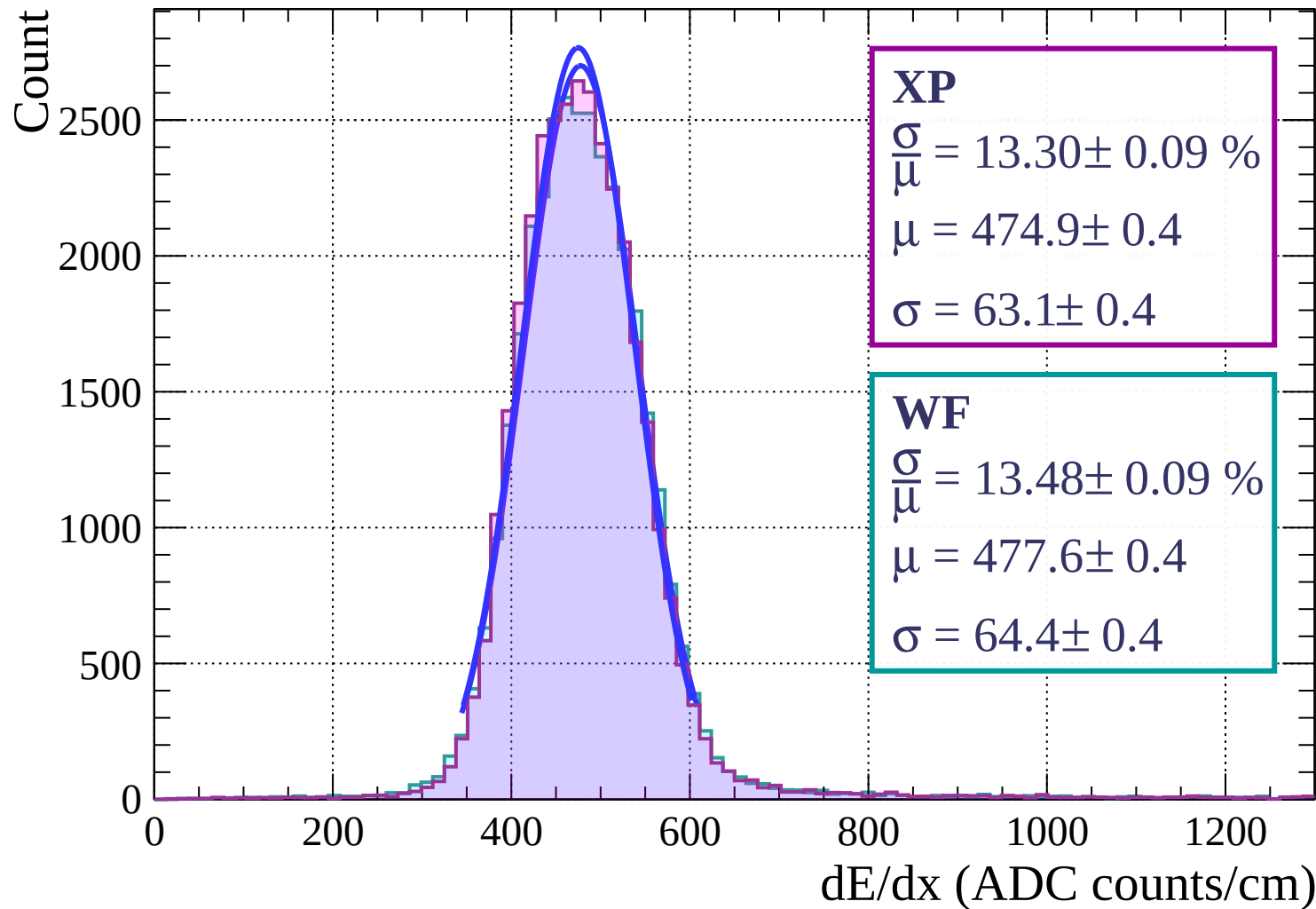
Count



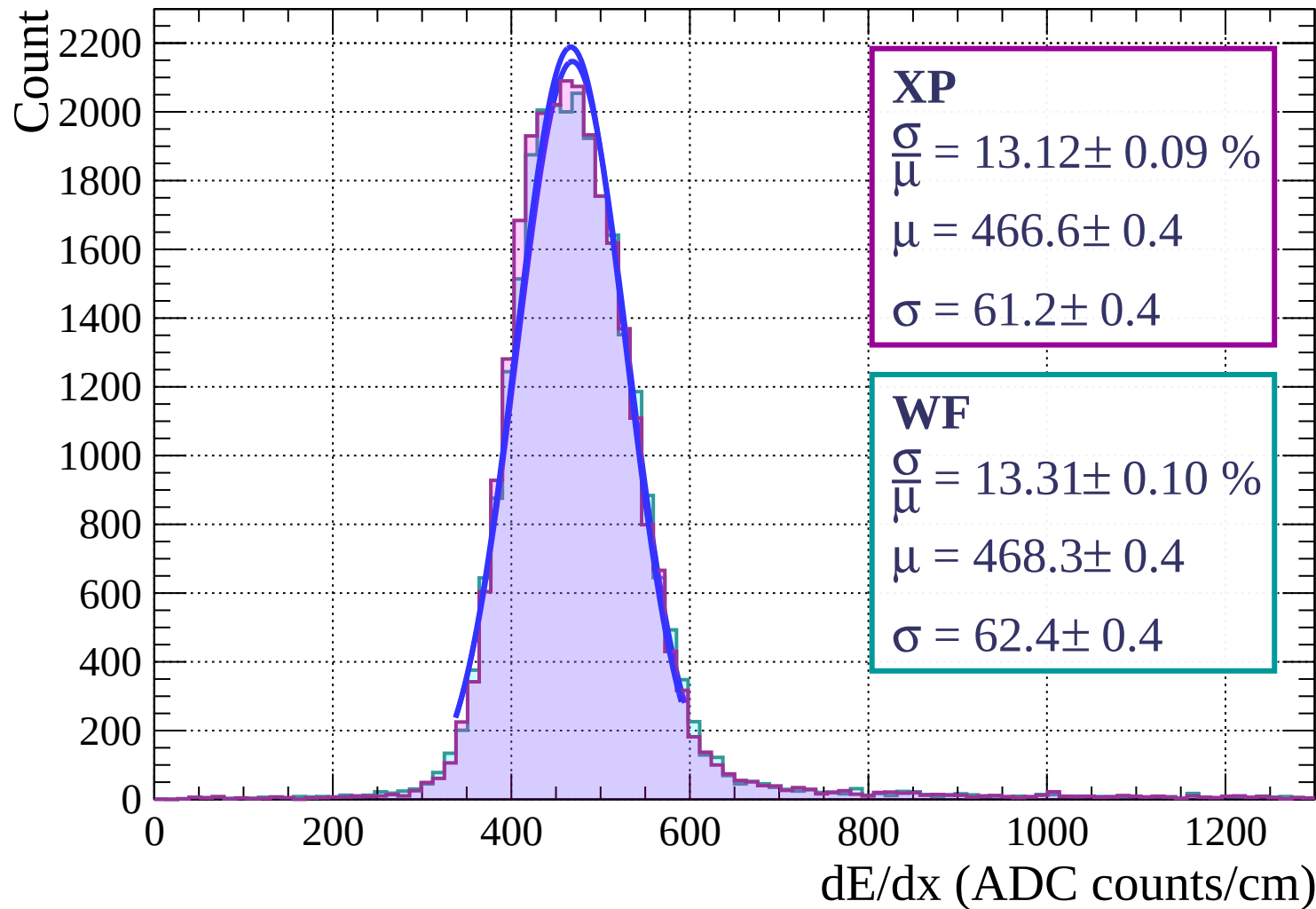
Energy loss | $0 < \theta < 3$



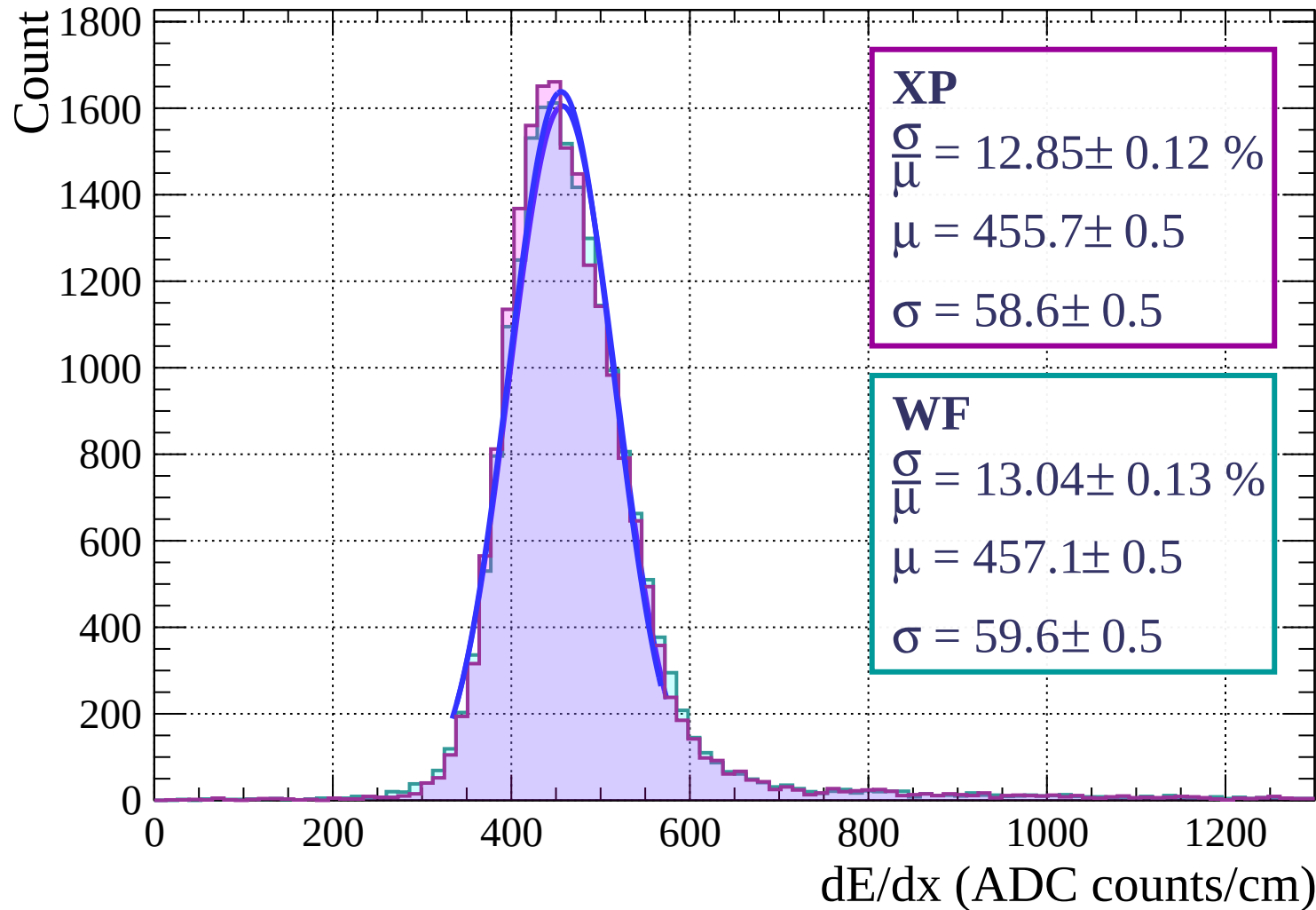
Energy loss | $3 < \theta < 6$



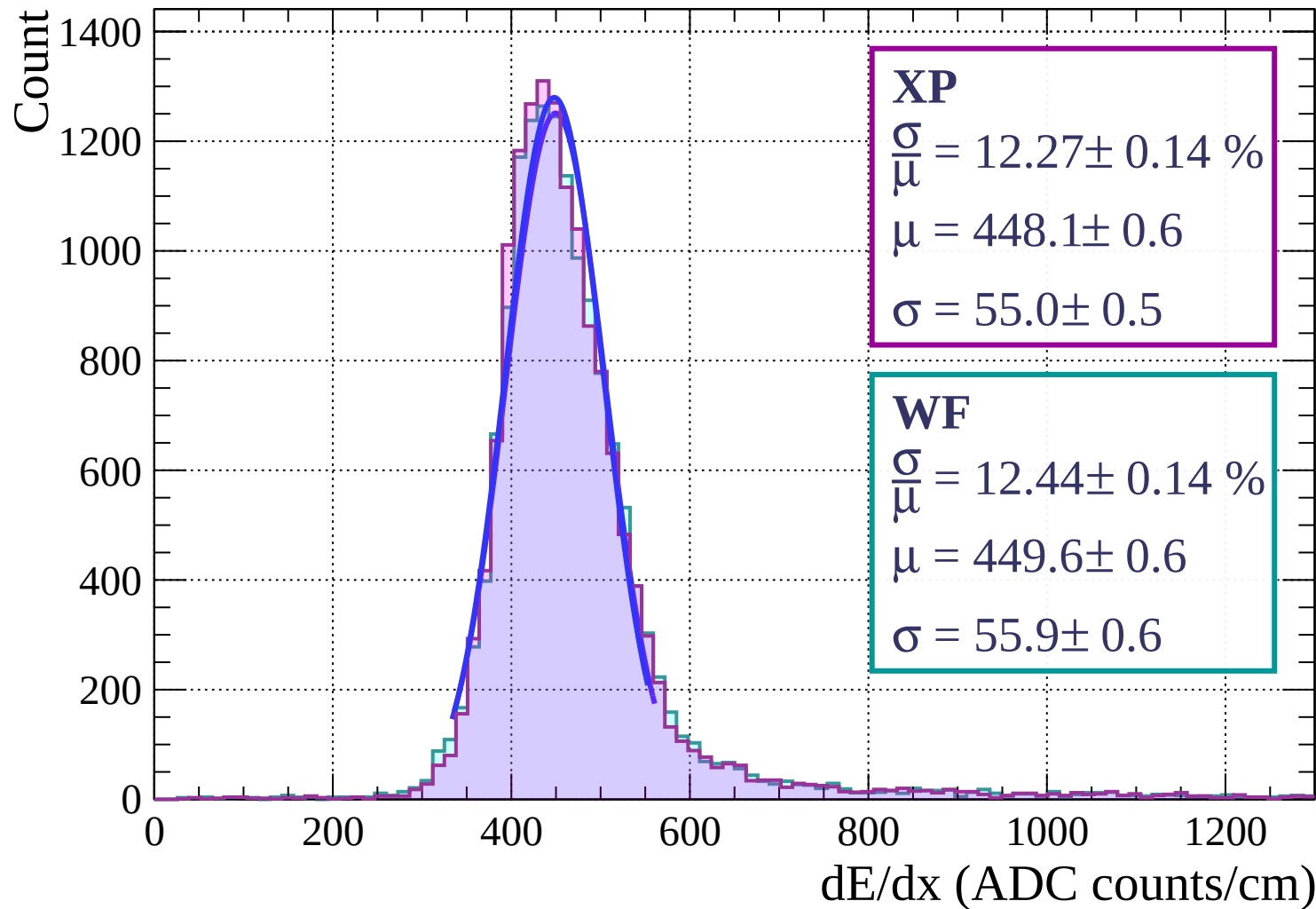
Energy loss | $6 < \theta < 9$



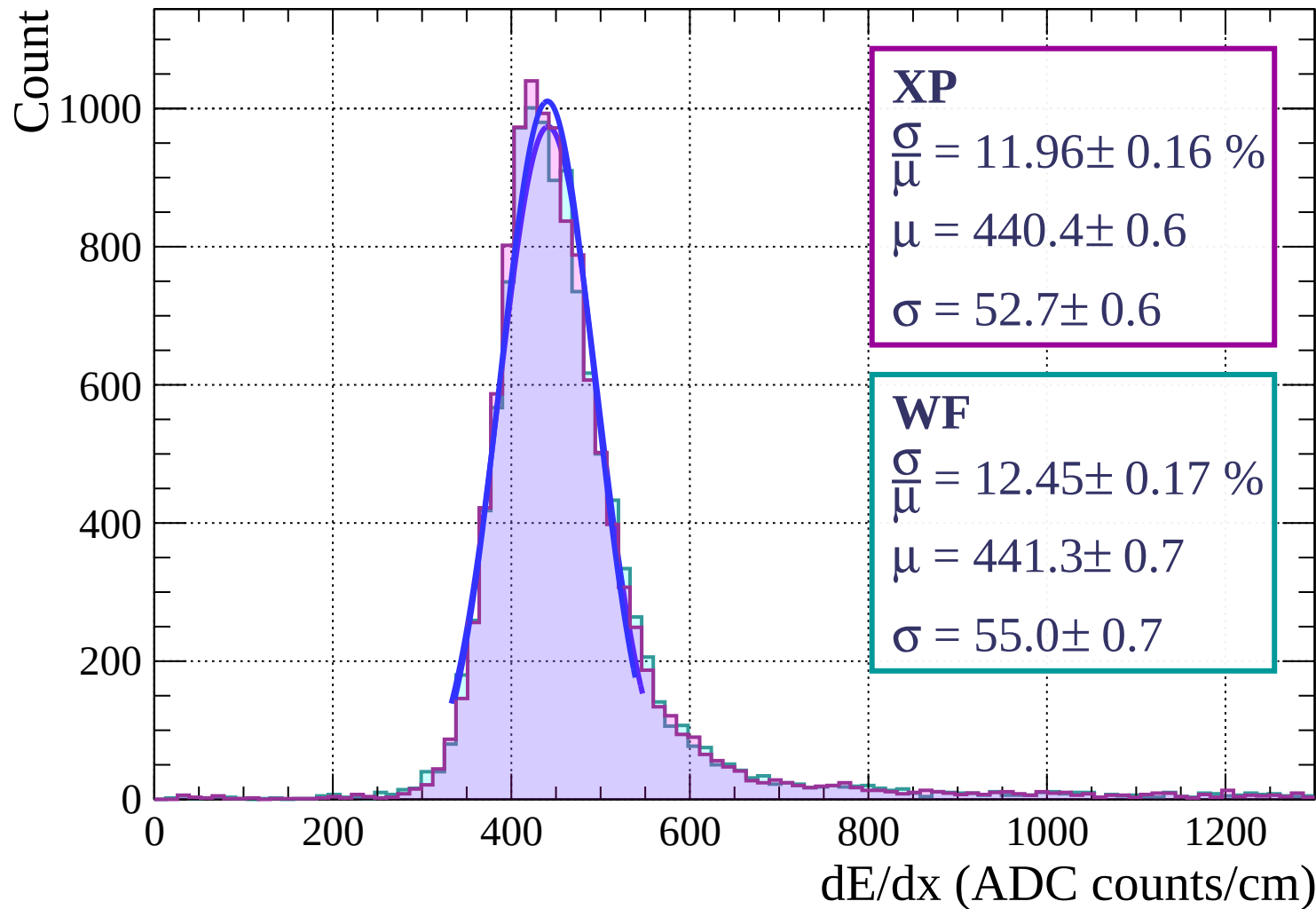
Energy loss | $9 < \theta < 12$



Energy loss | $12 < \theta < 15$

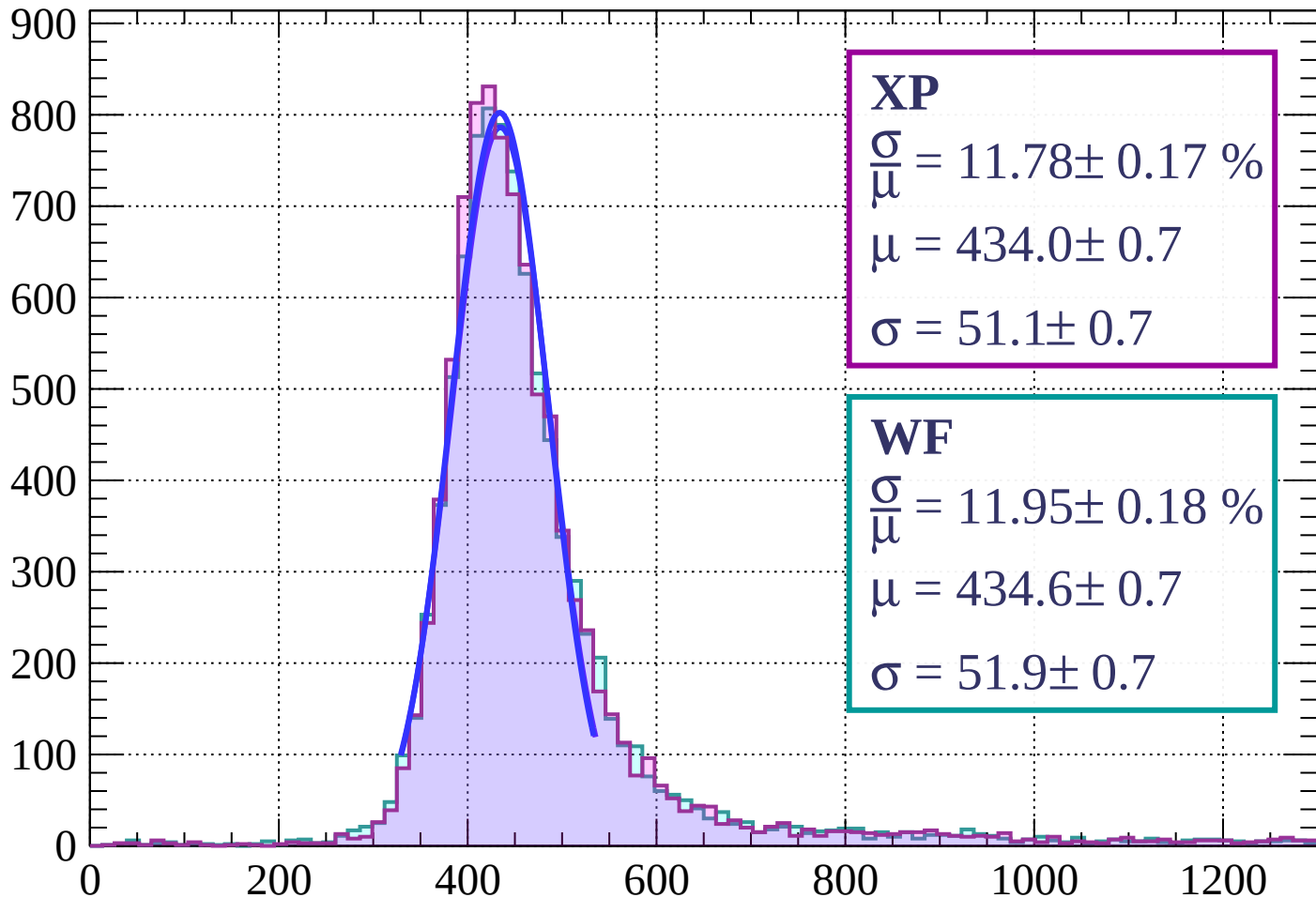


Energy loss | $15 < \theta < 18$



Energy loss | $18 < \theta < 21$

Count



XP

$$\frac{\sigma}{\mu} = 11.78 \pm 0.17 \%$$

$$\mu = 434.0 \pm 0.7$$

$$\sigma = 51.1 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 11.95 \pm 0.18 \%$$

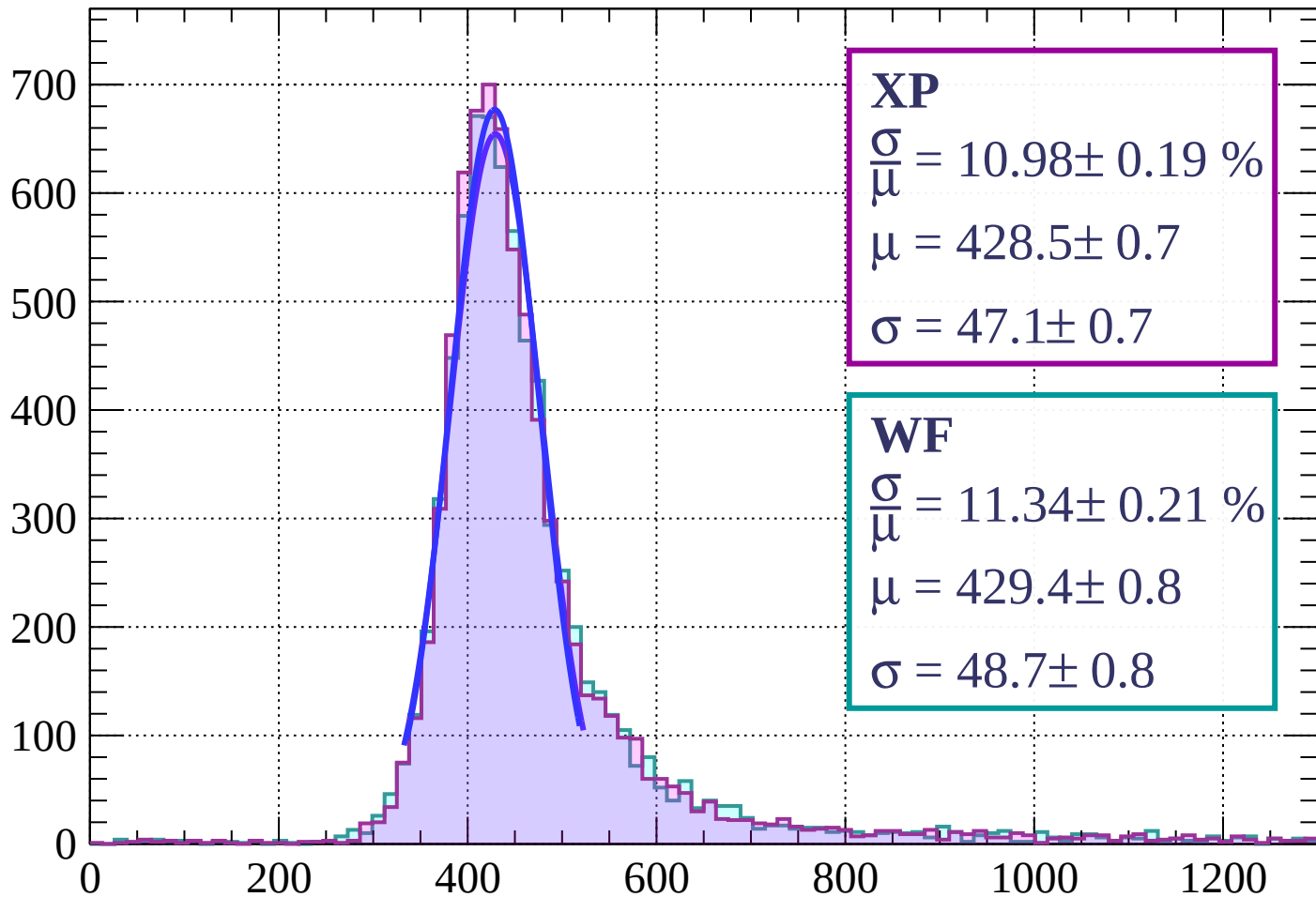
$$\mu = 434.6 \pm 0.7$$

$$\sigma = 51.9 \pm 0.7$$

dE/dx (ADC counts/cm)

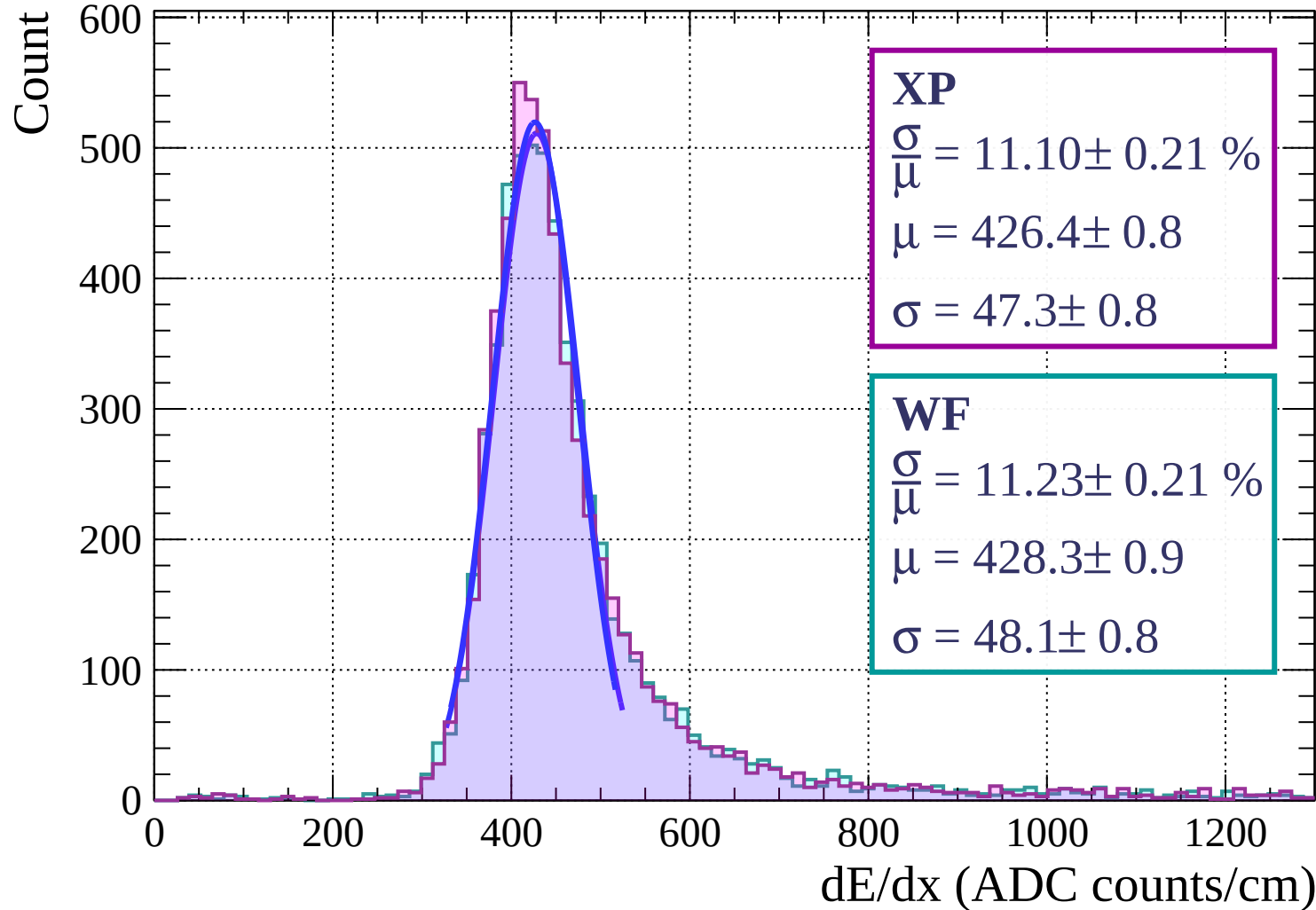
Energy loss | $21 < \theta < 24$

Count

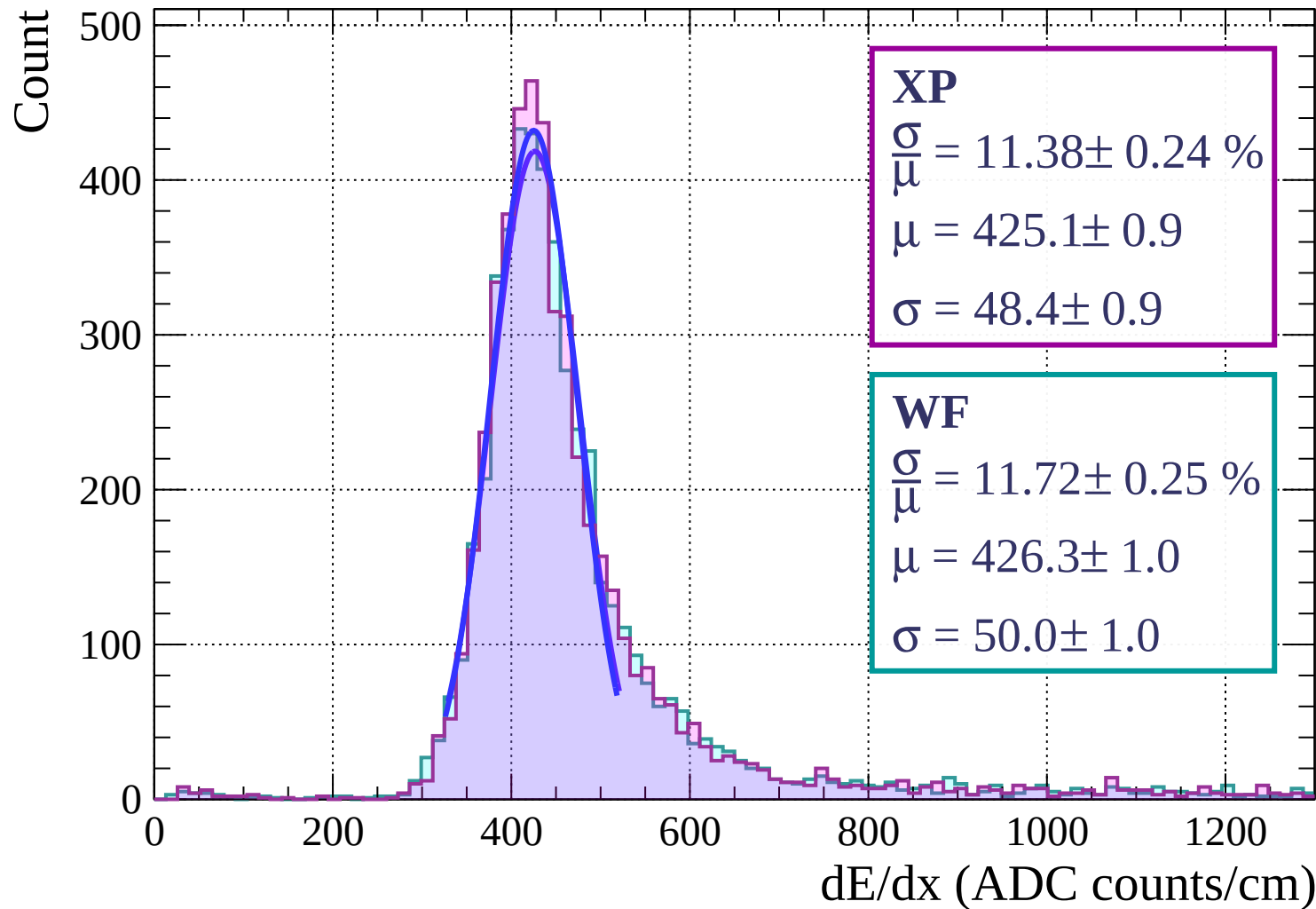


dE/dx (ADC counts/cm)

Energy loss | $24 < \theta < 27$

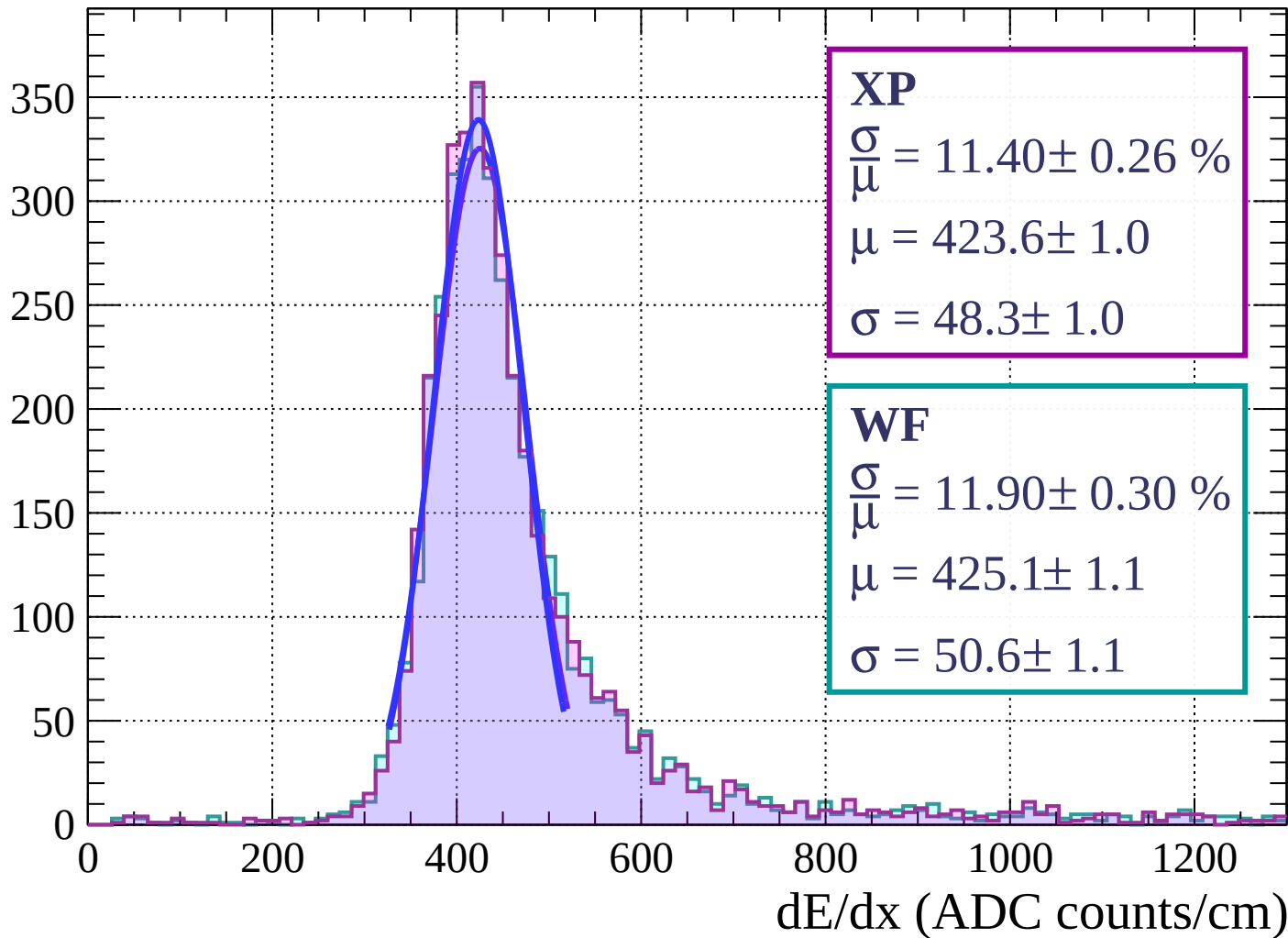


Energy loss | $27 < \theta < 30$

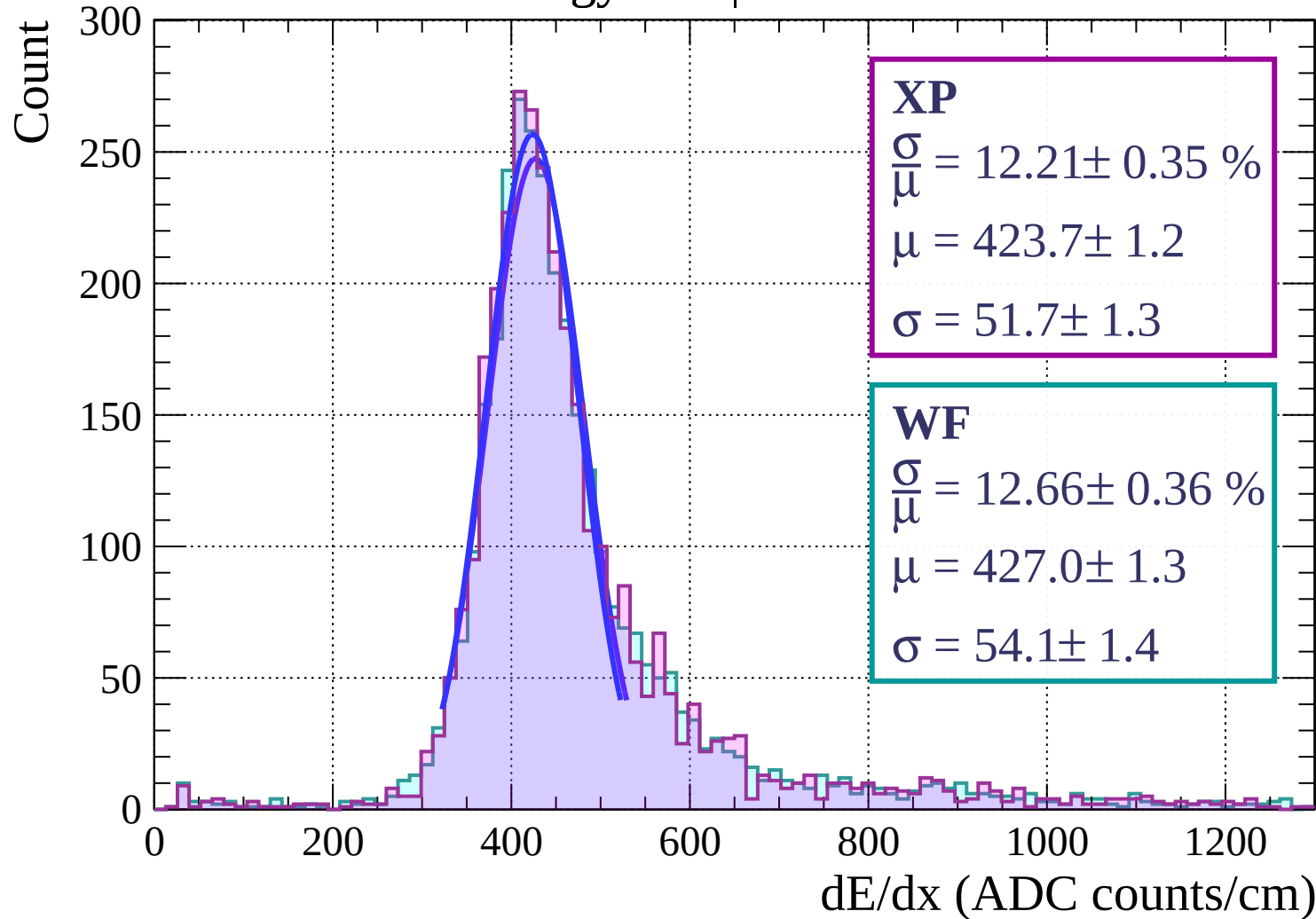


Energy loss | $30 < \theta < 33$

Count

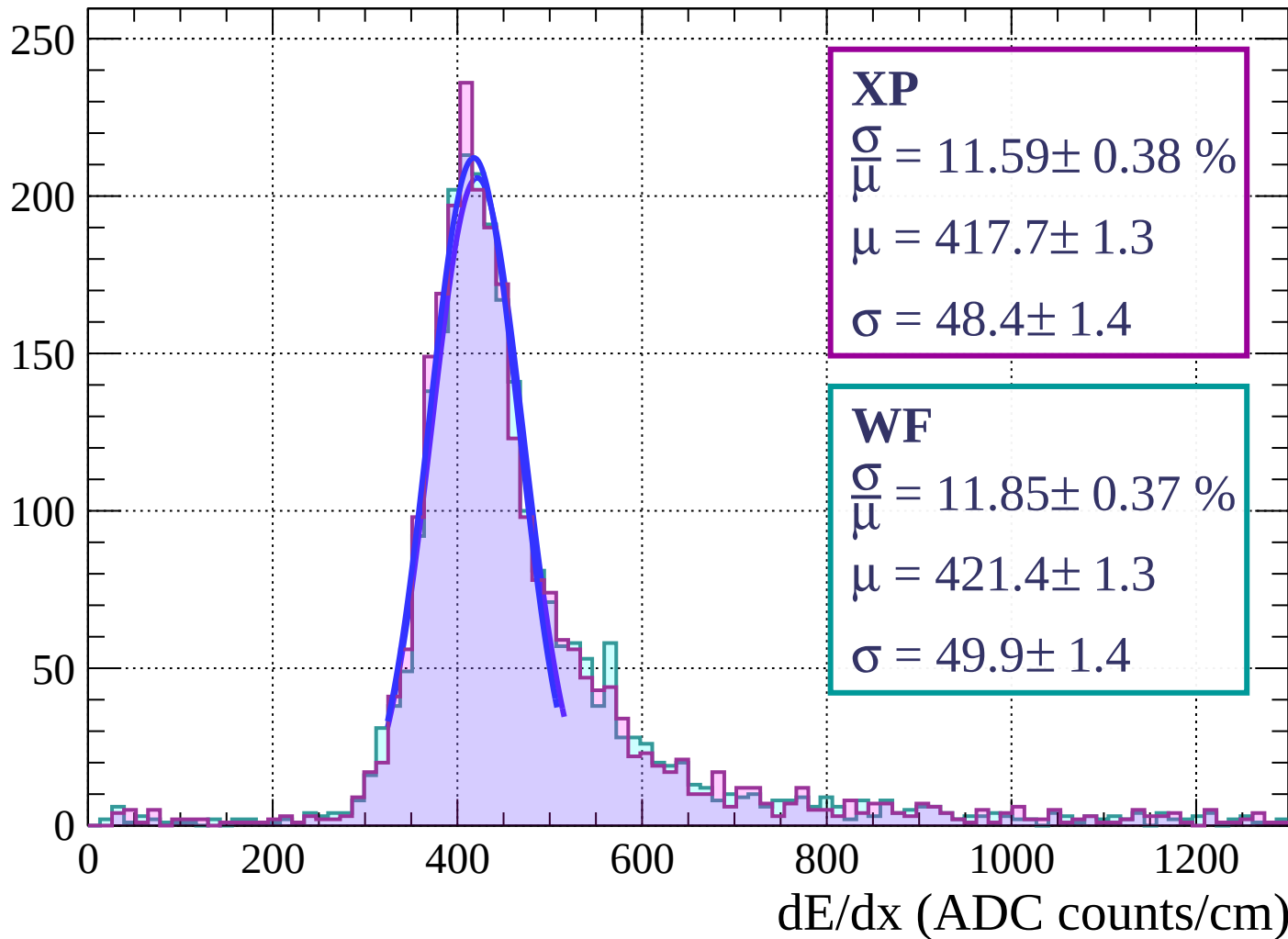


Energy loss | $33 < \theta < 36$



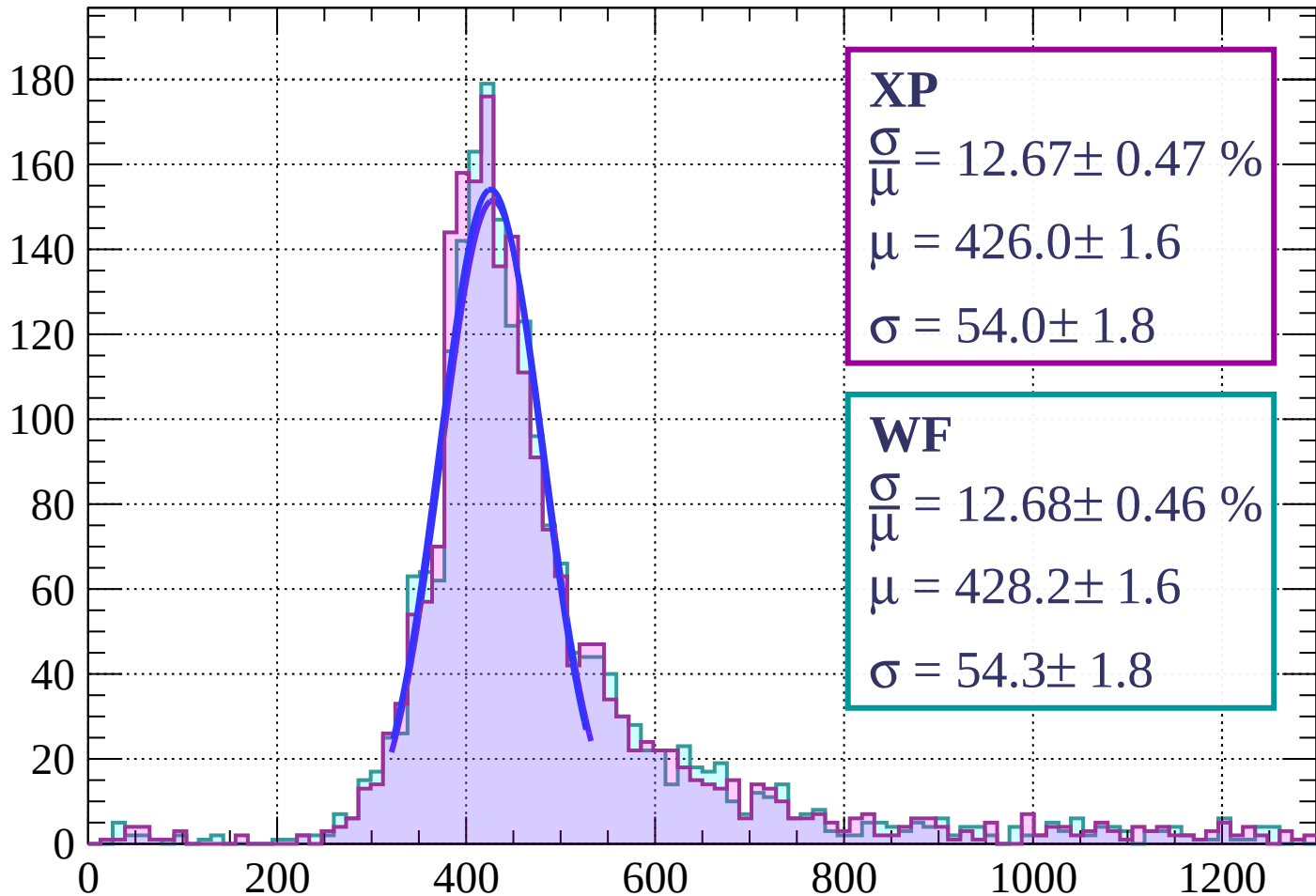
Energy loss | $36 < \theta < 39$

Count



Energy loss | $39 < \theta < 42$

Count



XP

$$\frac{\sigma}{\mu} = 12.67 \pm 0.47 \%$$

$$\mu = 426.0 \pm 1.6$$

$$\sigma = 54.0 \pm 1.8$$

WF

$$\frac{\sigma}{\mu} = 12.68 \pm 0.46 \%$$

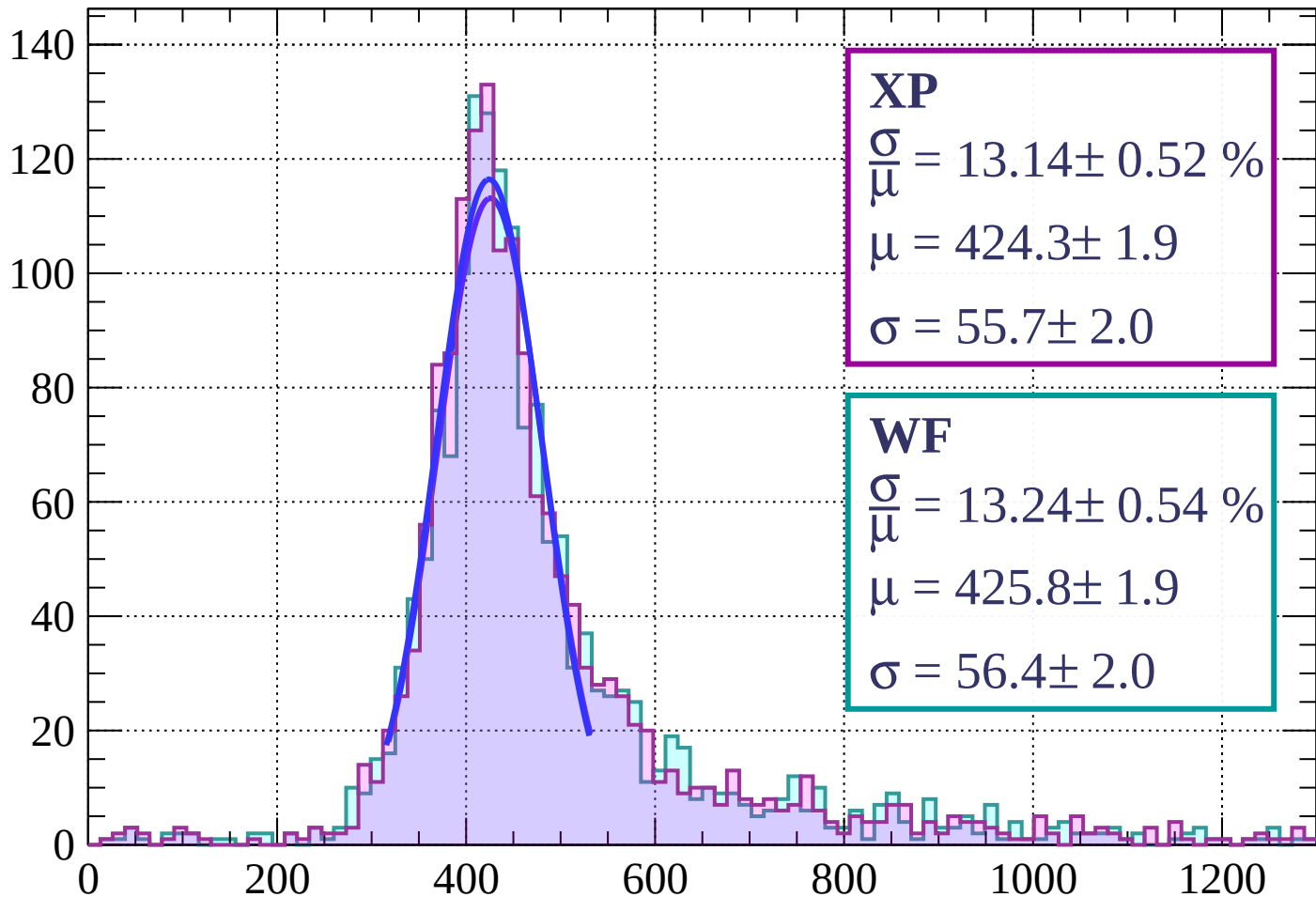
$$\mu = 428.2 \pm 1.6$$

$$\sigma = 54.3 \pm 1.8$$

dE/dx (ADC counts/cm)

Energy loss | $42 < \theta < 45$

Count



XP

$$\frac{\sigma}{\mu} = 13.14 \pm 0.52 \%$$

$$\mu = 424.3 \pm 1.9$$

$$\sigma = 55.7 \pm 2.0$$

WF

$$\frac{\sigma}{\mu} = 13.24 \pm 0.54 \%$$

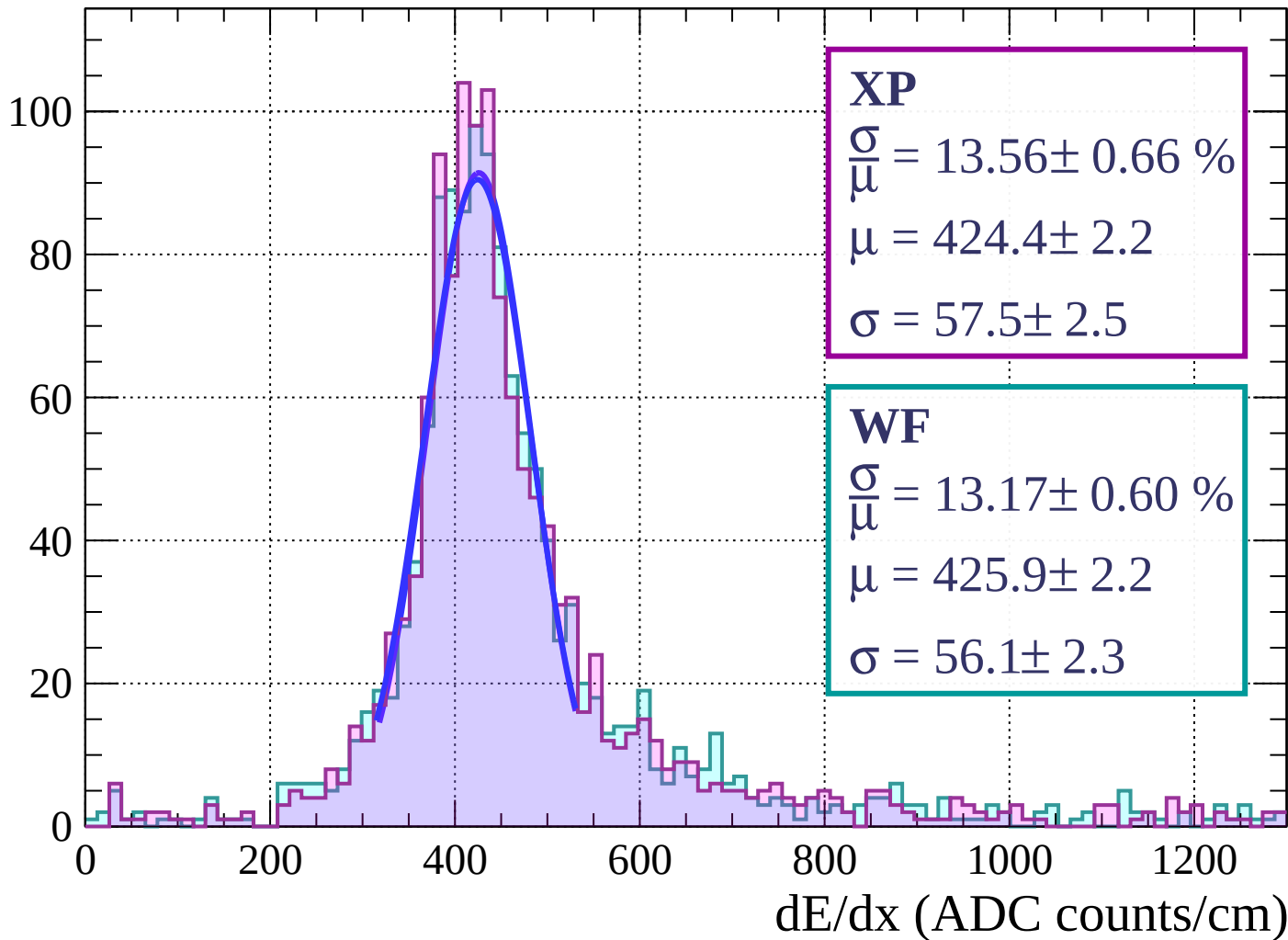
$$\mu = 425.8 \pm 1.9$$

$$\sigma = 56.4 \pm 2.0$$

dE/dx (ADC counts/cm)

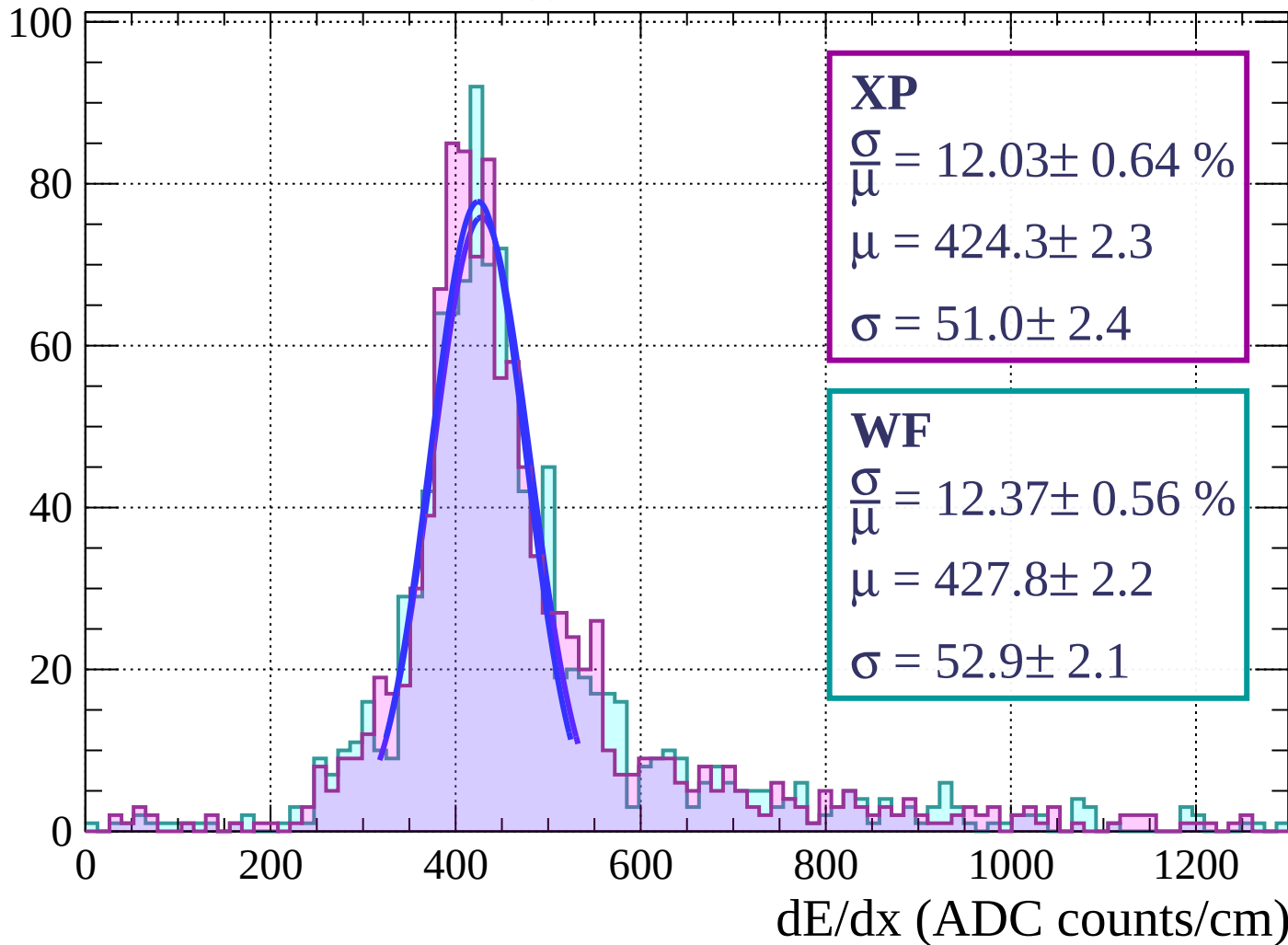
Energy loss | $45 < \theta < 48$

Count



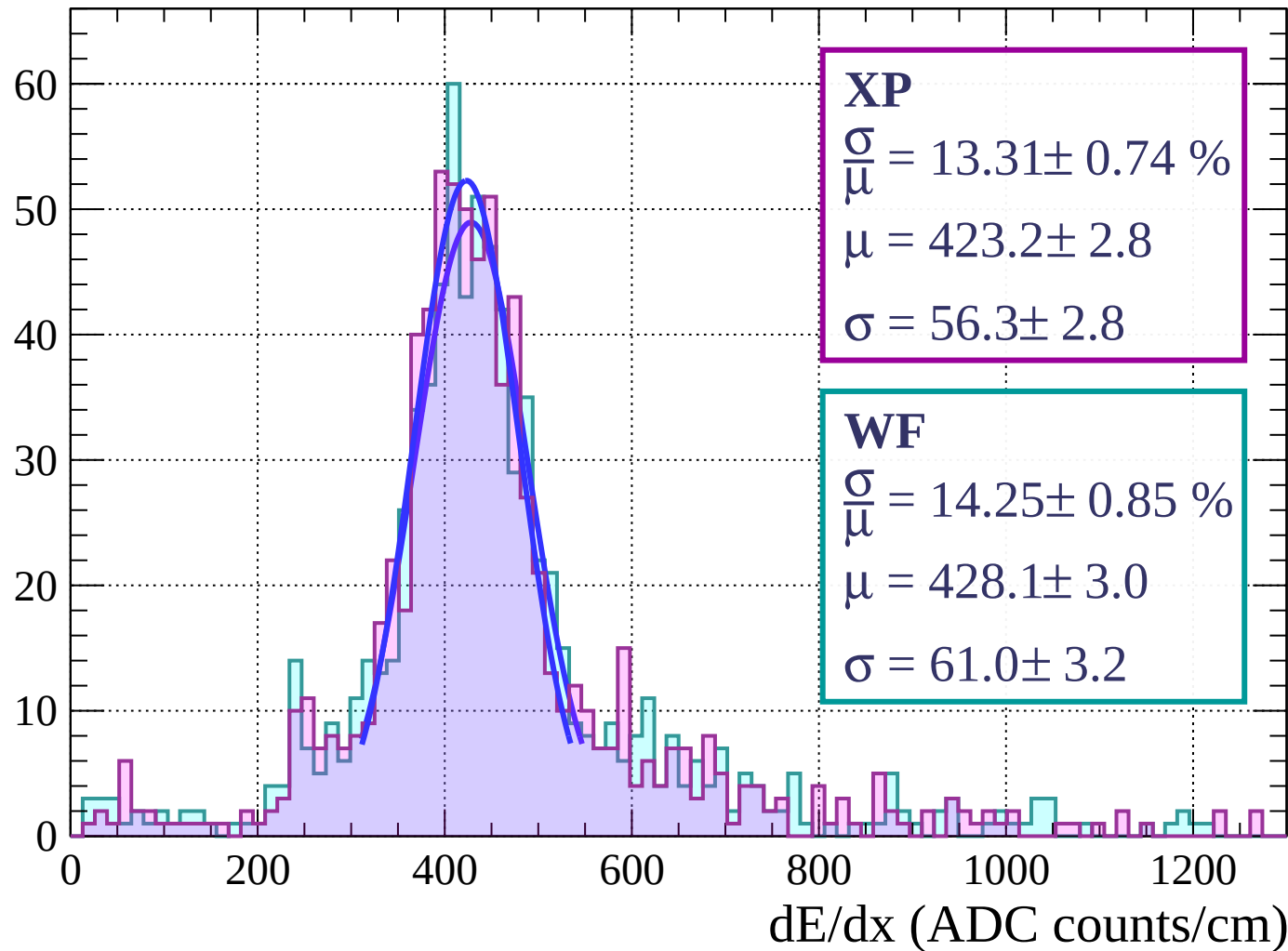
Energy loss | $48 < \theta < 51$

Count



Energy loss | $51 < \theta < 54$

Count



Energy loss | $54 < \theta < 57$

Count

XP

$$\frac{\sigma}{\mu} = 15.08 \pm 0.97 \%$$

$$\mu = 432.3 \pm 3.7$$

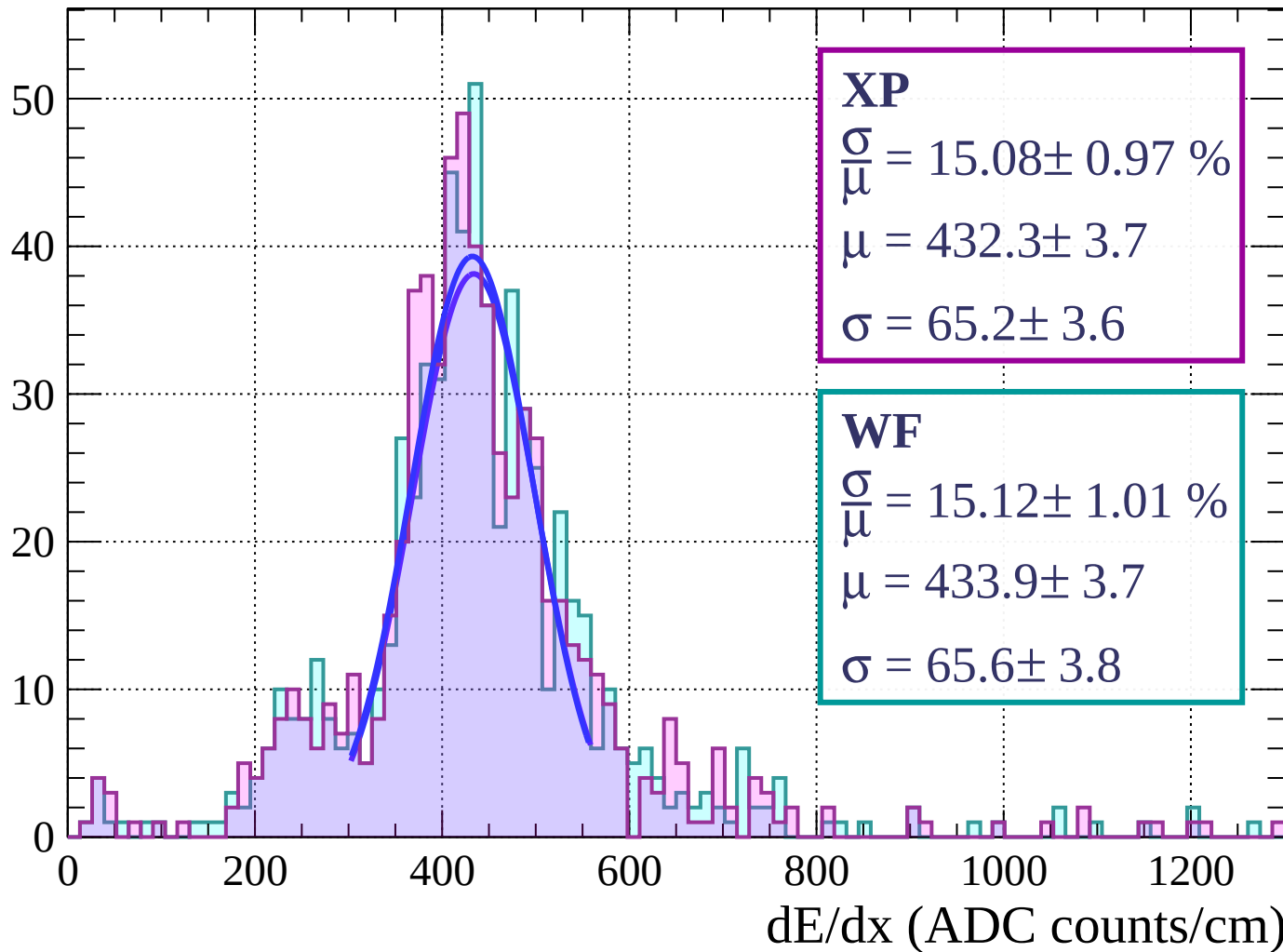
$$\sigma = 65.2 \pm 3.6$$

WF

$$\frac{\sigma}{\mu} = 15.12 \pm 1.01 \%$$

$$\mu = 433.9 \pm 3.7$$

$$\sigma = 65.6 \pm 3.8$$



Energy loss | $57 < \theta < 60$

Count

XP

$$\frac{\sigma}{\mu} = 14.11 \pm 1.09 \%$$

$$\mu = 418.2 \pm 4.0$$

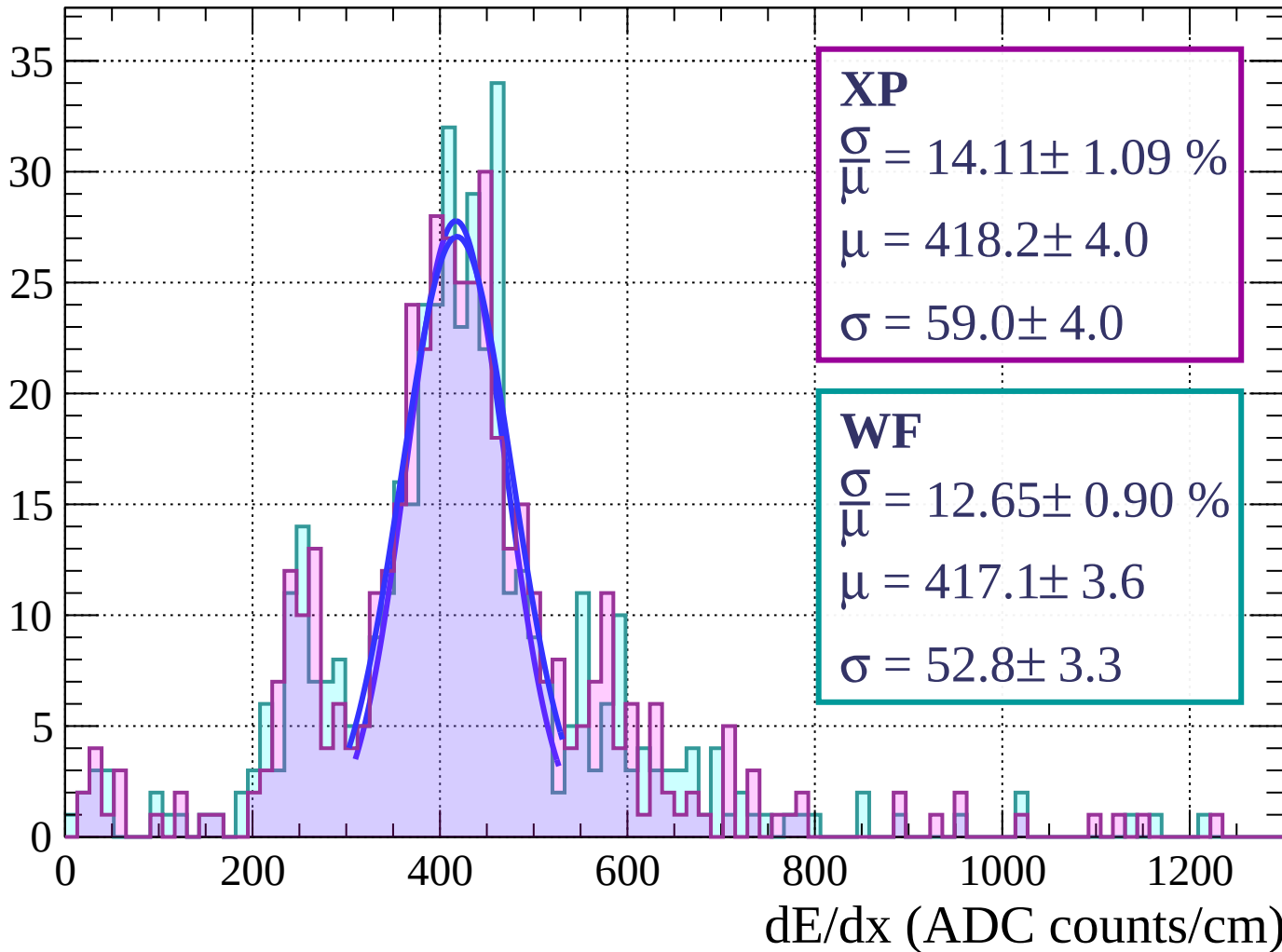
$$\sigma = 59.0 \pm 4.0$$

WF

$$\frac{\sigma}{\mu} = 12.65 \pm 0.90 \%$$

$$\mu = 417.1 \pm 3.6$$

$$\sigma = 52.8 \pm 3.3$$



Energy loss | $60 < \theta < 63$

Count

XP

$$\frac{\sigma}{\mu} = 16.21 \pm 1.83 \%$$

$$\mu = 419.4 \pm 5.3$$

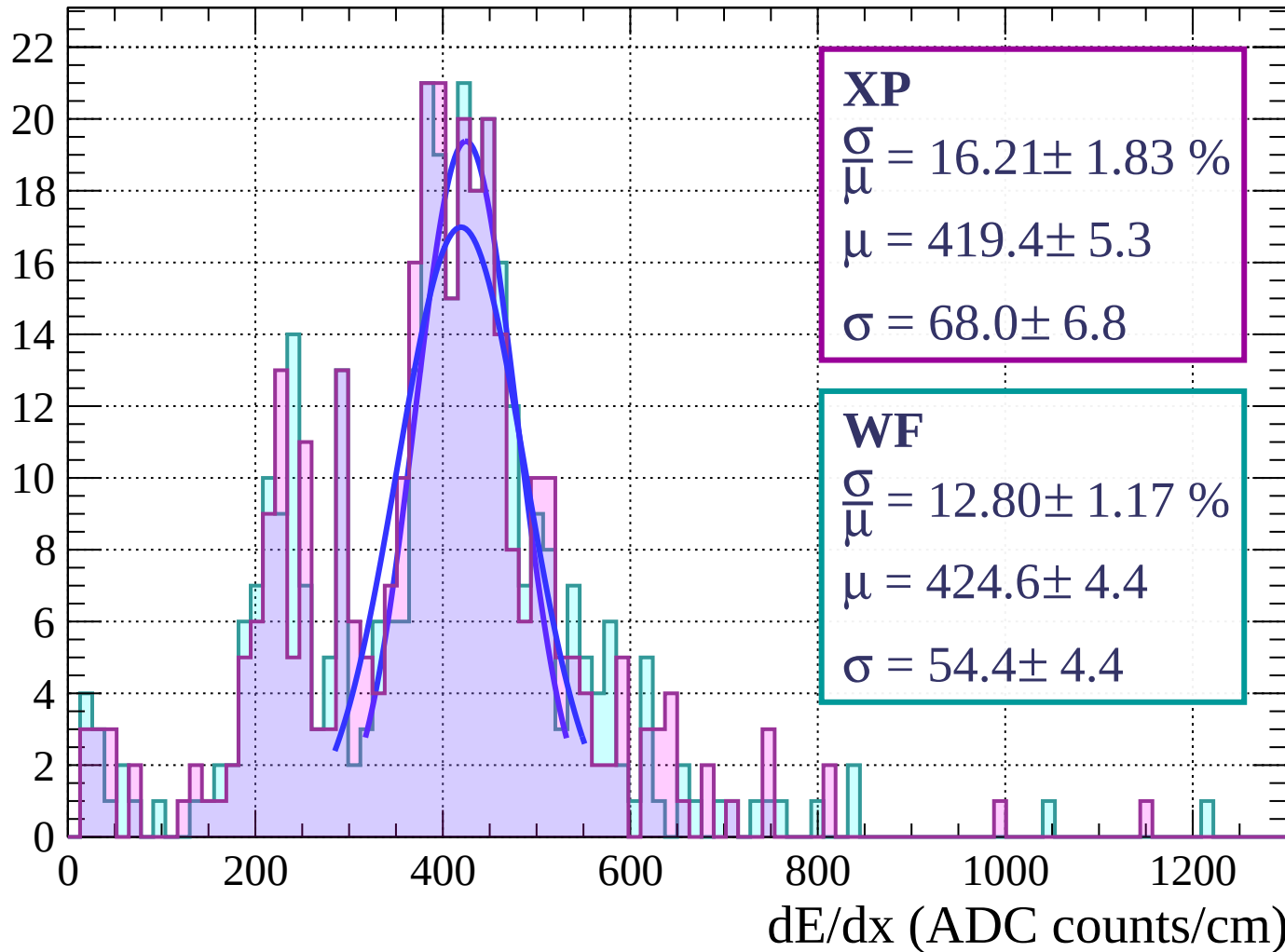
$$\sigma = 68.0 \pm 6.8$$

WF

$$\frac{\sigma}{\mu} = 12.80 \pm 1.17 \%$$

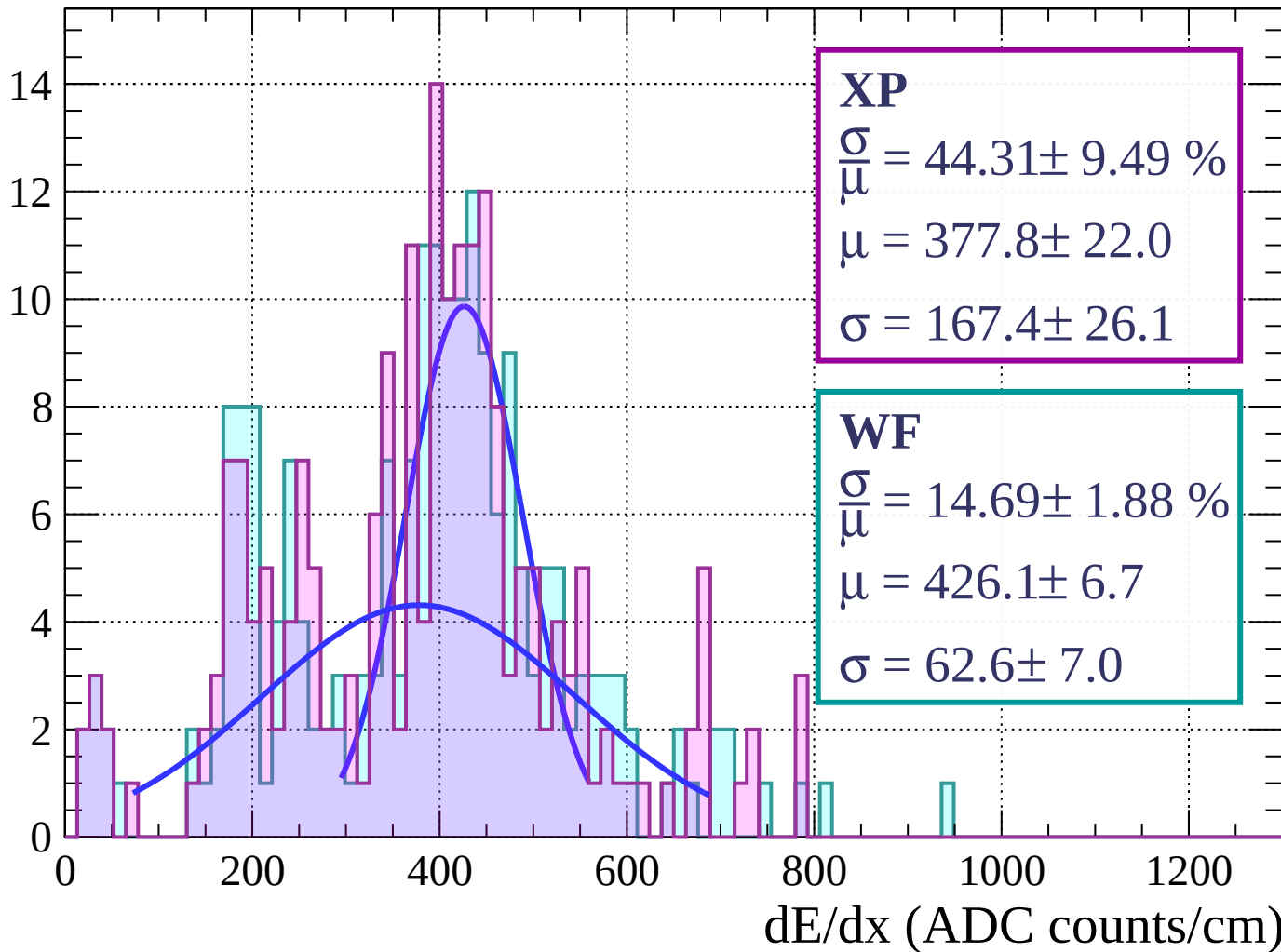
$$\mu = 424.6 \pm 4.4$$

$$\sigma = 54.4 \pm 4.4$$



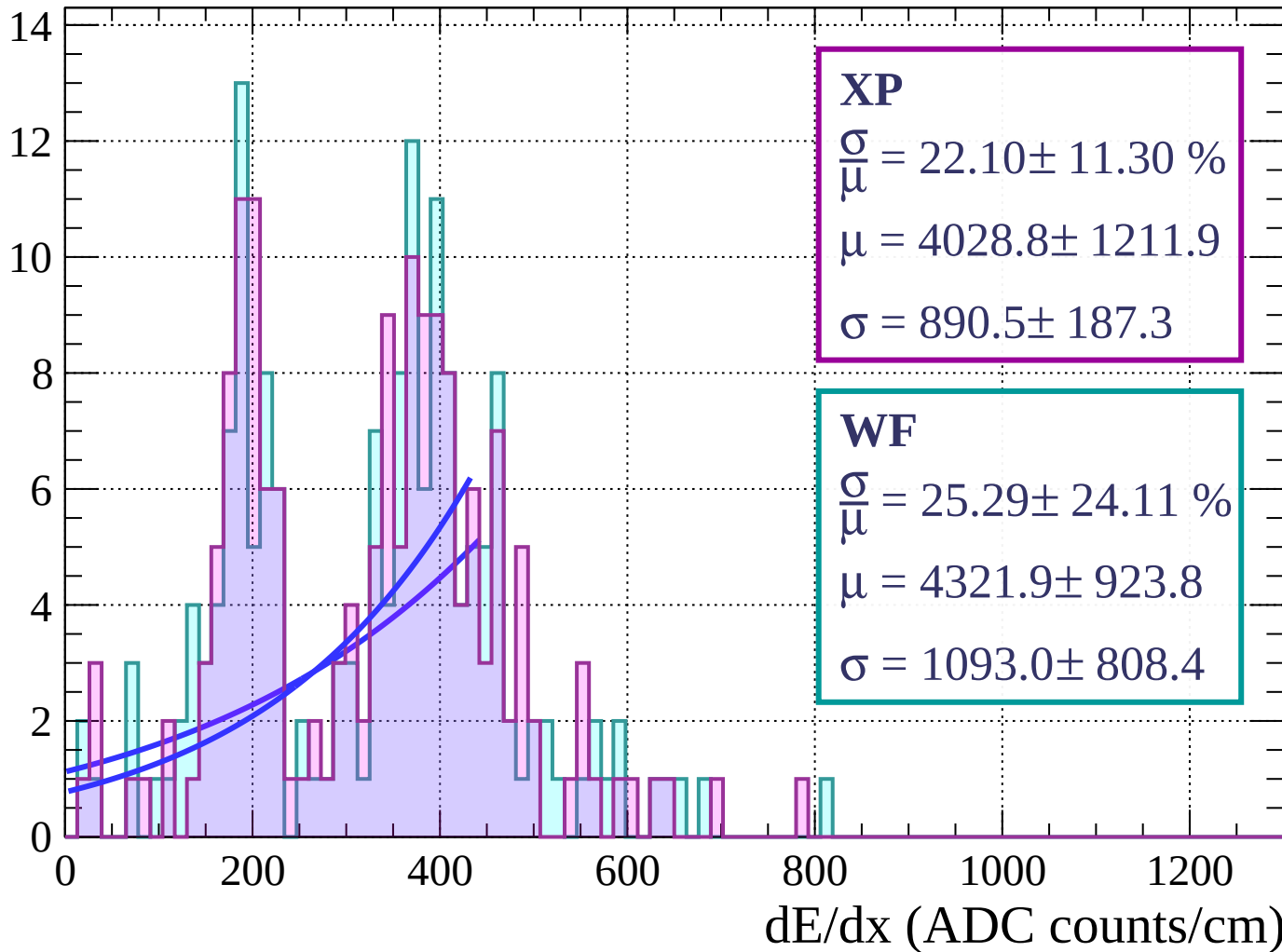
Energy loss | $63 < \theta < 66$

Count



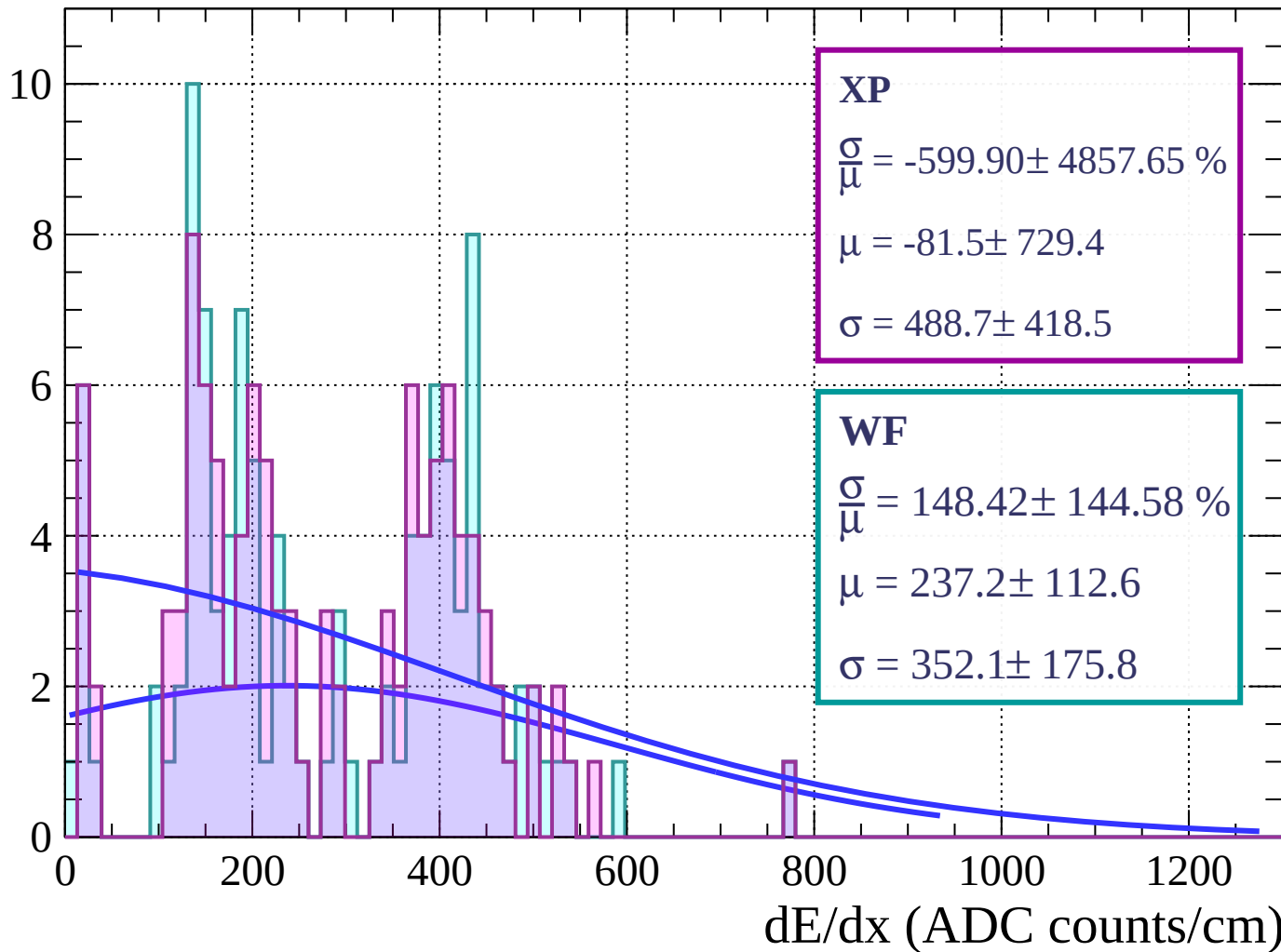
Energy loss | $66 < \theta < 69$

Count



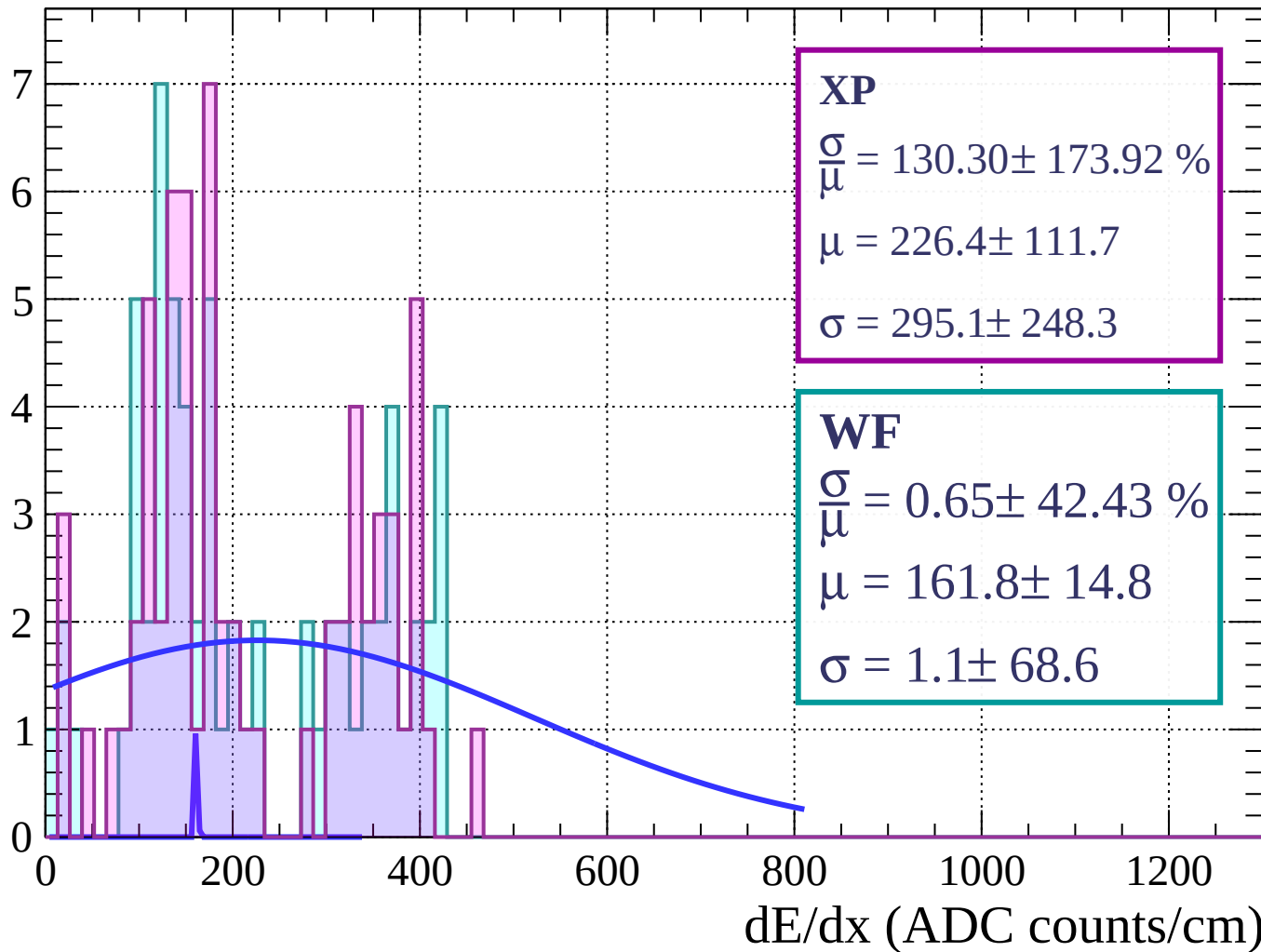
Energy loss | $69 < \theta < 72$

Count



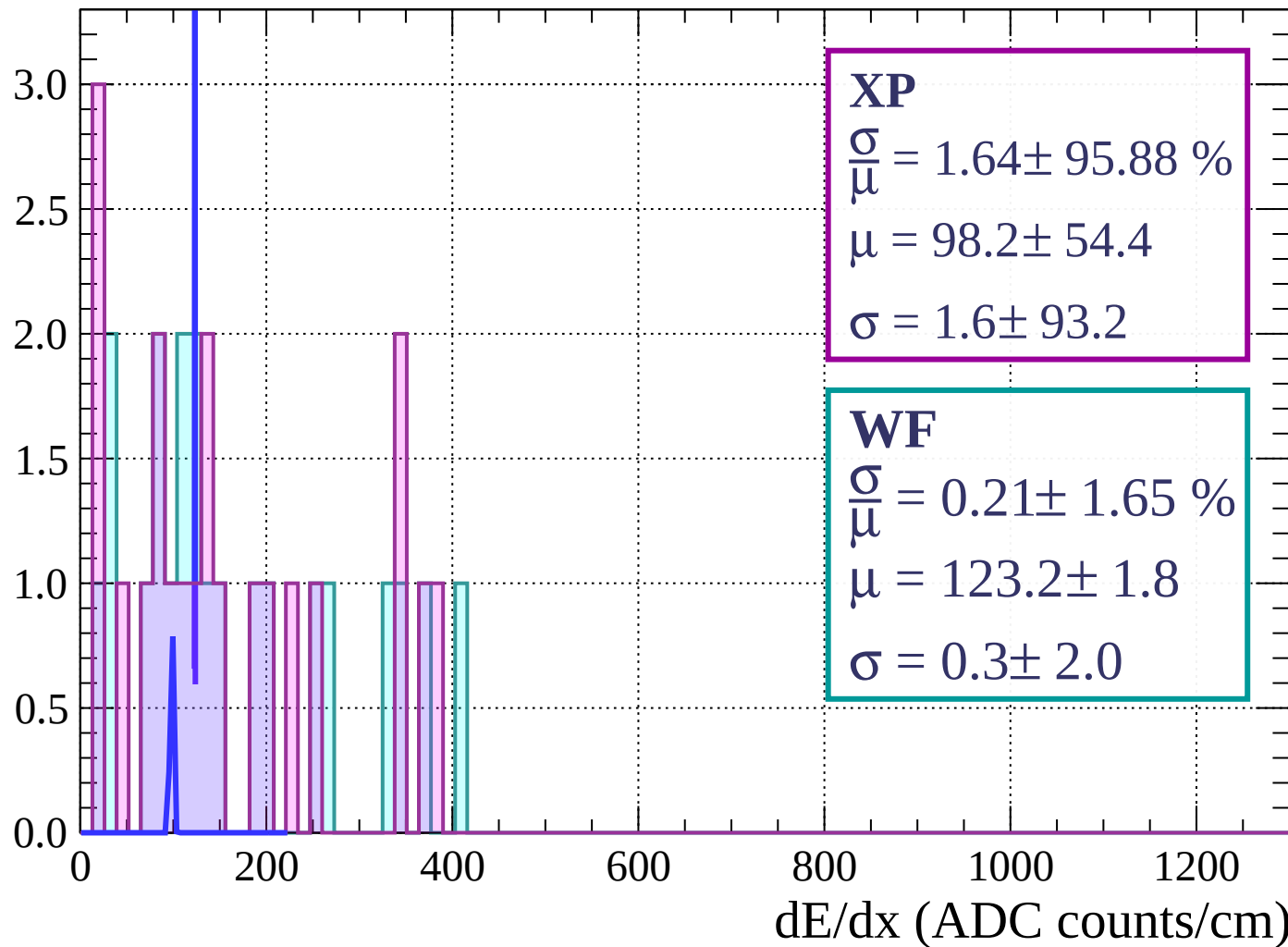
Energy loss | $72 < \theta < 75$

Count



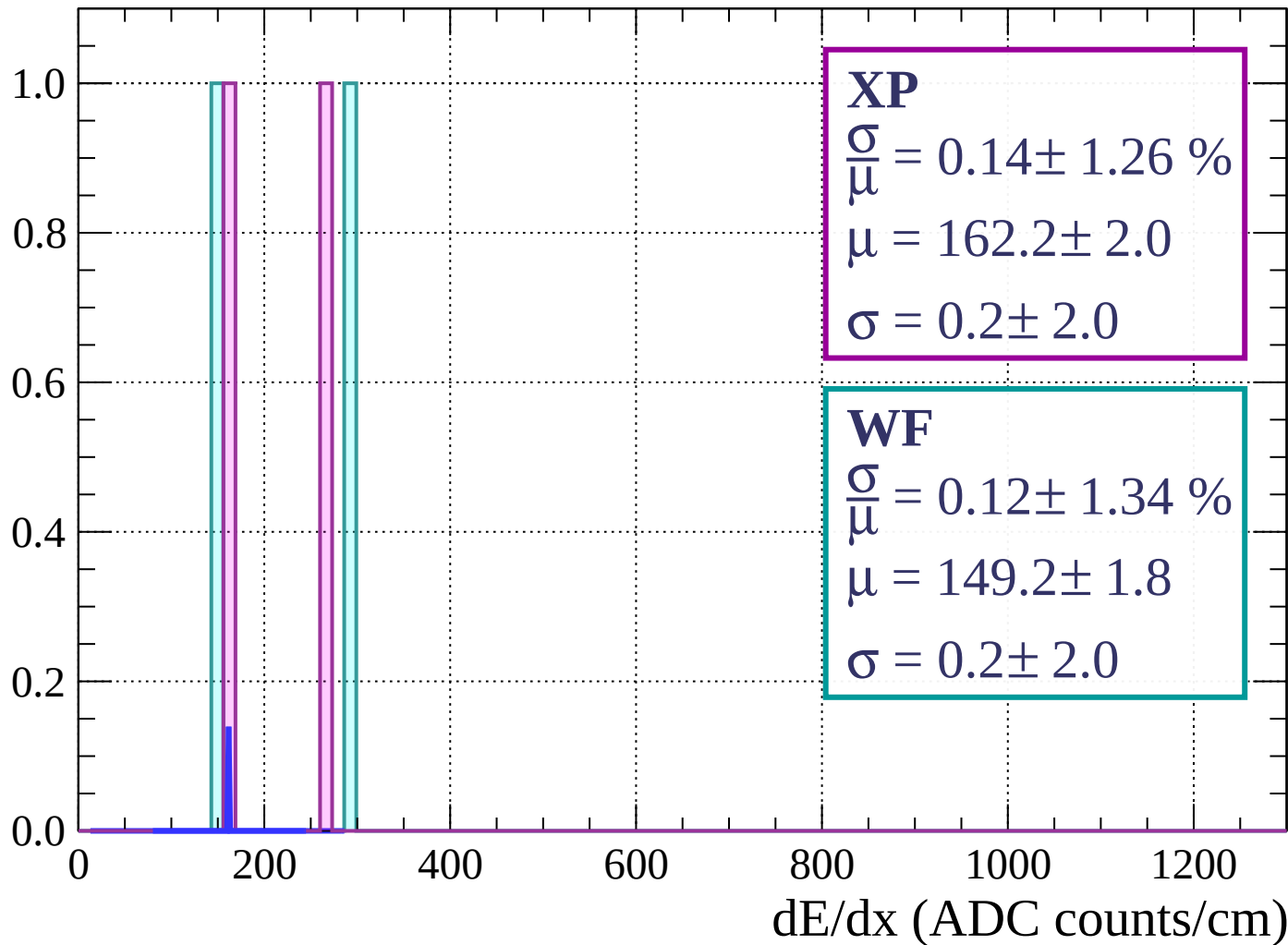
Energy loss | $75 < \theta < 78$

Count



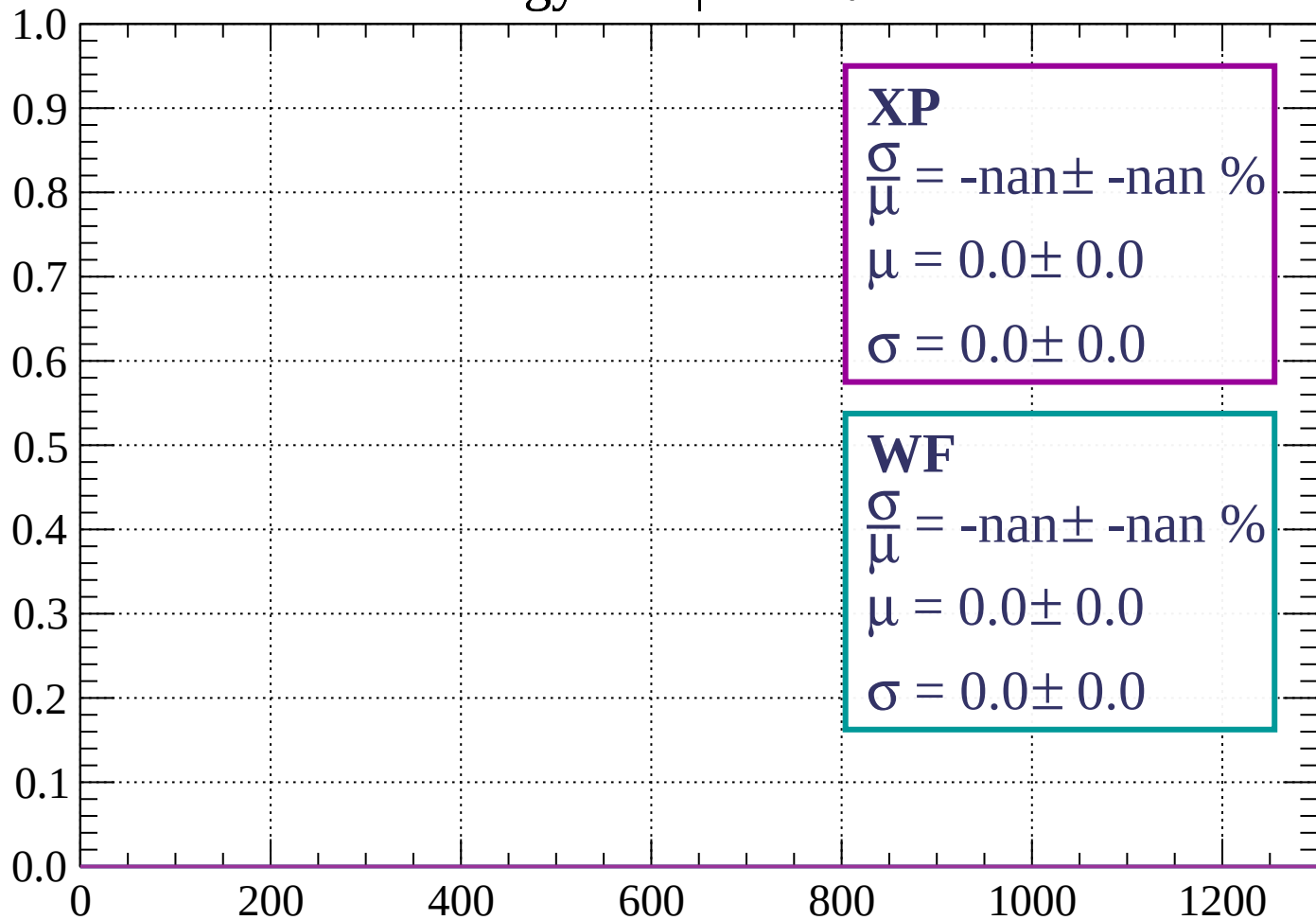
Energy loss | $78 < \theta < 81$

Count



Energy loss | $81 < \theta < 84$

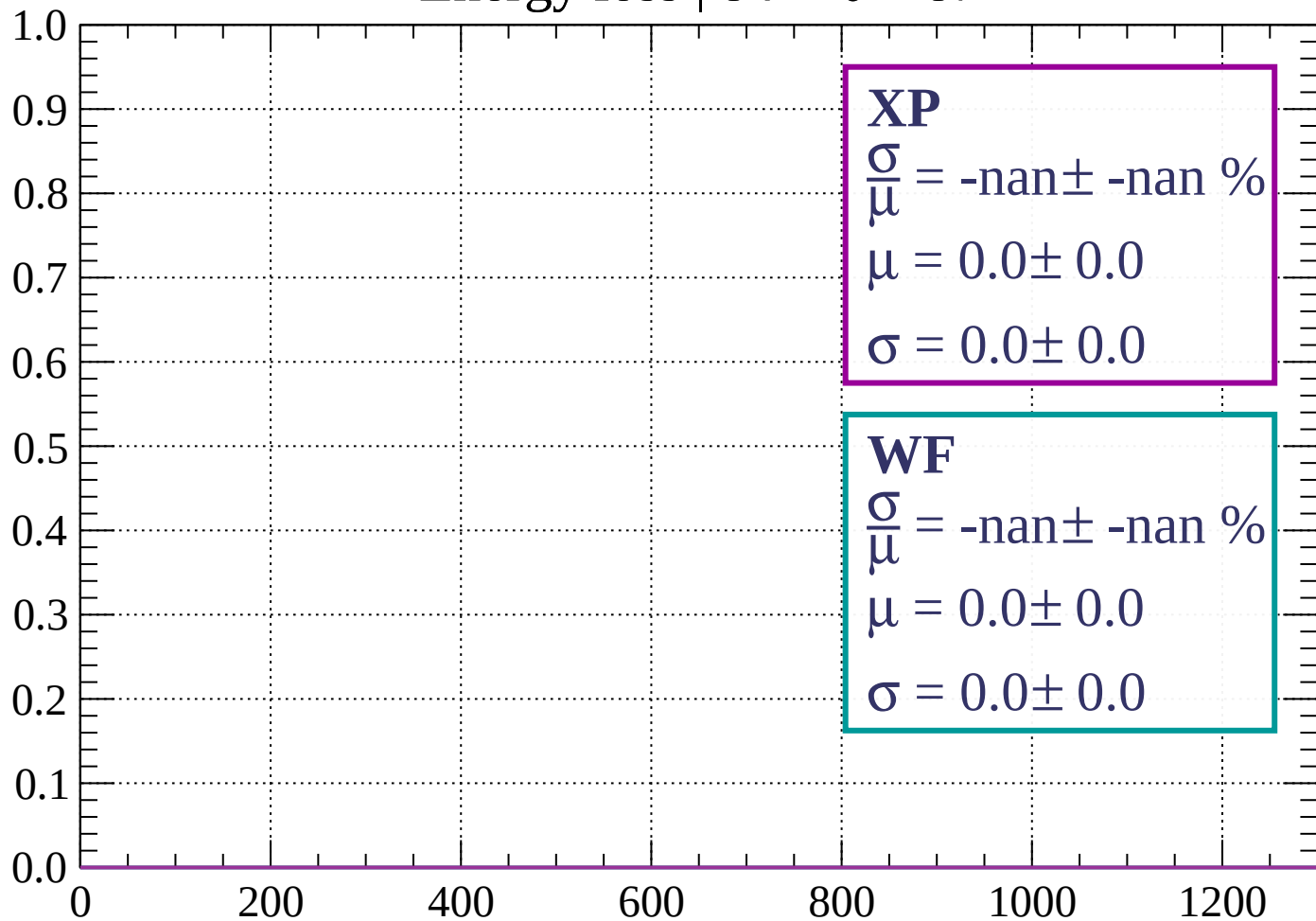
Count



dE/dx (ADC counts/cm)

Energy loss | $84 < \theta < 87$

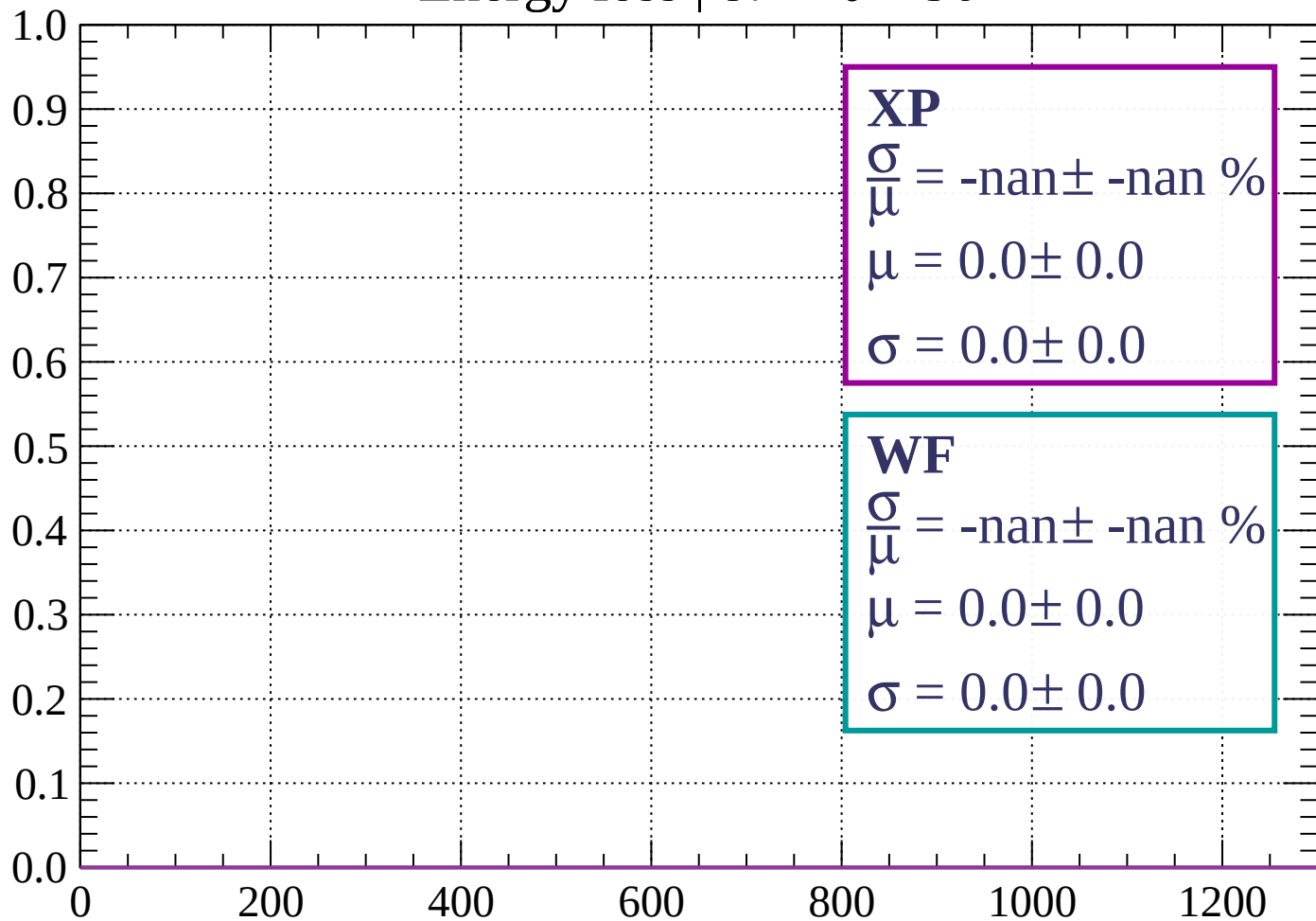
Count



dE/dx (ADC counts/cm)

Energy loss | $87 < \theta < 90$

Count



dE/dx (ADC counts/cm)

